

12. The strength of the magnetic field inside a current carrying long straight solenoid is :

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- (A) maximum at its centre.
- (B) minimum at its centre.
- (C) maximum at the ends and minimum at its centre.
- (D) uniform throughout its length.

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(D)/ Uniform throughout its length

17. *Assertion (A) :* In our homes we receive supply of electric power through a main supply. One of the wires in this supply, usually with red insulation, is called *live wire* and another wire with green insulation is called *neutral wire*.

Reason (R) : In our country, the potential difference between the live wire and the neutral wire is 220 volts.

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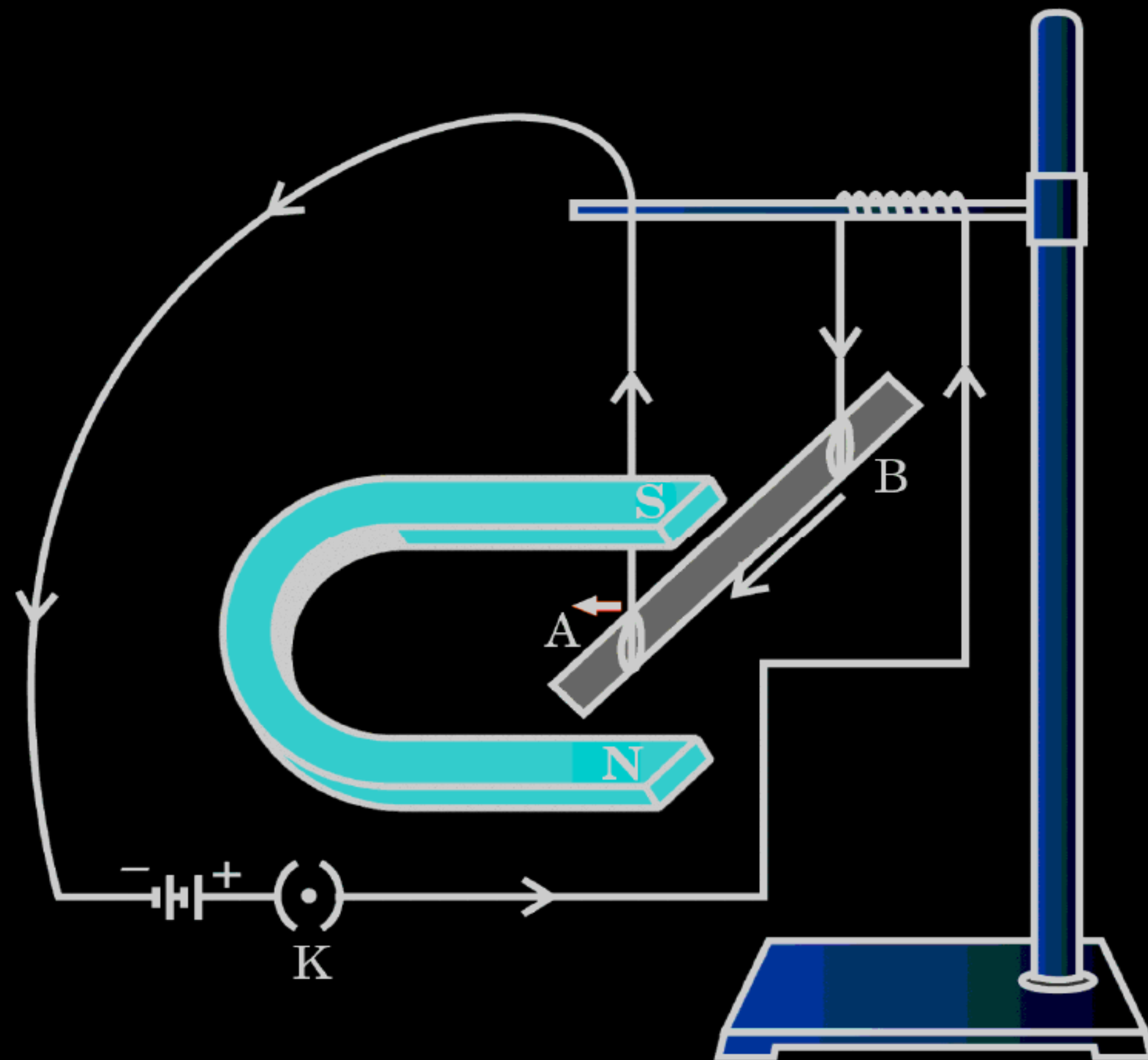
Reason (R) : In our country, the potential difference between the live wire and the neutral wire is 220 volts.

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(D)/ Assertion (A) is false, but Reason (R) is true.

33. As shown in the figure a small aluminum rod AB is suspended horizontally between the poles of a strong horseshoe magnet. This rod is also connected with a battery and a key. Study the arrangement shown.

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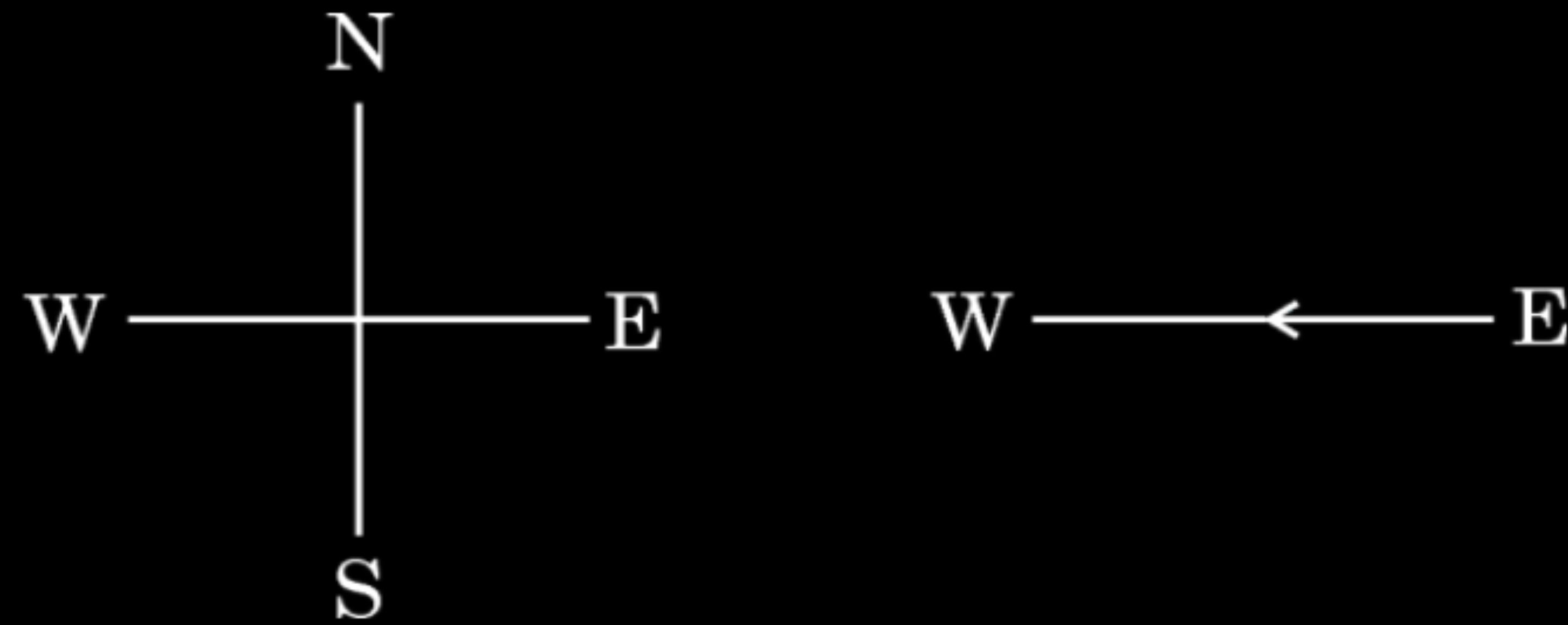


- a) State Fleming's left-hand rule.
- b) Apply Fleming's left-hand rule to determine :
 - (i) What is observed when a current is passed from B to A in the rod ?
 - (ii) What is observed when a current is passed from A to B in the rod ?
 - (iii) What is observed when the rod AB is aligned parallel to the magnetic field and current is passed through it from B to A ? Justify your answer in this case.

1+2

(a) According to this rule, stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor.	1
(b) (i) Rod is displaced towards left.	$\frac{1}{2}$
(ii) Rod is displaced towards right.	$\frac{1}{2}$
(iii) • No displacement is observed in rod. • This is because current and magnetic field are parallel, so there will be no force experienced by the conductor.	$\frac{1}{2}$ $\frac{1}{2}$

- 31.** (a) State the Right-hand thumb rule.
- (b) A steady current flows through a horizontal power line from east to west direction as shown in the figure. Apply right-hand thumb rule to determine the direction of magnetic field (i) at a point directly below it, and (ii) at a point directly above it. 1+2



(a) Imagine that you are holding a current carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field.	1
(b) In both (i) and (ii), magnetic field turns clockwise in plane perpendicular to the wire when viewed from the east end & anti-clockwise when viewed from the west end.	2

26. In a straight conductor, the current is flowing vertically downwards. State the pattern of the magnetic field lines around the conductor. Name the rule which helps us to determine the direction of the magnetic field lines in this case. 2

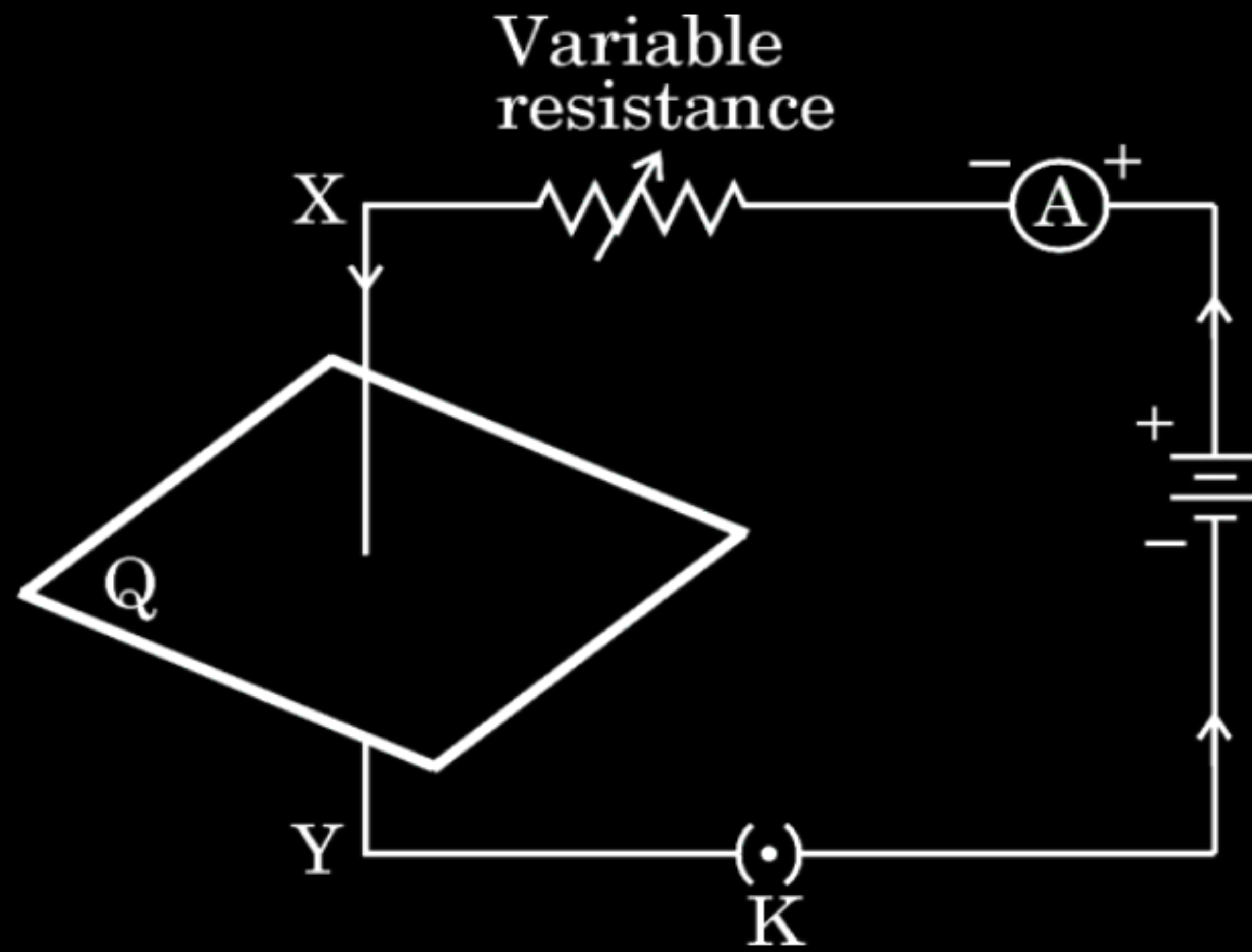
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26. In a straight conductor, the current is flowing vertically downwards. State the pattern of the magnetic field lines around the conductor. Name the rule which helps us to determine the direction of the magnetic field lines in this case. 2

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• The magnetic field lines are in the form concentric circles centred on the wire.	1
• Right Hand Thumb Rule.	1

34. (a) The given figure shows the current passing through the straight conductor XY.

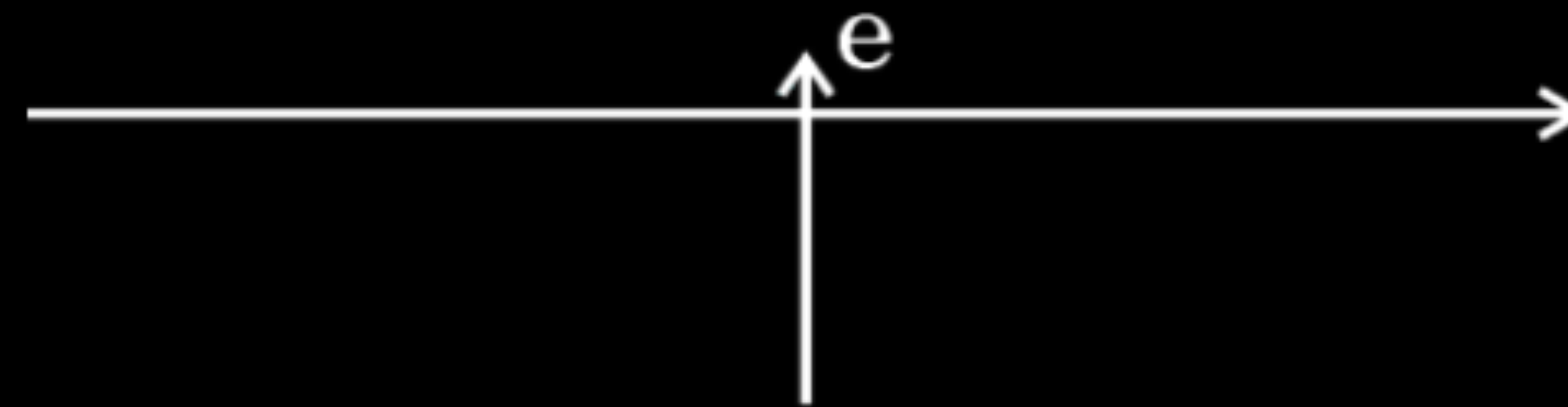


- Copy the diagram and draw the magnetic field lines when current flows from conductor X to Y.
- Name and state the rule used in determining the direction of the magnetic field lines in the situation given above.
- State Fleming's left hand rule. Using this rule, determine the direction of force applied on an electron entering a uniform magnetic field as shown in the figure.

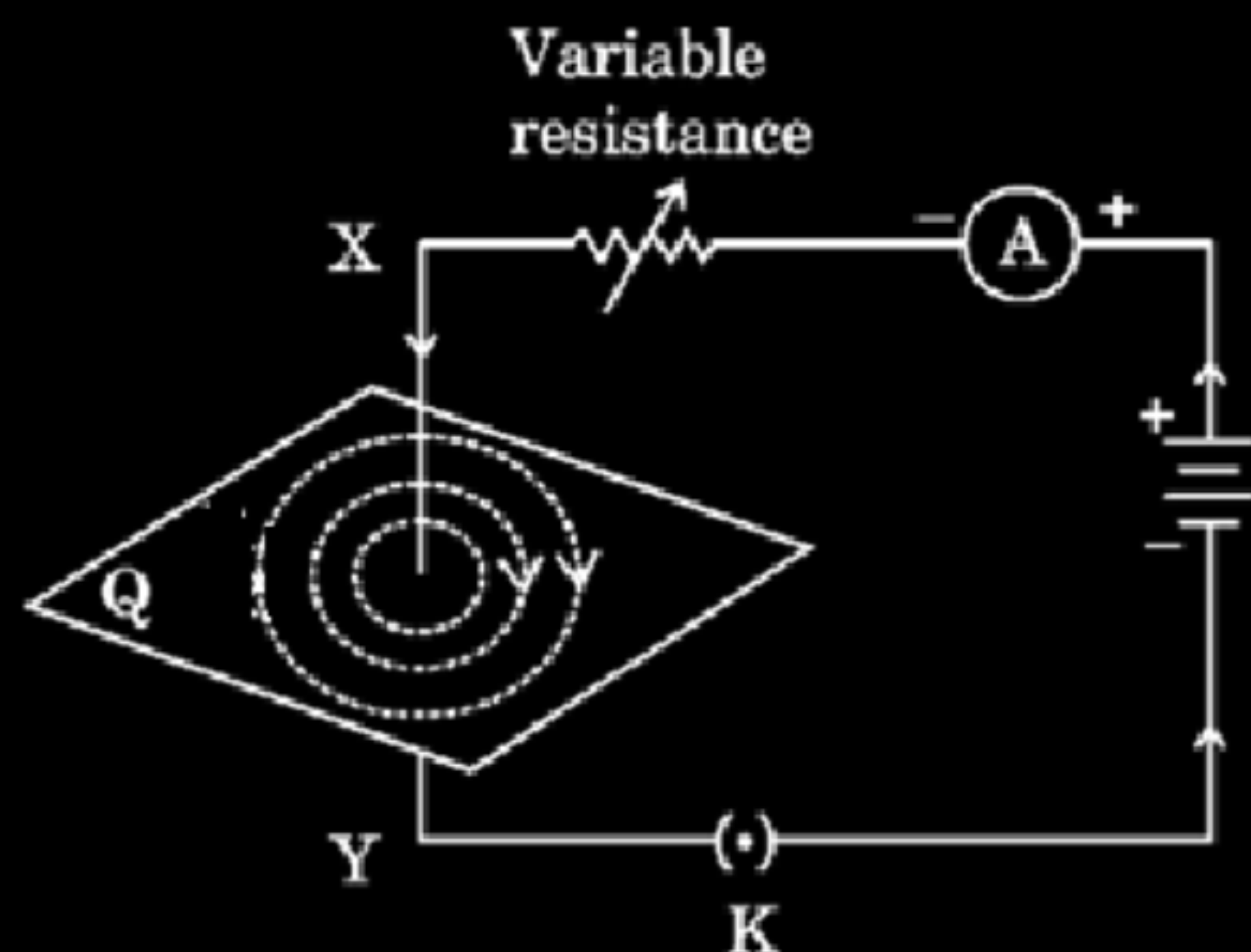
5



→ Magnetic field



(a) (i)



1½

(ii) Right hand thumb rule

Statement of the rule - Imagine holding a current carrying straight conductor in the right hand such that the thumb points towards the direction of current, then the fingers will wrap around the conductor in the direction of the field lines of the magnetic field.

½

1

(iii)

- According to Fleming's left-hand rule, stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor.
- Out of the plane/ upwards

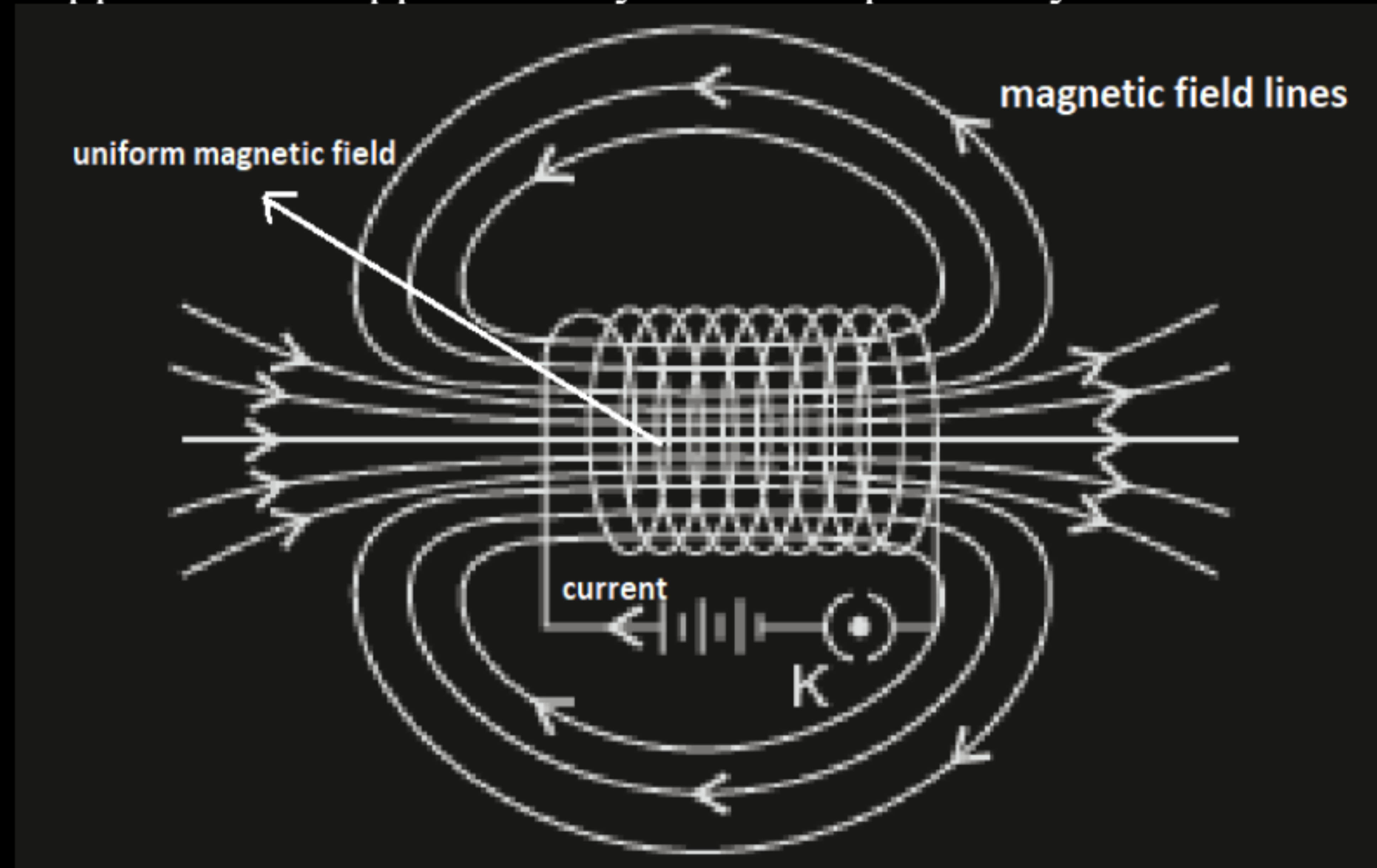
1

1

- (b) (i) Define the term solenoid. Draw the pattern of the magnetic field lines in and around a current carrying straight solenoid. Mark on the pattern the (i) direction of current, (ii) direction of field lines near the ends of the solenoid, and (iii) region where the magnetic field is uniform.
- (ii) How would you make an electromagnet using a current carrying solenoid ?

(b) (i) Solenoid is a coil of many turns of insulated copper wire wrapped closely in the shape of a cylinder.

1



Diagram

1½

Marking (i), (ii) and (iii)

½ × 3

(ii) By inserting a piece of magnetic material like soft iron inside the current carrying solenoid.

1

- 19.** *Assertion (A) :* No two magnetic field lines are found to cross each other.
 Reason (R) : The compass needle cannot point towards two directions
 at the point of intersection of two magnetic field lines.

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A / Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

39. In our homes, we receive the supply of electric power through a main supply also called mains, either supported through overhead electric poles or by underground cables. In our country the potential difference between the two wires (live wire and neutral wire) of this supply is 220 V.

(a) Write the colours of the insulation covers of the line wires through which supply comes to our homes. 1

(b) What should be the current rating of the electric circuit (220 V) so that an electric iron of 1 kW power rating can be operated ? 1

(c) (i) What is the function of the earth wire ? State the advantage of the earth wire in domestic electric appliances such as electric iron. 2

OR

(c) (ii) List two precautions to be taken to avoid electrical accidents. State how these precautions prevent possible damage to the circuit/appliance. 2

(a)	Live wire- Red Neutral wire- Black	$\frac{1}{2}$ $\frac{1}{2}$
(b)	Power, $P = 1 \text{ kW} = 1 \times 1000 \text{ W} = 1000 \text{ W}$ Voltage, $V = 220 \text{ V}$ Current drawn $I = ?$ $P = V \times I$ $I = \frac{1000 \text{ W}}{220 \text{ V}} = 4.54 \text{ A}$ Current rating should be of 5A.	$\frac{1}{2}$ $\frac{1}{2}$
(c)	(i) - The earth wire provides a low resistance conducting path for the current which ensures that any leakage of current to flow to the metallic body of the appliances, keeps its potential to that of the earth. - The user will not get an electric shock. OR	1 1
(c)	(ii) - Fuse wire - Earth wire - A fuse in a circuit prevents damage to the circuit due to overloading. - Earth wire prevents electric shock due to leakage of current.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

26. What are magnetic field lines. List two important properties of magnetic field lines.

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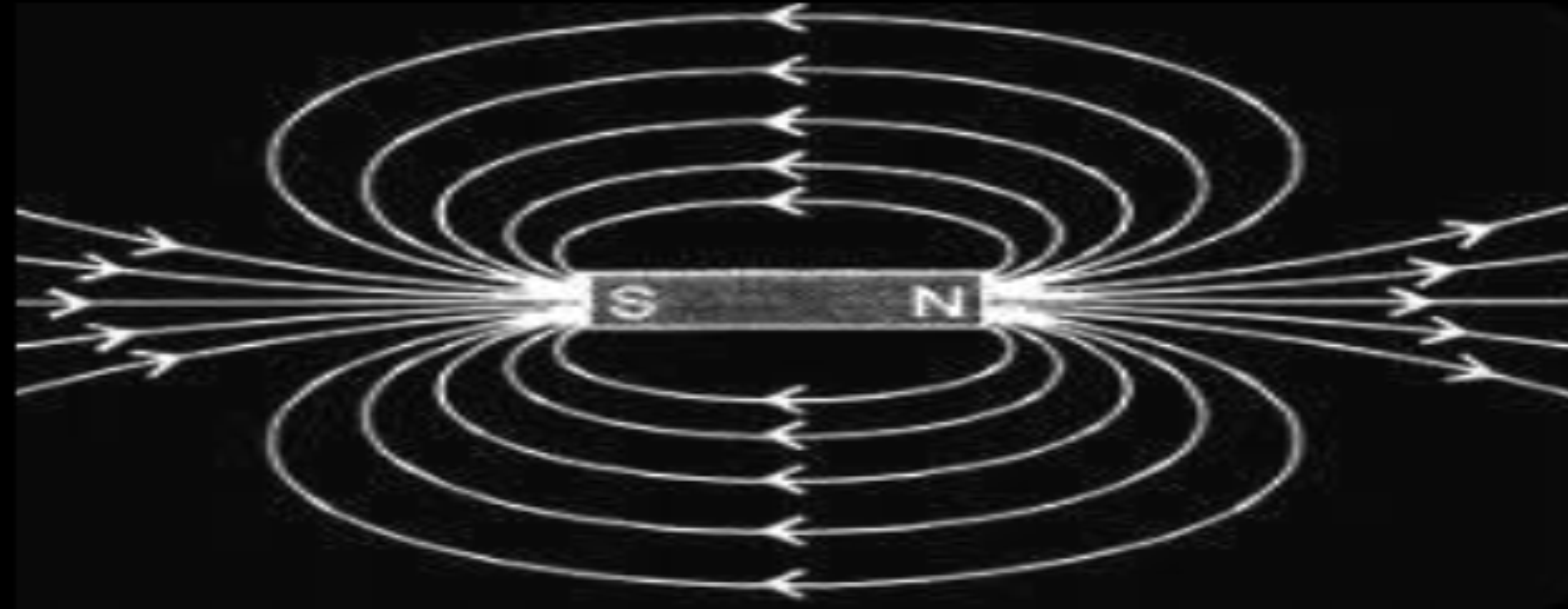
2

26. What are magnetic field lines. List two important properties of magnetic field lines.

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2

The lines representing magnetic field around a magnet. /



Properties:

No two field lines cross each other.

Field lines emerge from north pole and merge at south pole.

Field lines are closed curves.

The direction of the field lines inside the magnet is from its south pole to north pole.

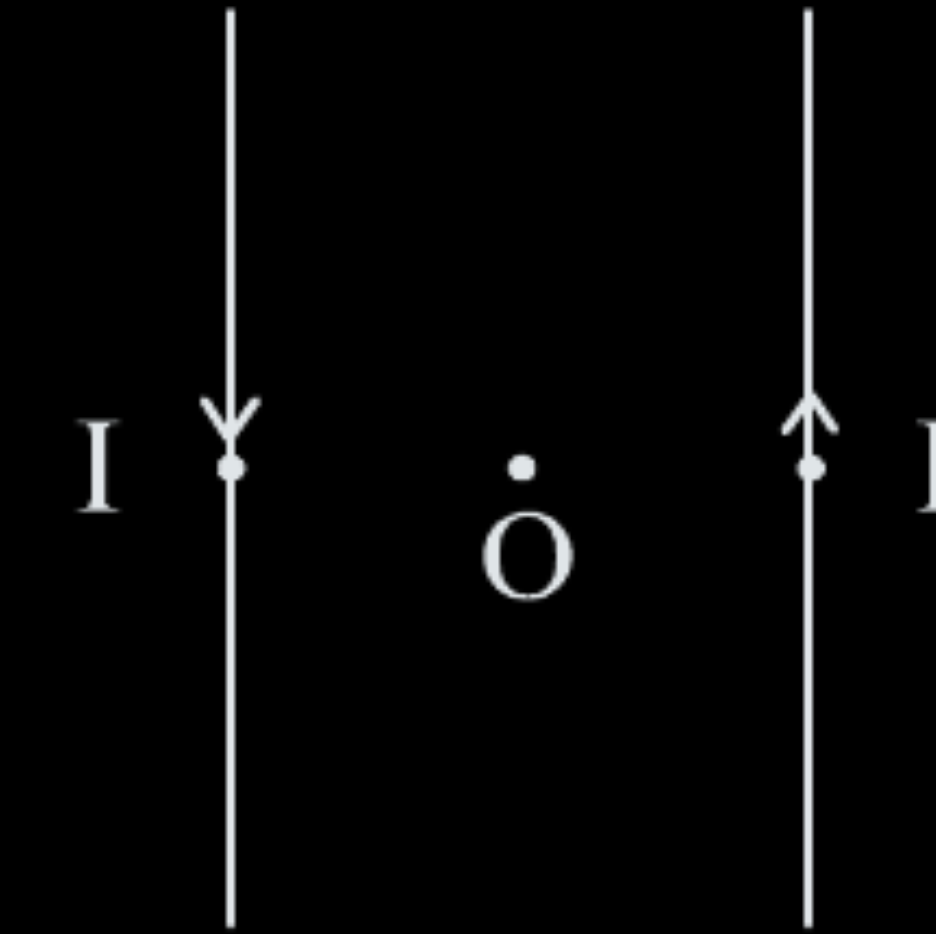
(any two properties)

1

$\frac{1}{2} + \frac{1}{2}$

36. (A) (i) Draw the pattern of the magnetic field lines for the two parallel straight conductors carrying current of same magnitude 'I' in opposite directions as shown. Show the direction of magnetic field at a point O which is equidistant from the two conductors. (Consider that the conductors are inserted normal to the plane of a rectangular cardboard.)

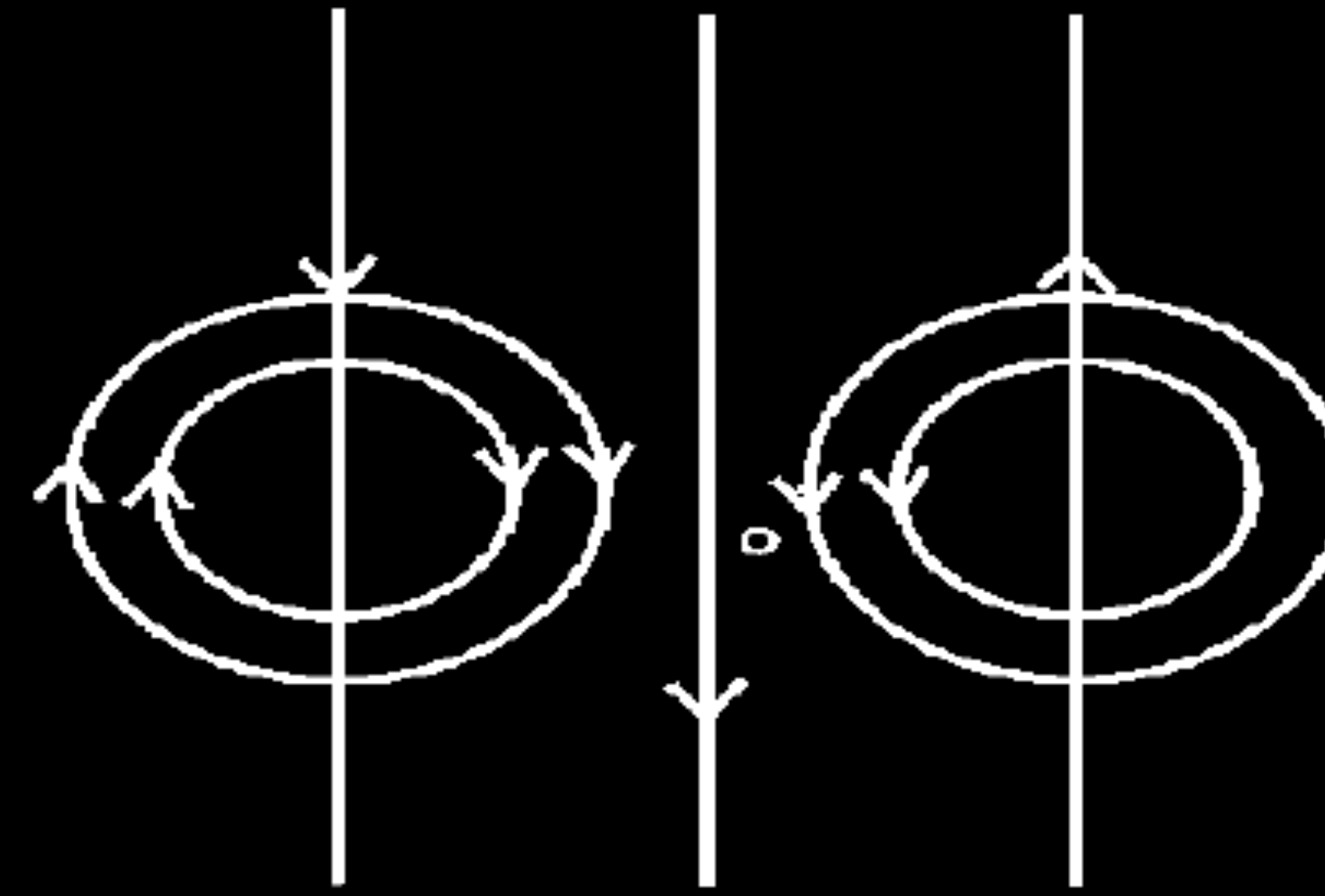
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- (ii) In our houses we receive A.C. electric power of 220 V. In electric iron or electric heater cables having three wires with insulation of three different colours – red, black and green are used to draw current from the mains.
- (a) What are these three different wires called? Name them colourwise.
 - (b) What is the potential difference between the red wire and the black wire?
 - (c) What is the role of the wire with green insulation in case of accidental leakage of electric current to the metallic body of an electrical appliance?

(A)

(i)



Correct Pattern

1

Correct direction

1

(ii) (a)

red wire : Live wire

black wire : Neutral wire

green wire : Earth wire

$\frac{1}{2} \times 3$

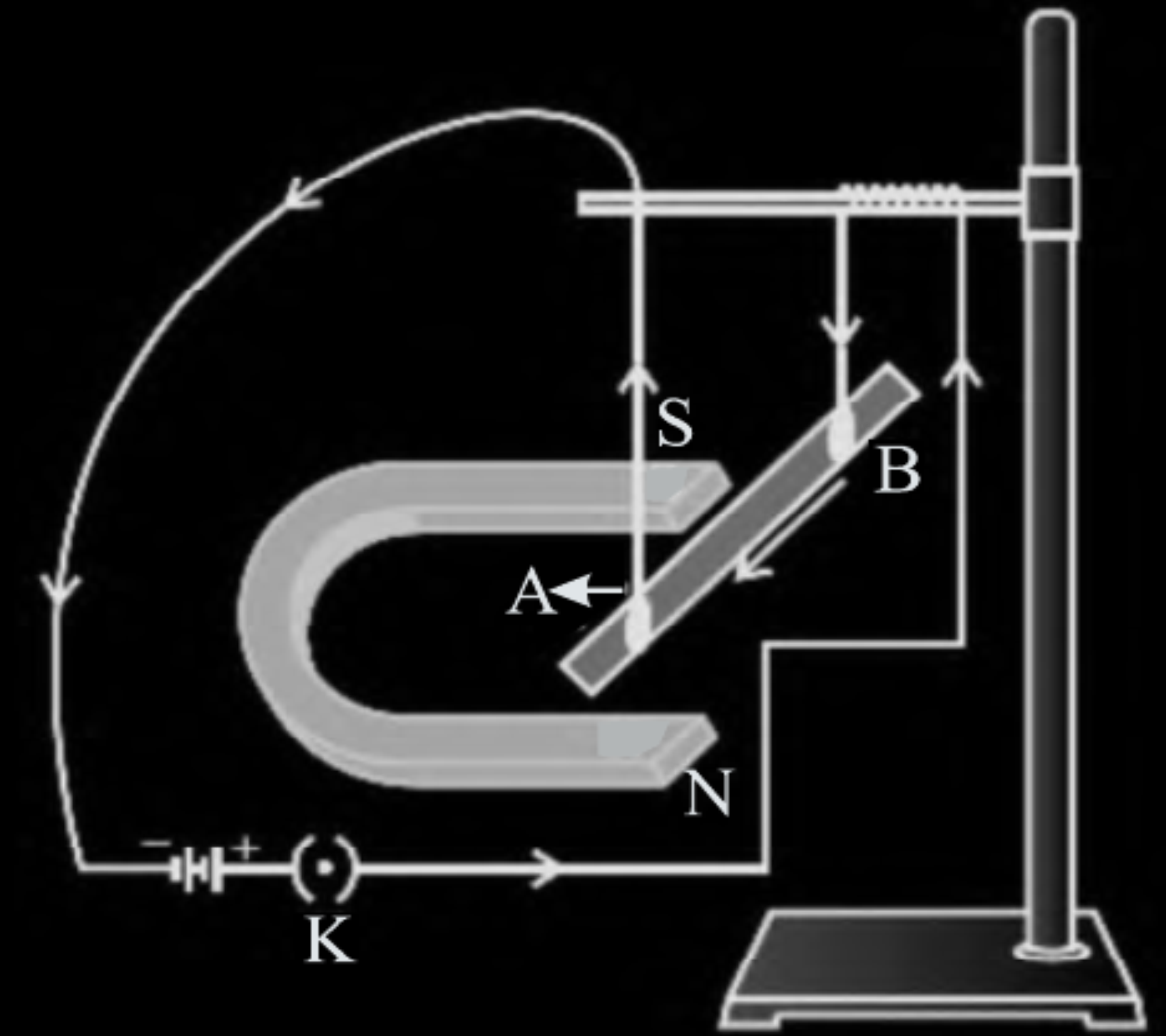
(b) 220 V

$\frac{1}{2}$

(c) This is used as a safety measure. It ensures that any leakage of the current to the metallic body of the appliance keeps its potential to that of the earth and the user may not get a severe electric shock.

1

- (B)** (i) By using the given experimental set-up. How can it be shown that :
- (a) a force is exerted on the current-carrying conductor AB when it is placed in a magnetic field.
 - (b) the direction of force can be reversed in two ways.



- (ii) When will the magnitude of the force be highest?
- (iii) State Fleming's left hand rule.

(B)

(i) (a) The conductor AB gets displaced.

1

(b)

- By reversing the direction of the current.

1+1

- By reversing the direction of the magnetic field.

(ii) When the direction of current is at right angles to the direction of the magnetic field.

1

26. State the role of an electric fuse, used in series with an electrical appliance. Why should in an electric circuit a fuse with defined rating not be replaced by one with a larger rating?

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2

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2

- | | |
|---|---|
| <ul style="list-style-type: none">• An electric fuse is a safety device used to prevent any damage to an electrical appliance due to short-circuiting and overloading of the electrical circuit. | 1 |
| <ul style="list-style-type: none">• If a fuse wire with defined rating is replaced by one with a larger rating, then the fuse wire will not melt and the electrical appliance will be damaged due to flow of unduly high current during short-circuiting and overloading. | 1 |

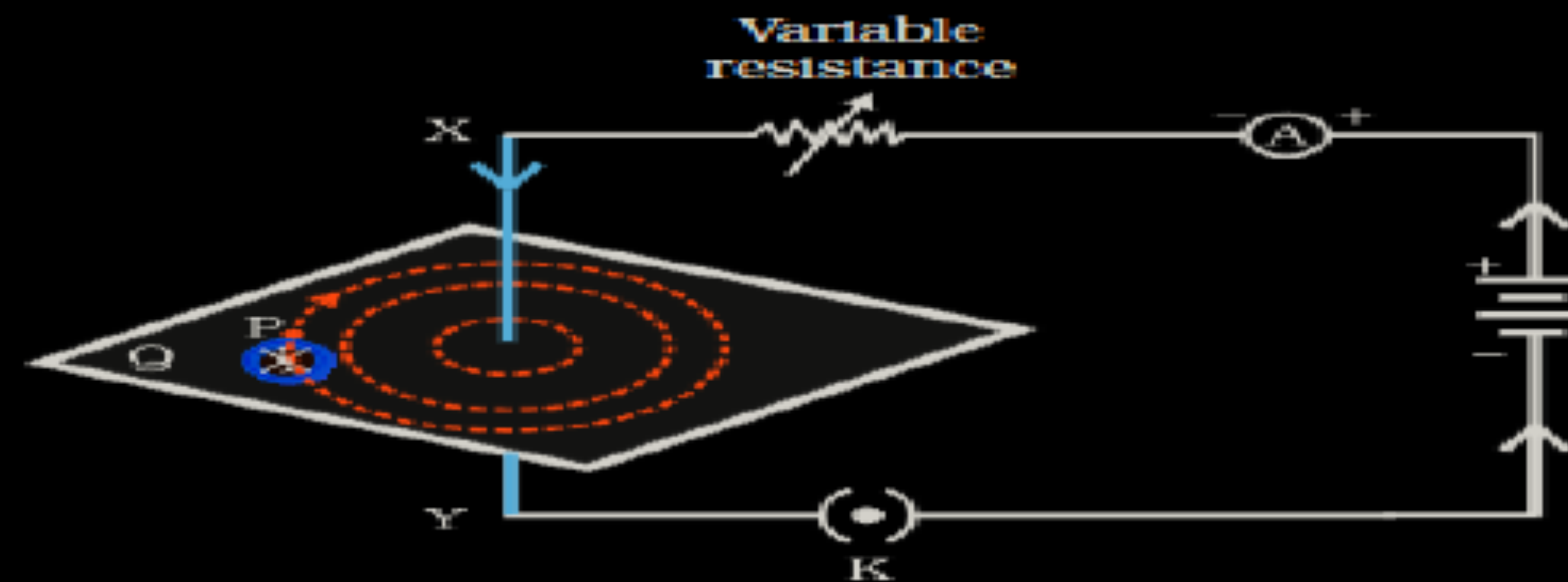
- 26.** Draw the pattern of magnetic field lines due to a current carrying straight conductor. Mark on it the direction of current in the conductor and the direction of the magnetic field developed. Name the rule that helps us to determine the direction of magnetic field lines in this case.

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2



- Right hand thumb rule.

1

1

15. The strength of the magnetic field produced inside a current carrying long straight solenoid does *not* depend upon :

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- (A) Area of cross-section of the tube of the solenoid
- (B) Number of turns per unit length in the solenoid
- (C) Direction of current flowing through the coil of the solenoid
- (D) Material of the core of the solenoid

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(C) /Direction of current flowing through the coil of the solenoid

20. *Assertion (A) :* Magnetic field lines around a bar magnet never intersect each other.

Reason (R) : Magnetic field produced by a bar magnet is a quantity that has both magnitude and direction.

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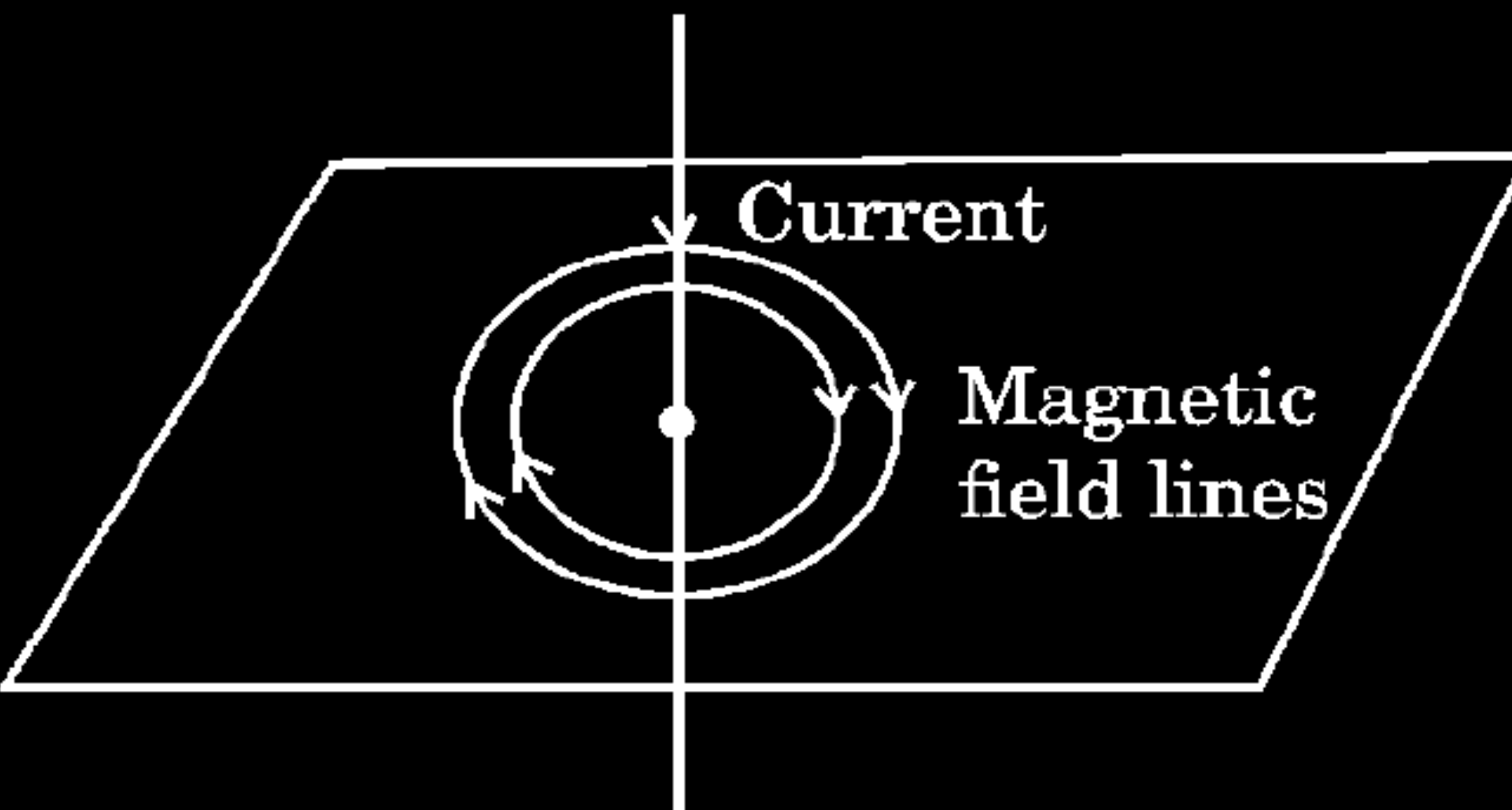
(B) /Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of Assertion (A).

- 32.** (a) Draw a labelled diagram to show the pattern of magnetic field lines on a horizontal white board due to a straight current carrying conductor passing perpendicular through its centre. If the direction of current in the conductor is vertically downwards, mark the direction of (i) current, and (ii) magnetic field lines.
- (b) State the right hand thumb rule and check whether the directions marked on the diagram are in accordance with this rule.

3

- 32.** (a) Draw a labelled diagram to show the pattern of magnetic field lines on a horizontal white board due to a straight current carrying conductor passing perpendicular through its centre. If the direction of current in the conductor is vertically downwards, mark the direction of (i) current, and (ii) magnetic field lines.
- (b) State the right hand thumb rule and check whether the directions marked on the diagram are in accordance with this rule.

3

(a) •  Diagram : 1 Labelling : 1 (b) Right hand thumb rule : Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the magnetic field lines.	2 1
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14. Which one of the following statements is ***not*** true about a bar magnet ?
- (A) It sets itself in north-south direction when suspended freely.
 - (B) It has attractive power for iron filings.
 - (C) It produces magnetic field lines.
 - (D) The direction of magnetic field lines inside a bar magnet is from its north pole to its south pole.

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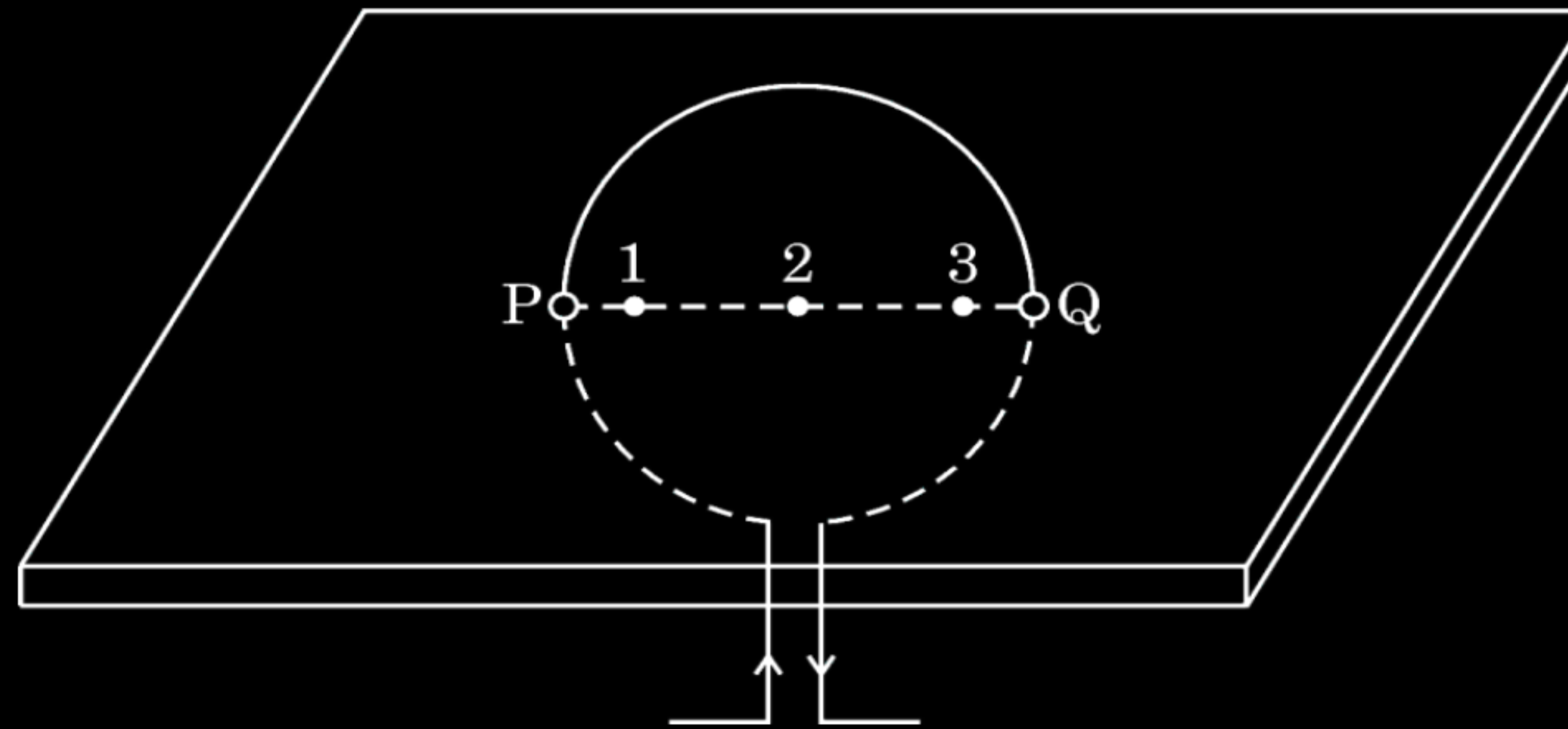
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(D) /The direction of magnetic field lines inside a bar magnet is from its north pole to its south pole.

32. Consider a rectangular cardboard having two holes P and Q through which a current carrying circular loop has been inserted as shown in the diagram.

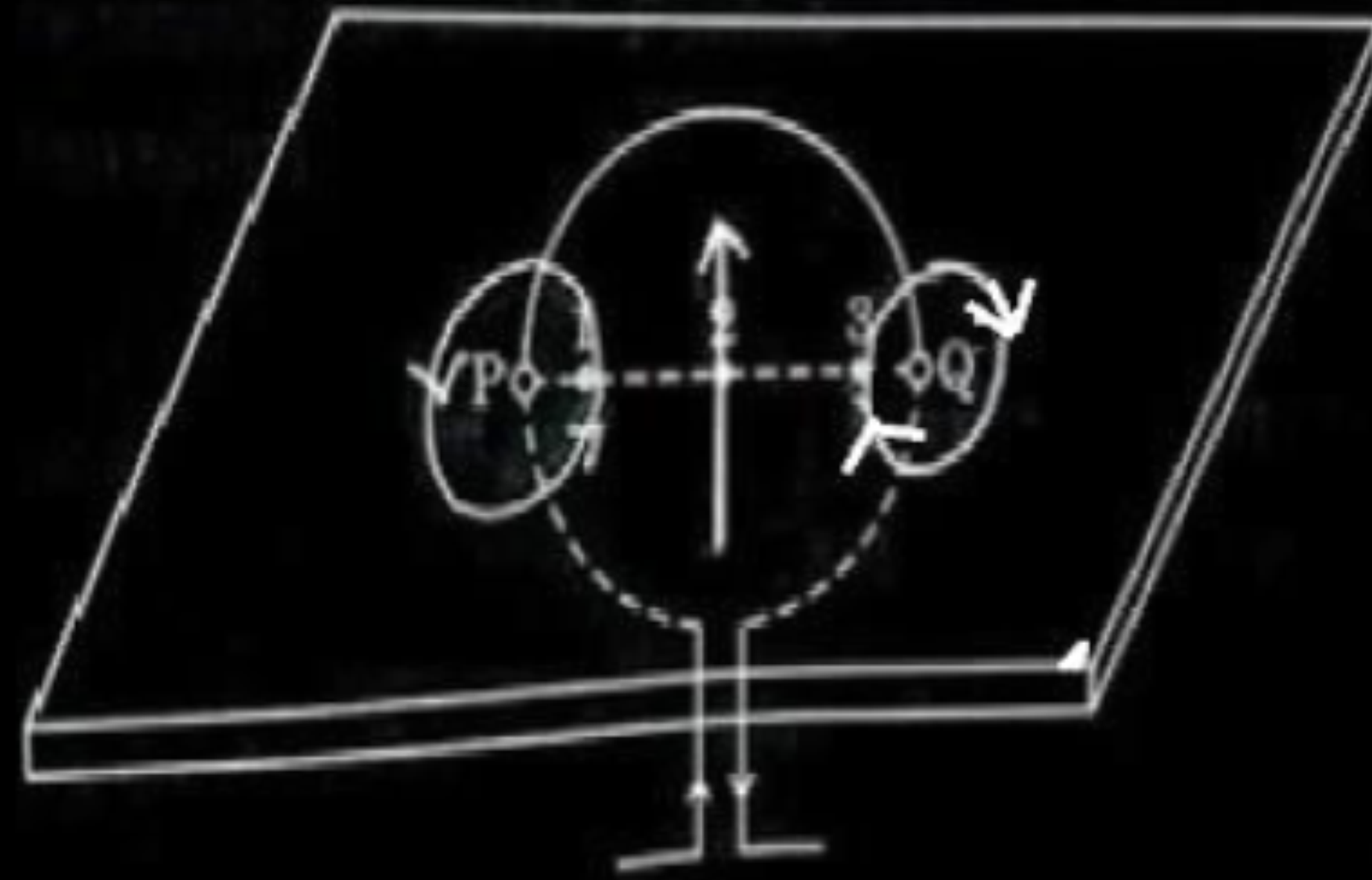
- (a) Make this diagram on your answer sheet and draw three magnetic field lines, one each passing through the points 1 (near P), 2 (at the centre of the loop) and 3 (near Q).

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- (b) List two factors on which the intensity of the magnetic field produced at the centre of the loop depends.
- (c) Name the rule you will apply to determine the direction of magnetic field produced due to a current carrying straight conductor.

(a)



Marking magnetic lines for points 1,2 and 3

(b)

- Strength of the current in the loop
- No. of turns of loop

(c) Right hand thumb rule

$\frac{1}{2} \times 3$

$\frac{1}{2} \times 2$

$\frac{1}{2}$

13. In domestic electric circuits, the colour of insulation covers of wires in the cables of electric iron/electric toaster is generally :

- (A) red for live wire, green for neutral wire and black for earth wire
- (B) red for live wire, black for neutral wire and green for earth wire
- (C) green for live wire, black for neutral wire and red for earth wire
- (D) green for live wire, red for neutral wire and black for earth wire

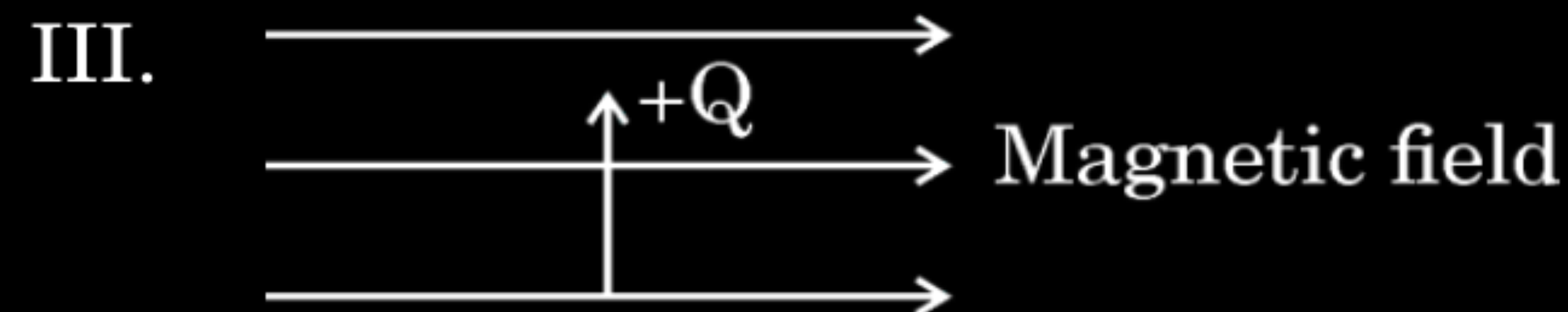
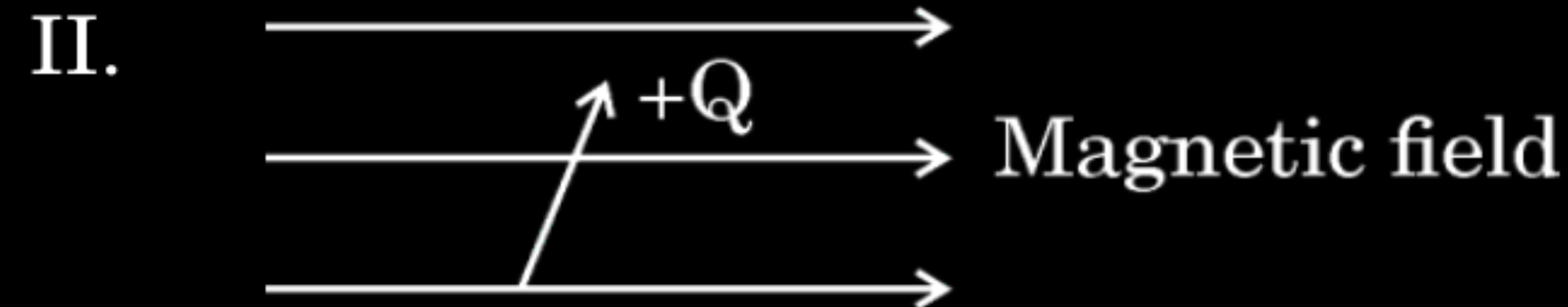
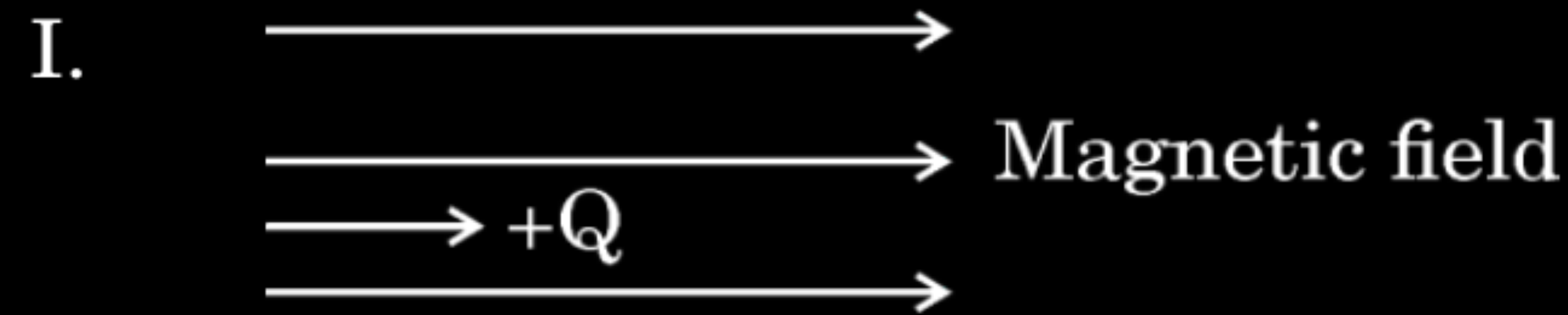
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- (B) red for live wire, black for neutral wire and green for earth wire
- (C) green for live wire, black for neutral wire and red for earth wire
- (D) green for live wire, red for neutral wire and black for earth wire

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(B) / Red for live wire, black for neutral wire and green for earth wire

- 32.** (a) Name and state the rule which determines the force on a current carrying conductor placed in a uniform magnetic field.
- (b) Consider the following three diagrams in which the entry of a positive charge (+Q) in a magnetic field is shown. Identify giving reason the case in which the force experienced by the charge is (i) maximum, and (ii) minimum.



(a) Fleming's Left hand rule Statement : Stretch the thumb, fore figure and the middle figure of your left hand such that they are mutually perpendicular. If the first finger points in the direction of field (F) and the second finger in the direction of current, then the thumb will point in the direction of the force acting on the conductor.	$\frac{1}{2}$
(b) (i) Maximum in Case III – Magnetic field and current/motion of charge are perpendicular to each other.	$\frac{1}{2}$
(ii) Minimum in Case I – Magnetic field and current/motion of charge are parallel to each other.	$\frac{1}{2}$

20. **Assertion (A) :** In the common domestic circuits the earth wire is connected to a metallic plate buried deep inside the earth. 1

Reason (R) : Earth wire ensures that any leakage of current to the metallic body of the appliance keeps its potential to that of the earth, so the user may not get a severe electric shock.

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Reason (R) : Earth wire ensures that any leakage of current to the metallic body of the appliance keeps its potential to that of the earth, so the user may not get a severe electric shock.

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(A)/ Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

34. (a) What are magnetic field lines ? How is the direction of magnetic field at a point determined ? Draw the pattern of magnetic field lines of the magnetic field produced by a current carrying circular loop. Mark on it the direction of (i) current and (ii) magnetic field lines.

Name the two factors on which the magnitude of the magnetic field due to a current carrying coil depends.

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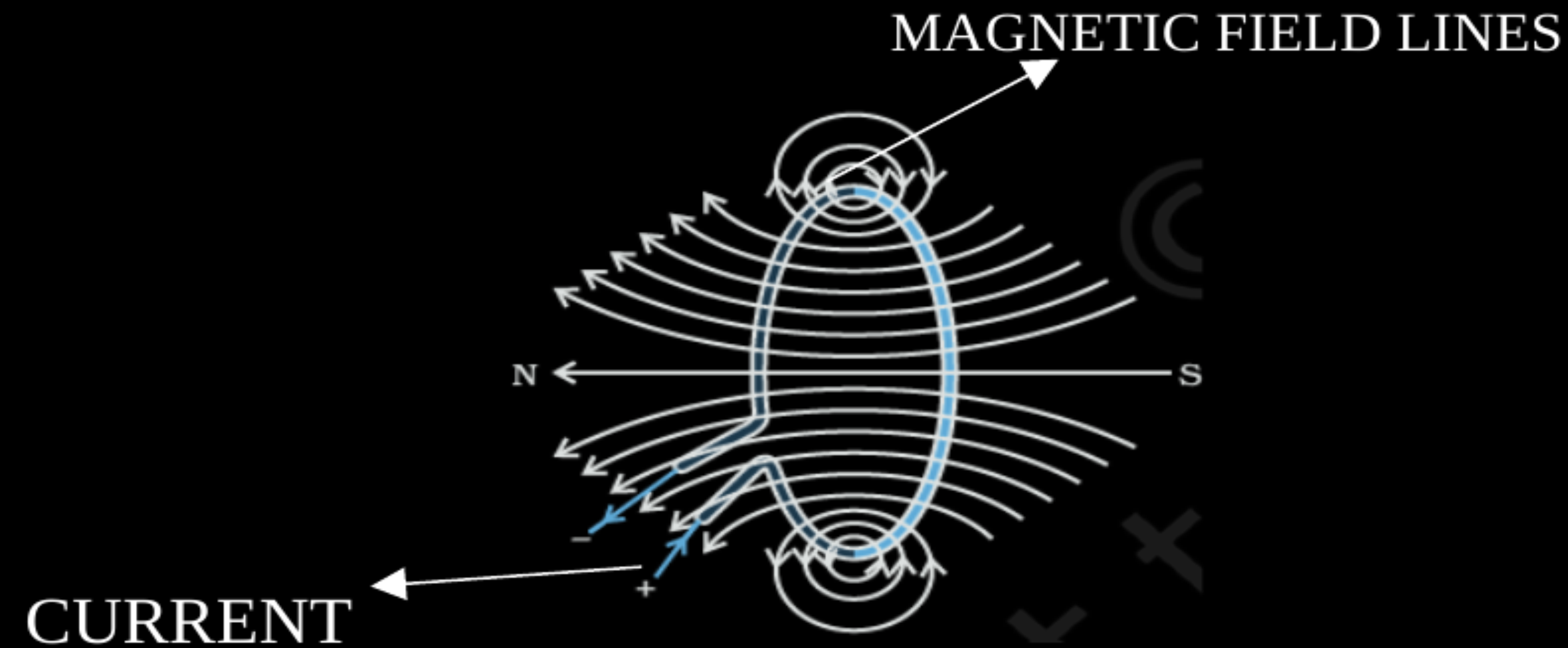
(a) Magnetic field lines are the imaginary lines around the magnet

1/2

- Magnetic field line directions at a point are determined by placing a small compass needle.

1/2

- Figure of magnetic field produced by a current carrying circular coil –



2

(1 Mark for each labelling)

- Amount of the electric current flowing through it.
- The number of turns in the circular coil
- The radius of circular coil

2

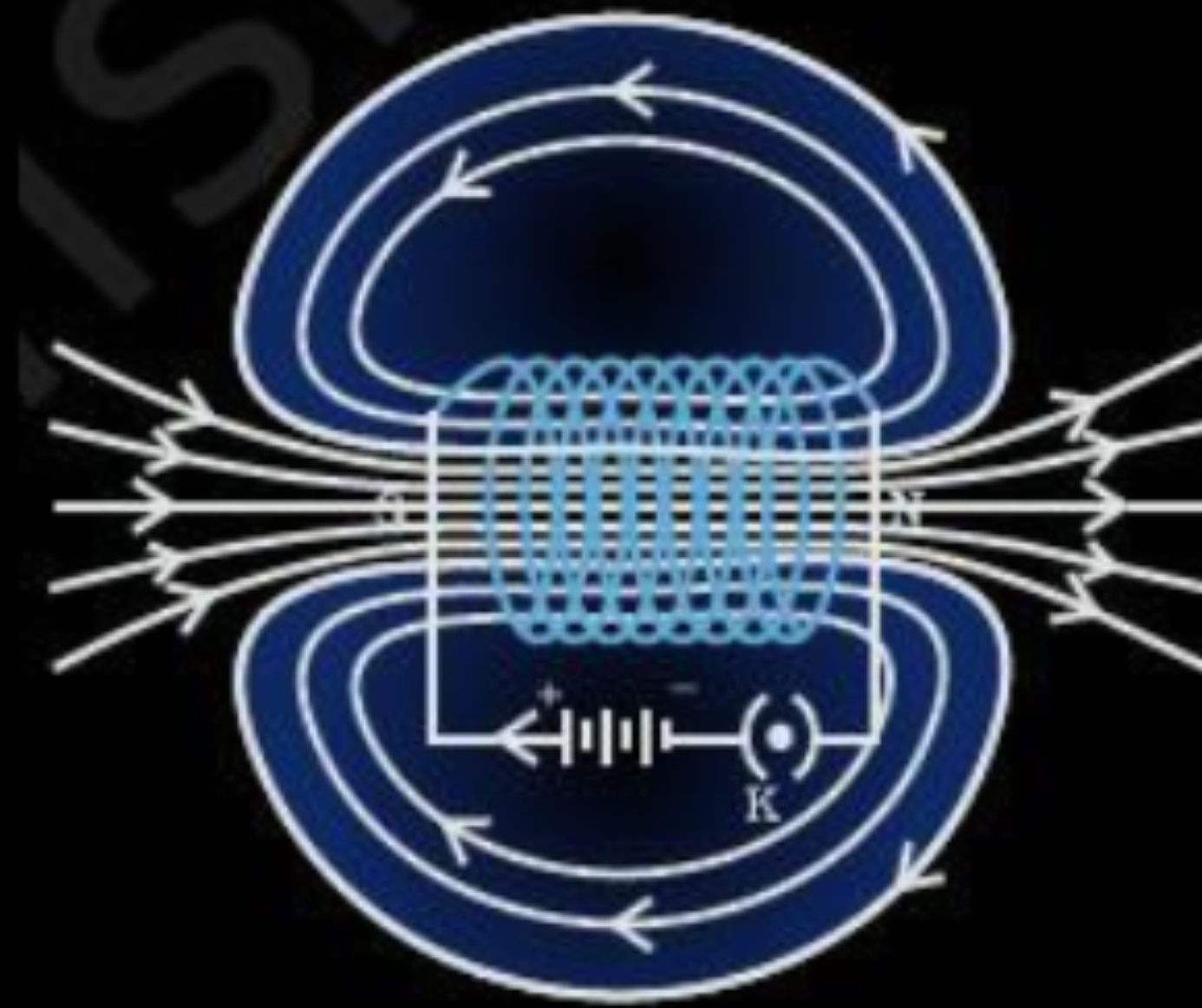
(Any two)

- (b) Why can't two magnetic field lines cross each other ? Draw magnetic field lines showing the direction of the magnetic field due to a current carrying long straight solenoid. State the conclusion which can be drawn from the pattern of magnetic field lines inside the solenoid.

Name any two factors on which the magnitude of the magnetic field due to this solenoid depends.

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(b) • There will be two directions of the field at the same point i.e. the point where the two field lines intersect which is not possible. / At the point of intersection, the compass needle would point towards two directions, which is not possible.



• Magnetic field lines are equidistant and parallel i.e. the magnetic field is the same at all points inside the solenoid.

- Number of turns
- Amount of current in solenoid,
- Core material inside the solenoid

(Any two)

1

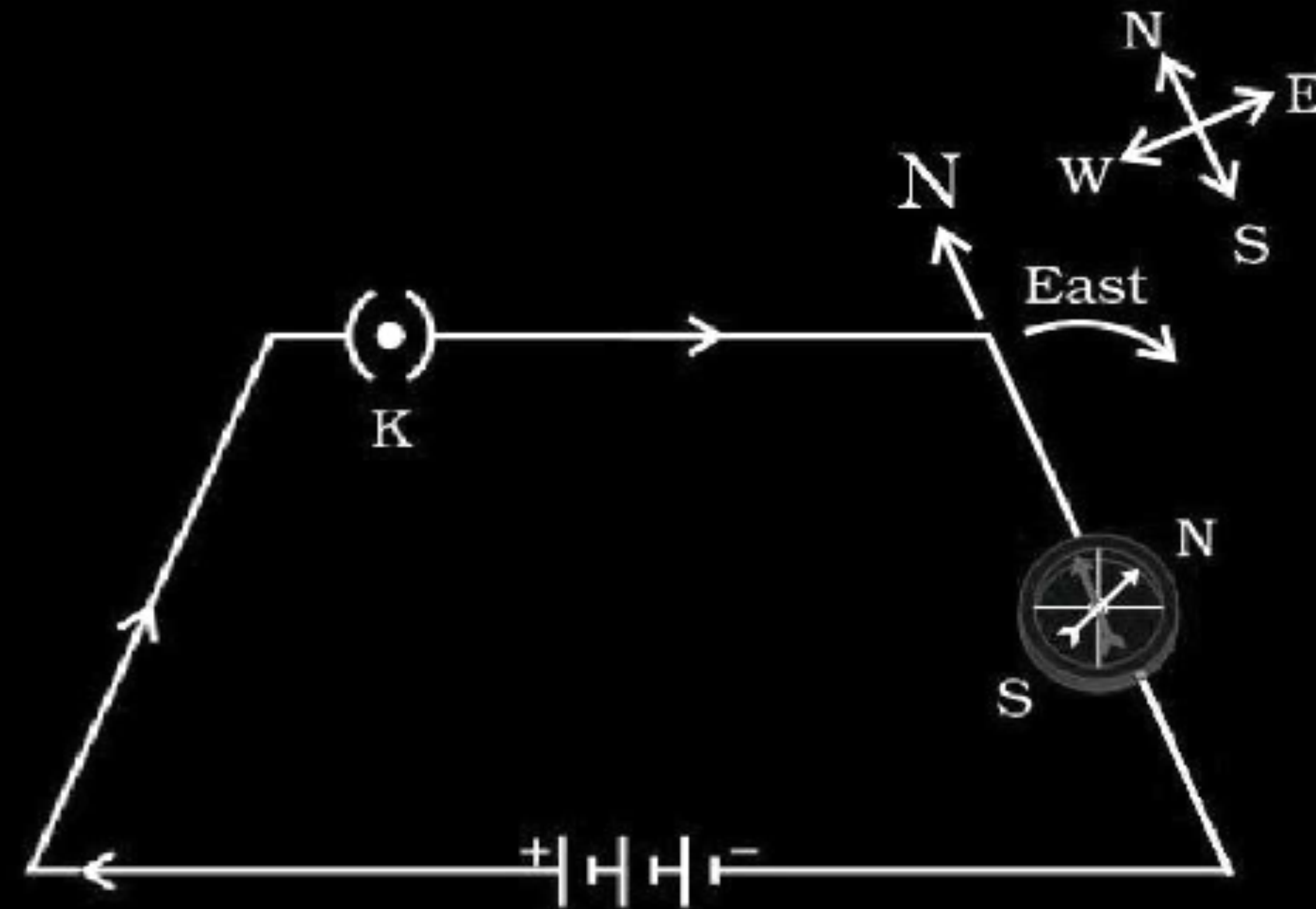
1

1

1x2

36. (a) Study the following electric circuit diagram and answer the questions that follow :

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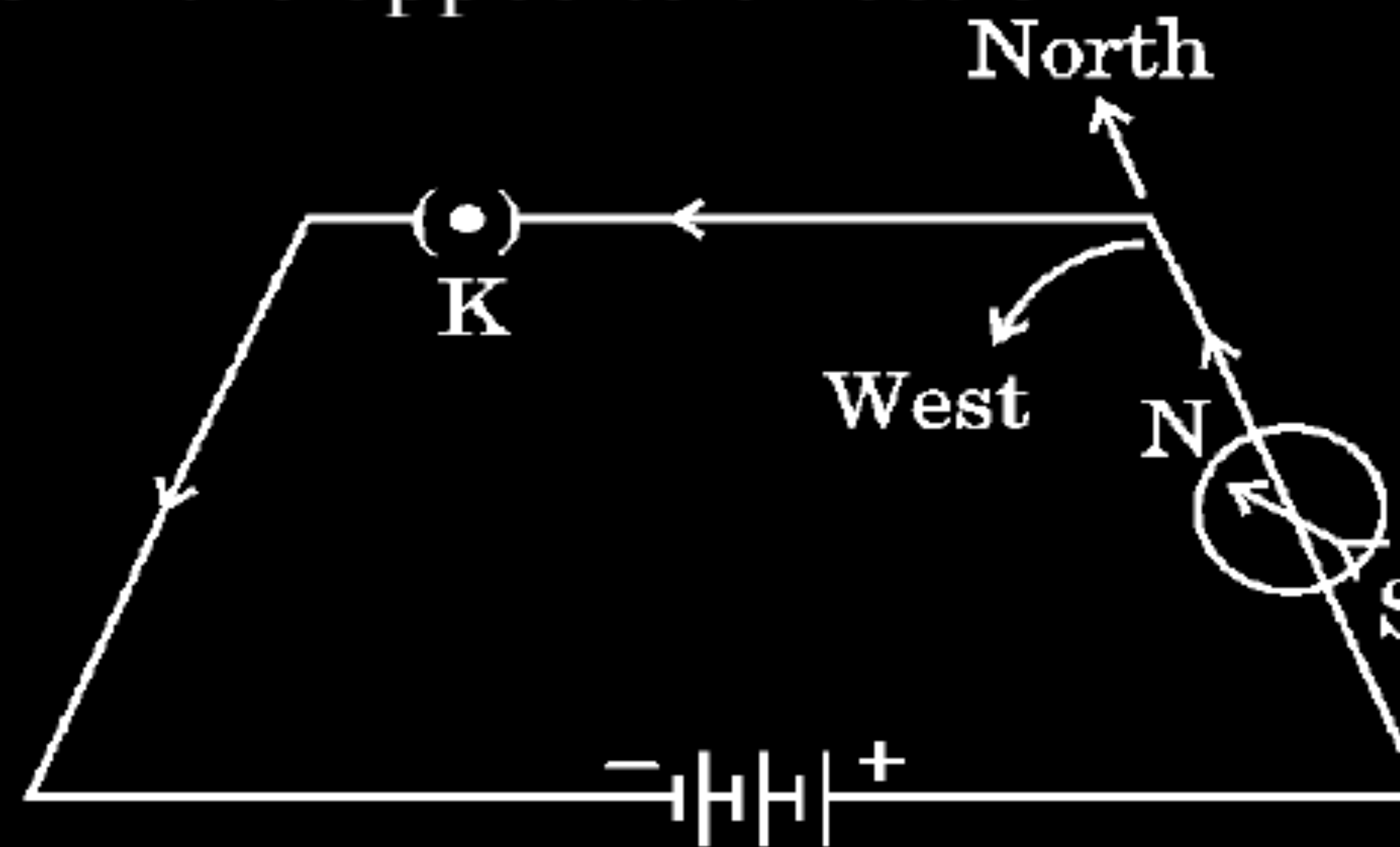


- (i) What does the circuit diagram shows ?
 - (ii) What will happen if the direction of current is reversed ? Justify your answer giving circuit diagram.
- (b) Name and state the rule to determine the direction of magnetic field associated with a straight current carrying conductor.

(a)

(i) Deflection of a compass needle, when placed near a current-carrying straight wire/the electric current through the conductor produces a magnetic effect

(ii) If the direction of the current is reversed, the compass needle deflects in the opposite direction



(b) Right hand thumb rule

Rule: Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of current, then the fingers will wrap around the conductor in the direction of the field lines of the magnetic field.

1

1

1

1

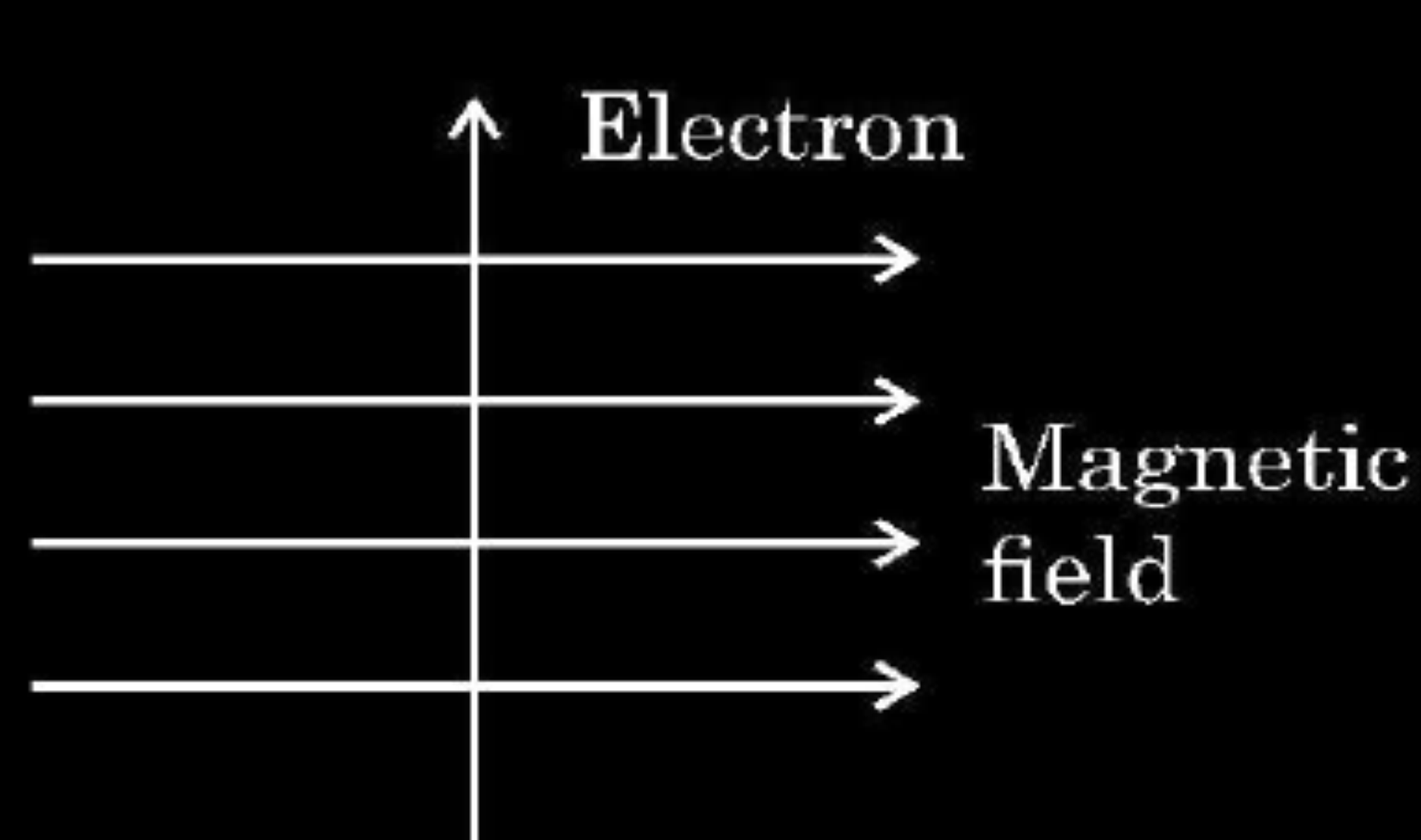
1

- (a) Draw the pattern of magnetic field lines of
- (i) a bar magnet
 - (ii) a current carrying solenoid

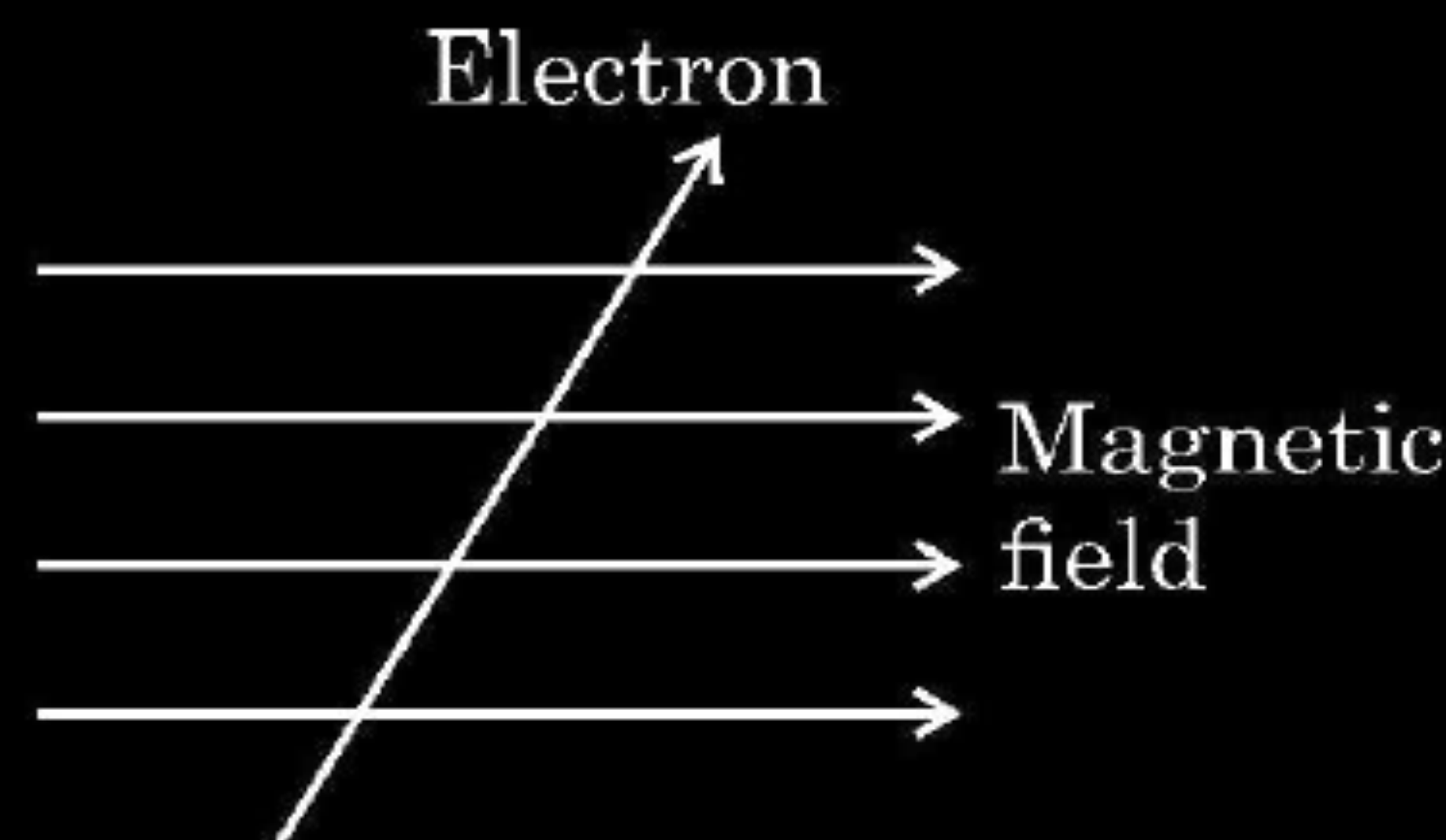
List two distinguishing features between the two magnetic fields.

- (b) Study the following three diagrams in which the entry of an electron in a magnetic field is shown. Identify the case in which the magnetic force experienced by the electron is (i) maximum, and (ii) minimum.

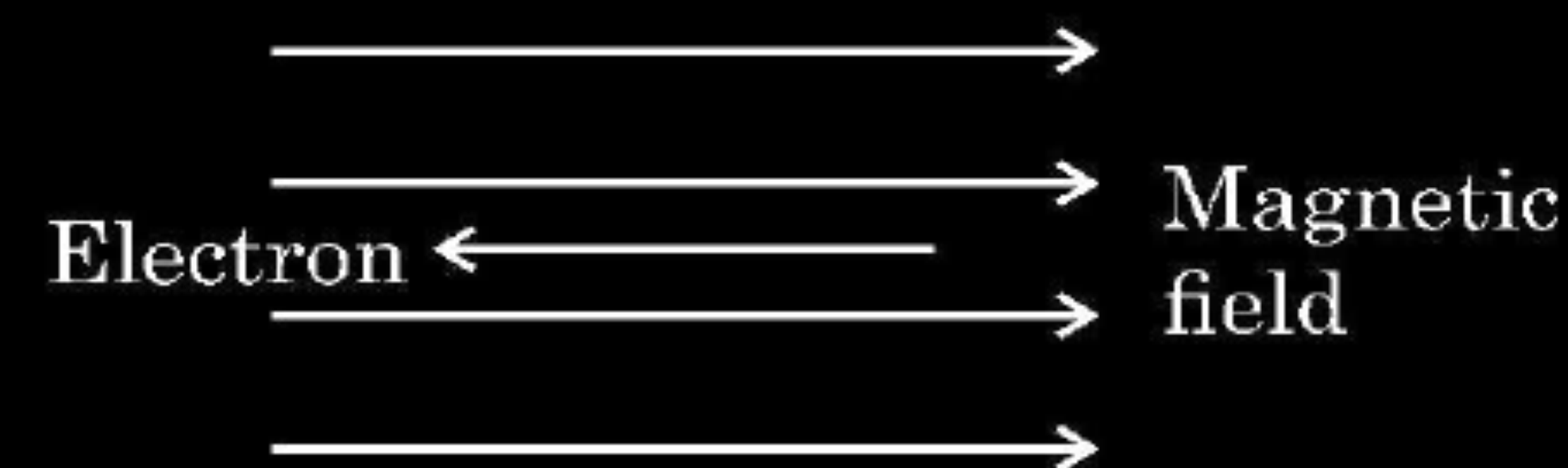
Give reason for your answers in each case.



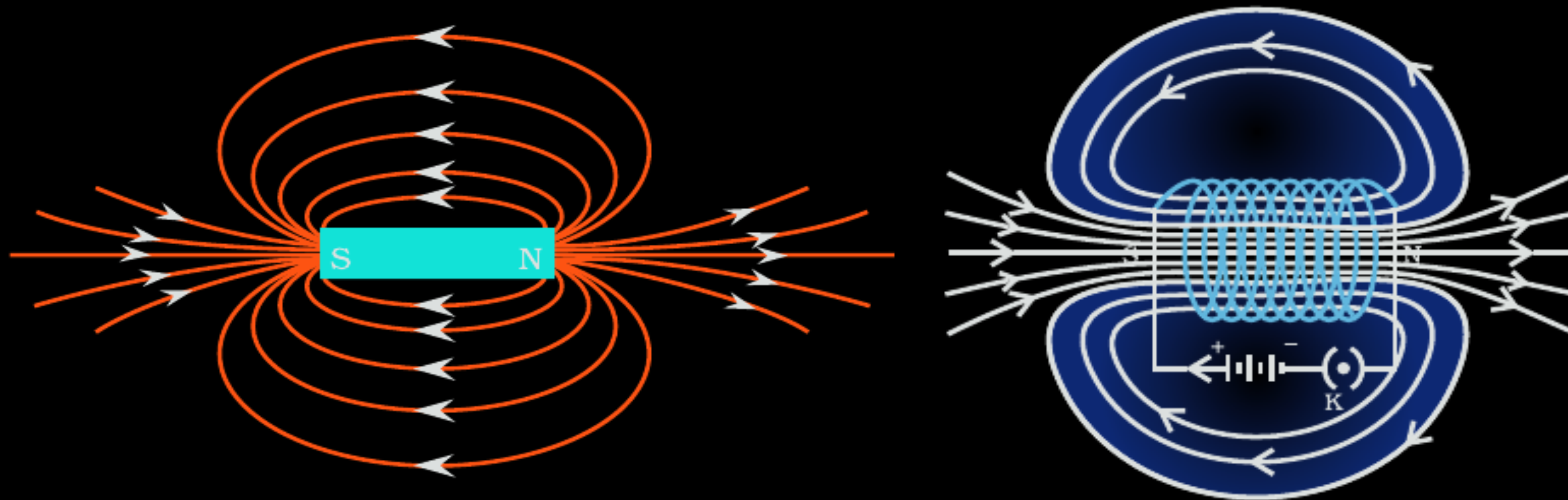
(A)



(B)



(C)



Two differences:

BAR MAGNET	CURRENT CARRYING SOLONEOID
Strength of magnetic field is fixed	Strength of magnetic field can be changed
Polarity is fixed	Polarity can be changed

(b) Magnetic force of the electron is :

(i)Maximum in Case A

electron moving perpendicular to magnetic field.

(ii)Minimum in Case C

electron moving in direction opposite to the direction of magnetic field

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

13. A steady current flows through a thick horizontal straight wire from east to west. The direction of the magnetic field produced at a point above the wire is towards

(A) North

(B) South

(C) East

(D) West

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(A)/ North

16. A uniform magnetic field
- (A) always exerts a force on a current carrying conductor.
 - (B) never exerts a force on a current carrying conductor.
 - (C) exerts a force only when current flows through a conductor along the direction of the field.
 - (D) exert a force only when current flows through a conductor at right angles to the field.

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(D)/ exerts a force only when current flows through a conductor at right angles to the field.

20. **Assertion (A)** : An electric fuse wire is always connected in parallel with the mainline.

Reason (R) : If an electric current larger than the rated value flows through the electric circuit, the fuse wire melts.

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20. **Assertion (A)** : An electric fuse wire is always connected in parallel with the mainline.

Reason (R) : If an electric current larger than the rated value flows through the electric circuit, the fuse wire melts.

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(D)/ Assertion (A) is false, but Reason (R) is true.

32. (a) (i) List two important safety measures commonly used in domestic electric circuits and the appliances such as electric irons/ovens.
- (ii) The power rating of an electric appliance is 220 V; 2 kW. If it is used in a room having electric circuit of current rating of 5 A, what result do you expect ? Give reason for your answer with necessary calculations.

3

32. (a) (i) List two important safety measures commonly used in domestic electric circuits and the appliances such as electric irons/ovens. 3
- (ii) The power rating of an electric appliance is 220 V; 2 kW. If it is used in a room having electric circuit of current rating of 5 A, what result do you expect ? Give reason for your answer with necessary calculations.

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(a)	(i)	• Electric fuse • Earthing/use earth wire	$\frac{1}{2} \times 2$
	(ii)	• $I = \frac{P}{V}$ $= \frac{2000W}{220V}$ $= 9.09A$	$\frac{1}{2}$
		• The appliance will stop working due to overloading.	$\frac{1}{2}$
			1

- (b) Name and state the rule to determine the direction of a
- (i) magnetic field produced around a straight conductor carrying a steady current.
 - (ii) force experienced by a current carrying straight conductor placed perpendicular to a uniform magnetic field.

- (b) Name and state the rule to determine the direction of a
- magnetic field produced around a straight conductor carrying a steady current.
 - force experienced by a current carrying straight conductor placed perpendicular to a uniform magnetic field.

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(b) (i) • Right hand thumb rule Imagine holding a current carrying straight conductor carrying in your right hand such that the thumb points toward the direction of current, then the fingers wrap around the conductor indicates the direction of the magnetic field lines.	$\frac{1}{2}$ 1
(ii) • Fleming's left hand rule Stretch the thumb, forefinger and the middle finger of your left hand mutually perpendicular to each other. If the forefinger points in the direction of magnetic field, middle finger in the direction of current, then the thumb gives the direction of motion or force on the current carrying conductor.	$\frac{1}{2}$ 1

18. **Assertion (A)** : The pattern of the magnetic field of a solenoid carrying a current is similar to that of a bar magnet.

Reason (R) : The pattern of the magnetic field around a current carrying conductor is independent of the shape of the conductor.

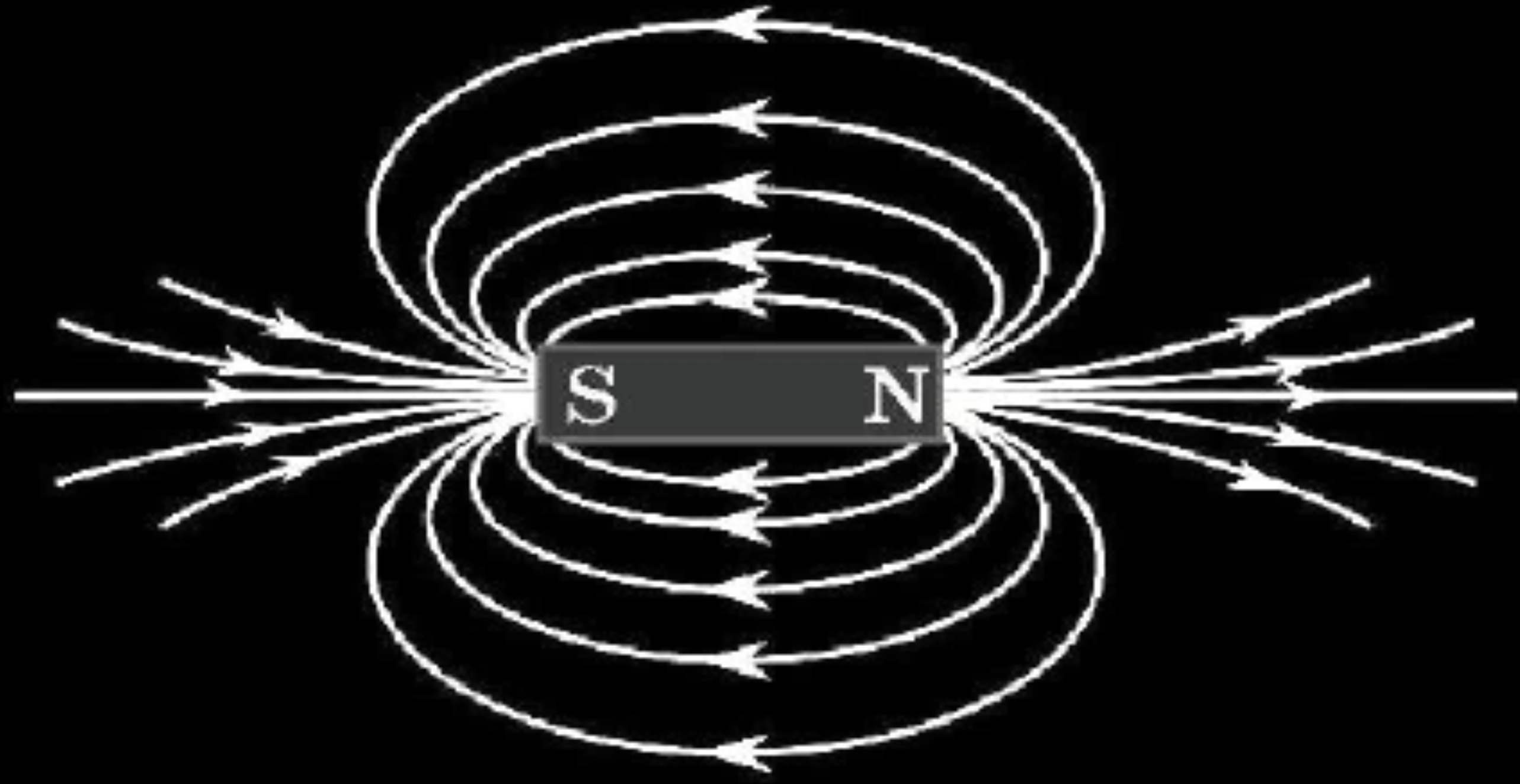
CBSE 2025

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Reason (R) : The pattern of the magnetic field around a current carrying conductor is independent of the shape of the conductor.

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(C) / Assertion (A) is true, but Reason (R) is false.



37. In order to obtain magnetic field lines around a bar magnet, a student performed an experiment using a magnetic compass and a bar magnet. The magnet was placed on a sheet of white paper fixed on a drawing board. Using magnetic needle he obtained on the paper a pattern of magnetic field lines (as shown below) around the bar magnet.

- (a) By convention, the field lines emerge from north pole and merge at south pole. Why ? Give reason.
- (b) State the relationship between strength of the magnetic field and the degree of closeness of the field lines.
- (c) (A) (i) No two field lines can ever intersect each other. Give reason.
(ii) The magnetic field in a given region is uniform. Draw a diagram to represent it.

OR

- (c) (B) Draw the pattern of the magnetic field lines through and around a current carrying solenoid. What does the pattern of field lines inside the solenoid represent ?

(a) The direction of the magnetic field is taken to be the direction in which a north pole of the compass needle moves inside it.	1
(b) Closer the field lines stronger is the magnetic field.	1
(c) (A)	
(i) It would mean that at the point of intersection, the compass needle would point towards two directions, which is not possible.	1

27. What is short circuiting ? State its possible causes. What is likely to happen if a domestic circuit gets short circuited ? Give reason for the justification of your answer.

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27. What is short circuiting ? State its possible causes. What is likely to happen if a domestic circuit gets short circuited ? Give reason for the justification of your answer.

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- Short circuiting occurs when the live wire and neutral wire of a domestic electric circuit come in direct contact with each other.
- Damaged insulation of the live wire and neutral wire, Fault in the electrical appliance/ overloading
(Any 2)
- Due to abrupt increase in the current, the fuse wire will melt and break the circuit.

33. (a) “The third wire of earth connection is very important in domestic electric appliances.” Justify this statement.
- (b) List two precautions to be taken to avoid the overloading of domestic electric circuits.

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33. (a) “The third wire of earth connection is very important in domestic electric appliances.” Justify this statement.
- (b) List two precautions to be taken to avoid the overloading of domestic electric circuits.

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(a) The third wire (earth wire) is a safety measure to ensure that in case of any leakage of current to the metallic body of the appliance, it keeps its potential to that of the Earth and the user may not get a severe shock.

(b)(i) Use of electric fuse of proper rating

(ii) Not connecting too many appliances to a single socket.

(any other)

14. When too many electrical appliances are connected to a single socket in a domestic circuit :

- (A) total resistance of the circuit increases.
- (B) current in the circuit decreases.
- (C) short circuiting occurs.
- (D) current in the circuit increases.

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14. When too many electrical appliances are connected to a single socket in a domestic circuit :

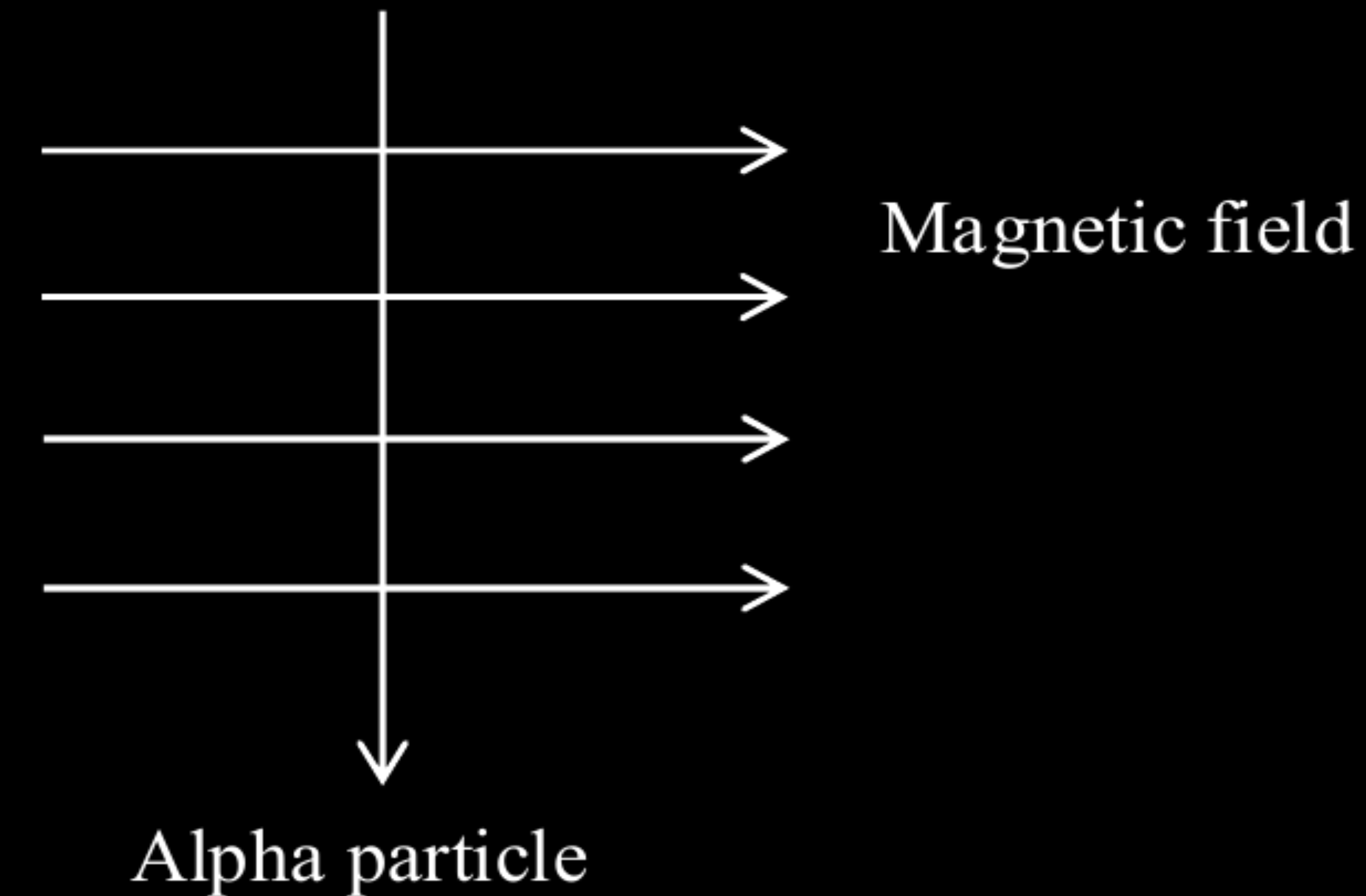
- (A) total resistance of the circuit increases.
- (B) current in the circuit decreases.
- (C) short circuiting occurs.
- (D) current in the circuit increases.

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(C)/(D)

Both may happen.

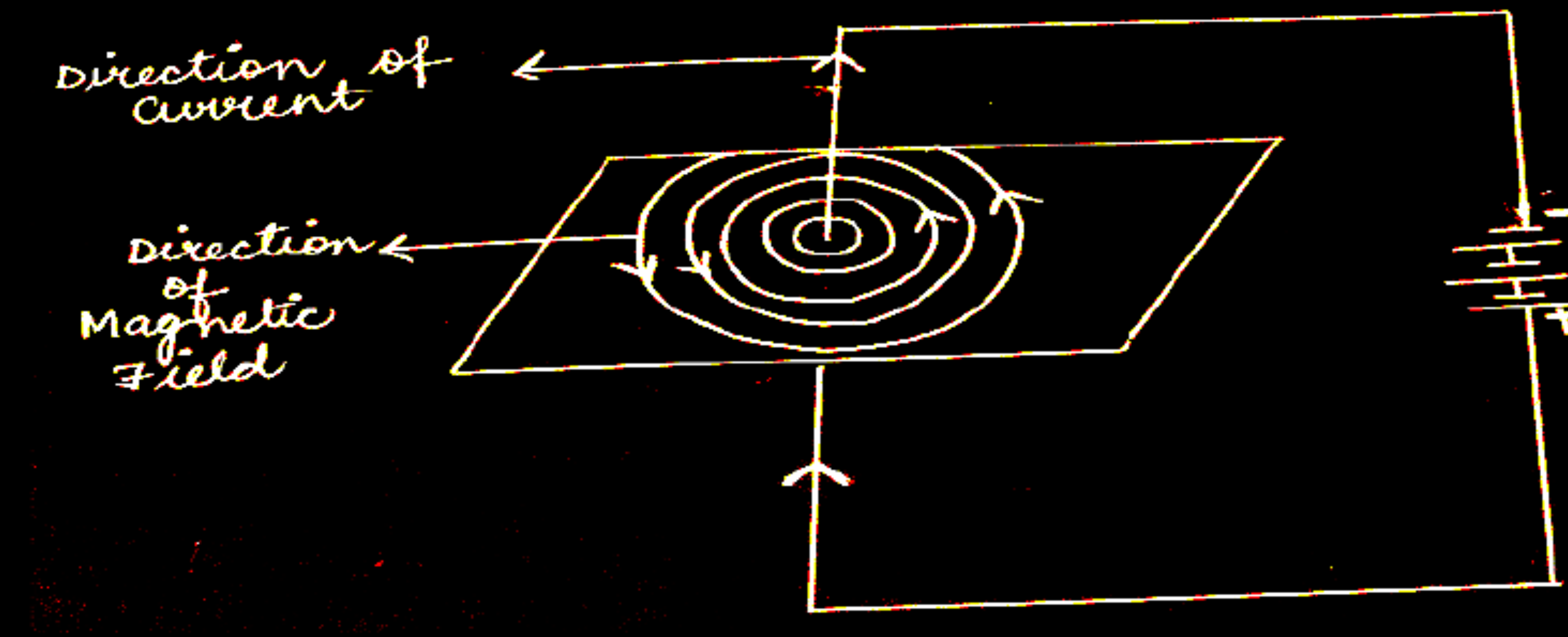
32. (a) Draw the pattern of magnetic field lines around a current carrying straight conductor through which current is flowing vertically upward. Mark the direction of the field lines corresponding to the direction of current in the conductor.



- (b) An alpha particle enters a uniform magnetic field at right angles to it as shown in the diagram.
- (i) Name the rule which can help in determining the direction of force acting on the alpha particle in this case.
- (ii) Apply this rule and find the direction of force on the alpha particle.

3

(a)



Diagram

Labelling of direction of current and corresponding field lines

(b)

(i) Fleming's Left Hand Rule

(ii) Out of the plane of paper

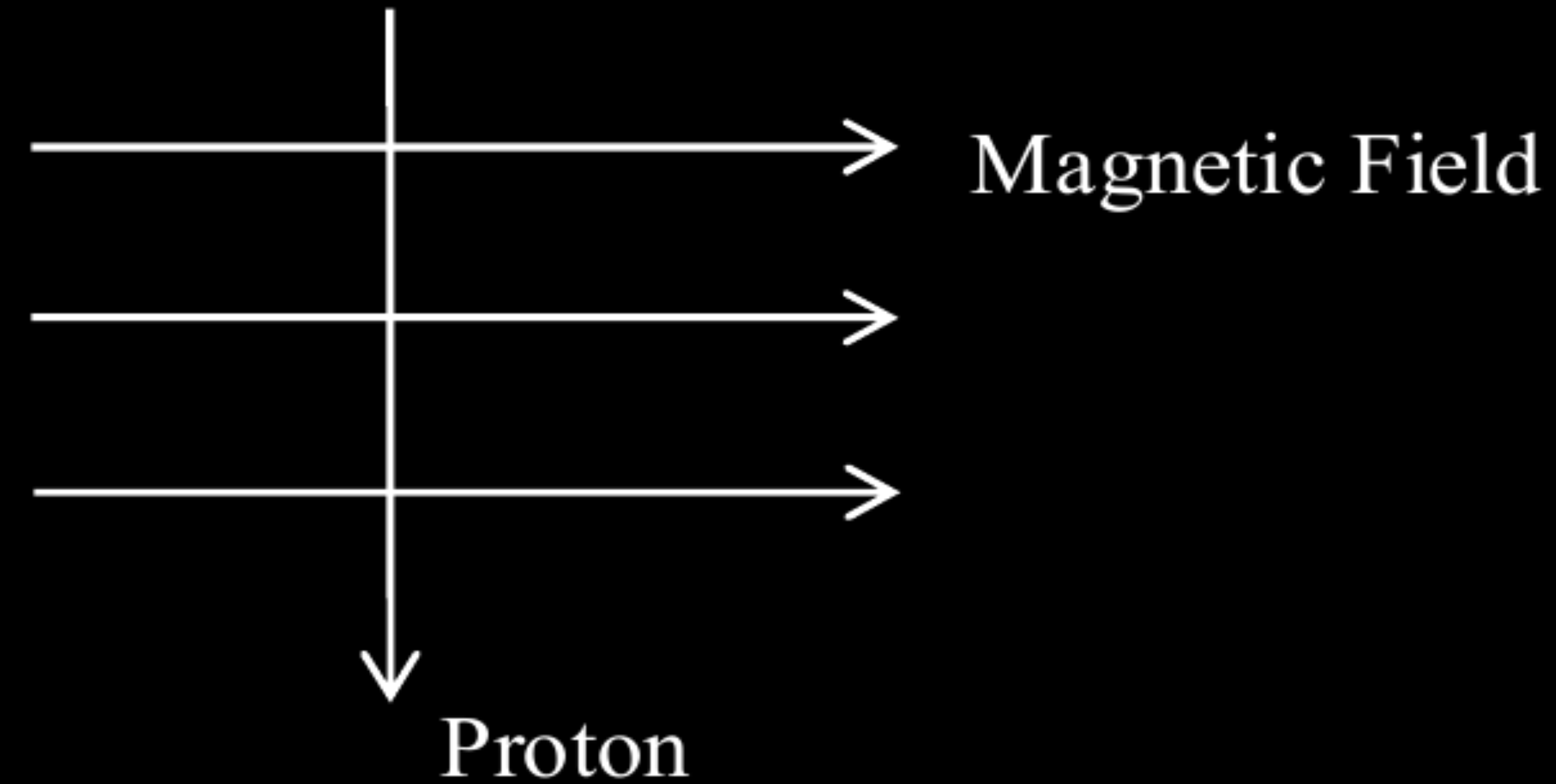
1

1

$\frac{1}{2}$

$\frac{1}{2}$

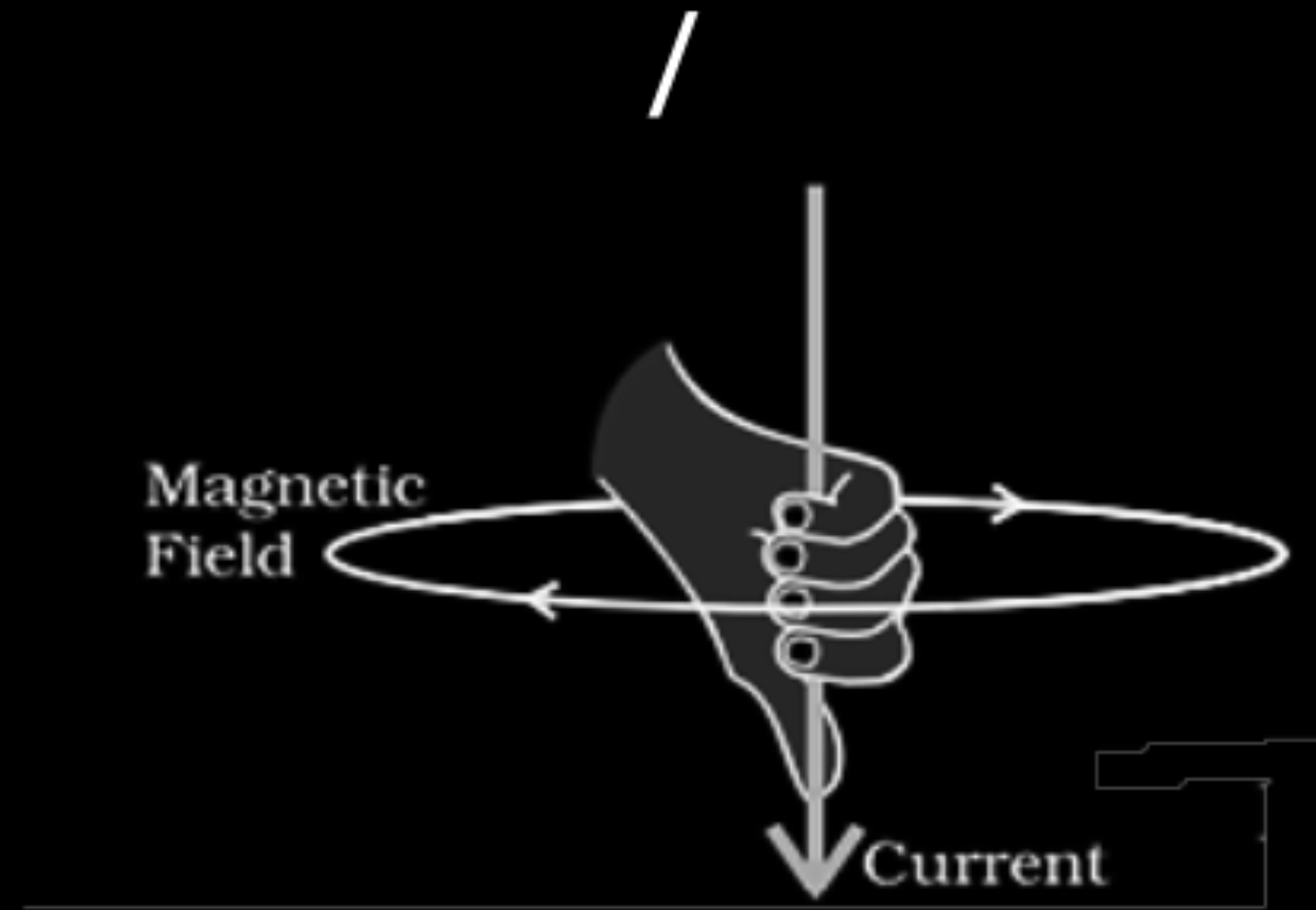
- 31.** (a) State (i) right-hand thumb rule, and (ii) Fleming's left-hand rule.
- (b) Using Fleming's left-hand rule determine the direction of force experienced by a proton, which enters vertically downwards in a uniform magnetic field acting horizontally from west to east as shown in the diagram.



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(a)

(i) Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field. This is known as the right-hand thumb rule.

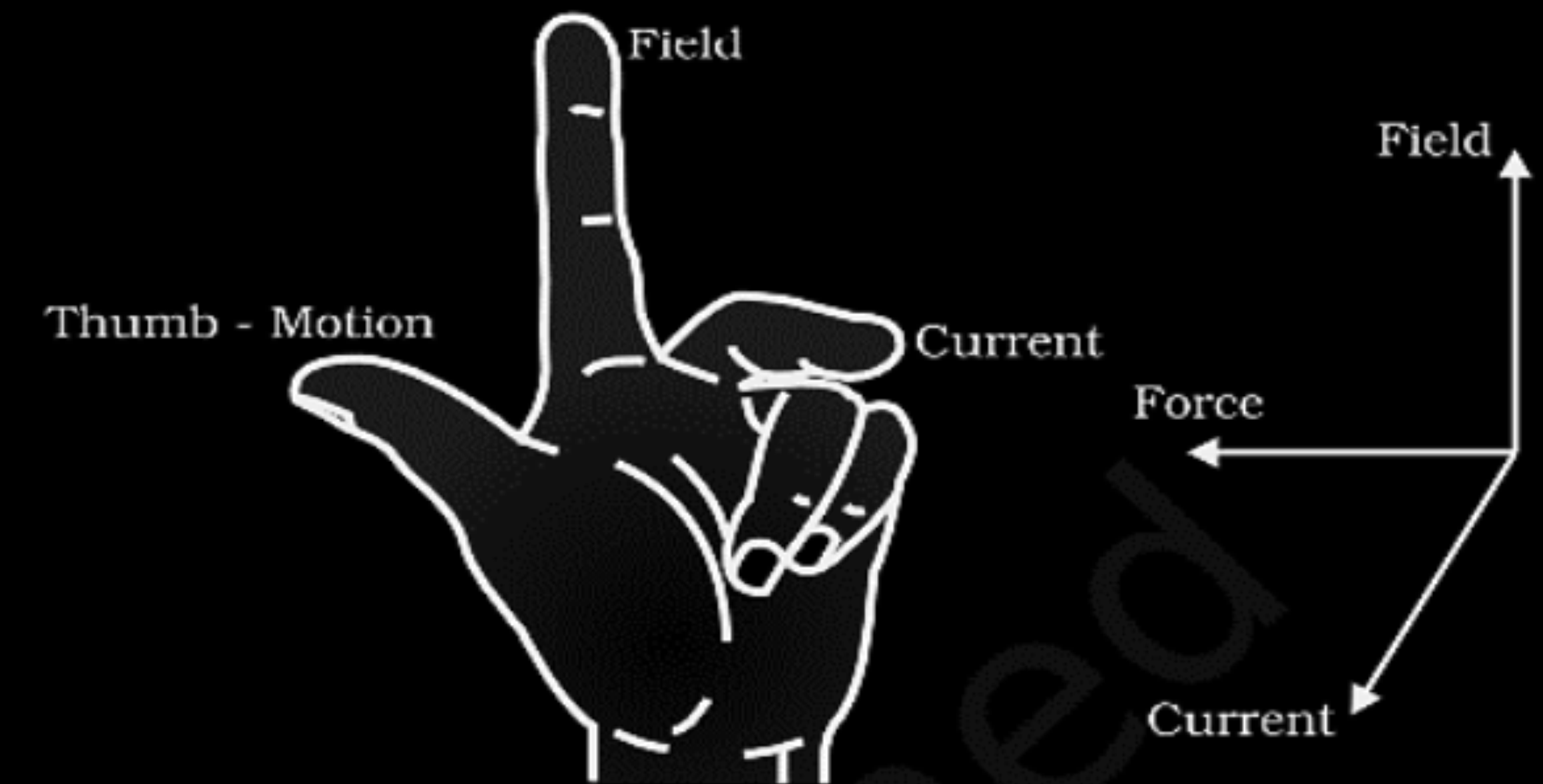


[Fig 12.7, page 200 of NCERT]

Note : Award full marks if explained diagrammatically.

(ii) Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor.

1



[Fig 12.13, page 203 of NCERT]

Note : Award full marks if explained diagrammatically.

11. The direction of magnetic field produced around a current carrying straight conductor is determined by using :

- | | |
|------------------------------|-------------------------------|
| (A) Fleming's left hand rule | (B) Left hand thumb rule |
| (C) Right hand thumb rule | (D) Fleming's right hand rule |

11. The direction of magnetic field produced around a current carrying straight conductor is determined by using :

- | | |
|------------------------------|-------------------------------|
| (A) Fleming's left hand rule | (B) Left hand thumb rule |
| (C) Right hand thumb rule | (D) Fleming's right hand rule |

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13. When a steady current flows through a long straight solenoid it behaves as a bar magnet having a north and a south pole. The strength of the magnetic field inside the solenoid is :

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- (A) zero everywhere.
- (B) uniform everywhere.
- (C) maximum at the ends and minimum at the centre.
- (D) minimum at the ends and maximum at the centre.

13. When a steady current flows through a long straight solenoid it behaves as a bar magnet having a north and a south pole. The strength of the magnetic field inside the solenoid is :

CBSE 2024

- (A) zero everywhere.
- (B) uniform everywhere.
- (C) maximum at the ends and minimum at the centre.
- (D) minimum at the ends and maximum at the centre.

(B)

17. *Assertion (A)* : Magnetic field lines never intersect each other.

Reason (R) : If two field lines intersect each other at a point then it would mean that at the point of intersection the resultant field has two directions, which is not possible.

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17. *Assertion (A)* : Magnetic field lines never intersect each other.

Reason (R) : If two field lines intersect each other at a point then it would mean that at the point of intersection the resultant field has two directions, which is not possible.

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(A)

- 33.** (a) How is the type of current used in household supply different from the current given by a battery of dry cells ?
- (b) Which one of the two currents mentioned above in (a) is considered to be more advantageous for the long distance transmission of electric power and why ?
- (c) How does the electric fuse prevent the electric circuit and the appliances from a possible damage due to short circuiting ?

3

(a) Current used in household supply is AC (alternating current) which changes its direction periodically with time whereas the current given by a battery of dry cells is DC (direct current) which does not change its direction with time.	1
(b) • Alternating current • Alternating current can be transmitted over long distances without much loss of energy.	$\frac{1}{2}$ $\frac{1}{2}$
(c) At the time of short circuiting, current in the circuit exceeds the rated value of the fuse, hence the fuse melts and breaks the circuit.	1

14. The pattern of the magnetic field produced inside a current carrying solenoid is :

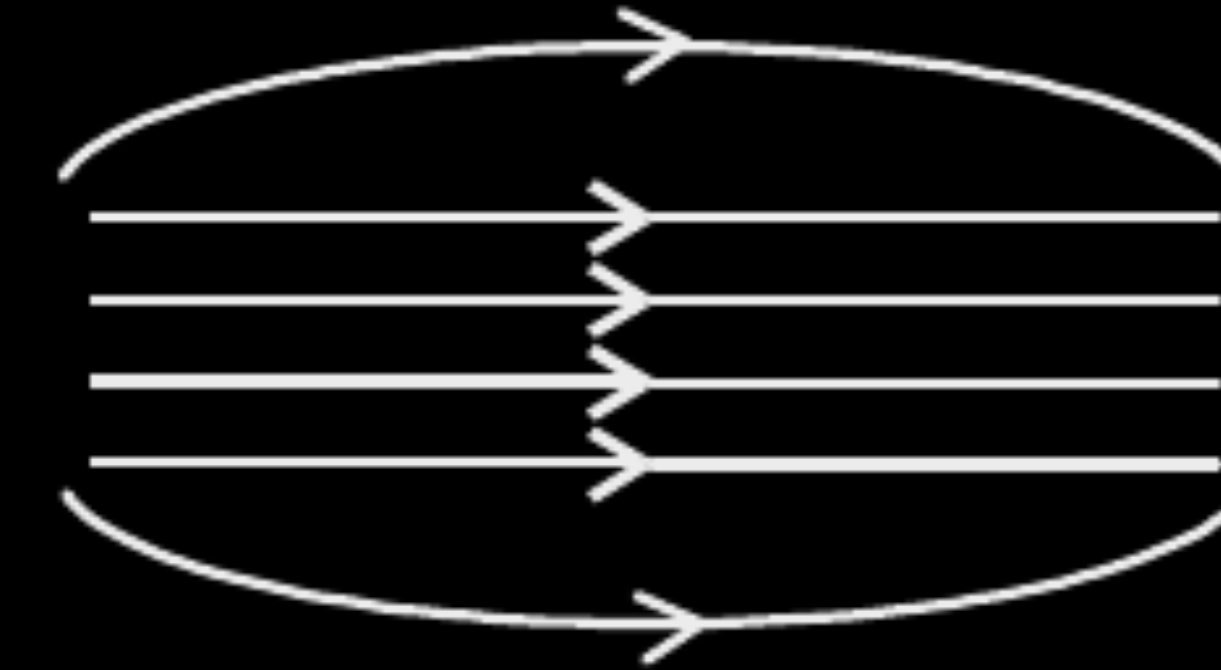
1



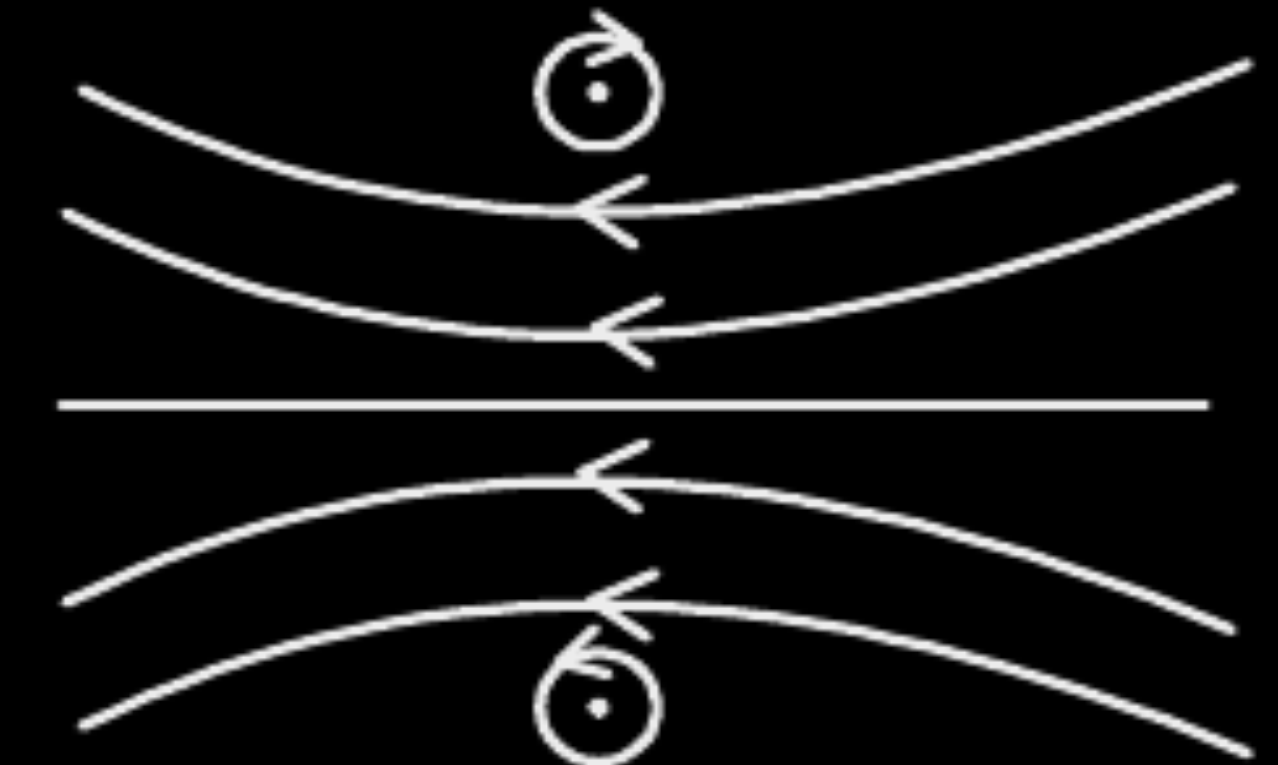
(a)



(b)



(c)



(d)

14. The pattern of the magnetic field produced inside a current carrying solenoid is :

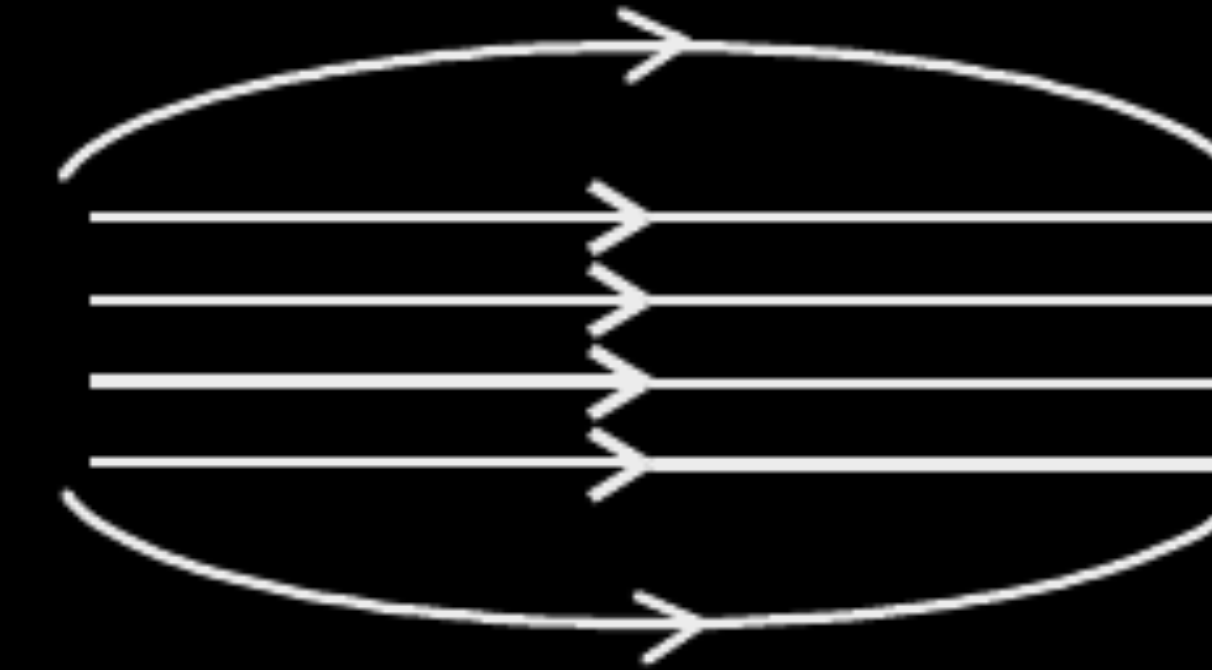
1



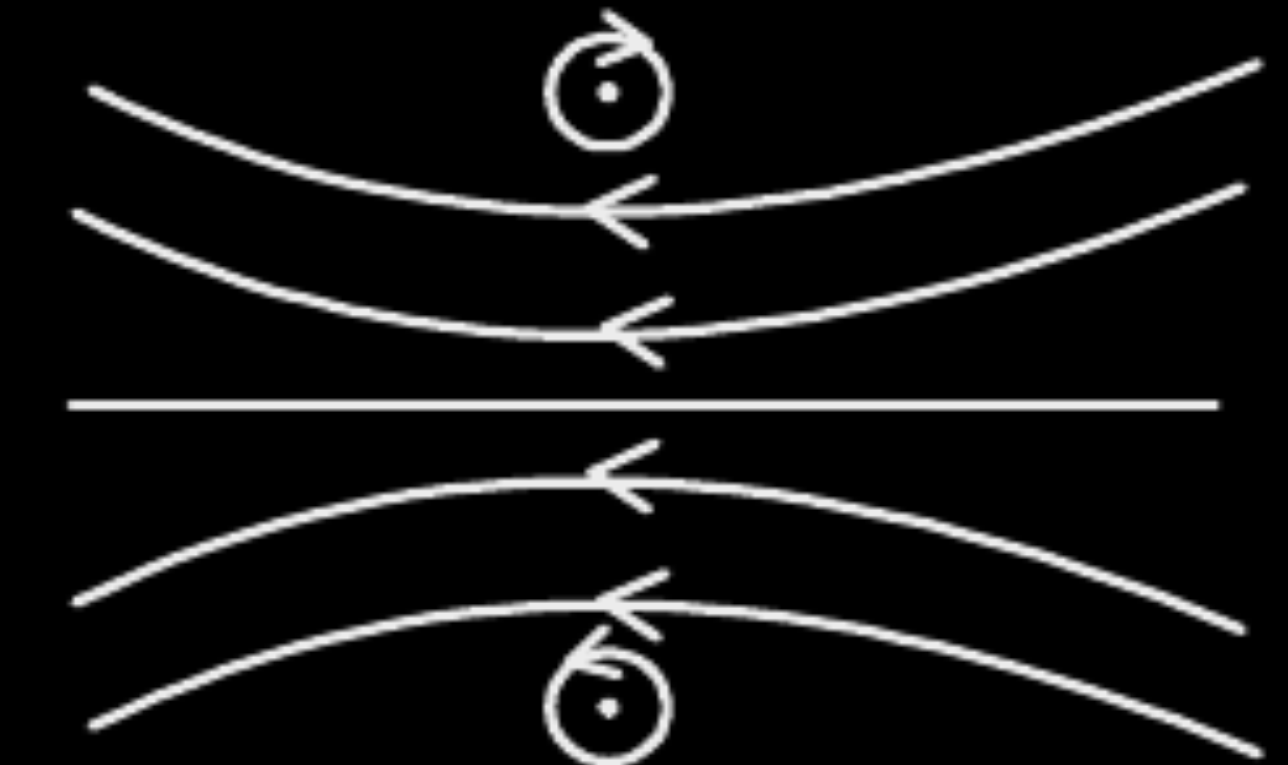
(a)



(b)



(c)



(d)

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(a) /



- 26.** (i) Two magnetic field lines do not intersect each other. Why ? **2**
- (ii) How is a uniform magnetic field in a given region represented ?
Draw a diagram in support of your answer.

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26. (i) Two magnetic field lines do not intersect each other. Why ? 2
- (ii) How is a uniform magnetic field in a given region represented ?
 Draw a diagram in support of your answer. CBSE 2024

(i) If they intersect then at the point of intersection, there would be two directions of magnetic field or compass needle would point towards two directions, which is not possible.	1
(ii) Uniform magnetic field is represented by equidistant parallel straight lines	$\frac{1}{2}$
	$\frac{1}{2}$

32. Name and state the rule to determine the direction of a :

3

- (i) magnetic field produced around a current carrying straight conductor.
- (ii) force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it.

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32. Name and state the rule to determine the direction of a :

3

- (i) magnetic field produced around a current carrying straight conductor.
- (ii) force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it.

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(i)	• Right - Hand Thumb Rule • If the wire carrying current is held in our right hand such that the Thumb points towards the Direction of Current, then the fingers wrap around the conductor in the direction of field lines of the magnetic field.	$\frac{1}{2}$ 1
(ii)	• Fleming's Left - Hand Rule • Stretch the thumb, forefinger and middle finger of left hand mutually perpendicular to each other, such that first finger points in the direction of Magnetic Field, second finger in the direction of Current, then thumb in the direction of motion or force acting on the conductor.	$\frac{1}{2}$ 1

- 33.** (i) Name two safety measures commonly used in electric circuits and appliances. **3**
- (ii) The power rating of an electric oven is 220 V; 2 kW. If it is used in a domestic electric circuit of current rating of 5A, what result do you expect ? Justify your answer with necessary calculations.

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- 33.** (i) Name two safety measures commonly used in electric circuits and appliances. **3**
- (ii) The power rating of an electric oven is 220 V; 2 kW. If it is used in a domestic electric circuit of current rating of 5A, what result do you expect ? Justify your answer with necessary calculations.

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(i) • Electric fuse and Earth wire

$\frac{1}{2} + \frac{1}{2}$

(ii)

$$I = P/V$$

$\frac{1}{2}$

$$= \frac{2000 \text{ W}}{220 \text{ V}} = 9.09 \text{ A}$$

$\frac{1}{2}$

Since, current drawn by the oven is greater than the rated value of current, ($9.09\text{A} > 5\text{A}$), the fuse wire melts/ the electric oven stops working.

1

16. Study the following statements :

- (a) A fuse in a circuit prevents damage to the circuit due to overloading.
- (b) Total resistance in a circuit increases due to overloading.
- (c) During short circuiting the current in the circuit abruptly increases.
- (d) In order that each appliance has same current, they are connected in parallel to each other.

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The correct statements are

- | | |
|-----------------|----------------------|
| (A) (a) and (b) | (B) (b) and (d) |
| (C) (a) and (c) | (D) (a), (c) and (d) |

16. Study the following statements :

1

- (a) A fuse in a circuit prevents damage to the circuit due to overloading.
- (b) Total resistance in a circuit increases due to overloading.
- (c) During short circuiting the current in the circuit abruptly increases.
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The correct statements are

- | | |
|-----------------|----------------------|
| (A) (a) and (b) | (B) (b) and (d) |
| (C) (a) and (c) | (D) (a), (c) and (d) |

(C) / (a) and (c)

18. **Assertion (A)** : The deflection of a compass needle placed near a current carrying wire decreases when the magnitude of an electric current in the wire is increased.

1

Reason (R) : Strength of the magnetic field at a point due to a current carrying conductor increases on increasing the current in the conductor.

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1

Reason (R) : Strength of the magnetic field at a point due to a current carrying conductor increases on increasing the current in the conductor.

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(D)/ (A) is false, but (R) is true.

32. Define the term solenoid. Draw the pattern of magnetic field lines around a current carrying solenoid. State how this magnetic field can be used to magnetise a piece of magnetic material, like soft iron.

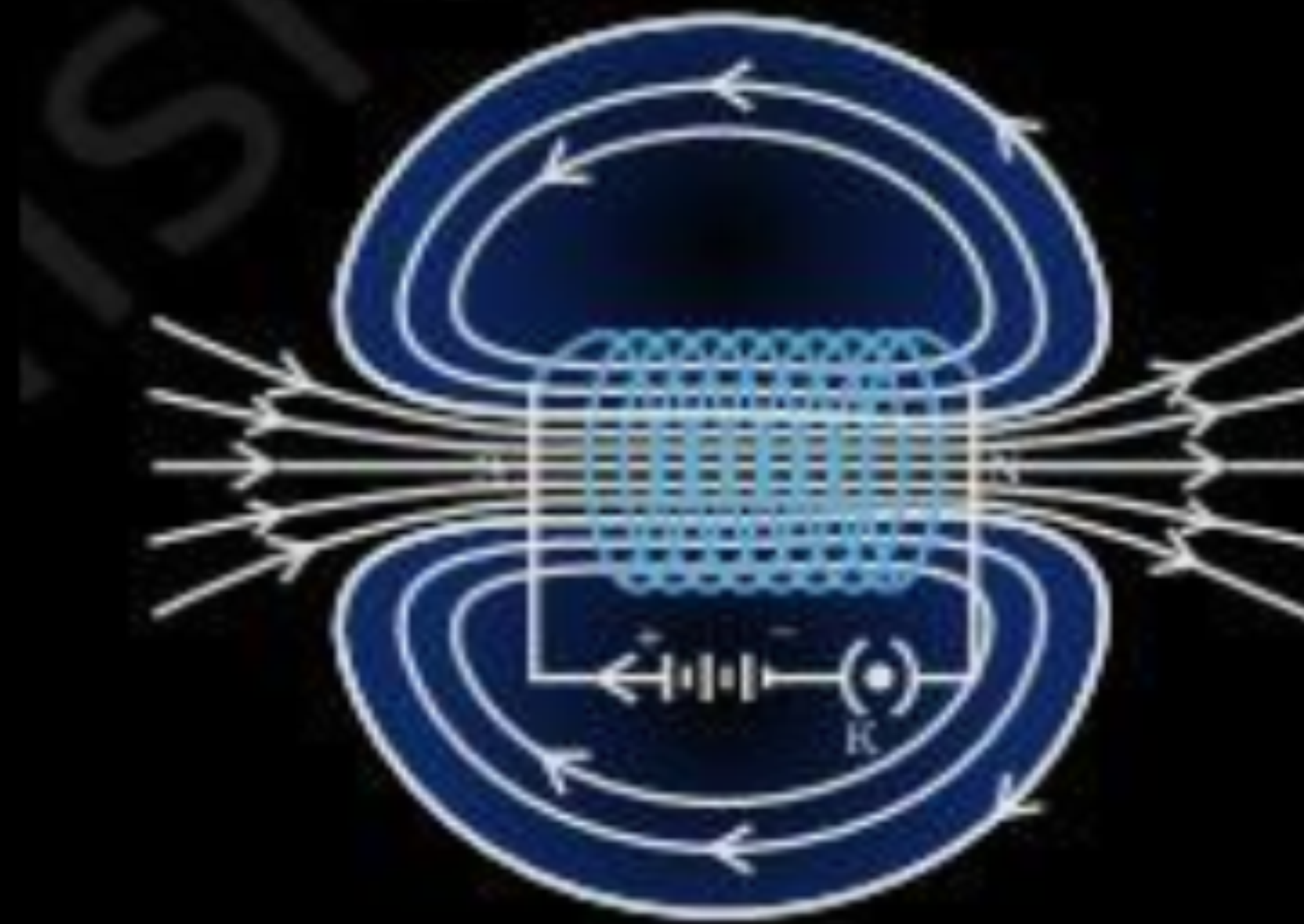
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32. Define the term solenoid. Draw the pattern of magnetic field lines around a current carrying solenoid. State how this magnetic field can be used to magnetise a piece of magnetic material, like soft iron.

CBSE 2024

3

- Solenoid : A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder.



Refer figure 12.10 page 201 NCERT

(Note: Deduct $\frac{1}{2}$ mark if direction of magnetic field lines not shown)

- A strong magnetic field produced inside a solenoid can be used to magnetise a piece of soft iron kept inside it.

38. When electric current flows in a purely resistive circuit electrical energy gets fully converted into heat energy. The amount of heat produced (H) in the circuit is found to be directly proportional to (i) the square of current (I^2) (ii) the resistance (R) of the conductor and (iii) the time (t) for which current flows. In other words $H = I^2Rt$. Electrical devices such as electric fuse, electric heater, electric iron etc. are all based on this effect called heating effect of electric current.

4

(a) List two properties of heating elements.

1

(b) List two properties of electric fuse.

1

(c) Name the principle on which an electric fuse works. Explain how a fuse wire is capable of saving electrical appliances from getting damaged due to accidentally produced high currents.

2

OR

(c) The power of an electric heater is 1100 W. If the potential difference between the two terminals of the heater is 220 V, find the current flowing in the circuit. What will happen to an electric fuse of rating 5 A connected in this circuit ?

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2

• Higher resistivity than pure metals • Do not oxidise (burn) at high temperature.	1/2 , 1/2
(c) • Higher resistivity than pure metals • Low melting point.	1/2 , 1/2
(c) • Heating effect of electric current	1
• When high current flows in the circuit accidentally, the fuse wire melts and breaks the circuit .	1
OR	
(c) P = 1100 W; V = 220 V, I = ?	
$P = VI$	1/2
$I = \frac{P}{V} = \frac{1100\text{ W}}{220\text{ V}} = 5\text{A}$	1/2
• No effect on the fuse of 5A rating.	1

13. Strength of magnetic field produced by a current carrying solenoid **DOES**
NOT depend upon :

1

- (A) number of turns in the solenoid
- (B) direction of the current flowing through it
- (C) radius of solenoid
- (D) material of core of the solenoid

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13. Strength of magnetic field produced by a current carrying solenoid **DOES**
NOT depend upon :

1

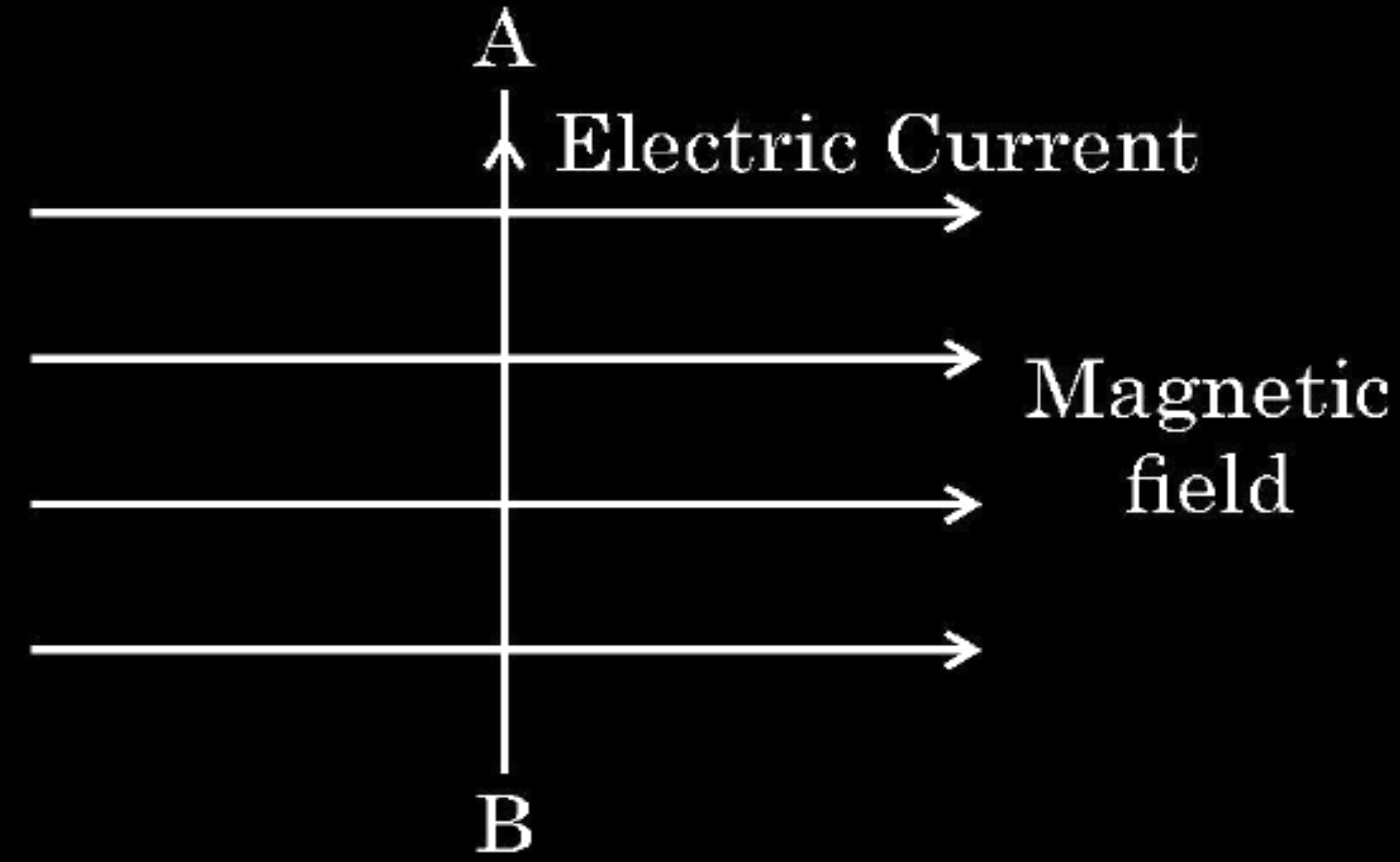
- (A) number of turns in the solenoid
- (B) direction of the current flowing through it
- (C) radius of solenoid
- (D) material of core of the solenoid

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(B) /direction of current flowing through it.

32. (a) State Fleming's left hand rule. Apply this rule to determine the direction of force experienced by a straight current carrying conductor AB placed in a uniform magnetic field as shown.

3



- (b) What will happen to an electron which enters in the same field in the same direction in which the current is flowing in the conductor AB ? Give reason to justify your answer.

(a) Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor.	1
Direction of force vertically downwards/Into the page	$\frac{1}{2}$
(b) Electron will experience force in vertically upward direction/ will deflect out of the page.	$\frac{1}{2}$
Reason: Direction of flow of electrons is opposite to the direction of current in AB.	1

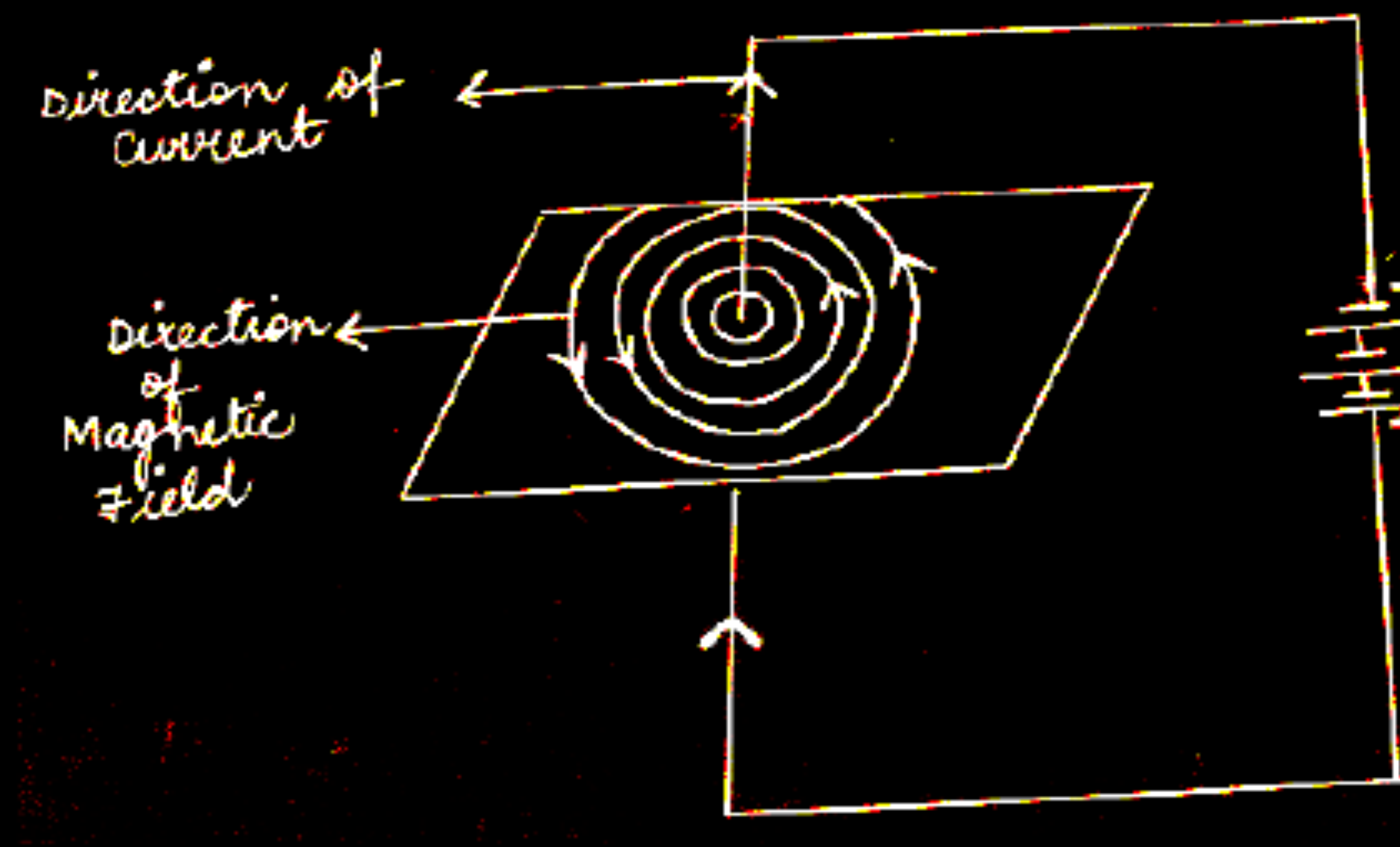
32. Draw a diagram to show the pattern of magnetic field lines on a horizontal sheet of paper due to a straight conductor passing through its centre and carrying current vertically upwards. Mark on it (i) the direction of current in the conductor and (ii) the corresponding magnetic field lines. State right hand thumb rule and check whether the directions marked by you are in accordance with this rule or not.

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32. Draw a diagram to show the pattern of magnetic field lines on a horizontal sheet of paper due to a straight conductor passing through its centre and carrying current vertically upwards. Mark on it (i) the direction of current in the conductor and (ii) the corresponding magnetic field lines. State right hand thumb rule and check whether the directions marked by you are in accordance with this rule or not.

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3



1

Direction of Current
Direction of magnetic field lines

$\frac{1}{2}$

$\frac{1}{2}$

• **Right-Hand Thumb Rule :**

When a current-carrying straight conductor is being held in right-hand such that the thumb points towards the direction of current, then fingers will wrap around the conductor in the direction of the magnetic field lines.

1

24. (a) Draw a labelled diagram to show the pattern of magnetic field lines produced due to a current carrying straight conductor. Mark on it the direction of current in the conductor and the direction of magnetic field lines.

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2

24. (a) Draw a labelled diagram to show the pattern of magnetic field lines produced due to a current carrying straight conductor. Mark on it the direction of current in the conductor and the direction of magnetic field lines.

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2

(a)

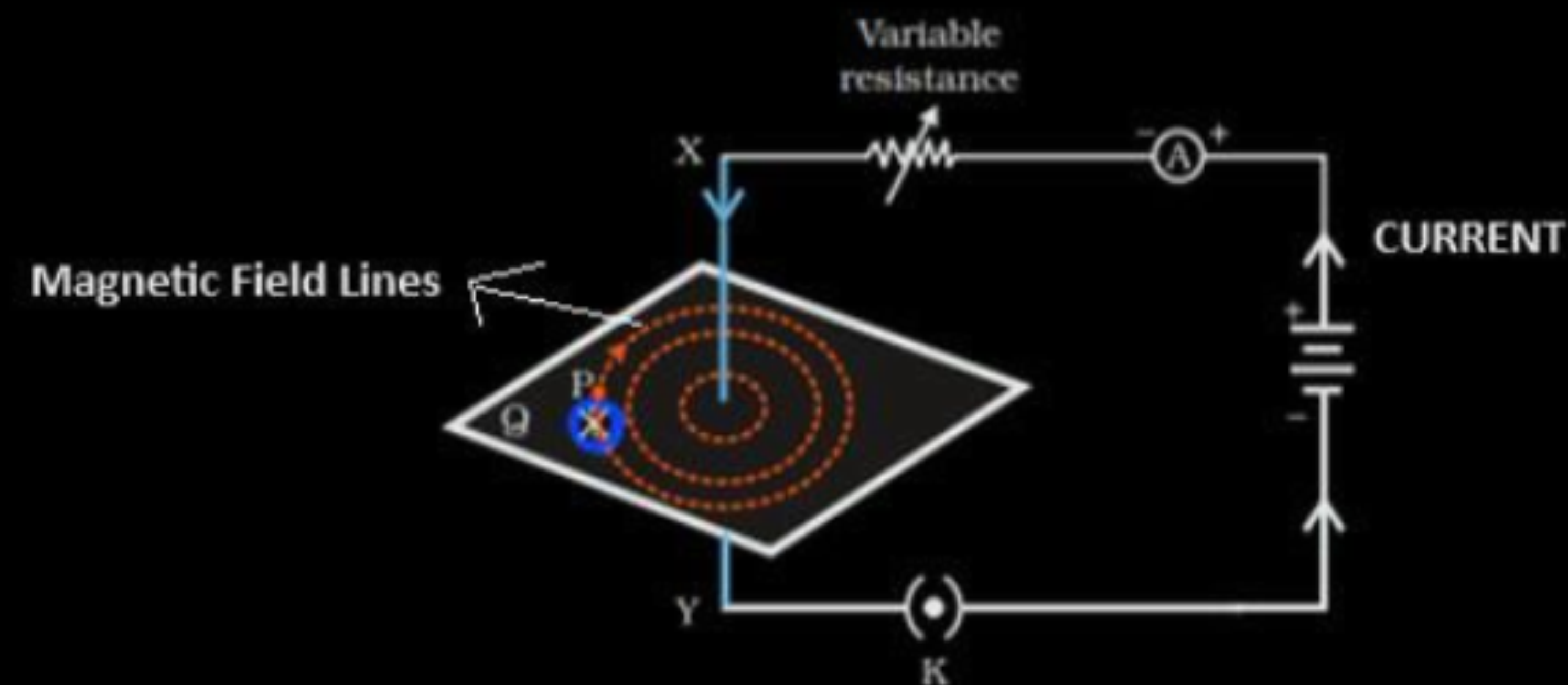


Fig.12.6(a), on page 199-NCERT

Diagram:
Directions of current and magnetic field:

1
 $\frac{1}{2} + \frac{1}{2}$

24. (b) Name the device used to magnetise a piece of magnetic material.
Draw a labelled diagram to show the arrangement used for the
magnetisation of a cylinder made of soft iron.

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2

24. (b) Name the device used to magnetise a piece of magnetic material.
Draw a labelled diagram to show the arrangement used for the magnetisation of a cylinder made of soft iron.

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2

(b)

- Permanent magnet / Current carrying solenoid/ Electromagnet
-

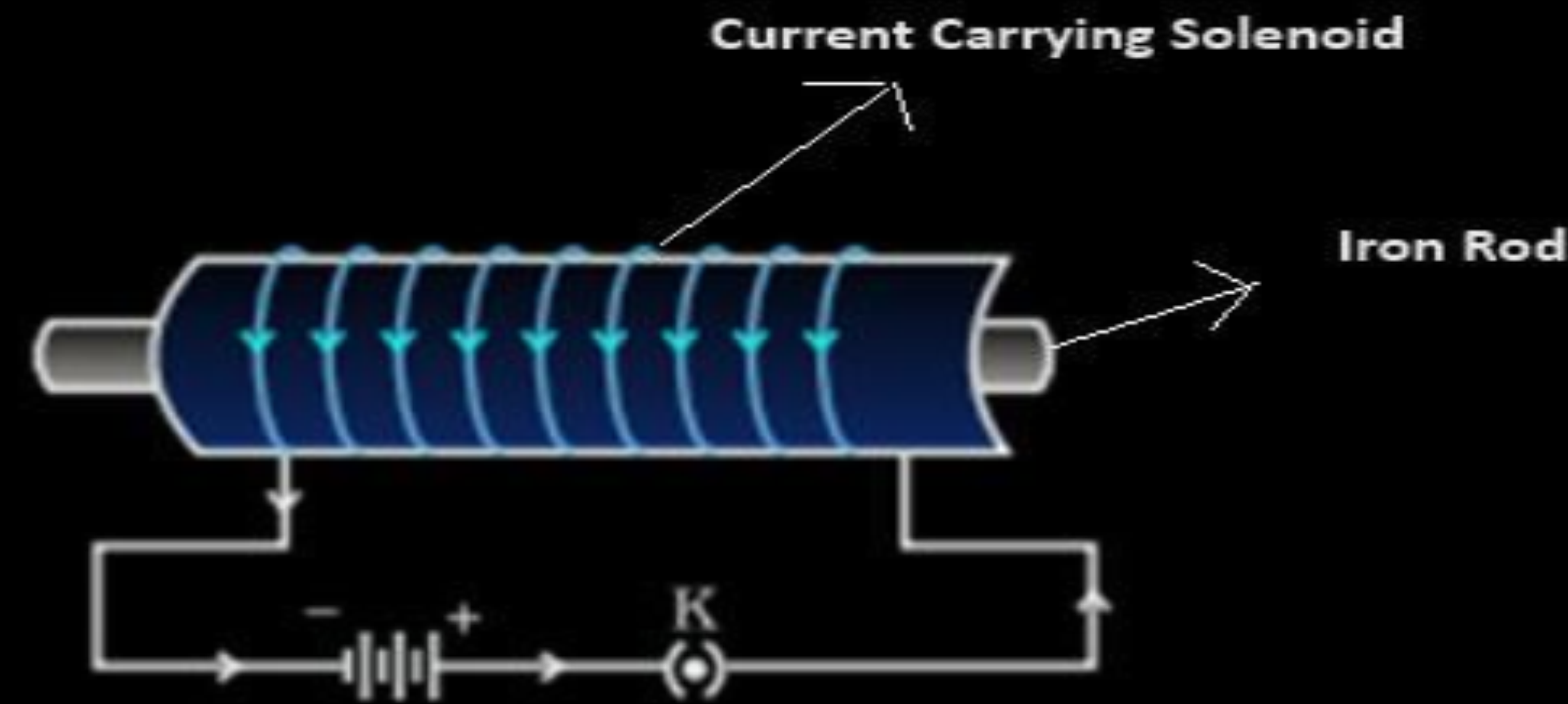


Fig-12.11, page no.201-NCERT

Diagram:
Labelling:

$\frac{1}{2}$

1
 $\frac{1}{2}$

31. Explain in brief the function of an electric fuse in a domestic circuit. An electric heater of current rating 3 kW; 220 V is to be operated in an electric circuit of rating 5 A. What is likely to happen when the heater is switched 'ON' ? Justify your answer with necessary calculation.

3

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31. Explain in brief the function of an electric fuse in a domestic circuit. An electric heater of current rating 3 kW; 220 V is to be operated in an electric circuit of rating 5 A. What is likely to happen when the heater is switched 'ON' ? Justify your answer with necessary calculation. 3

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- It prevents damage to the appliances and the electrical circuit from overloading and short circuiting.

-

Here $P = 3 \text{ kW} = 3000 \text{ W}$, $V = 220 \text{ V}$, $I = ?$

$$P = V I$$

$$I = \frac{P}{V} = \frac{3000 \text{ W}}{220 \text{ V}} = 13.63 \text{ A}$$

$13.63 \text{ A} > \text{Rating of fuse } 5 \text{ A}$, therefore fuse wire will melt and break the circuit.

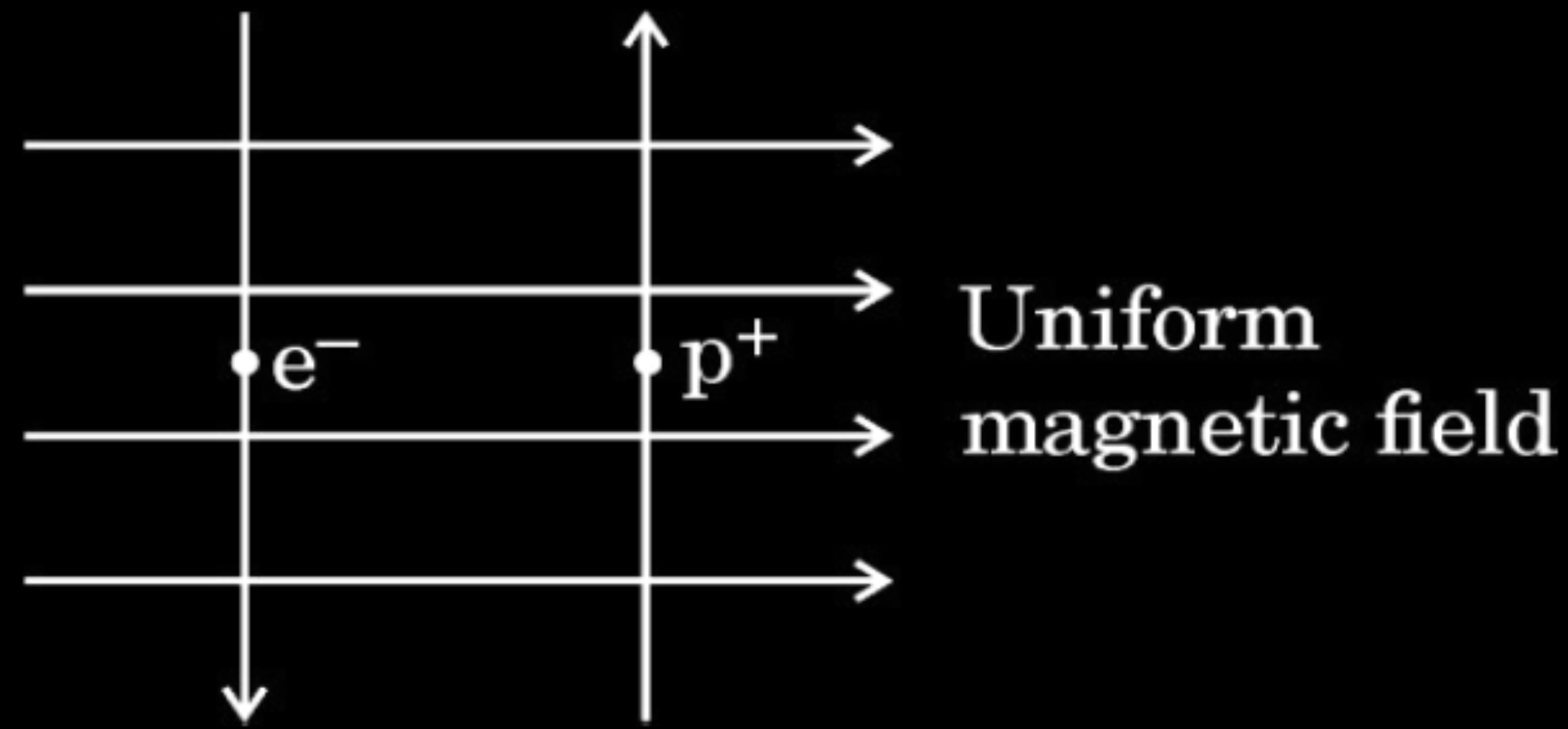
1

$\frac{1}{2}$

1

$\frac{1}{2}$

12.

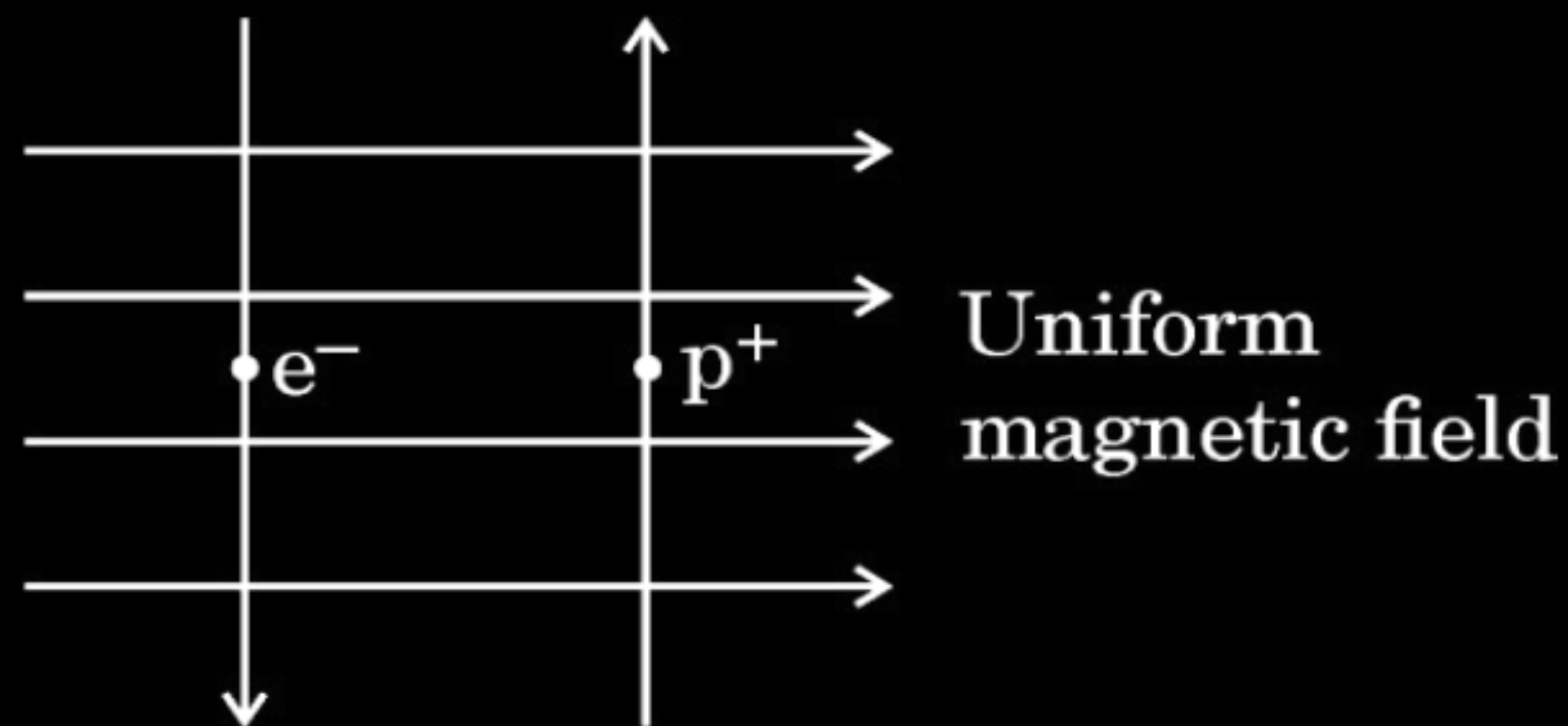


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A uniform magnetic field exists in the plane of paper as shown in the diagram. In this field, an electron (e^-) and a positron (p^+) enter as shown. The electron and positron experience forces :

- (A) both pointing into the plane of the paper.
- (B) both pointing out of the plane of the paper.
- (C) pointing into the plane of the paper and out of the plane of the paper respectively.
- (D) pointing out of the plane of the paper and into the plane of the paper respectively.

12.



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A uniform magnetic field exists in the plane of paper as shown in the diagram. In this field, an electron (e^-) and a positron (p^+) enter as shown. The electron and positron experience forces :

- (A) both pointing into the plane of the paper.
- (B) both pointing out of the plane of the paper.
- (C) pointing into the plane of the paper and out of the plane of the paper respectively.
- (D) pointing out of the plane of the paper and into the plane of the paper respectively.

(A) / both pointing into the plane of the paper.

16. The current carrying device which produces a magnetic field similar to that of a bar magnet is :

(A) A straight conductor

(B) A circular loop

(C) A solenoid

(D) A circular coil

16. The current carrying device which produces a magnetic field similar to that of a bar magnet is :

(A) A straight conductor

(B) A circular loop

(C) A solenoid

(D) A circular coil

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(C) / A solenoid

- 32.** A 2000 W heater has a resistance of about $25\ \Omega$, whereas a 100 W bulb has a resistance of $500\ \Omega$. When 220 V is applied on these, then which of the two
- (a) can carry large currents ?
 - (b) may be used with an electrical circuit having 1.0 A rating ?
 - (c) will be fitted with a 15 A electric board and not with a 5 A electric board ?

Justify your answer in each case.

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- 32.** A 2000 W heater has a resistance of about $25\ \Omega$, whereas a 100 W bulb has a resistance of $500\ \Omega$. When 220 V is applied on these, then which of the two
- (a) can carry large currents ?
 - (b) may be used with an electrical circuit having 1.0 A rating ?
 - (c) will be fitted with a 15 A electric board and not with a 5 A electric board ?

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Justify your answer in each case.

3

(a) 2000 W heater , For heater, $I_1 = \frac{P}{V} = 9.09\text{ A}$; For Bulb, $I_2 = \frac{P}{V} = 0.45\text{ A}$ $I_1 > I_2$	$\frac{1}{2}$ $\frac{1}{2}$
(b) 100 W bulb $I_2 = \frac{P}{V} = 0.45\text{ A}$ As it draws only 0.45 A which is less than 1 A.	$\frac{1}{2}$ $\frac{1}{2}$
(c) 2000 W heater $I_1 = \frac{P}{V} = 9.09\text{ A}$ As the current drawn is 9.09 A which is higher than 5.0 A.	$\frac{1}{2}$ $\frac{1}{2}$

32. “Earth wire is a safety measure in domestic electric circuits.” Justify this statement explaining its role in case of accidental leakage of electric appliances.

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3

- | | |
|--|---|
| • Earthing is used as a safety measure, especially for those appliances that have a metallic body which is connected to the earth wire. | 1 |
| • It provides a low-resistance conducting path for the current. | 1 |
| • Thus, it ensures that any leakage of current to the metallic body of the appliance keeps its potential to that of the earth, and the user may not get a severe electric shock. | 1 |

- 32.** (a) What happens when a bundle of wires of soft iron is placed inside the coil of a solenoid carrying a steady current ? Name the device obtained. Why is it called so ?
- (b) Draw the magnetic field lines inside a current carrying solenoid. What does this pattern of magnetic field lines indicate ? CBSE 2024 3

32. (a) What happens when a bundle of wires of soft iron is placed inside the coil of a solenoid carrying a steady current ? Name the device obtained. Why is it called so ?

(b) Draw the magnetic field lines inside a current carrying solenoid.

What does this pattern of magnetic field lines indicate ?

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3

(a)

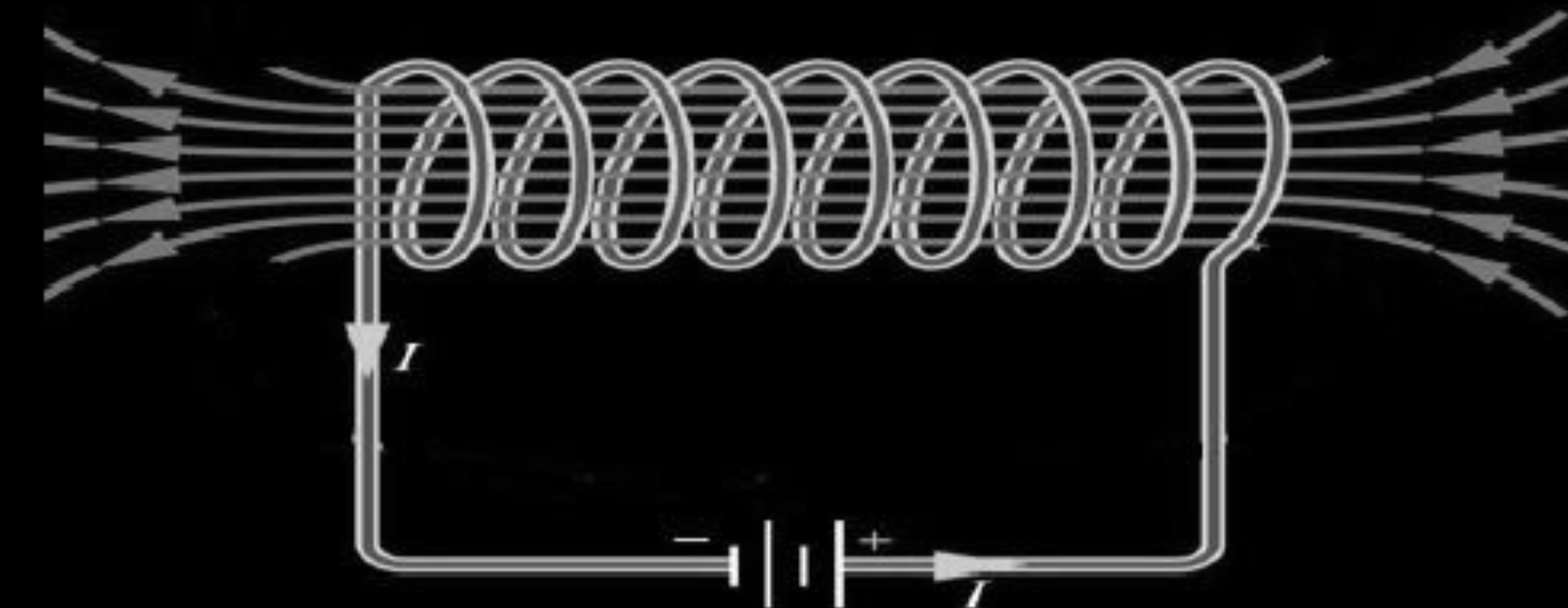
- It gets magnetised
- Electromagnet.
- It behaves as a magnet only when current passes through the solenoid.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(b)



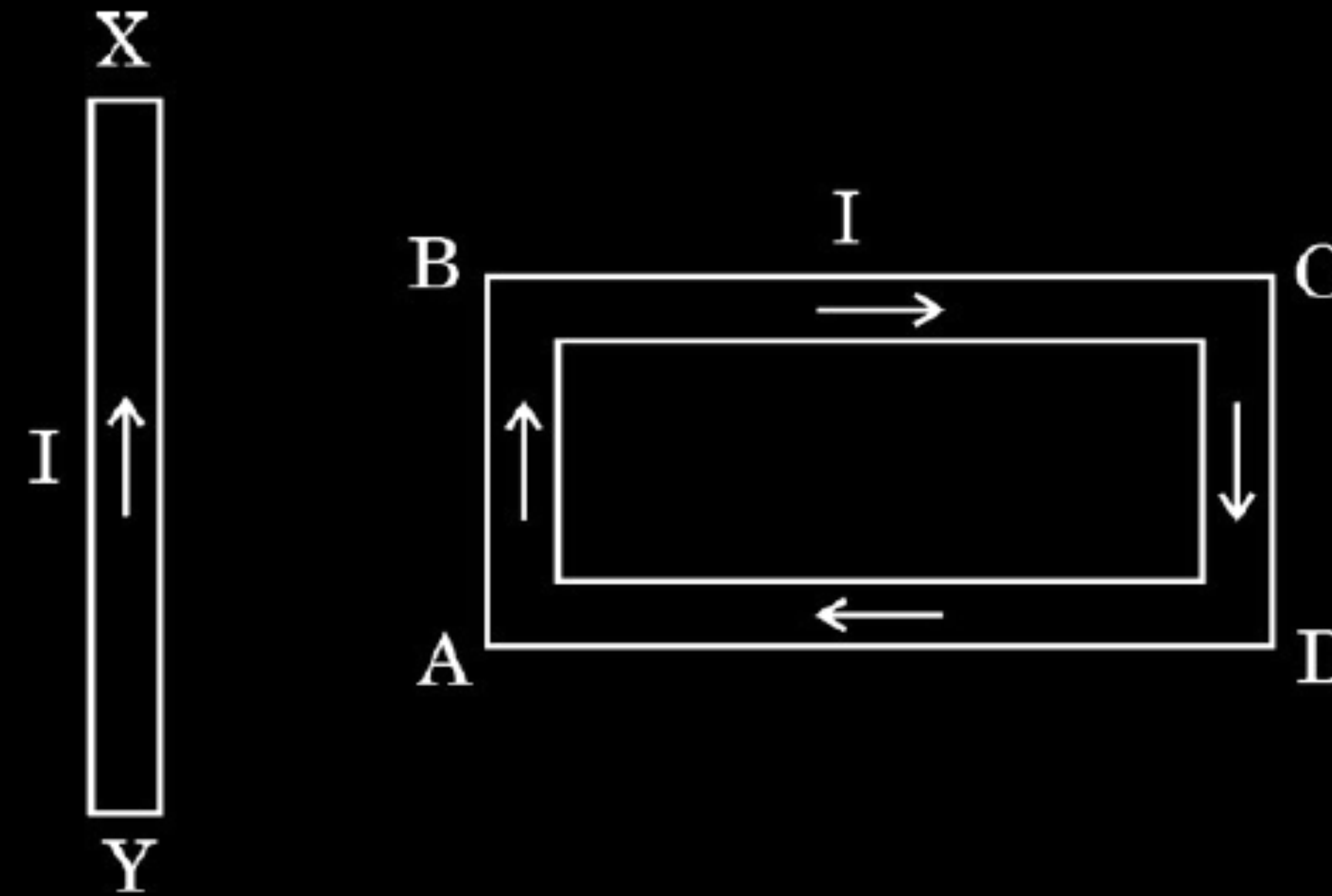
(Any one diagram)

1

- This pattern indicates that the magnetic field is uniform.

$\frac{1}{2}$

13. A rectangular loop ABCD carrying a current I is situated near a straight conductor XY, such that the conductor is parallel to the side AB of the loop and is in the plane of the loop. If a steady current I is established in the conductor as shown, the conductor XY will



CBSE 2024

- (A) remain stationary.
- (B) move towards the side AB of the loop.
- (C) move away from the side AB of the loop.
- (D) rotate about its axis.

18. *Assertion (A) :* Magnetic field lines never intersect each other.

Reason (R) : If they intersect, then at the point of intersection, the compass needle would point towards two directions, which is not possible.

CBSE 2024

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(A)/ Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

- 29.** A student fixes a sheet of white paper on a drawing board. He places a bar magnet in the centre of it. He sprinkles some iron filings uniformly around the bar magnet. Then he taps the drawing board gently and observes that the iron filings arrange themselves in a particular pattern.
- (a) Why do iron filings arrange in a particular pattern ?
 - (b) What does the crowding of iron filings at the ends of the magnet indicate ?
 - (c) What do the lines, along which the iron filings align, represent ?
 - (d) If the student places a cardboard horizontally in a current carrying solenoid and repeats the above activity, in what pattern would the iron filings arrange ? State the conclusion drawn about the magnetic field based on the observed pattern of the lines.

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3

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(a) Because a magnetic field exists around the bar magnet	$\frac{1}{2}$
(b) Strength of the magnetic field is maximum near the poles of the magnet	1
(c) The lines represent the magnetic field lines	$\frac{1}{2}$
(d) Equidistant parallel lines, magnetic field inside the solenoid is uniform	$\frac{1}{2} + \frac{1}{2}$

- 32.** (a) How is a solenoid prepared ? Differentiate between a circular coil and a solenoid.
- (b) Draw the pattern of the magnetic field lines inside a current carrying solenoid. What does this pattern indicate ?

CBSE 2024

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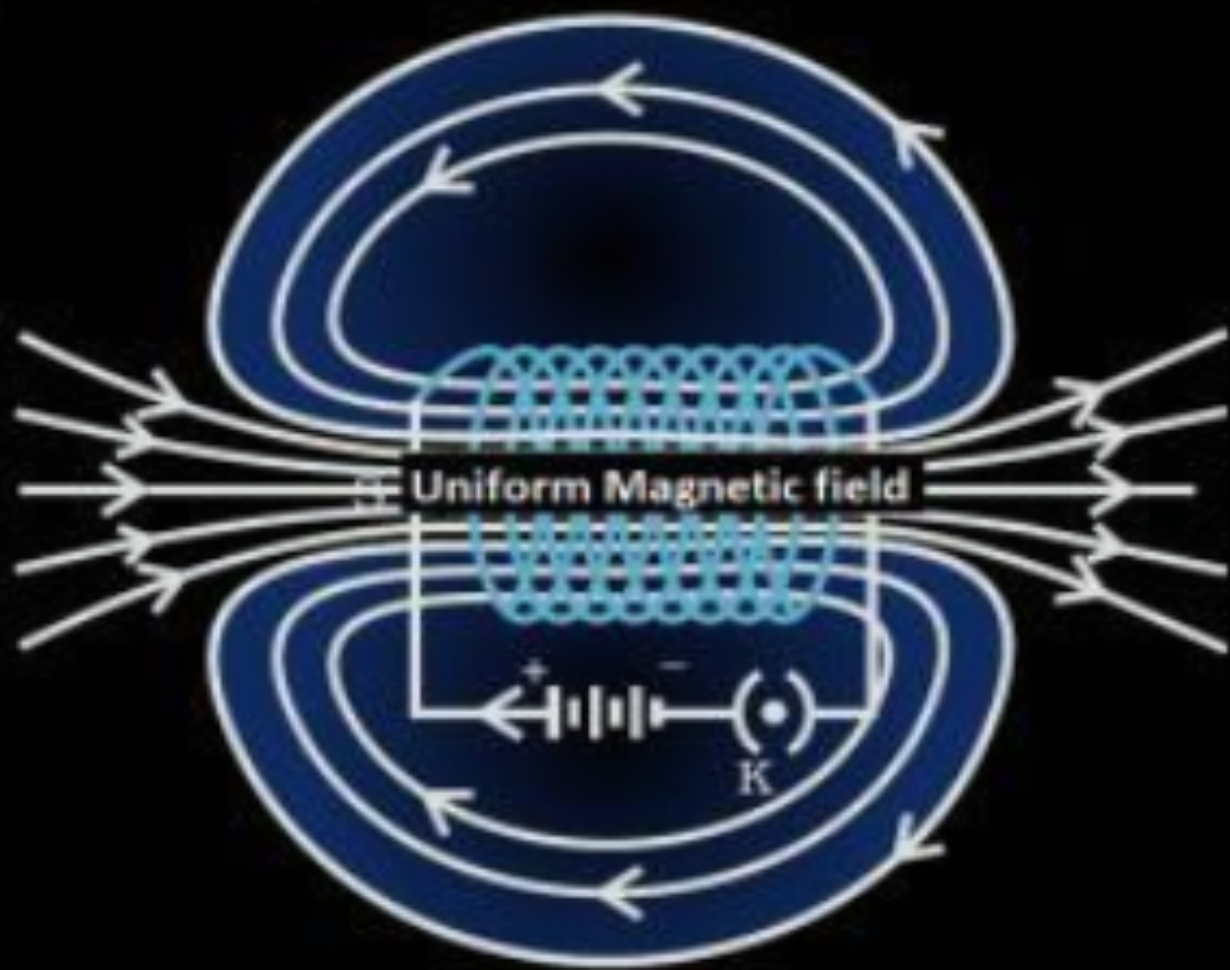
CBSE 2024

3

(a) A solenoid is made by winding insulated copper wire over a non-conducting cylindrical tube closely and tightly wound in the shape of a cylindrical spring.

Solenoid	Circular coil
Length of solenoid is much greater as compared to its diameter.	Its diameter is much greater as compared to its length.
Strong magnetic field	Weak magnetic field

(b)



- Field lines are in the form of parallel straight lines which indicates that the field is uniform.

17. *Assertion (A) :* The intensity of the magnetic field produced at the centre of a current carrying circular coil increases on increasing the number of turns in the coil.

Reason (R) : The current that flows in each turn of the coil has the same magnitude and same direction and the magnetic field, due to each turn, just adds up.

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(A) /Both (A) and (R) are true and (R) is the correct explanation of (A)

22. Write the shape of the magnetic field produced by a current carrying straight conductor. State the rule to find the direction of this field.

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22. Write the shape of the magnetic field produced by a current carrying straight conductor. State the rule to find the direction of this field.

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2

- | | |
|---|---|
| • Concentric circles around the current carrying straight conductor. | 1 |
| • Right hand thumb rule: Imagine that you are holding the current carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field. | 1 |

27. (a) (i) How does an electric fuse prevent the circuit and the appliances from possible damage due to short circuiting or overloading ?

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(ii) How is the type of current used in household supply different from the one given by dry cells or storage batteries ?

(iii) For distant transmission of electric power, an alternating current (ac) is considered to be advantageous over direct current (dc). Why ?

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CBSE 2024

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(iii) For distant transmission of electric power, an alternating current (ac) is considered to be advantageous over direct current (dc). Why ?

3

(a)

(i) the current increases and the fuse melts which breaks the circuit. So, the appliance is saved from the damage

1

(ii) Current used in household is alternating current (ac) whereas current from the dry cells / batteries are direct current (dc).

1

(iii) The distant transmission loss of electric power can be minimized by alternating current (ac).

1

(or any other alternate answer)

- (b) (i) A straight conductor is suspended horizontally with its axis perpendicular to the magnetic field of a strong horseshoe magnet. The conductor gets displaced towards the right when a steady current is passed through it. What will happen to the displacement of the conductor if the :
- (1) Polarity of the horseshoe magnet is reversed ?
 - (2) Direction of current in the conductor is reversed ?
 - (3) Current through the conductor is increased ?
- (ii) Name and state the rule for determining the direction of force on a current carrying conductor placed in a magnetic field.

(b) (i) (1) The conductor will get displaced towards the left (2) The conductor will get displaced towards the left. (3) Displacement of the conductor towards the right will increase (ii) • Fleming's left hand rule • Stretch the thumb, forefinger and the middle finger of your left hand such that they are mutually perpendicular to each other. If the forefinger points in the direction of magnetic field, and the middle finger points in the direction of electric current, then the thumb will point in the direction of motion or the force on the conductor.	$\frac{1}{2} \times 3$ $\frac{1}{2}$ 1
---	---

15. According to the International colour coding, wires used in electrical appliances which require earthing are :

- (a) Live is brown; Neutral is blue and Earth is black
- (b) Live is brown; Neutral is blue and Earth is green
- (c) Live is red; Neutral is green and Earth is blue
- (d) Live is red; Neutral is green and Earth is black

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CBSE 2023 C

- 16.** From a power station, the power is transmitted to distant places in the form of AC at a very high voltage because :
- (a) electricity is less dangerous at high voltages.
 - (b) electricity is generated at high voltage.
 - (c) transmission is faster at high voltage.
 - (d) at high voltage, the loss of energy during transmission is less.

CBSE 2023 C

- 16.** From a power station, the power is transmitted to distant places in the form of AC at a very high voltage because :
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CBSE 2023 C

20. *Assertion (A) :* A current carrying straight solenoid, when suspended freely rests in geographical north-south direction.

Reason (R) : Earth behaves as a huge magnet which has the north pole towards the geographical south, whereas the current carrying solenoid behaves like a bar magnet.

CBSE 2023 C

20. *Assertion (A) :* A current carrying straight solenoid, when suspended freely rests in geographical north-south direction.

Reason (R) : Earth behaves as a huge magnet which has the north pole towards the geographical south, whereas the current carrying solenoid behaves like a bar magnet.

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- 36.** (a) What is a solenoid ? When a current is passed through a long, hollow and straight solenoid, a magnetic field is produced. Where is the strength of the magnetic field produced maximum and what is the pattern of the magnetic field at this place ?
- (b) State Fleming's Left Hand Rule. Apply this rule to solve the following problem :
A straight conductor is placed in east-west direction in a uniform magnetic field. If on passing current from east to west, the conductor moves towards north, what is the direction of magnetic field ?

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(a)

- Solenoid –A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder.
- Inside the solenoid.
- Parallel straight magnetic field lines.

1

1

1

(b)

- Fleming's Left Hand Rule: stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular to each other. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor. /

1



- Upwards.

1

12. The shape of magnetic field lines produced (i) inside a solenoid (ii) around a straight conductor, both carrying current of the same magnitude are, respectively :

- (a) (i) straight, (ii) circular
- (b) (i) circular, (ii) circular
- (c) (i) straight, (ii) straight
- (d) (i) circular, (ii) straight

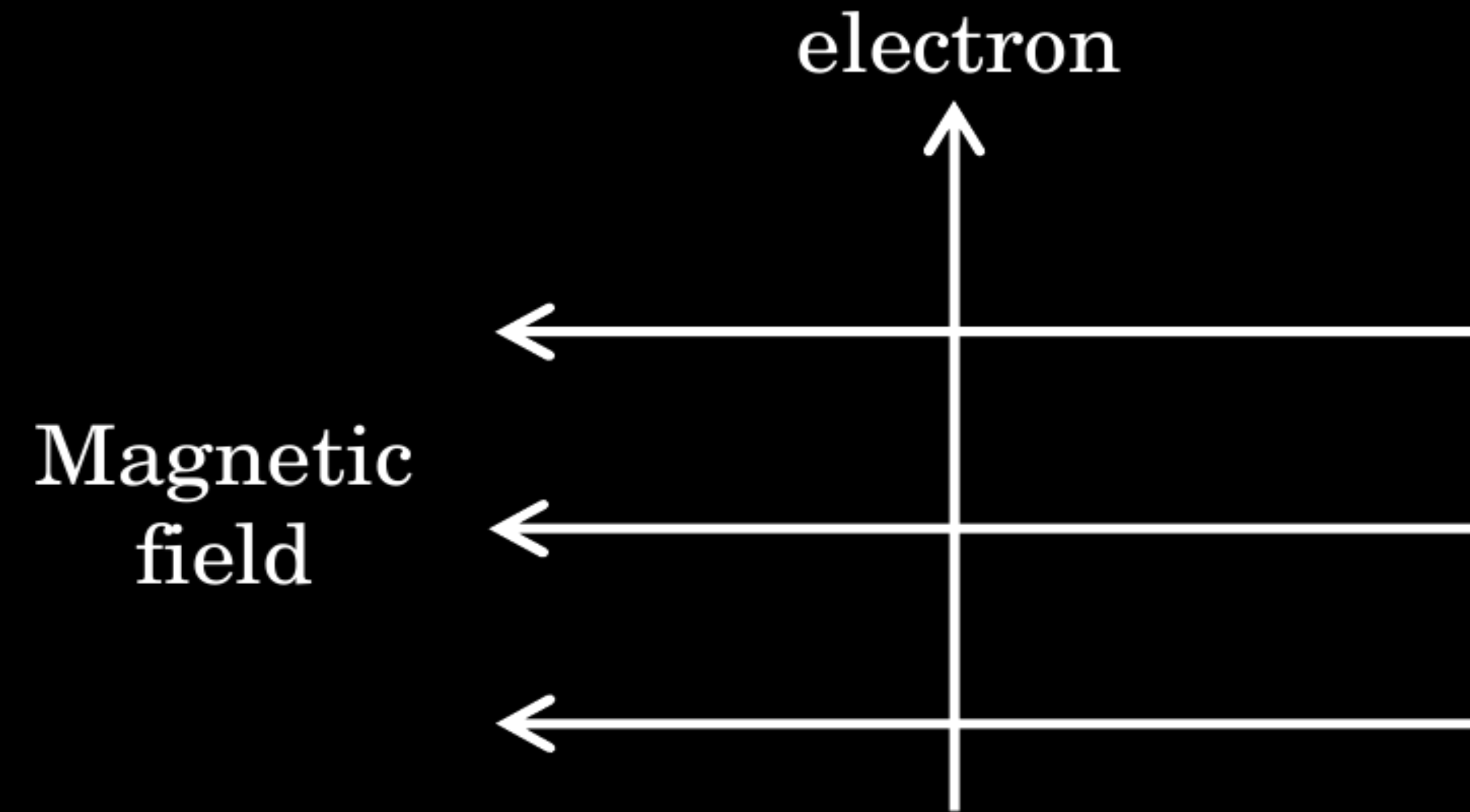
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CBSE 2023 C

- 25.** (a) State Fleming's left-hand rule.
- (b) An electron enters a magnetic field at right angles to it as shown in the following diagram. Use Fleming's left-hand rule to determine the direction of force experienced by the electron.

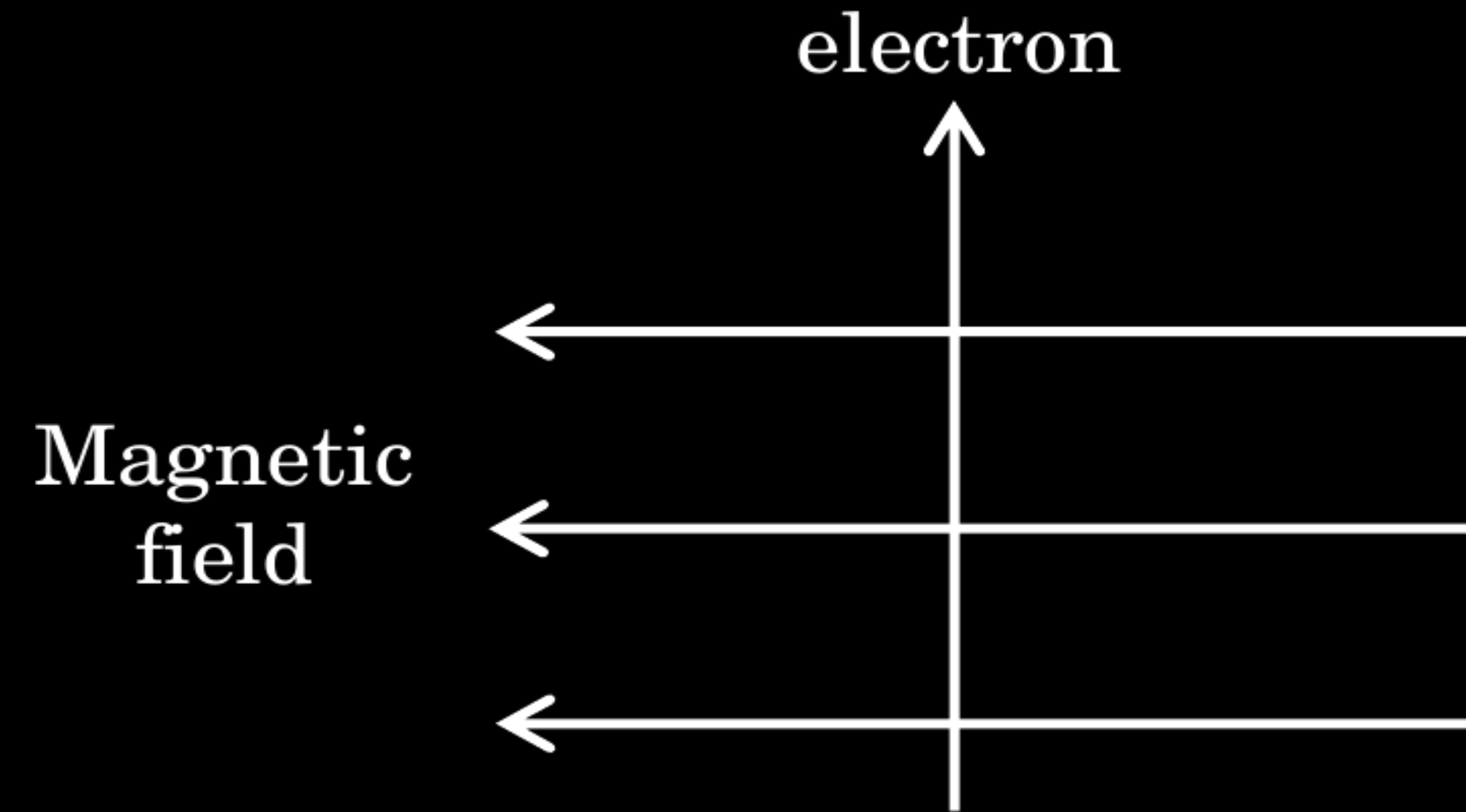
2



CBSE 2023 C

25. (a) State Fleming's left-hand rule.
- (b) An electron enters a magnetic field at right angles to it as shown in the following diagram. Use Fleming's left-hand rule to determine the direction of force experienced by the electron.

2



CBSE 2023 C



- | | |
|---|---|
| (a) | |
| <ul style="list-style-type: none"> According to Fleming's left hand rule, stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular to each other. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor. / | 1 |
| (b) Into the plane of the paper (downward) | 1 |

- 30.** (a) (i) Differentiate between a solenoid and a circular coil.
- (ii) Explain how a solenoid can be made in a school laboratory.
- (iii) Write one use of the strong magnetic field produced inside a current carrying solenoid.

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- 30.** (a) (i) Differentiate between a solenoid and a circular coil.
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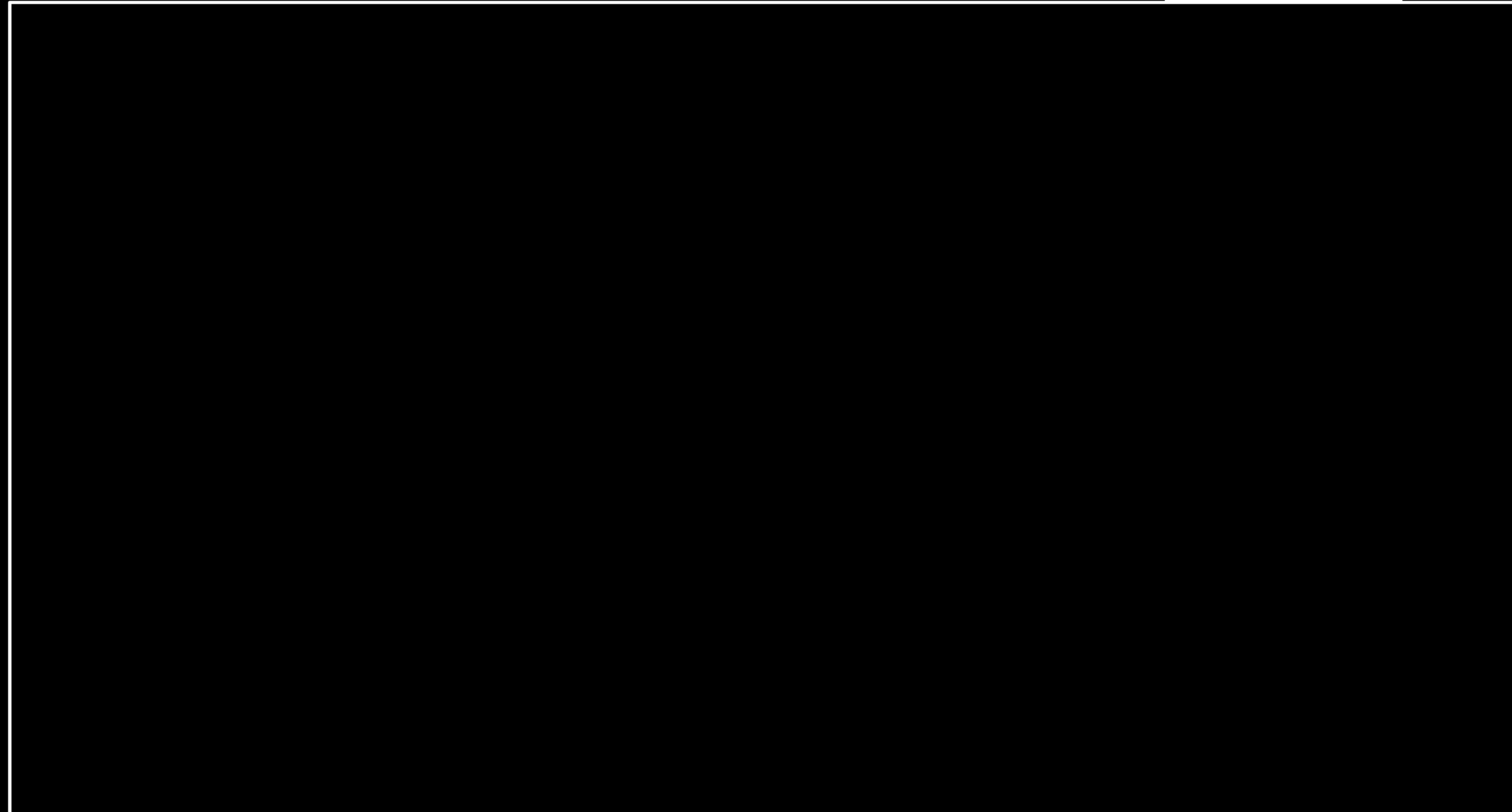
3

(a)	
(i) Solenoid : A coil of many turns of insulated copper wire wrapped closely in the shape of cylinder. Circular coil : Straight wire bent in the form of circular loop with many turns.	1
(ii) By taking a non-conducting cylindrical tube and winding a long, insulated copper wire tightly over it in the shape of a spring such that the turns are closely placed and lie side by side.	1
(iii) To magnetise a piece of magnetic material like soft iron / To make an electromagnet (any one)	1

- (b) With the help of a labelled circuit diagram, illustrate the pattern of the magnetic field lines of the magnetic field produced around a straight current carrying conductor. Explain how, with the help of right-hand thumb rule, we can determine and mark the direction of magnetic field lines due to a current.

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3



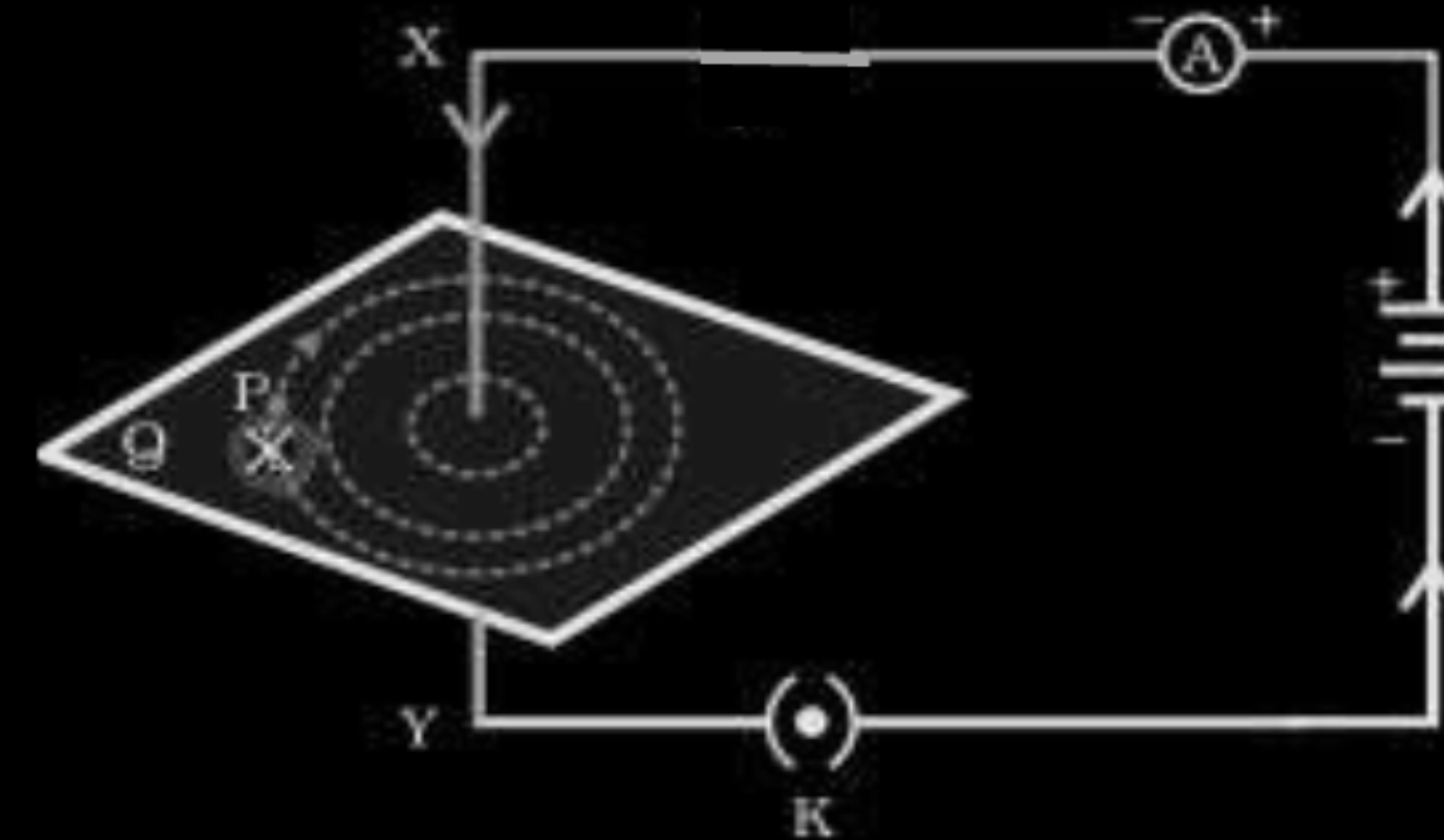
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CBSE 2023 C

3

(b)

-



Diagram

1

Labelling of magnetic field

1

(Please check the direction of magnetic field lines corresponding to the direction of current in the conductor.)

- Right hand Thumb Rule : Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field.

1

25. Differentiate between direct and alternating current. Name the type of current produced by the power plants in our country. Also state its frequency.

2

CBSE 2023 C

25. Differentiate between direct and alternating current. Name the type of current produced by the power plants in our country. Also state its frequency.

2

CBSE 2023 C

- | | |
|---|---------------|
| • DC always flow in one direction while AC reverses its direction periodically. | 1 |
| • AC /Alternating Current | $\frac{1}{2}$ |
| • 50 Hz | $\frac{1}{2}$ |

- 25.** What will happen if an electric oven of rating 3 kW; 220 V is operated in a domestic electric circuit (220 V) that has a current rating of 10 A ? Give reason to justify your answer.

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2

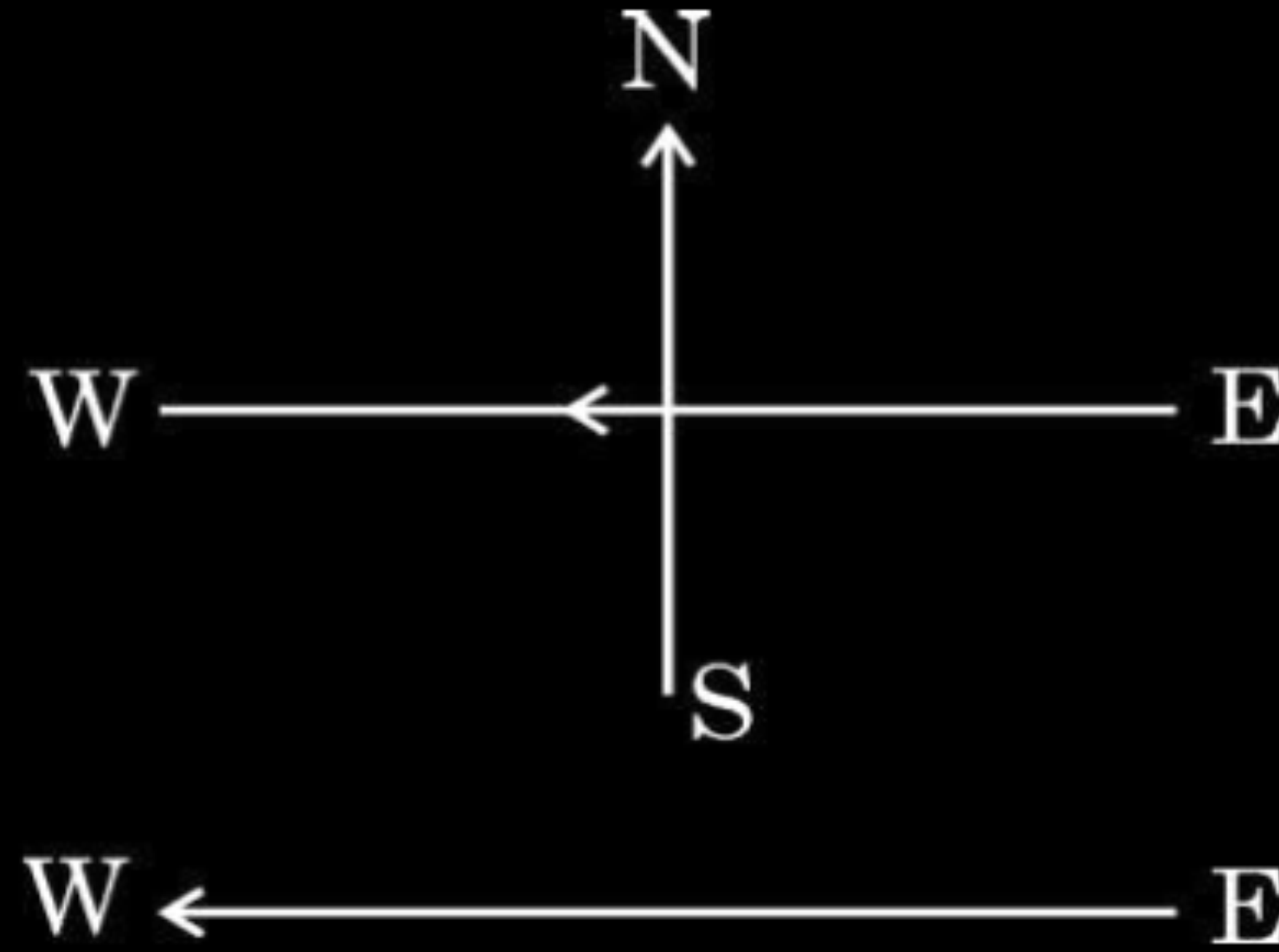
- 25.** What will happen if an electric oven of rating 3 kW; 220 V is operated in a domestic electric circuit (220 V) that has a current rating of 10 A ? Give reason to justify your answer.

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2

Short circuiting / fuse will blow or melt / break in circuit. (any one)	1
Justification: $I = \frac{P}{V} = \frac{3000 \text{ W}}{220 \text{ V}} = 13.6 \text{ A}$ As the current rating is only 10 A and the oven draws a current of 13.6 A, which is more than the current rating. Hence the oven will stop working.	1

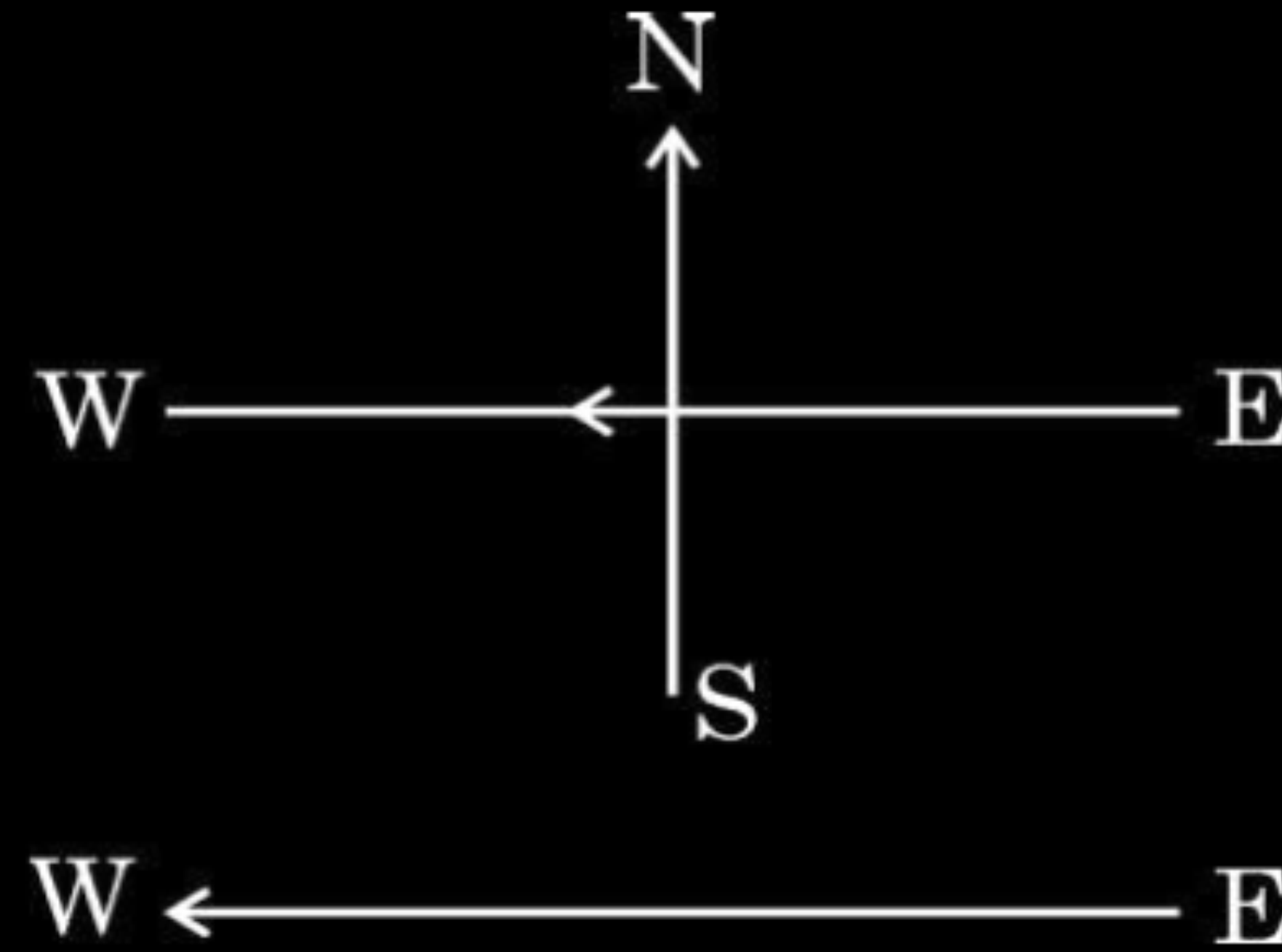
13. A constant current flows in a horizontal wire in the plane of the paper from east to west as shown in the figure. The direction of the magnetic field will be north to south at a point :



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- (a) directly above the wire.
- (b) directly below the wire.
- (c) located in the plane of the paper on the north side of the wire.
- (d) located in the plane of the paper on the south side of the wire.

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- (a) directly above the wire.
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- 15.** For a current in a long straight solenoid, N and S poles are created at the two ends. Among the following statements, the *incorrect* statement is :
- (a) The magnetic field lines inside the solenoid are in the form of straight lines, which indicates that the magnetic field is uniform at all points inside the solenoid.
 - (b) The strong magnetic field produced inside the solenoid can magnetize the soft iron placed inside it.
 - (c) The pattern of the magnetic field associated with a current carrying solenoid is different from the pattern of the magnetic field around a bar magnet.
 - (d) The N and S poles exchange positions when the direction of current through the solenoid is reversed.

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 - (d) The N and S poles exchange positions when the direction of current through the solenoid is reversed.

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(c)

19. *Assertion (A):* The magnetic field lines around a current carrying straight wire do not intersect each other.

Reason (R): The magnitude of the magnetic field produced at a given point increases as the current through the wire increases.

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19. *Assertion (A):* The magnetic field lines around a current carrying straight wire do not intersect each other.

Reason (R): The magnitude of the magnetic field produced at a given point increases as the current through the wire increases.

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- 31.** (a) (i) A straight cylindrical conductor is suspended with its axis perpendicular to the magnetic field of a horse-shoe magnet. The conductor gets displaced towards left when a current is passed through it. What will happen to the displacement of the conductor if the
- (1) current through it is increased ?
 - (2) horse-shoe magnet is replaced by another stronger horse-shoe magnet ?
 - (3) direction of current through it is reversed ?
- (ii) Name and state the rule for determining the direction of force on a current carrying conductor in a magnetic field.

3

- (a) (i) (1) Increased.
(2) Increased.
(3) The direction of displacement is reversed
- (ii) • Flemings left-hand rule
• According to this rule, stretch the thumb, forefinger and middle finger of left hand such that they are mutually perpendicular. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor.

$\frac{1}{2}$

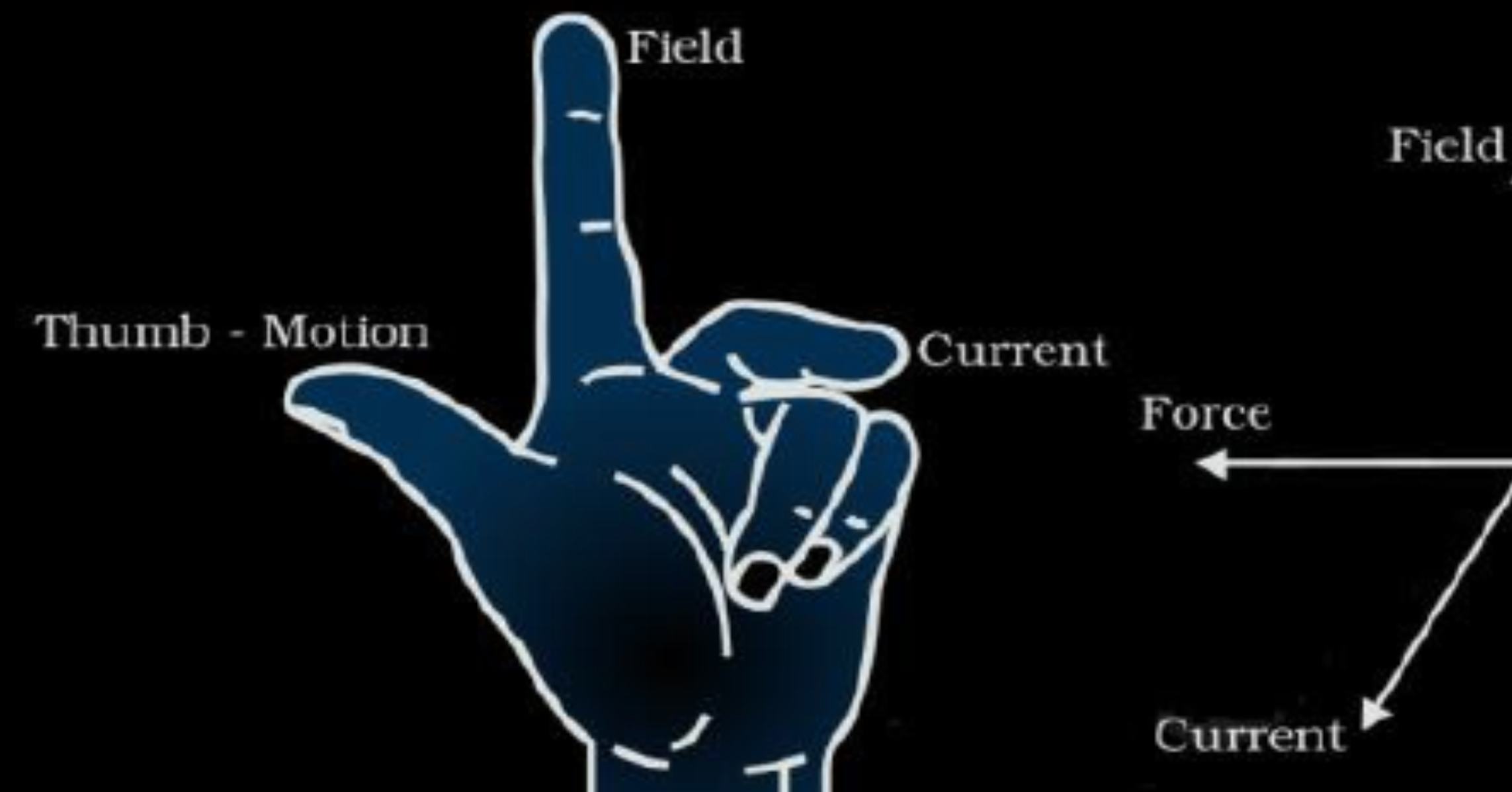
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

Alternate answer of the statement

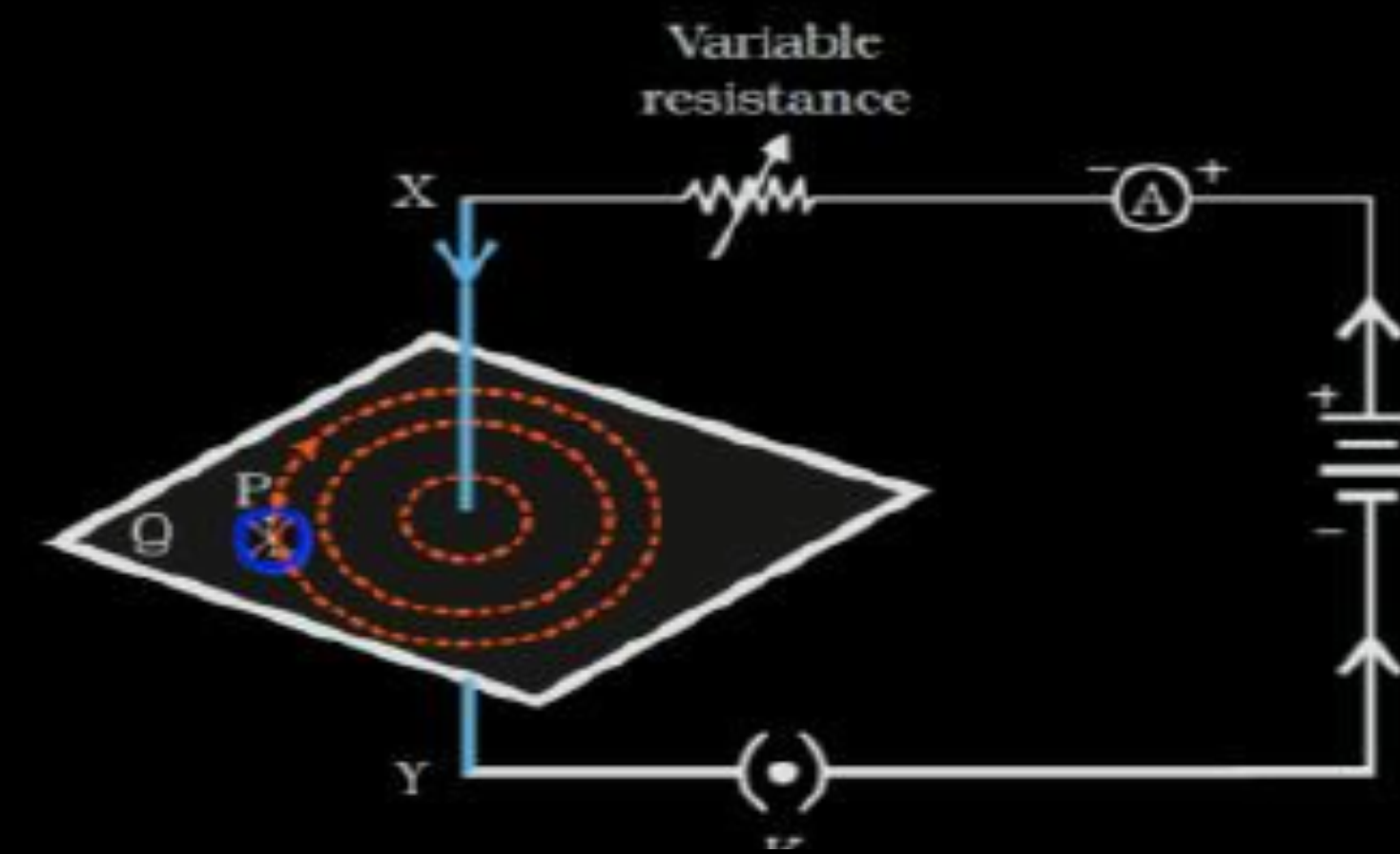


All the physical quantities mentioned in the diagram should be perpendicular to each other.

- (b) Draw the pattern of the magnetic field produced around a vertical current carrying straight conductor passing through a horizontal cardboard. Mark the direction of current and the magnetic field lines. Name and state the rule which is used to determine the direction of magnetic field associated with a current carrying conductor.

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(b) •



- Pattern
- Direction of current
- Magnetic field direction

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

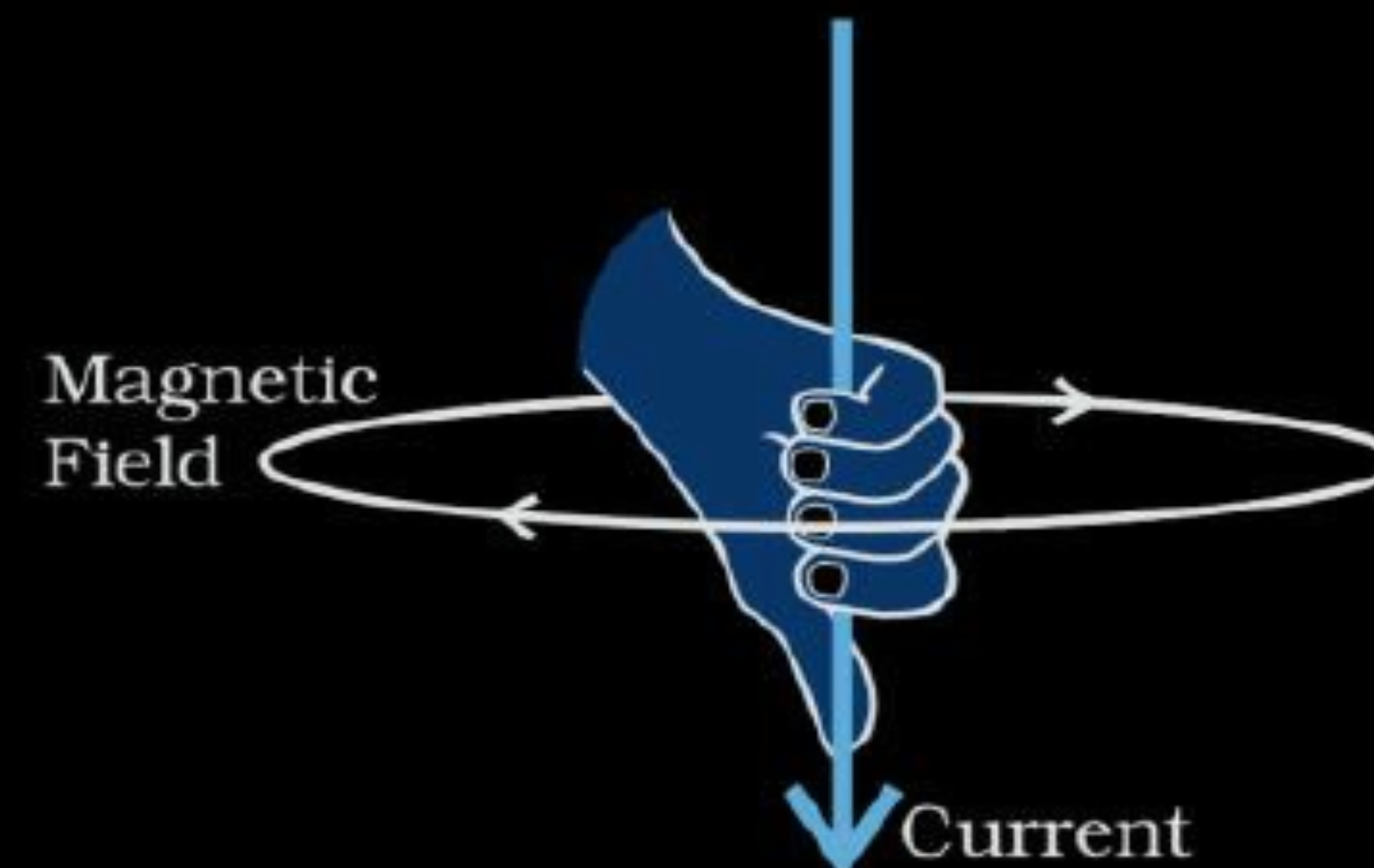
(Note: If the variable resistor and ammeter are not drawn, do not deduct marks.)

- Right hand thumb Rule
- Imagine that we are holding a current carrying straight conductor in our right hand such that the thumb points towards the direction of current. Then our finger will wrap around the conductor in the direction of the field lines of the magnetic field.

$\frac{1}{2}$

1

Alternate answer of the statement:



14. An electric kettle consumes 1 kW of electric power when operated at 220 V. The minimum rating of the fuse wire to be used for it is

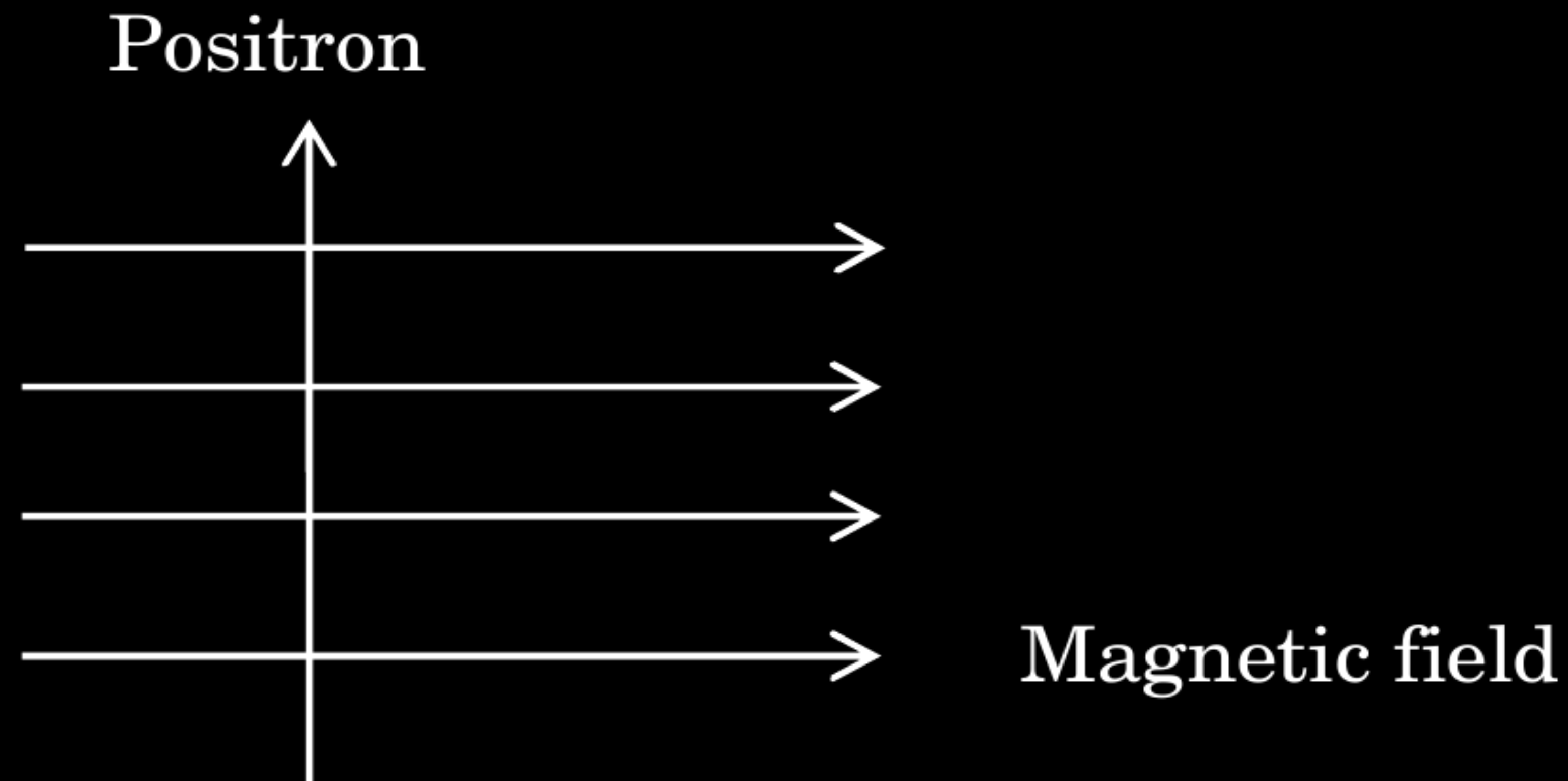
- (a) 1 A
- (b) 2 A
- (c) 4 A
- (d) 5 A

14. An electric kettle consumes 1 kW of electric power when operated at 220 V. The minimum rating of the fuse wire to be used for it is

- (a) 1 A
- (b) 2 A
- (c) 4 A
- (d) 5 A

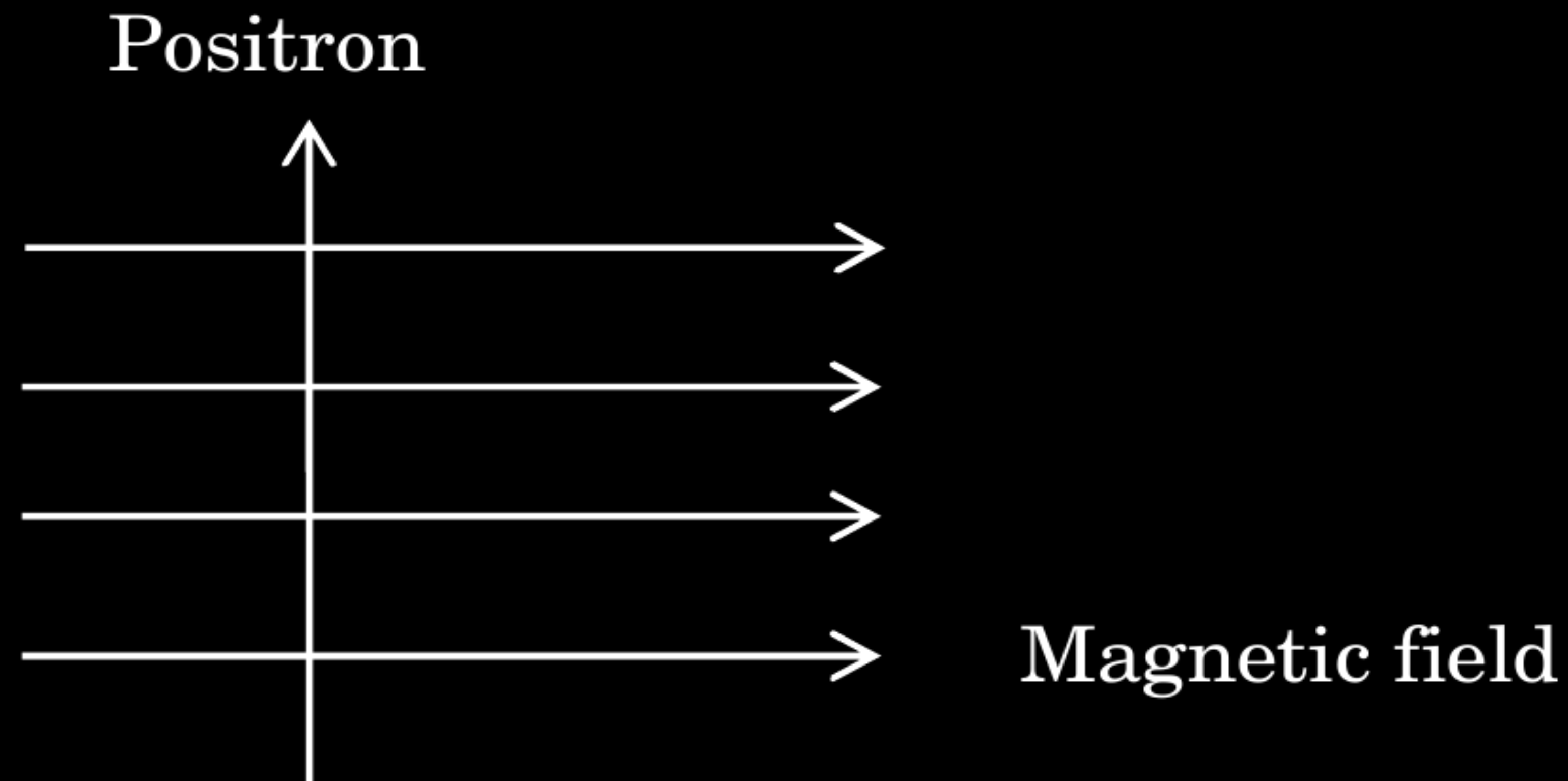
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15. A positron enters a uniform magnetic field at right angles to it as shown. The direction of force experienced by the positron will be :



- (a) to the right
- (b) to the left
- (c) into the page
- (d) out of the page

15. A positron enters a uniform magnetic field at right angles to it as shown. The direction of force experienced by the positron will be :



- (a) to the right
- (b) to the left
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- (d) out of the page

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CBSE 2023

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- 32.** (a) List two factors on which the magnitude of magnetic field produced by a current carrying straight conductor depends.
- (b) State the rule which determines the direction of magnetic field in the above case.
- (c) Draw the pattern of magnetic field lines produced in this case.

CBSE 2023

- 32.** (a) List two factors on which the magnitude of magnetic field produced by a current carrying straight conductor depends.
- (b) State the rule which determines the direction of magnetic field in the above case.
- (c) Draw the pattern of magnetic field lines produced in this case.

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3

(a) Two factors:

- (i) Current passing through the conductor
- (ii) Distance of the point from the conductor

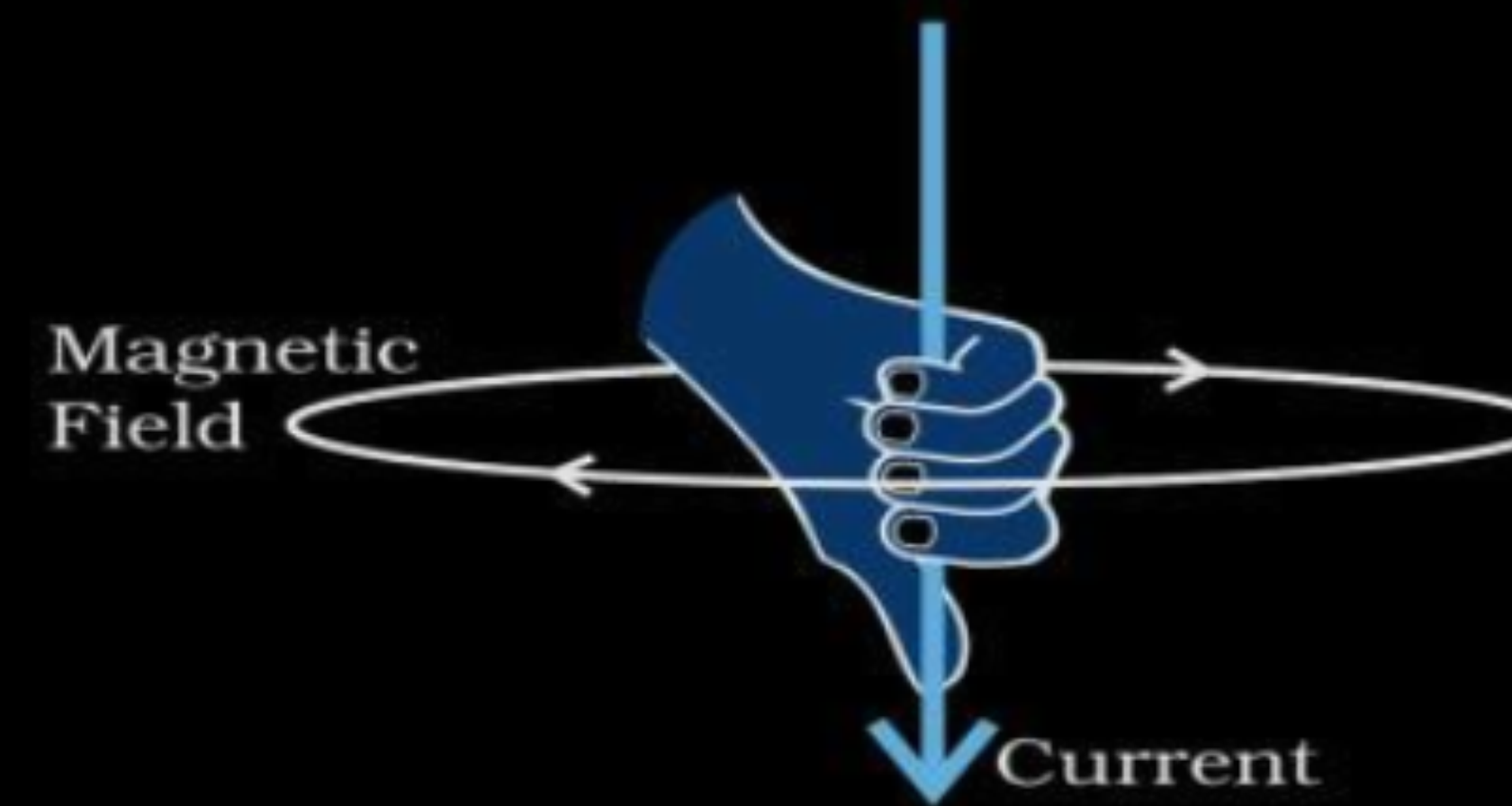
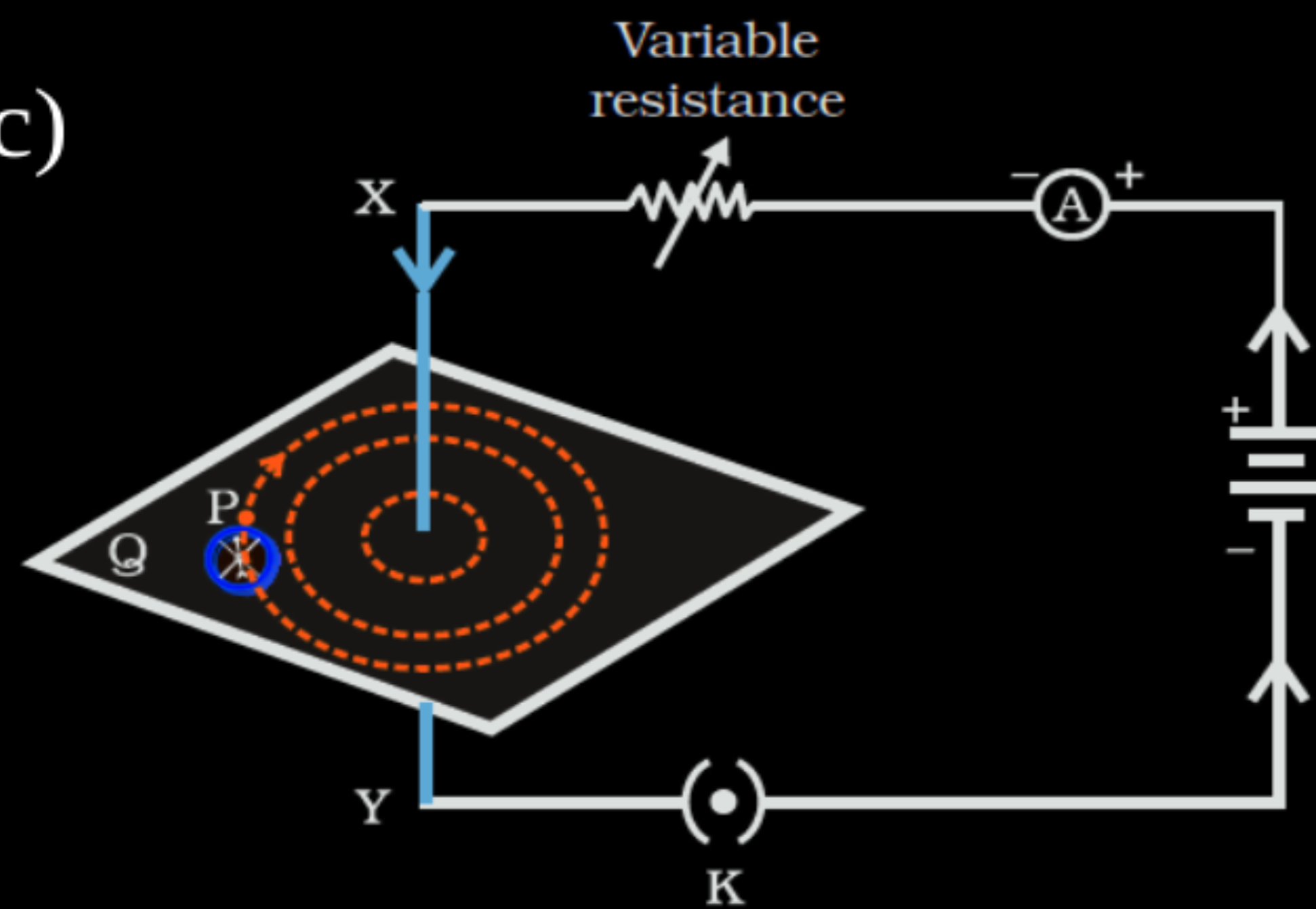
$\frac{1}{2}$

$\frac{1}{2}$

(b) Statement of Right-hand thumb rule- If we stretch the thumb of our right hand in the direction of current, then the direction in which the fingers will curl is the direction of magnetic field lines.

1

(c)



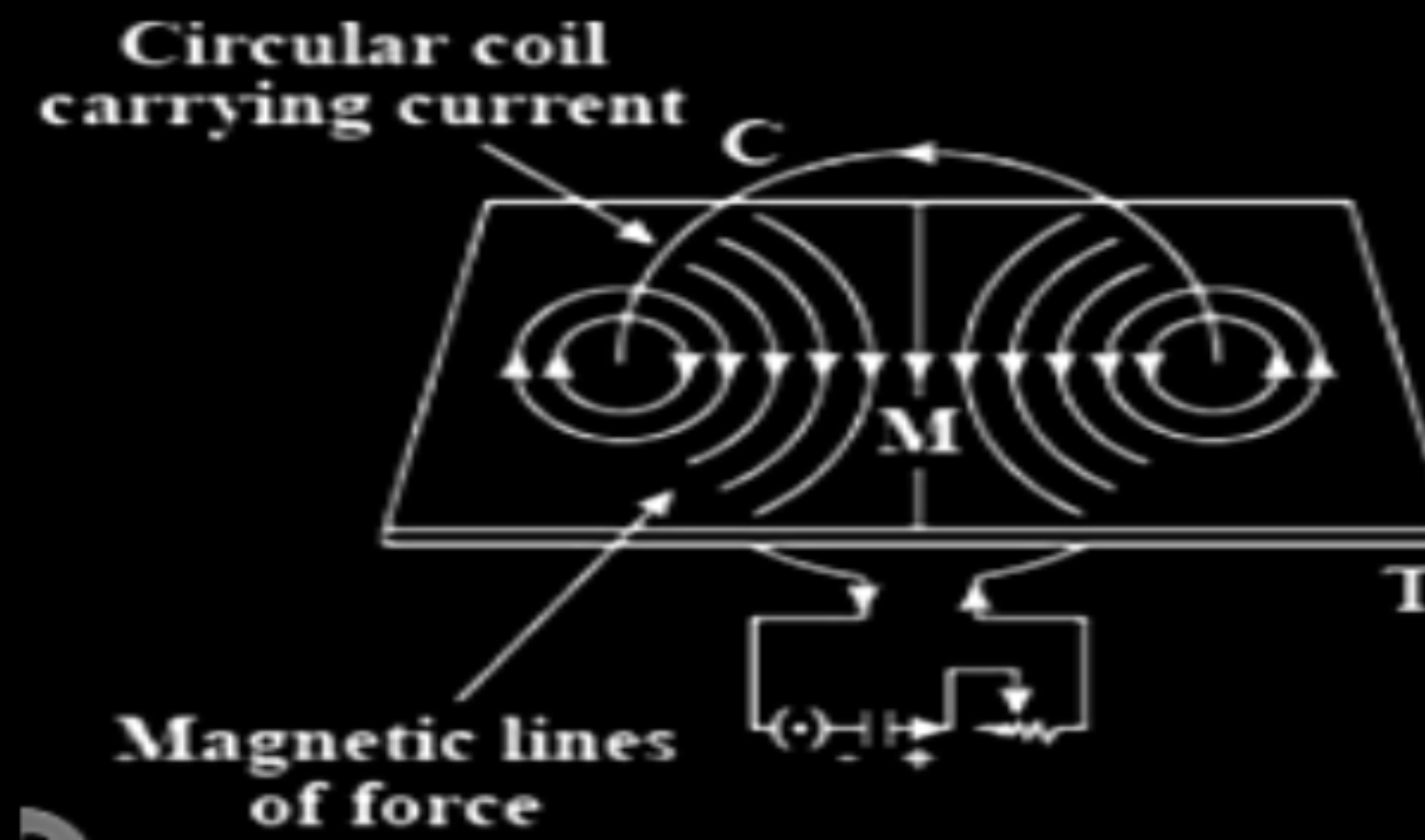
1

- 25.** Consider a circular loop of thick copper wire lying horizontally on a table. Let the current pass through the loop anticlockwise. Draw the magnetic field lines to show the direction and the pattern of the magnetic field inside and outside the loop.

25. Consider a circular loop of thick copper wire lying horizontally on a table. Let the current pass through the loop anticlockwise. Draw the magnetic field lines to show the direction and the pattern of the magnetic field inside and outside the loop.

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2



Direction of current / Polarity of cell

$\frac{1}{2}$

Direction of Magnetic field lines

$\frac{1}{2}$

Pattern of magnetic field lines.

1

32. What is a solenoid ? Draw a diagram to show the pattern of magnetic field around a current carrying solenoid. Name the region of uniform magnetic field.

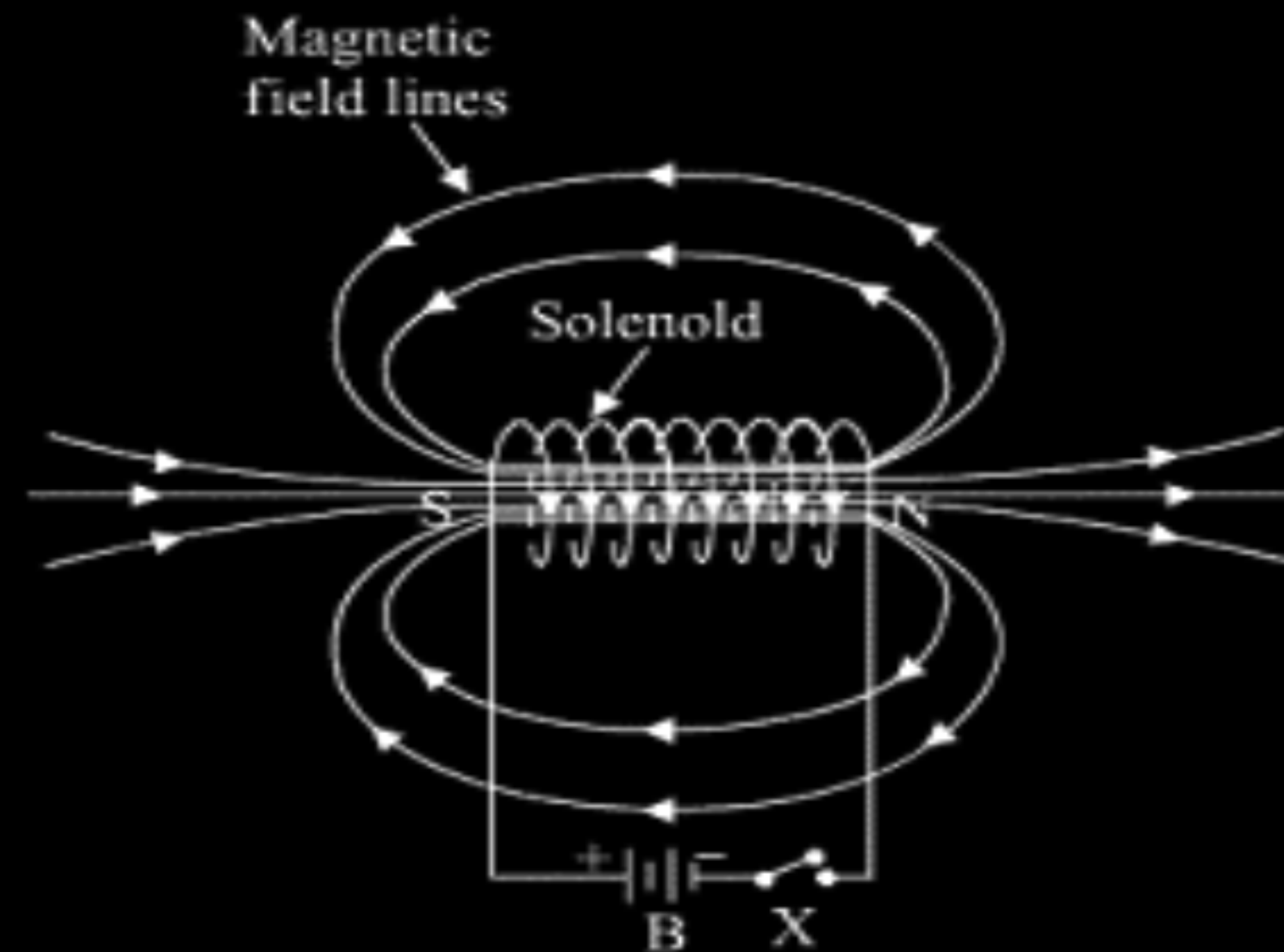
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32. What is a solenoid ? Draw a diagram to show the pattern of magnetic field around a current carrying solenoid. Name the region of uniform magnetic field.

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3

- A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder is called a solenoid.



(Deduct $\frac{1}{2}$ mark if direction of current or magnetic field not marked)

- Field is uniform inside the solenoid.

1

1

1

- 32.** Explain the meaning of overloading of an electrical circuit. List two possible causes due to which overloading may occur in household circuits. Write one preventive measure that should be taken to avoid overloading of domestic circuits.

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- 32.** Explain the meaning of overloading of an electrical circuit. List two possible causes due to which overloading may occur in household circuits. Write one preventive measure that should be taken to avoid overloading of domestic circuits.

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• Overloading : Overloading of an electrical circuit happens when an excessive amount of electric current flows through the wires.	1
• Two causes : (i) Contact between live and neutral wire/ short circuiting (ii) Connecting too many appliances to a single socket. (iii) Accidental hike in supply voltage	$\frac{1}{2}$ $\frac{1}{2}$
• Preventive measure : (i) To use wires of proper insulation/proper rating (ii) Not connecting too many appliances to a single socket. (iii) fuse wires	(Any two points) (Any other) (Any one) 1

15. An electron projected towards east is deflected towards south by a uniform magnetic field. The direction of the magnetic field is :

(a) upwards

(b) downwards

(c) towards north

(d) towards west

15. An electron projected towards east is deflected towards south by a uniform magnetic field. The direction of the magnetic field is :

(a) upwards

(b) downwards

(c) towards north

(d) towards west

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(b)

16. A student has a long straight thick copper wire. If he passes a current in this wire he would get a magnetic field whose pattern must be :

- (a) radial lines originating from the wire.
- (b) concentric circles centred around the wire.
- (c) straight lines parallel to the wire.
- (d) straight lines perpendicular to the wire.

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20. *Assertion (A) :* Magnetic fields interact with moving charges and not with charges at rest.

Reason (R) : Charges produce magnetic field only when they move.

CBSE 2023

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Reason (R) : Charges produce magnetic field only when they move.

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(a)

32. What is a solenoid ? How does a current carrying solenoid behave when suspended freely and why ? Give its one main use.

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3

- | | |
|---|---|
| • Solenoid –A coil of many circular turns of insulated copper wire wrapped closely in the shape of cylinder. | 1 |
| • A current carrying solenoid behaves like a strong bar magnet and when suspended freely, it stays in the geographical north-south direction. | 1 |
| • Used in making permanent magnets/electromagnets. | 1 |

13. The magnetic field inside a long straight current carrying solenoid :

1

- (a) is zero.
- (b) decreases as we move towards its end.
- (c) increases as we move towards its end.
- (d) is same at all points.

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13. The magnetic field inside a long straight current carrying solenoid :

1

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(d)

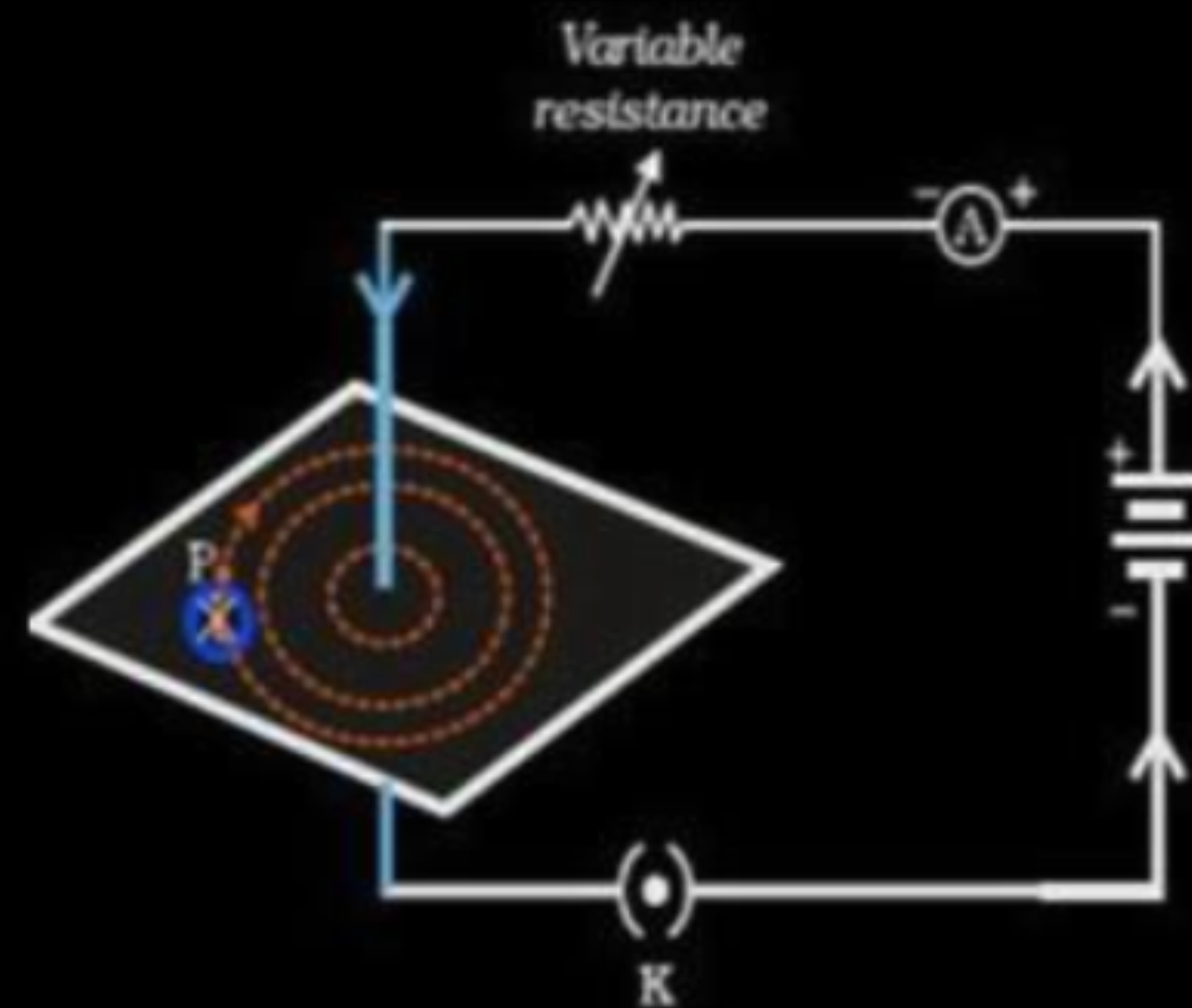
26. Draw magnetic field lines produced around a straight current carrying conductor passing through a cardboard. How will the strength of the magnetic field change when the point where magnetic field is to be determined is moved away from the conductor ?

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26. Draw magnetic field lines produced around a straight current carrying conductor passing through a cardboard. How will the strength of the magnetic field change when the point where magnetic field is to be determined is moved away from the conductor ?

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2



- Decreases

Note- if ammeter and rheostat are not drawn → do not deduct marks.

1

1

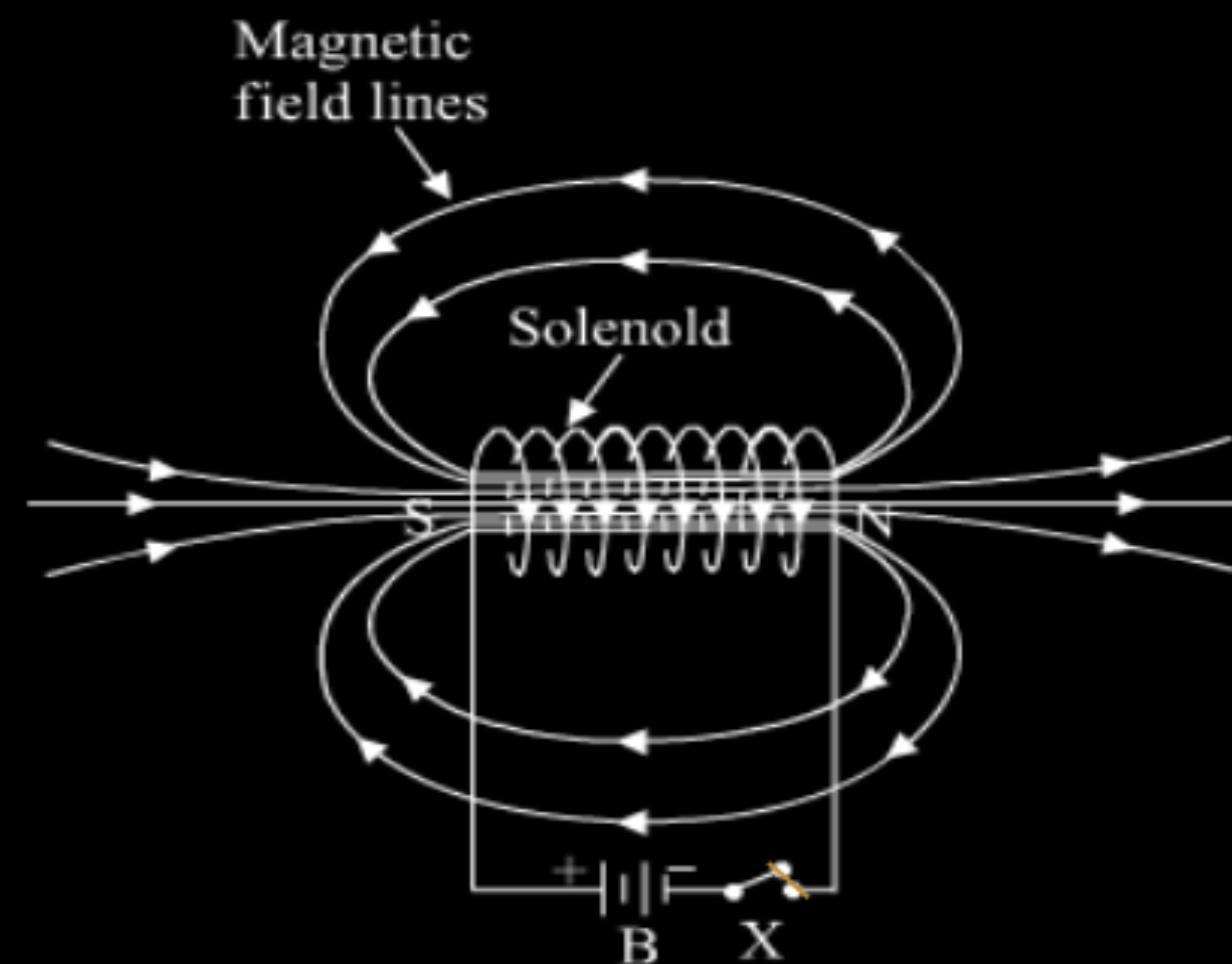
27. What is a solenoid ? When does a solenoid behave as a magnet ? Draw the pattern of the magnetic field produced inside it showing the directions of the magnetic field lines.

27. What is a solenoid ? When does a solenoid behave as a magnet ? Draw the pattern of the magnetic field produced inside it showing the directions of the magnetic field lines.

3

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- A solenoid is a coil of many turns of insulated copper wires wrapped closely in the shape of a cylinder
- When electric current is passed through it



(Deduct $\frac{1}{2}$ mark if direction of current or magnetic field is not marked.)

23. State the rule to determine the direction of a (a) magnetic field produced around a straight conductor carrying current and (b) force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it.

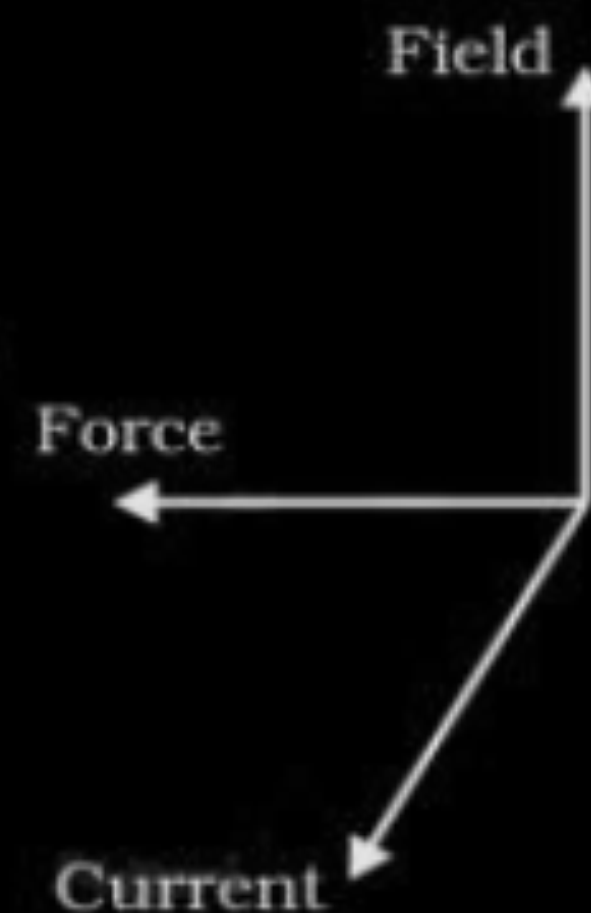
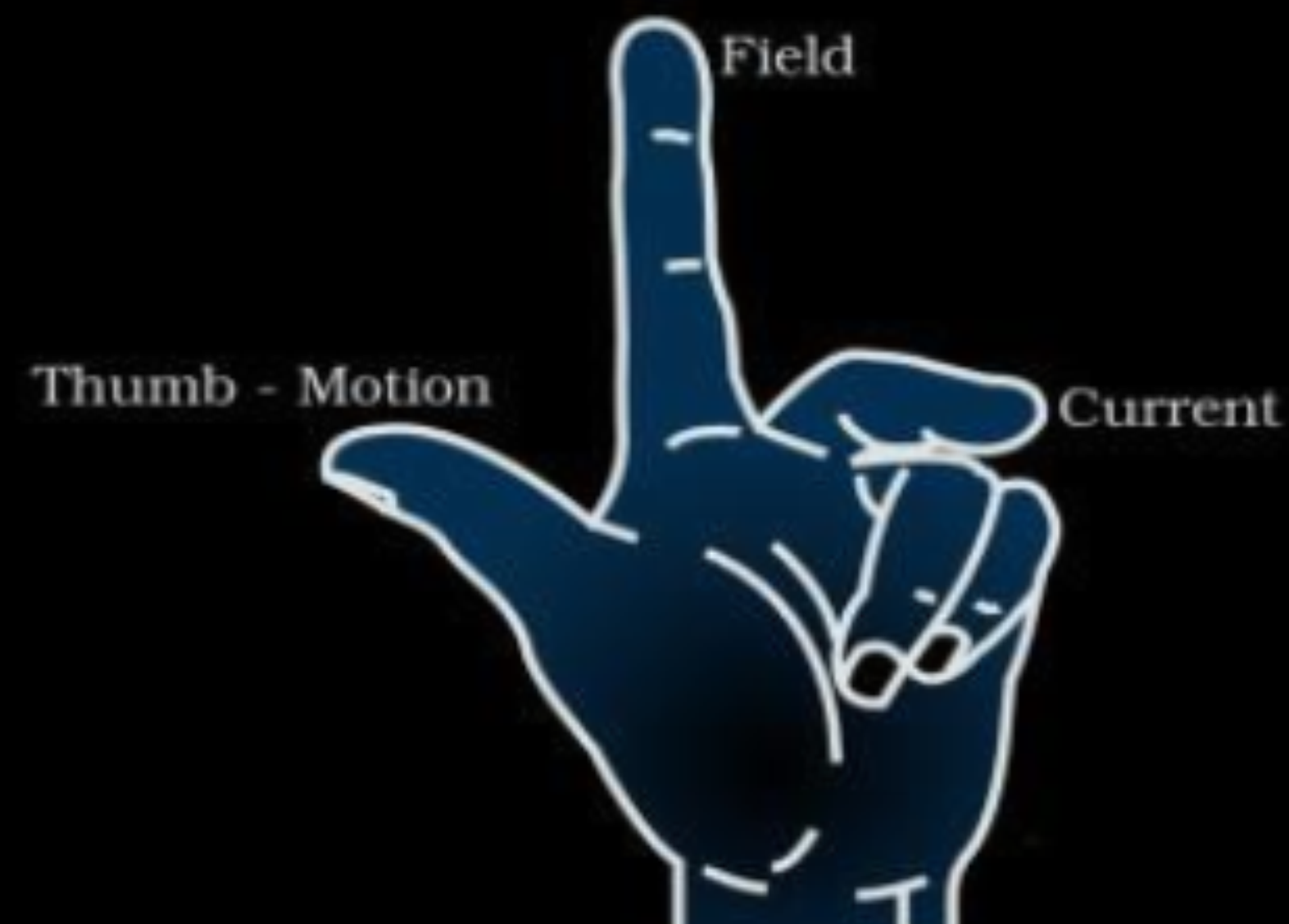
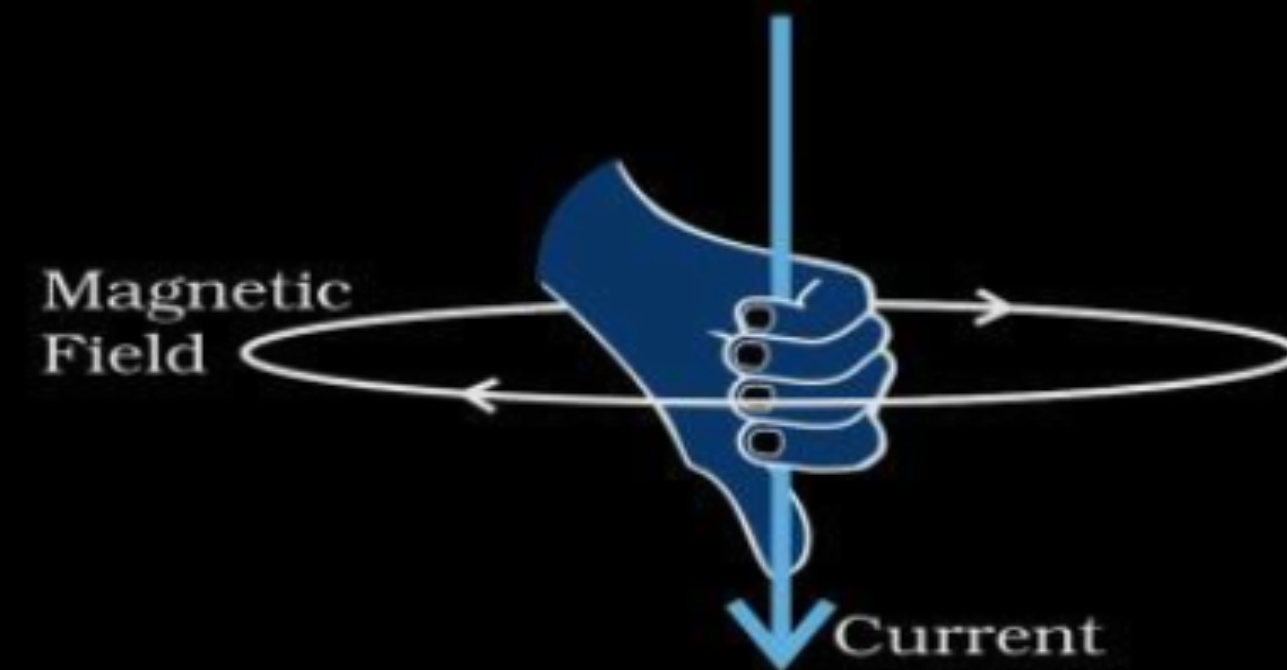
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23. State the rule to determine the direction of a (a) magnetic field produced around a straight conductor carrying current and (b) force experienced by a current carrying straight conductor placed in a magnetic field which is perpendicular to it.

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2

Alternate answer of the statement



(a)**Right-Hand Thumb Rule** : Hold the current carrying conductor in right hand, such that thumb indicates direction of current, then the fingers will wrap around conductor in the direction of field lines of the magnetic field.

(b)**Fleming's Left Hand Rule**: Stretch the forefinger, middle finger, and thumb of the left hand such that they are mutually perpendicular to each other. If the forefinger indicates the direction of the magnetic field, the middle finger indicates the direction of current, then the thumb points in the direction of motion or force acting on the conductor.

All the physical quantities mentioned in the diagram are mutually perpendicular to each other.

1

1

13. In domestic electric circuits the wiring with 15 A current rating is for the electric devices which have

1

- (a) higher power ratings such as geyser.
- (b) lower power ratings such as fan.
- (c) metallic bodies and low power ratings.
- (d) non-metallic bodies and low power ratings.

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13. In domestic electric circuits the wiring with 15 A current rating is for the electric devices which have

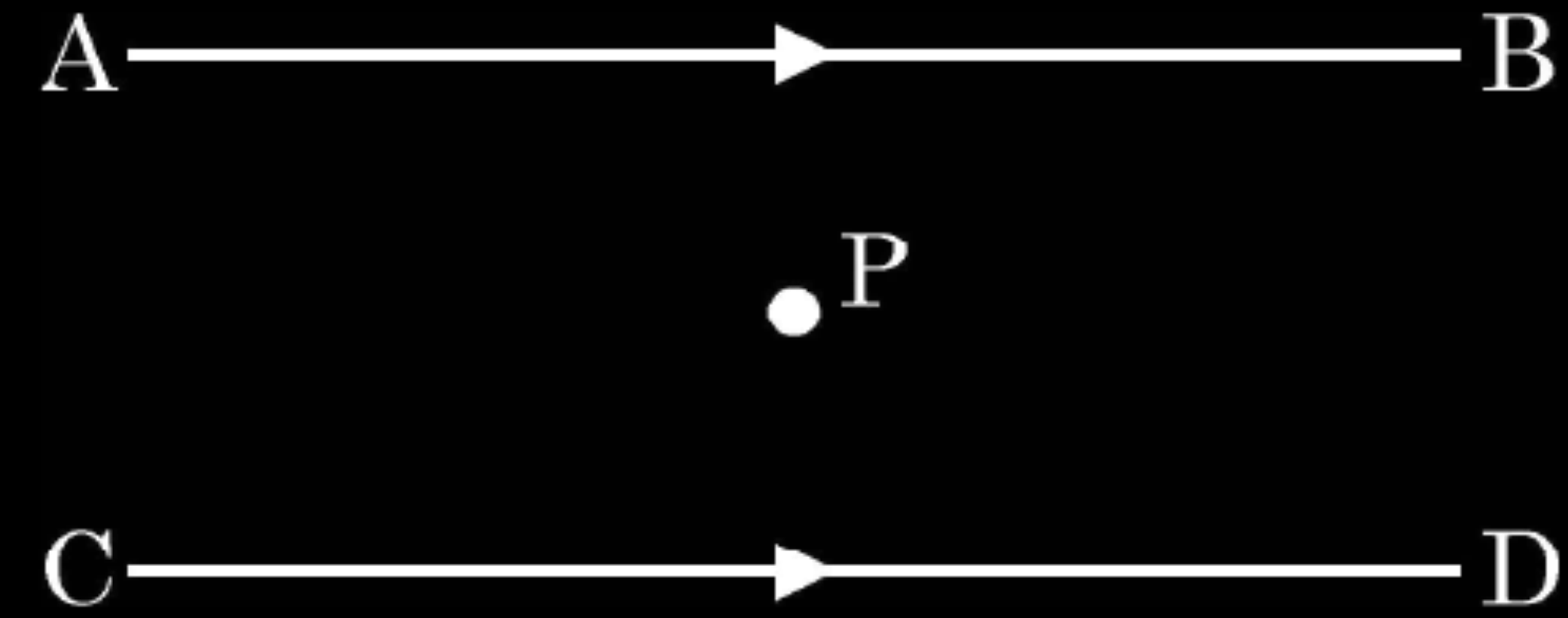
1

- (a) higher power ratings such as geyser.
- (b) lower power ratings such as fan.
- (c) metallic bodies and low power ratings.
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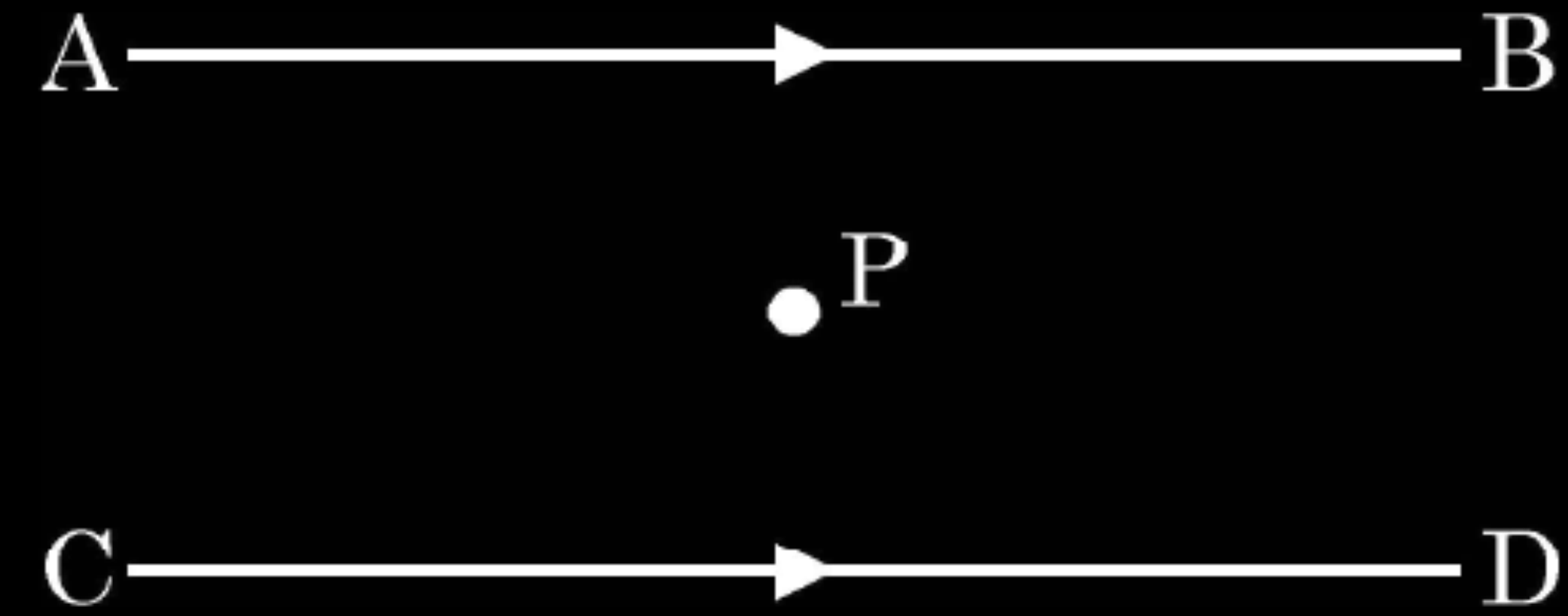
(a)

15. The resultant magnetic field at point 'P' situated midway between two parallel wires (placed horizontally) each carrying a steady current I is **1**



- (a) in the same direction as the current in the wires.
- (b) in the vertically upward direction.
- (c) zero
- (d) in the vertically downward direction.

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(c)

19. **Assertion (A)** : The strength of the magnetic field produced at the centre of a current carrying circular coil increases on increasing the number of turns in it.

1

Reason (R) : The current in each circular turn has the same direction and the magnetic field due to each turn then just adds up.

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19. **Assertion (A)** : The strength of the magnetic field produced at the centre of a current carrying circular coil increases on increasing the number of turns in it.

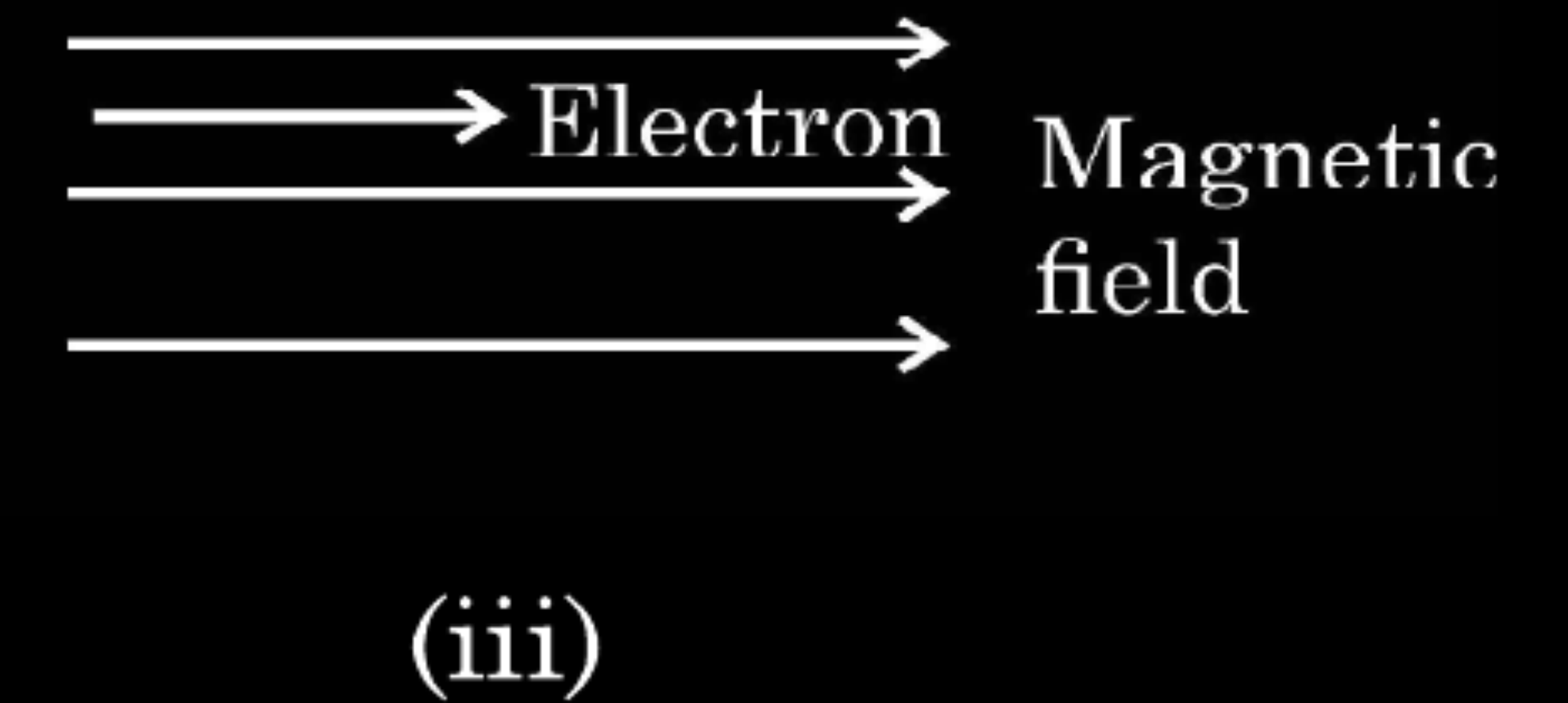
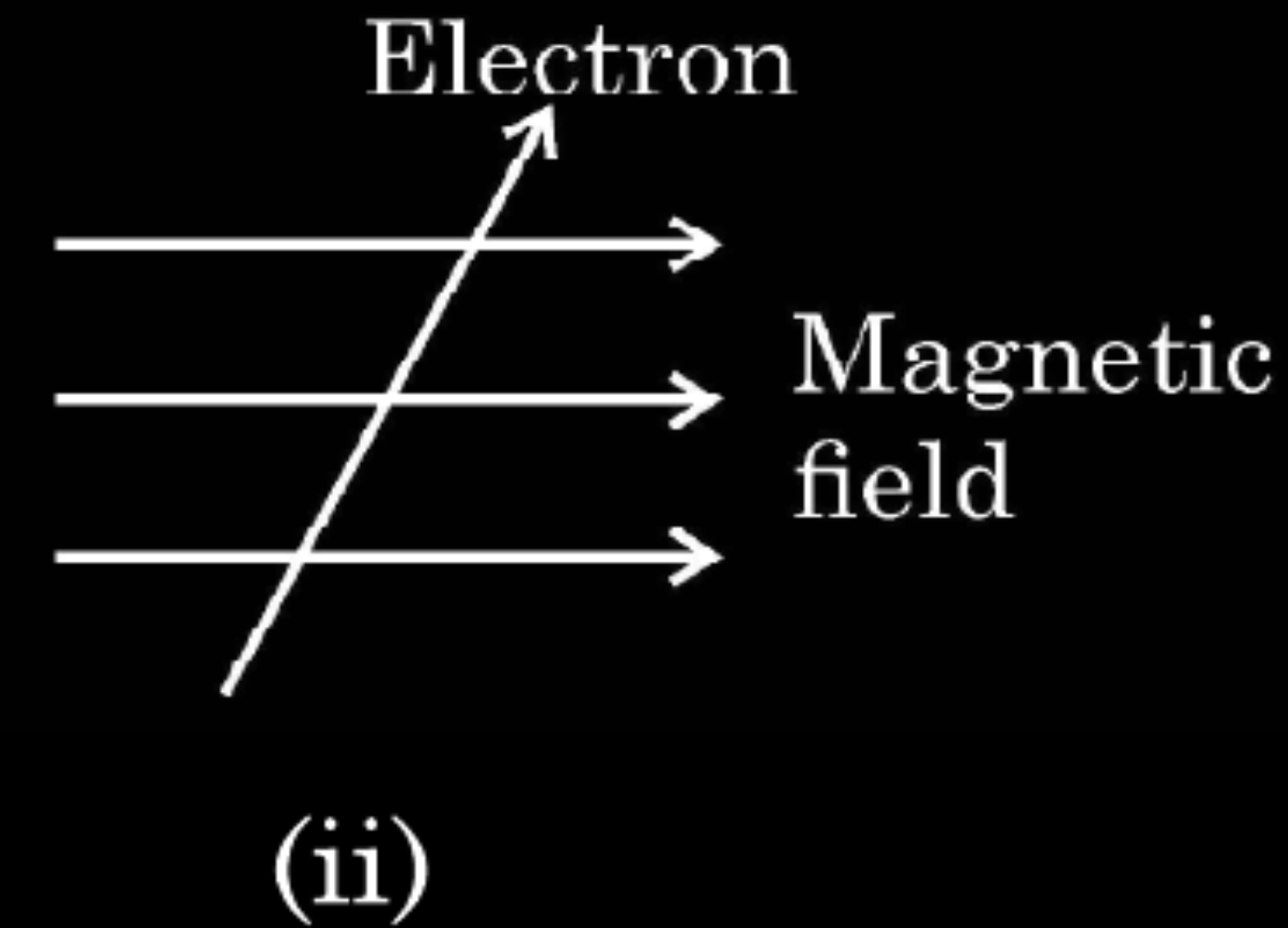
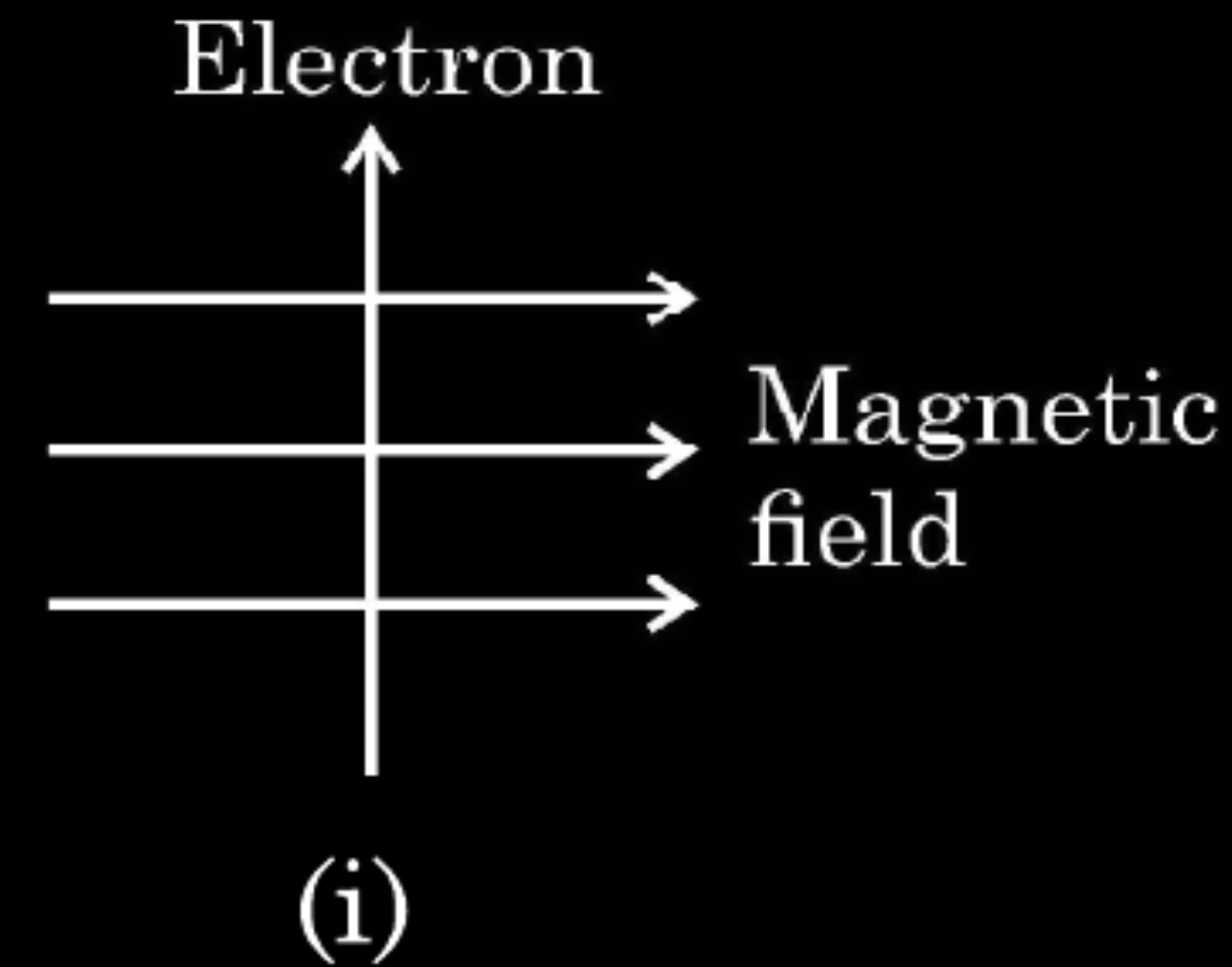
1

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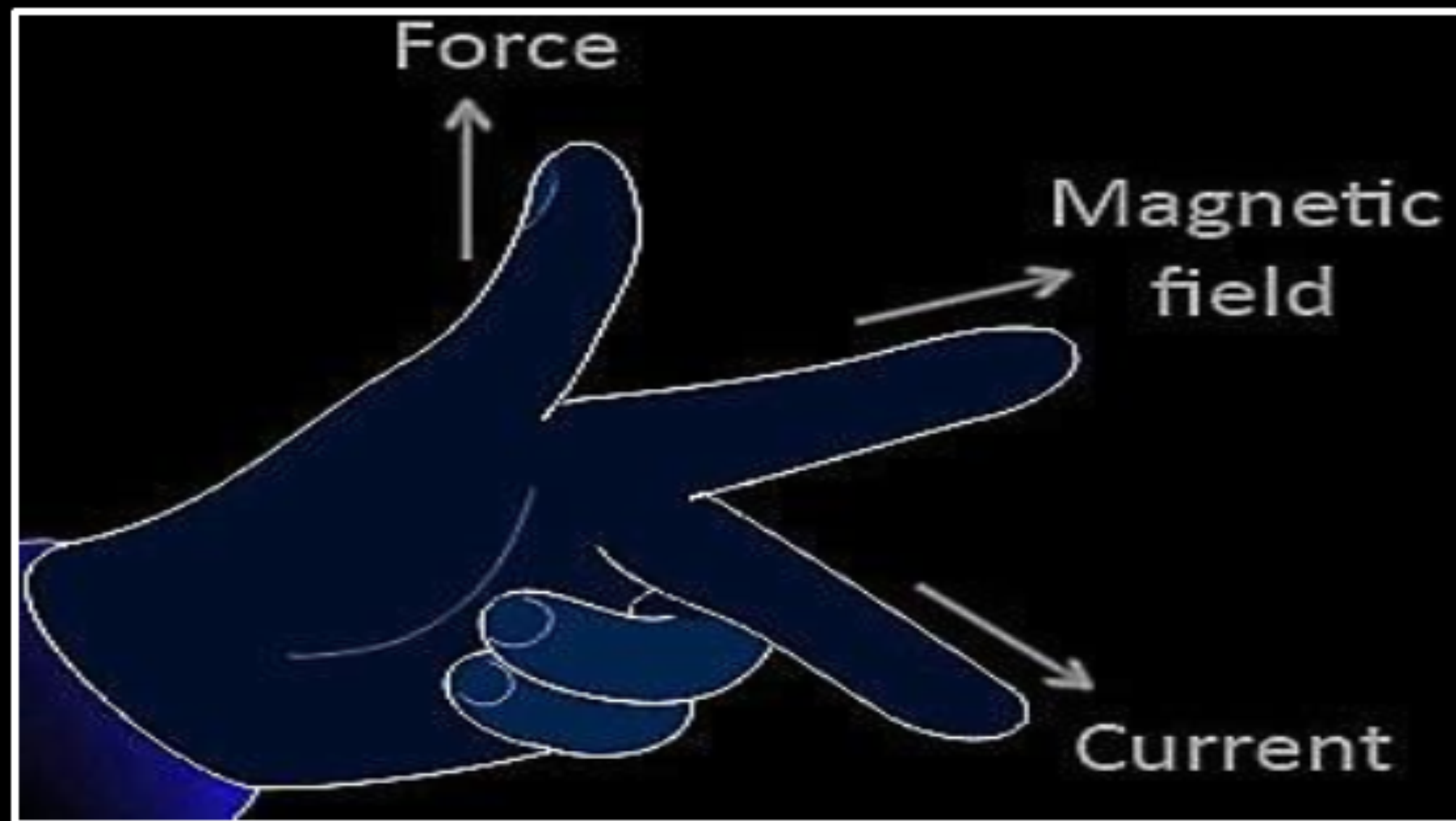
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(a)

31. (a) (i) State the rule used to find the force acting on a current carrying conductor placed in a magnetic field.
- (ii) Given below are three diagrams showing entry of an electron in a magnetic field. Identify the case in which the force will be (1) maximum and (2) minimum respectively. Give reason for your answer.



(a) (i) Flemings left-hand rule:



Stretch the forefinger, the central finger and the thumb of your left hand in mutually perpendicular directions. If the forefinger shows the direction of the magnetic field and the central finger that of the current, then the thumb will point towards the direction of motion of the conductor or direction of force /

(ii) (1) Force on electron is maximum in Fig (i) because the direction of motion of electron/current is at right angle/perpendicular to that of magnetic field.

(2) Force on electron is minimum in Fig (iii) because the electron is moving along / parallel to the direction of magnetic field

1

$\frac{1}{2}, \frac{1}{2}$

- (b) (i) Draw the pattern of magnetic field lines of
- (1) a current carrying solenoid
 - (2) a bar magnet
- (ii) List two distinguishing features between the two fields.

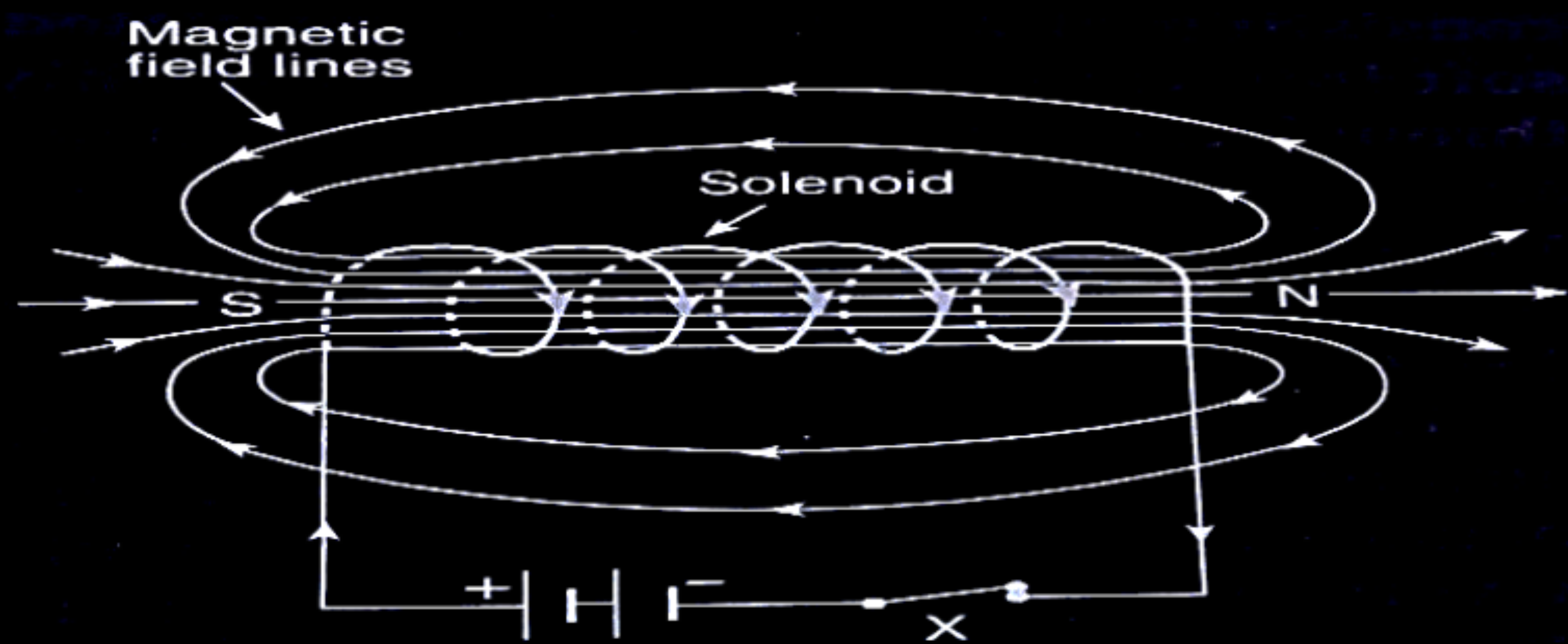
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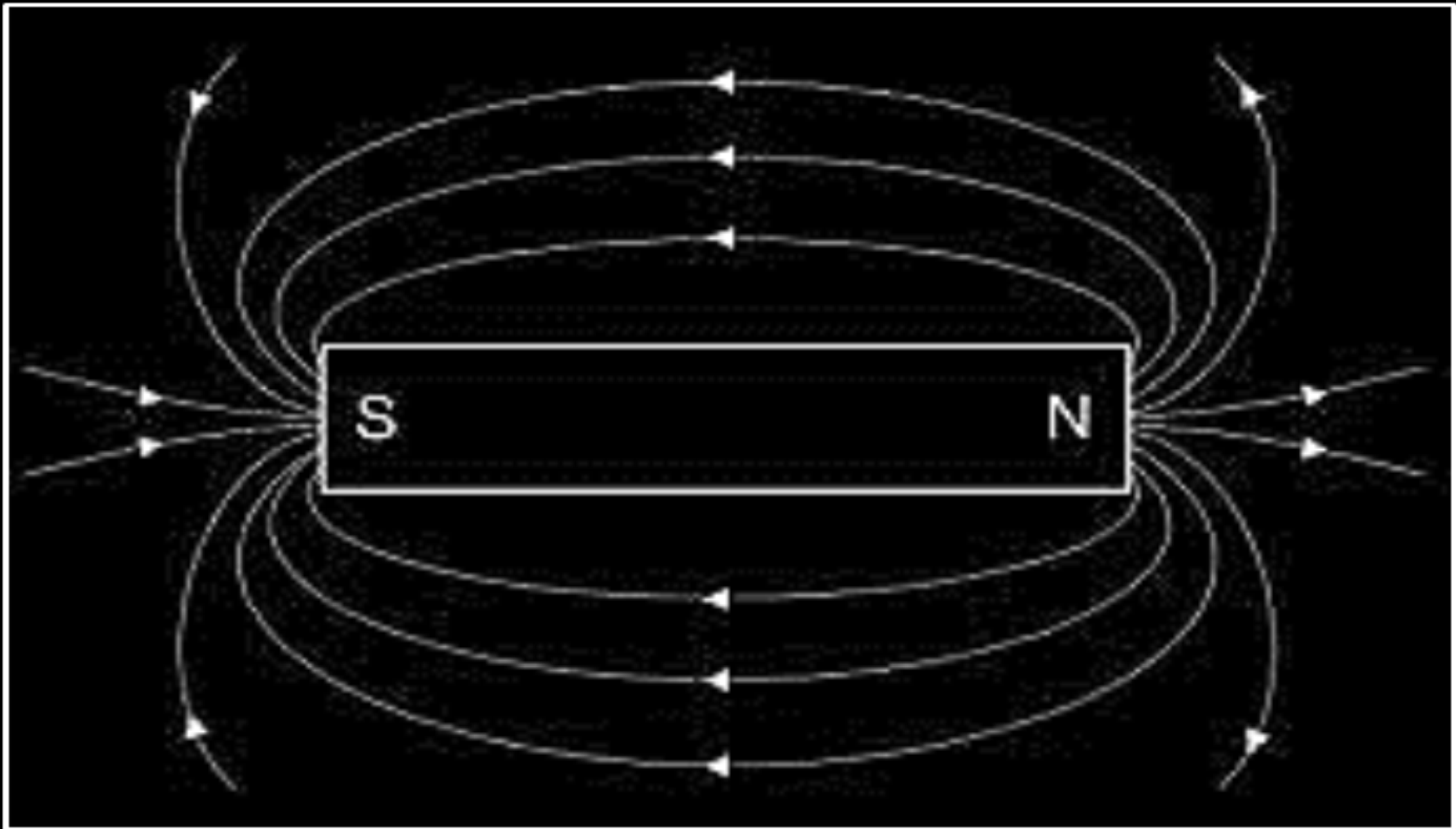
(b) (i) (1)



Magnetic field lines of a current carrying solenoid

1

(2)



Magnetic field lines of a bar magnet

1

(ii)

Magnetic field of a solenoid	Magnetic field of a bar magnet
1. The strength of the magnetic field can be changed by changing the current.	1. The strength of the magnetic field for a bar magnet cannot be changed.
2. The direction of magnetic field can be reversed by reversing the direction of current.	2. The direction of magnetic field for a bar magnet cannot be changed.
3. It is a temporary magnetic field.	3. It is a permanent magnetic field.

(Any two)

1/2+1/2

14. An electric iron of 1500 W, 200 V and a flash light of 500 W, 200 V are used in homes. The rating of fuse to be used should be

1

(a) 5 A

(b) 10 A

(c) 15 A

(d) 20 A

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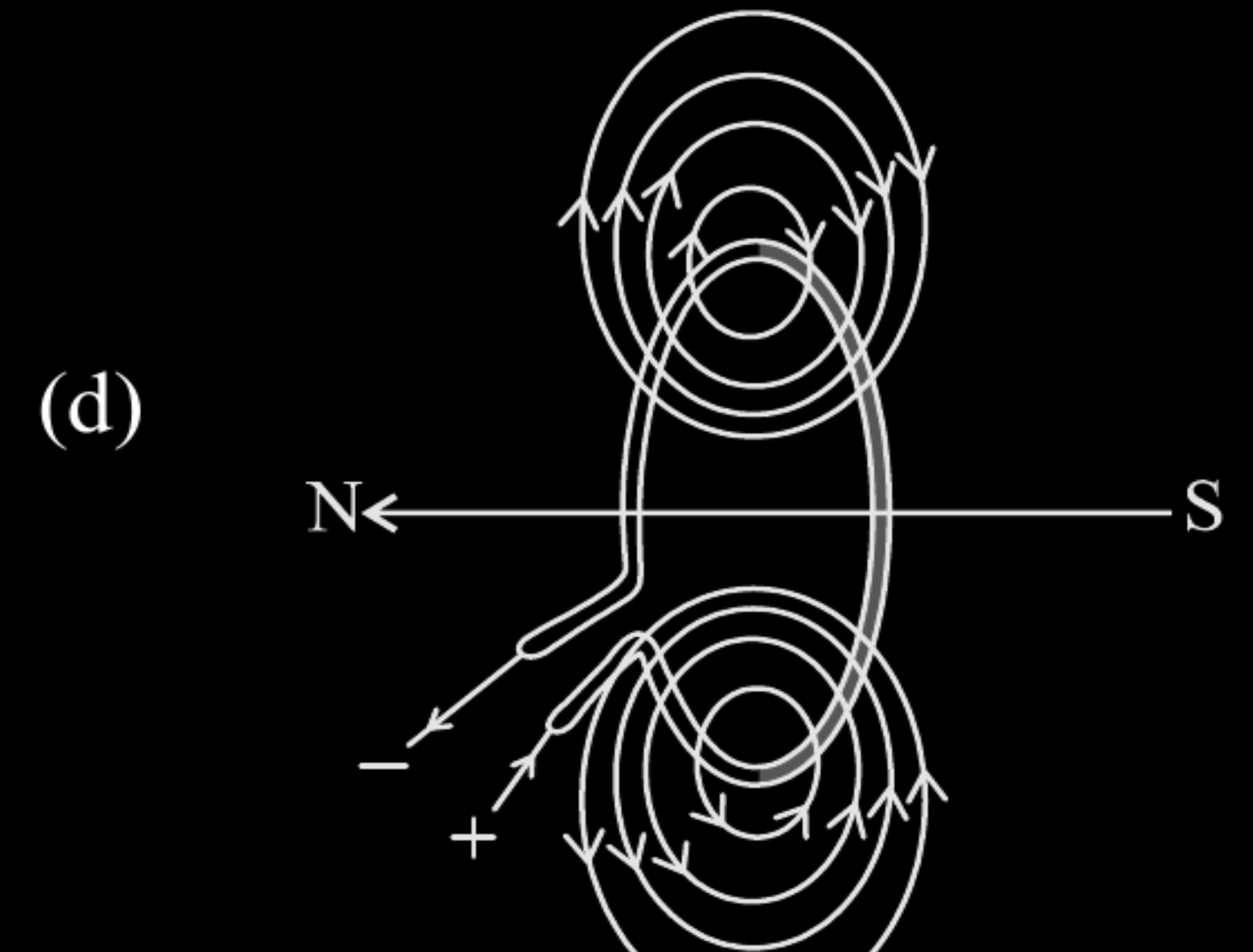
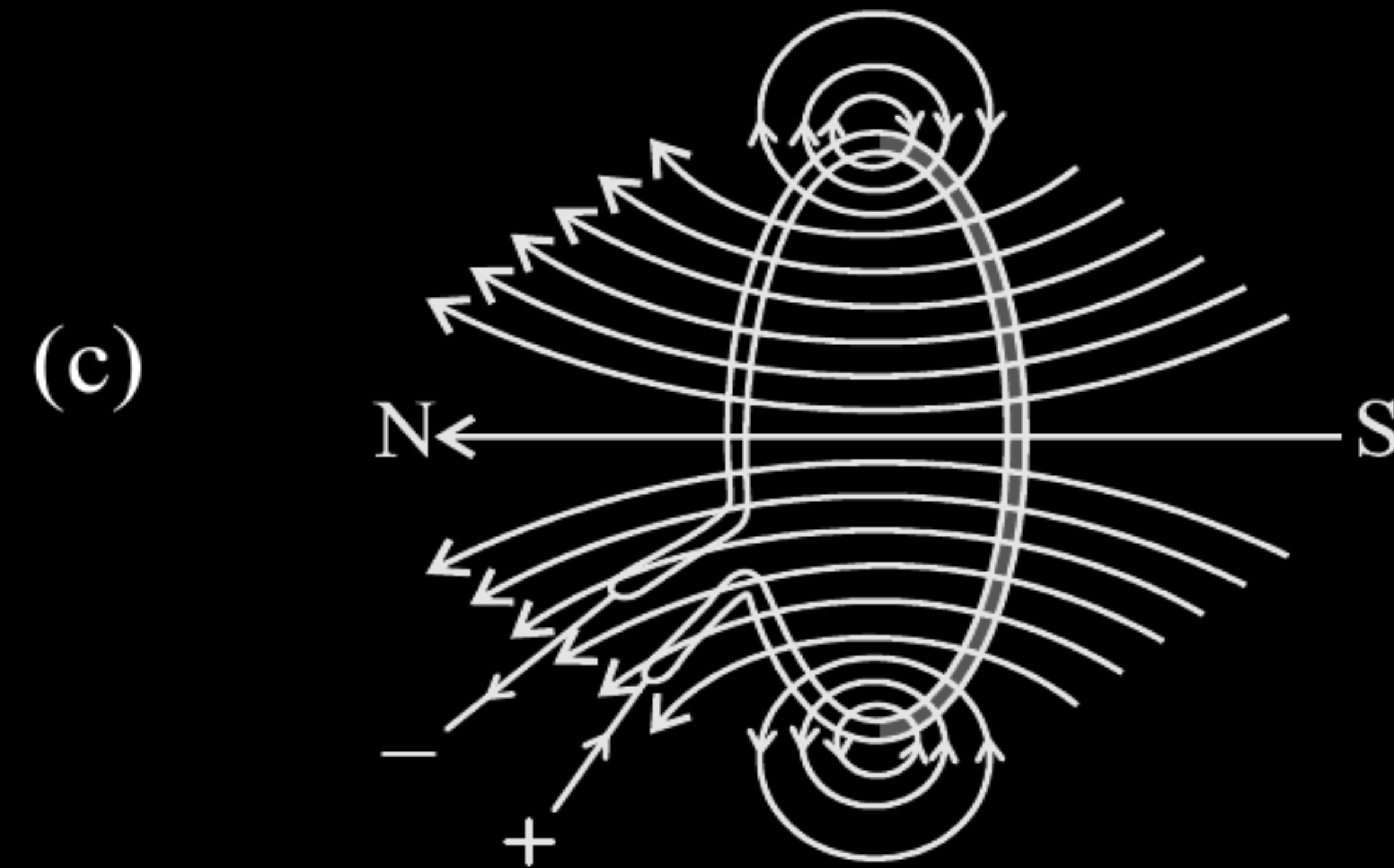
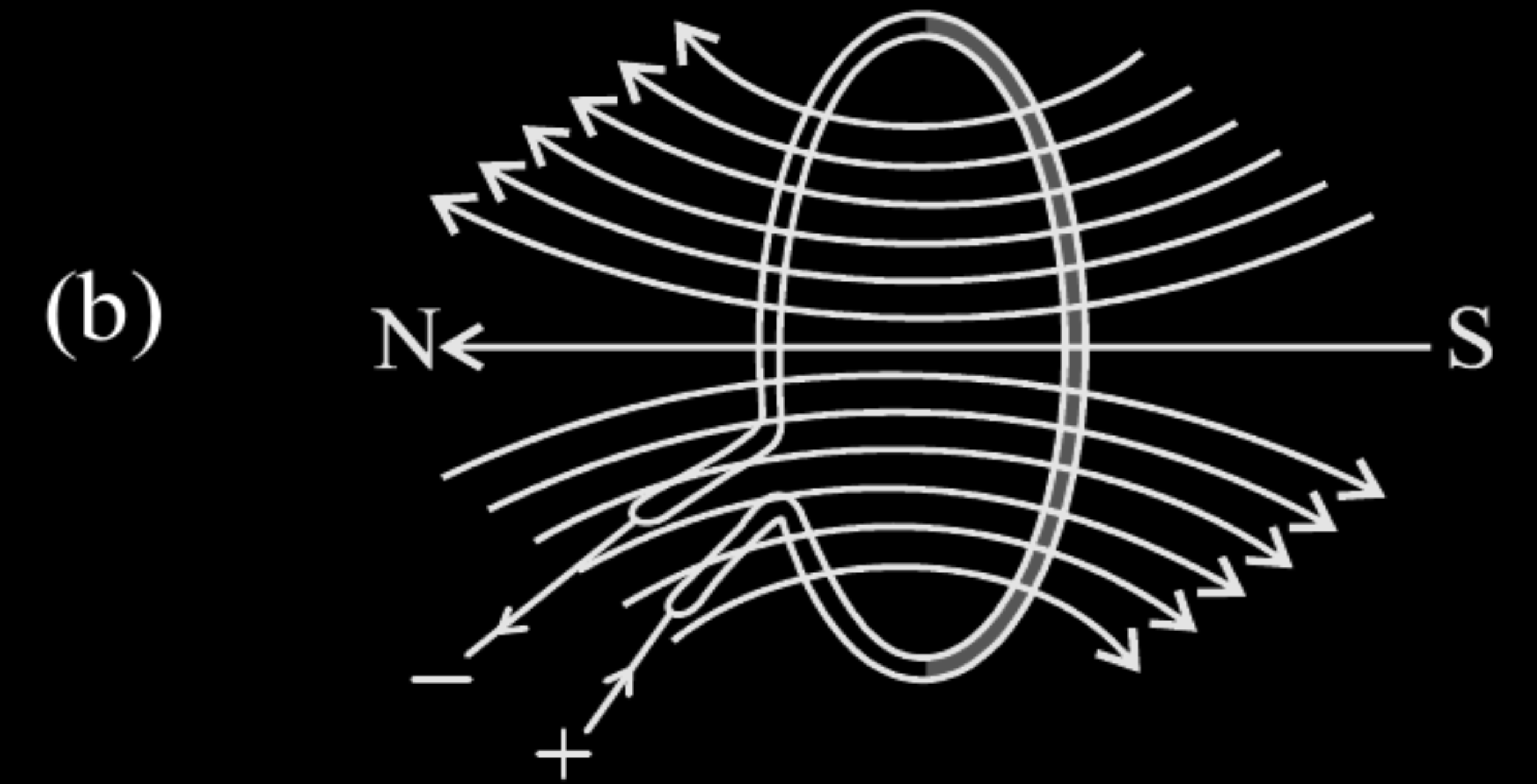
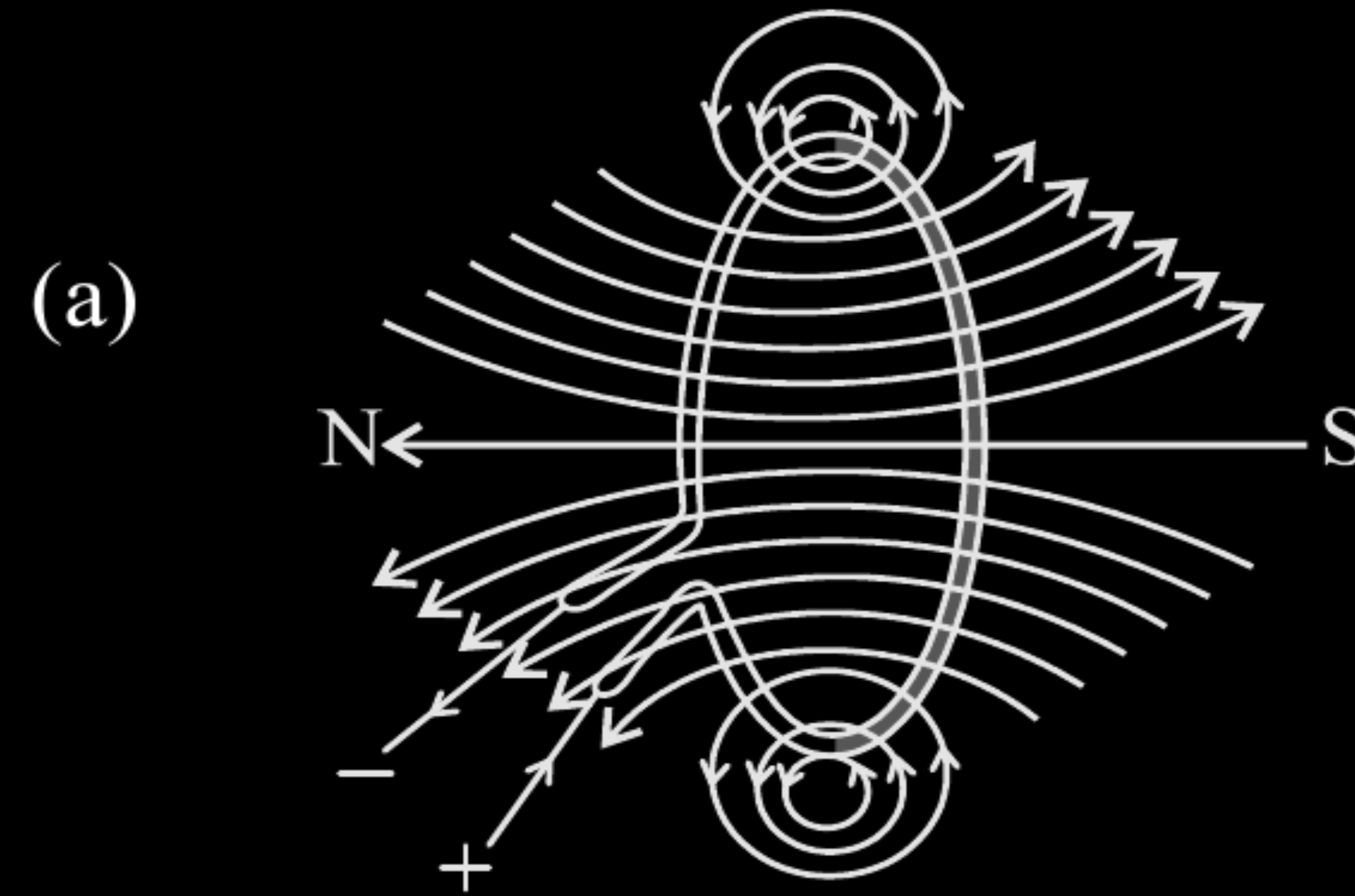
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(b)

13. The correct pattern of magnetic field lines of the field produced by a current carrying circular loop is :

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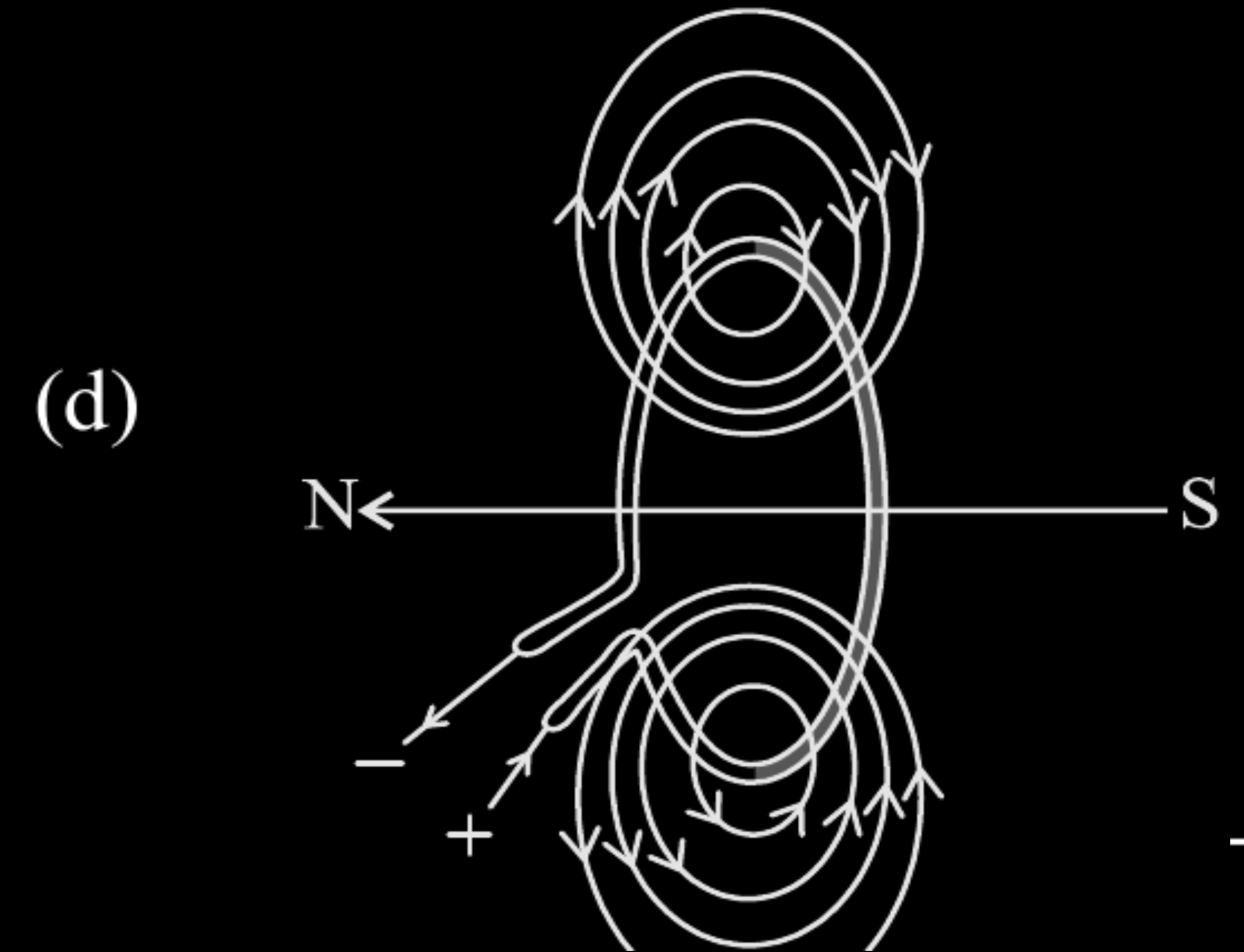
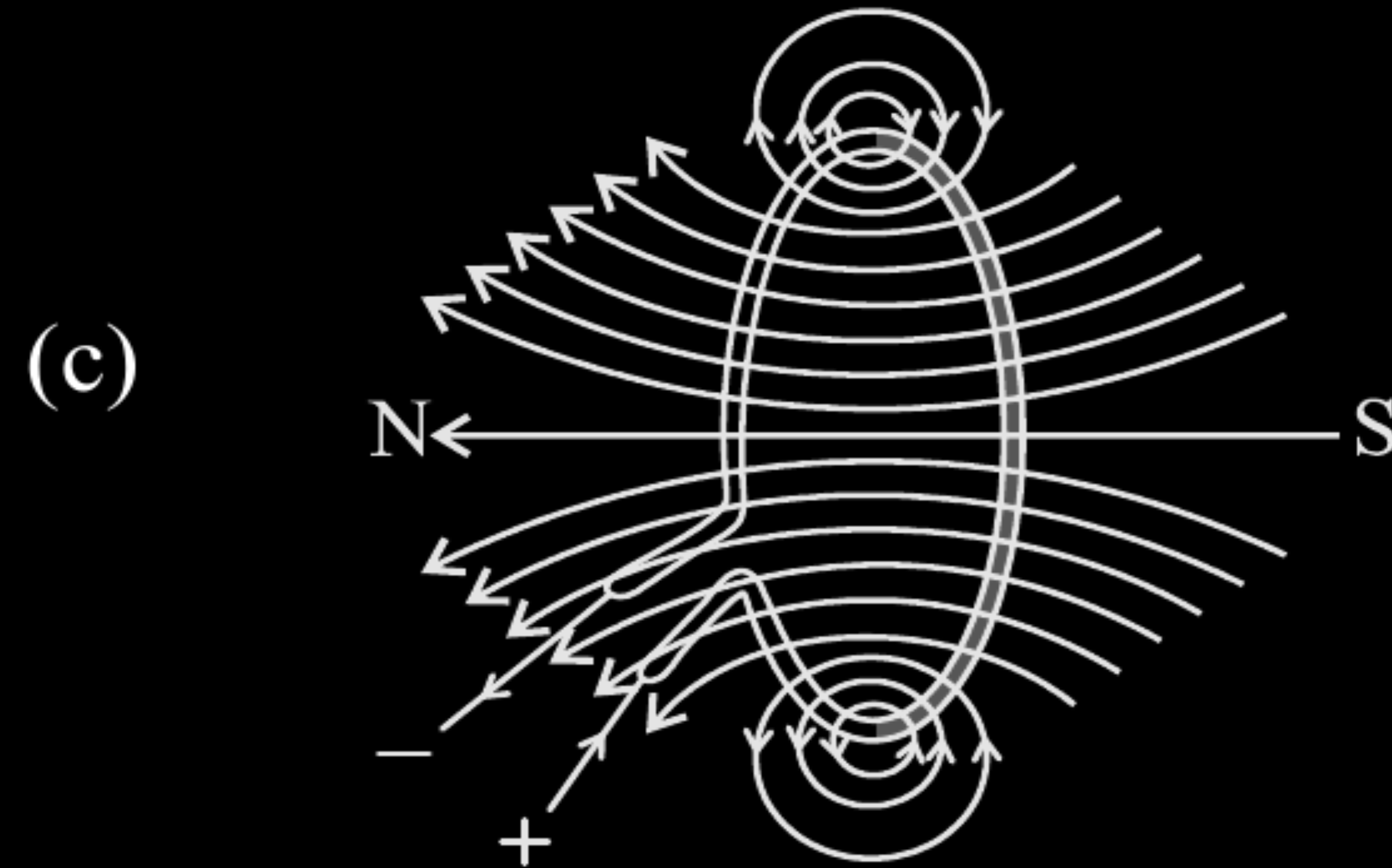
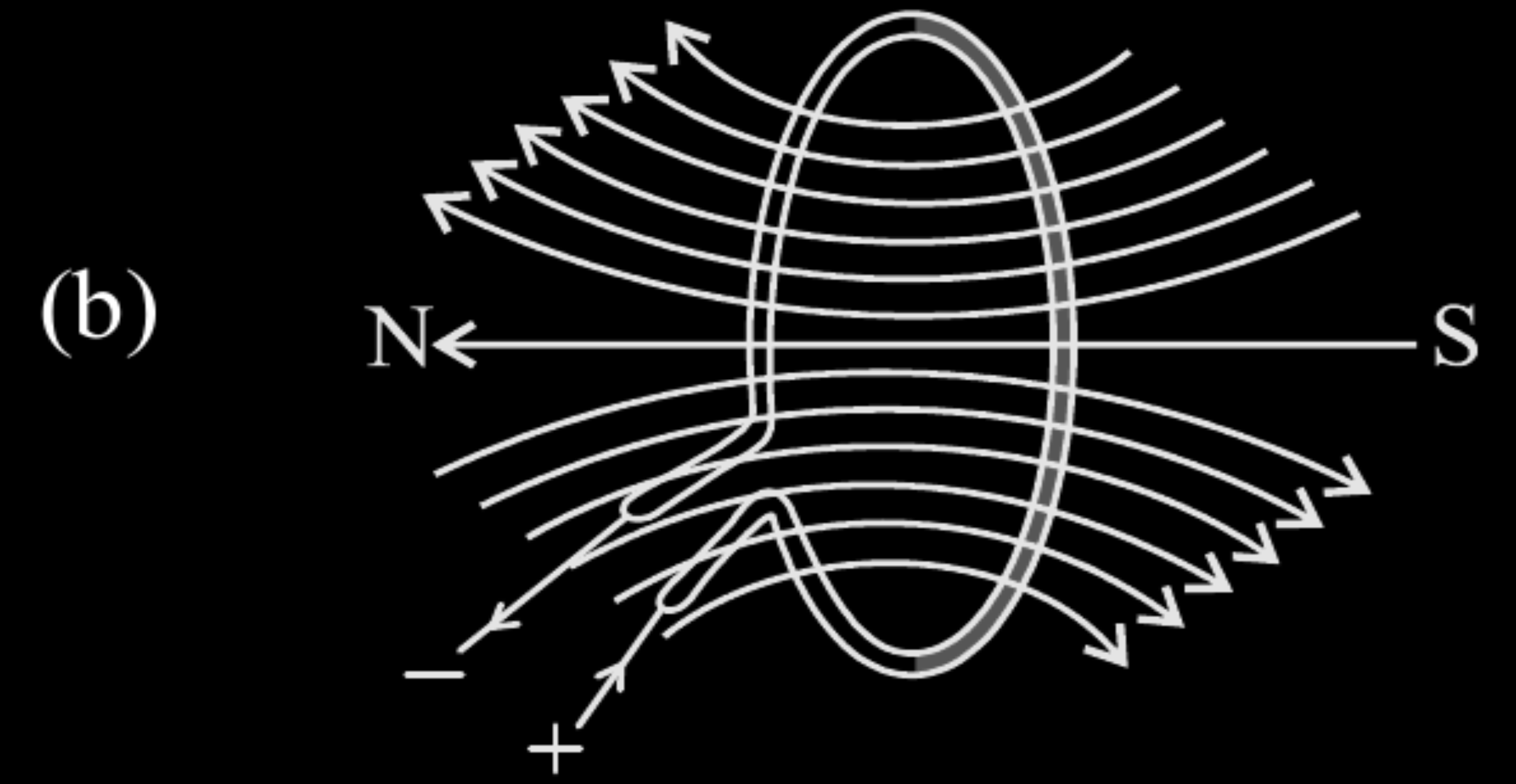
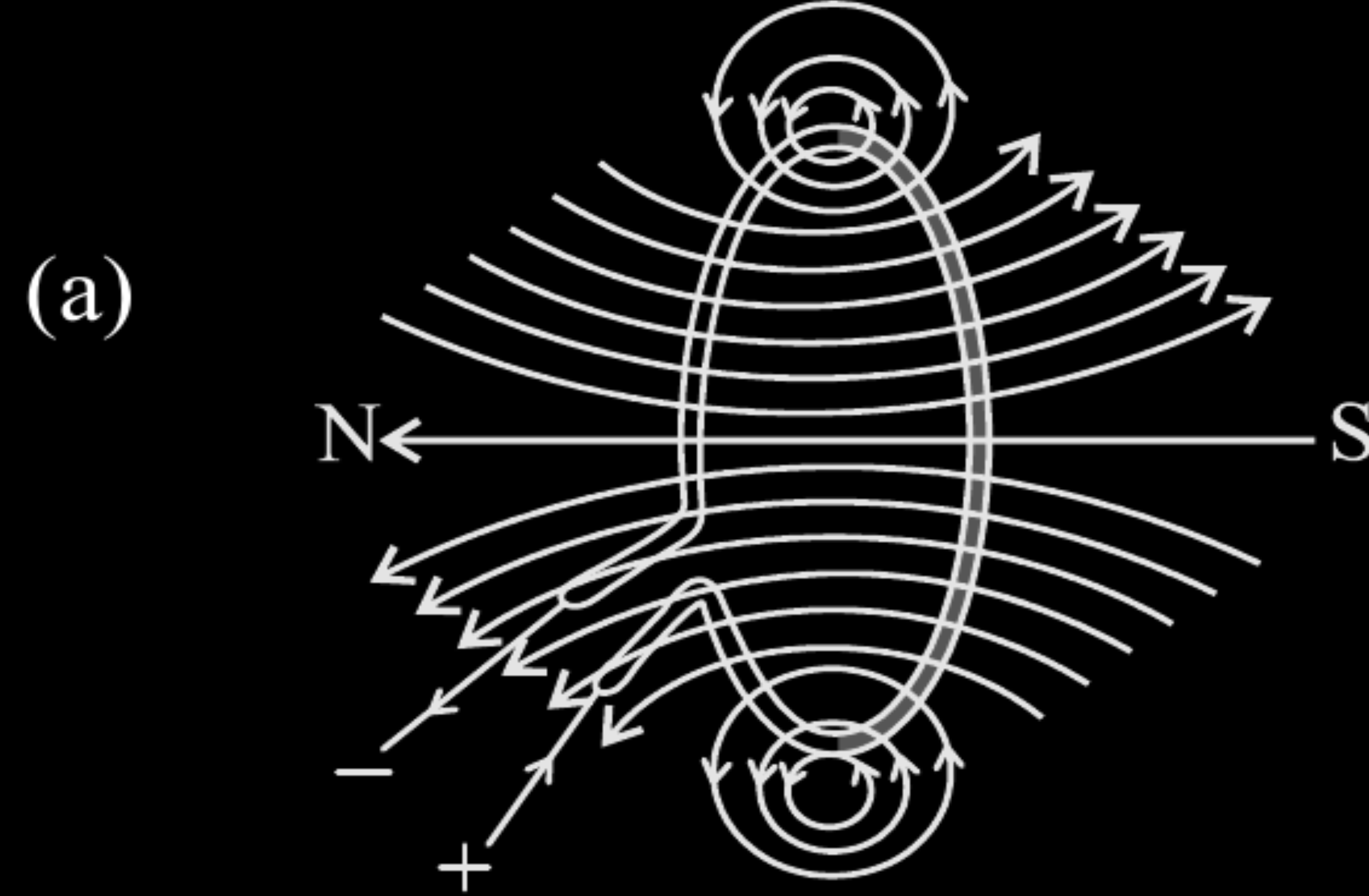
1



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CBSE 2023

1

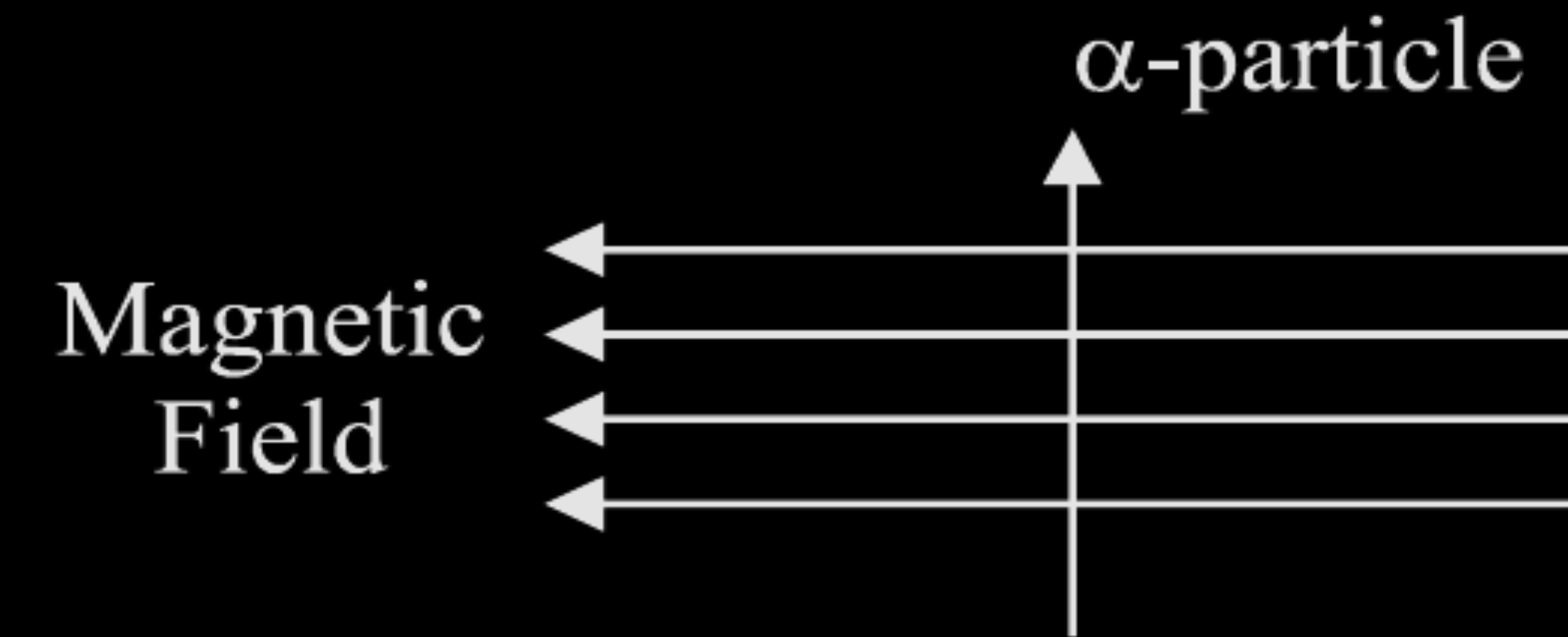


(c)

15. An alpha particle enters a uniform magnetic field as shown. The direction of motion of the alpha particle is :

1

- (a) towards right
- (b) towards left
- (c) into the page
- (d) out of the page

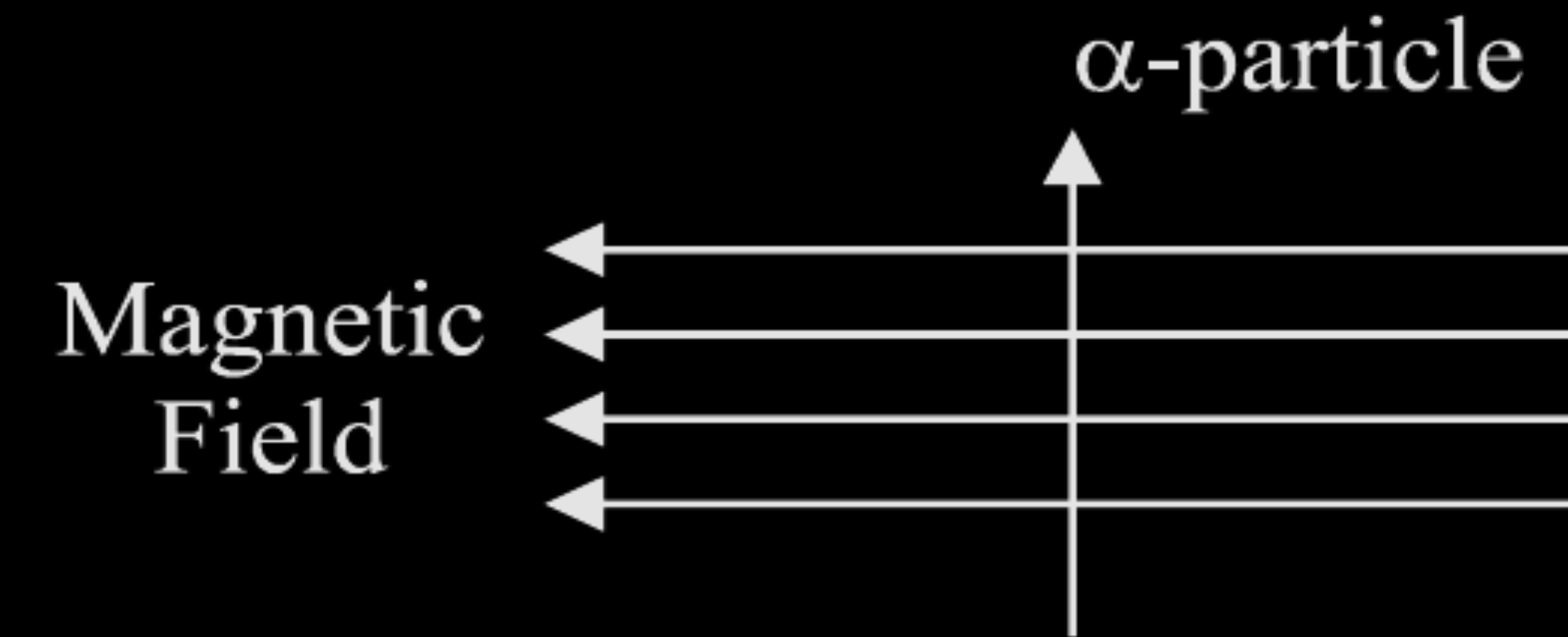


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- (d) out of the page



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(d)

19. Assertion (A) : A current carrying straight conductor experiences a force when placed perpendicular to the direction of magnetic field.

Reason (R) : The net charge on a current carrying conductor is always zero.

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19. Assertion (A) : A current carrying straight conductor experiences a force when placed perpendicular to the direction of magnetic field.

Reason (R) : The net charge on a current carrying conductor is always zero.

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1

(b)

- 31. (A)** (i) Why is an alternating current (A.C.) considered to be advantageous over direct current (D.C.) for the long distance transmission of electric power ? **3**
- (ii) How is the type of current used in household supply different from the one given by a battery of dry cells ?
- (iii) How does an electric fuse prevent the electric circuit and the appliances from a possible damage due to short circuiting or overloading.

- 31. (A)** (i) Why is an alternating current (A.C.) considered to be advantageous over direct current (D.C.) for the long distance transmission of electric power ? **3**
- (ii) How is the type of current used in household supply different from the one given by a battery of dry cells ?
- (iii) How does an electric fuse prevent the electric circuit and the appliances from a possible damage due to short circuiting or overloading.

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(A) (i) Alternating current can be transmitted over long distances without much loss of electric energy.

1

(ii) Household supply – Alternating current (AC)
Battery of Dry cell – Direct current (DC)

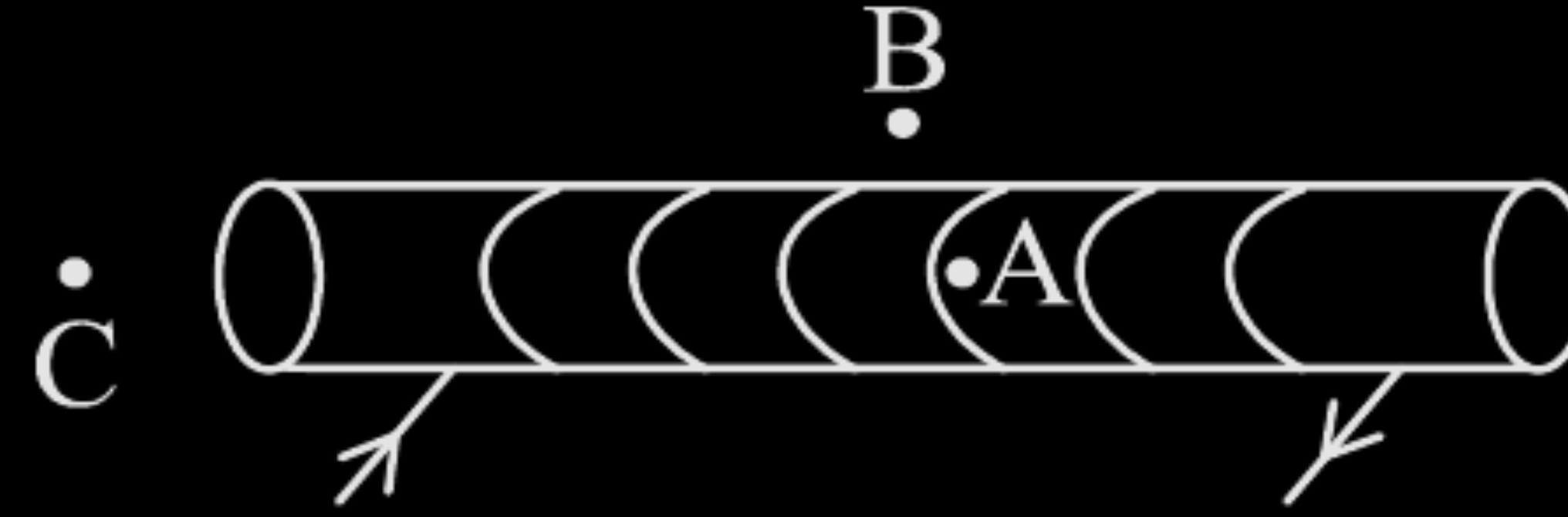
$\frac{1}{2}$

$\frac{1}{2}$

(iii) It melts and breaks the circuit when a current of higher value than its rating flows through it.

1

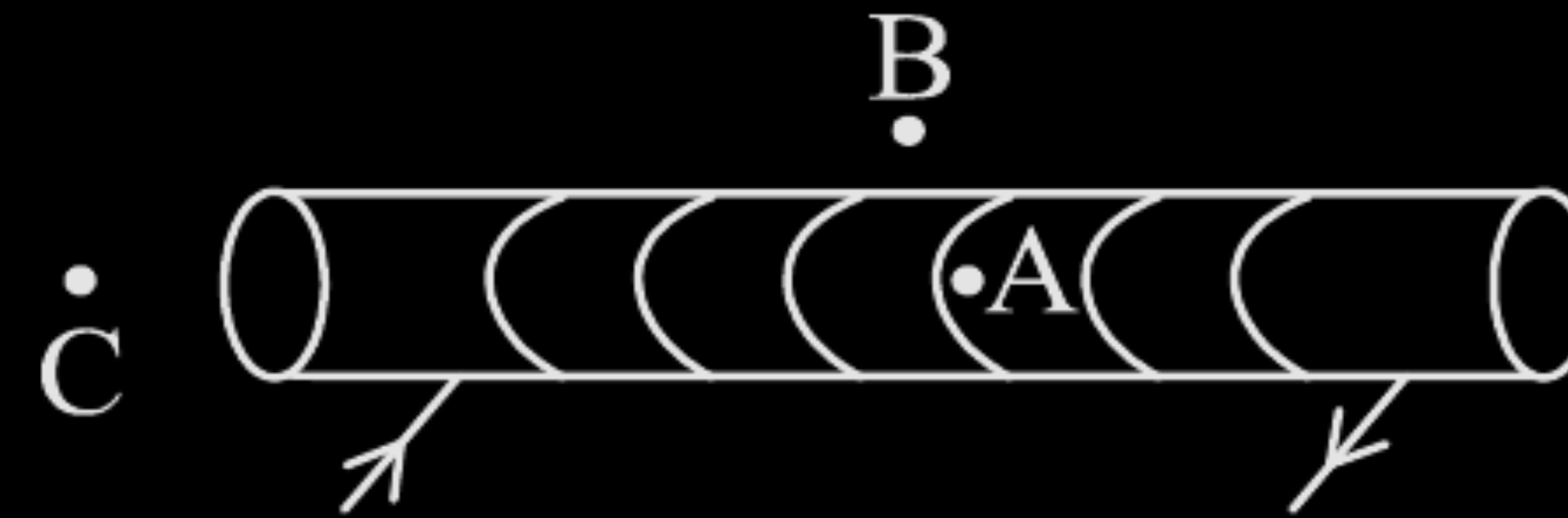
- (B)** For the current carrying solenoid as shown, draw magnetic field lines and give reason to explain that out of the three points A, B and C, at which point the field strength is maximum and at which point it is minimum ?



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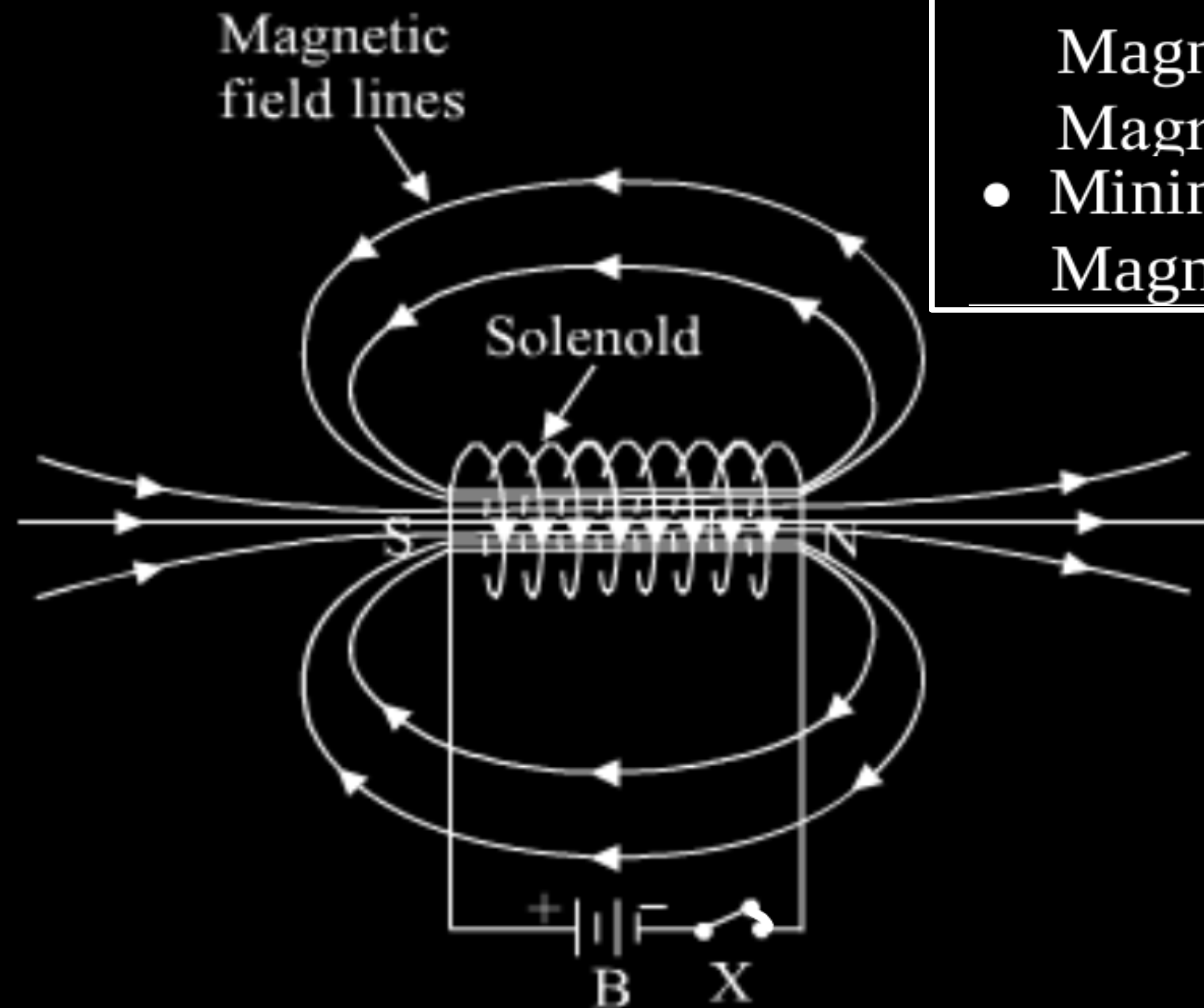
- (B) For the current carrying solenoid as shown, draw magnetic field lines and give reason to explain that out of the three points A, B and C, at which point the field strength is maximum and at which point it is minimum ?

3



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- Maximum at A
Magnetic field lines are crowded. /
Magnetic field adds up due to 'n' number of turns of a solenoid.
- Minimum at B
Magnetic field lines are far apart.

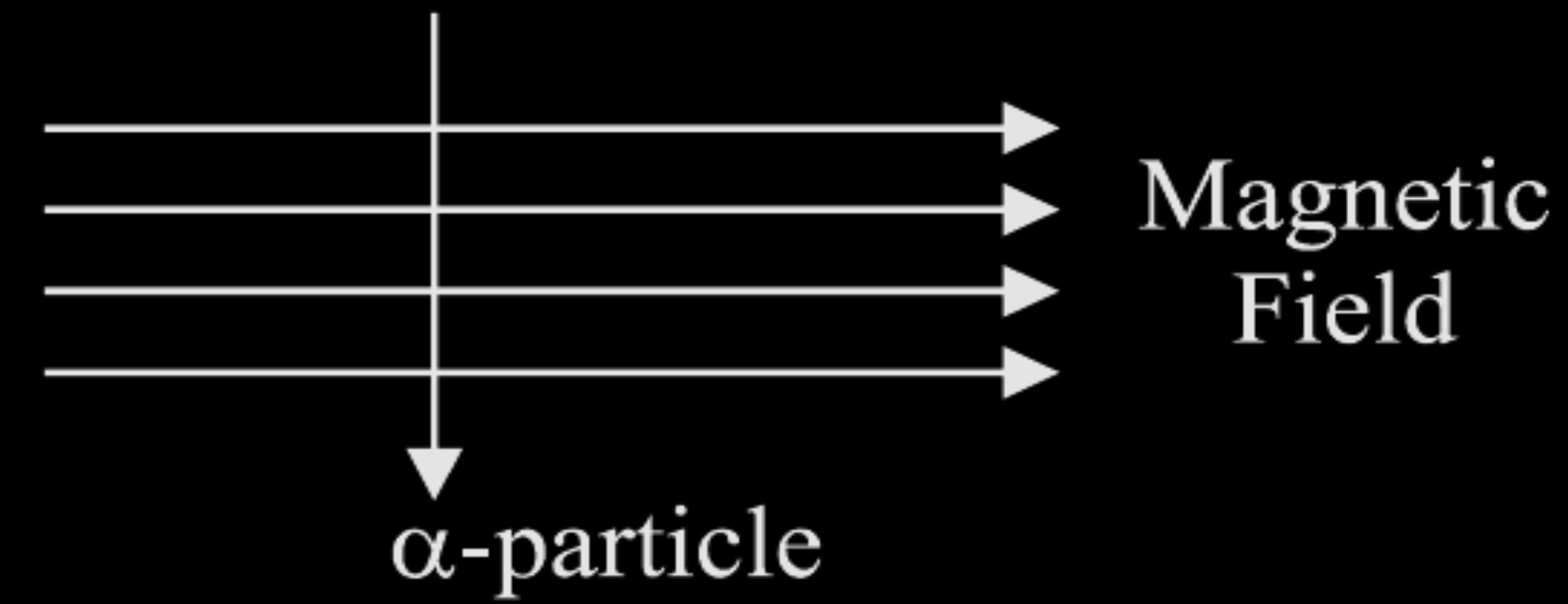


[Deduct ½ mark if direction of current or magnetic field is not marked]

13. An alpha particle enters a uniform magnetic field as shown. The direction of force experienced by the alpha particle is :

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- (b) towards left
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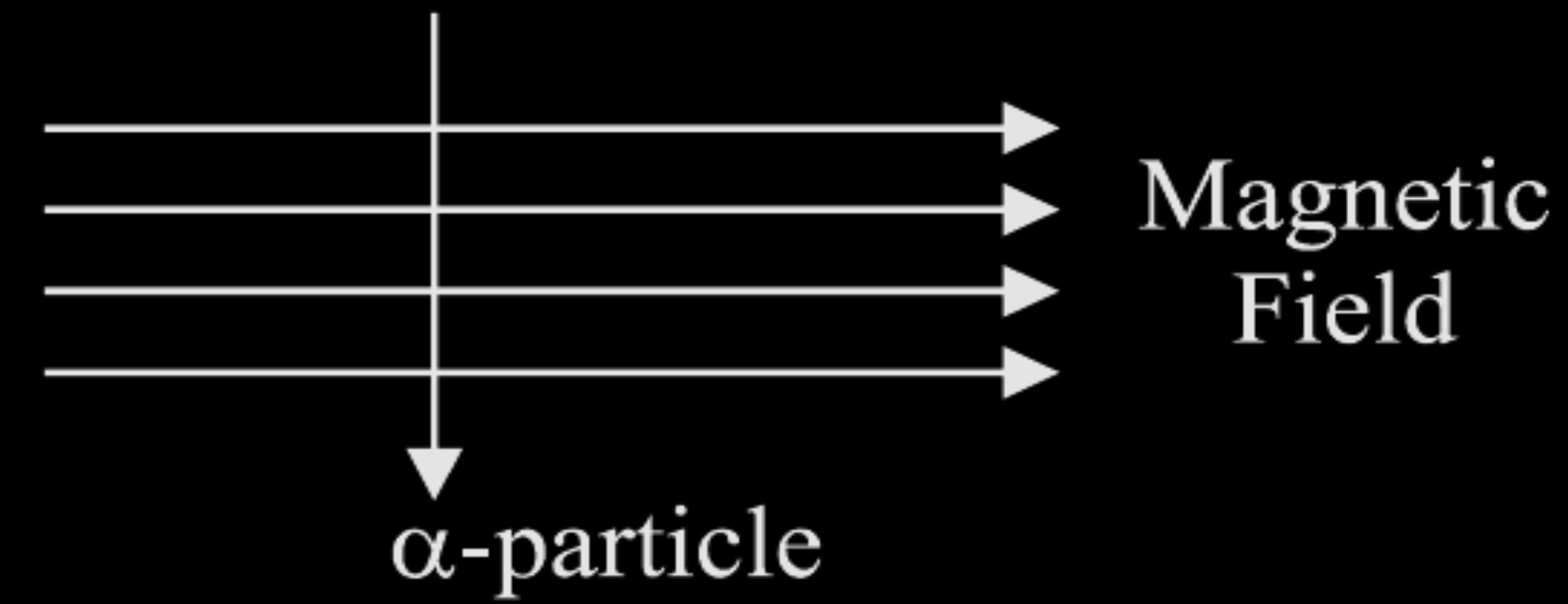


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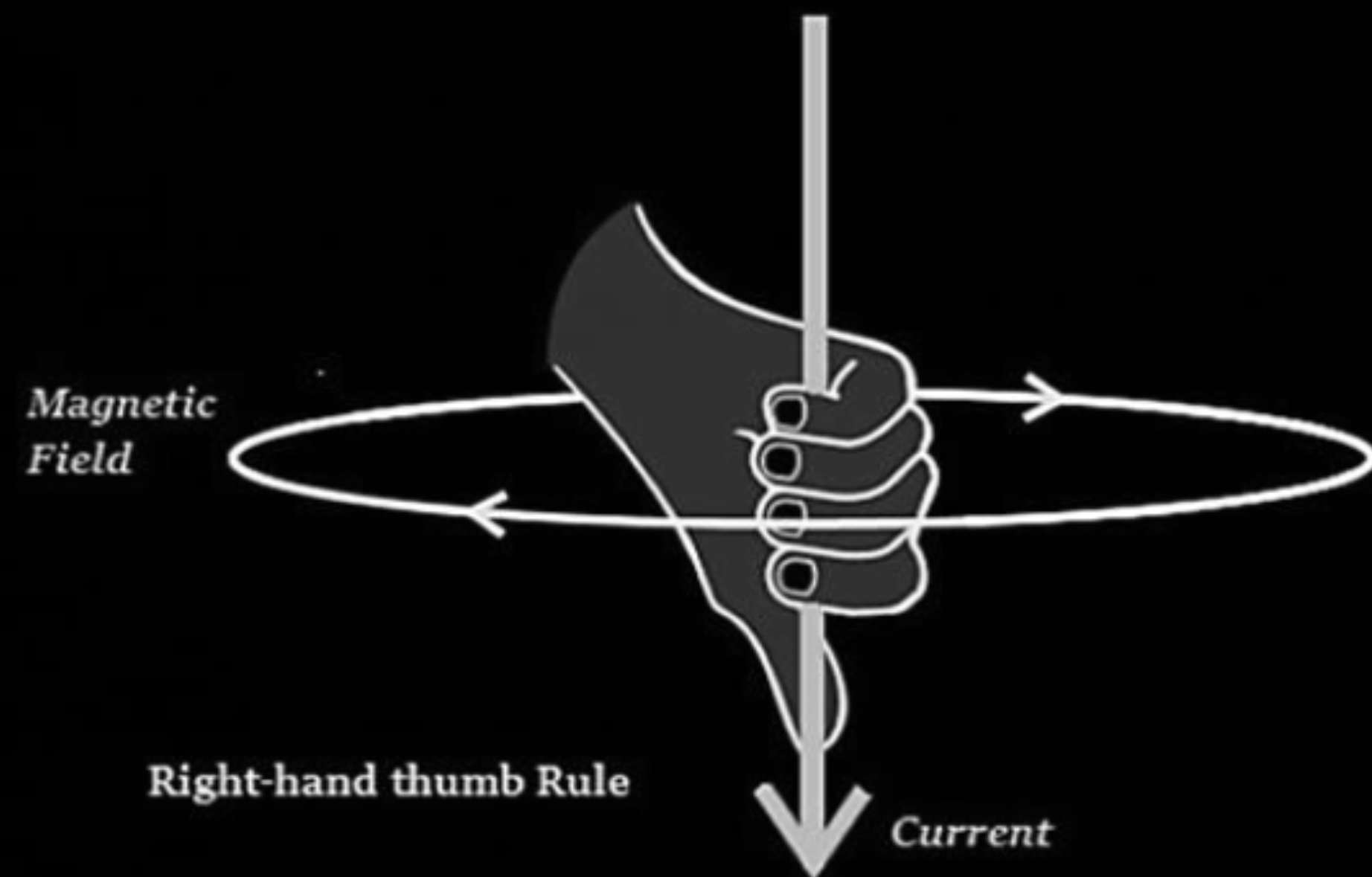
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- 5.** (a) (i) A magnetic compass shows a deflection when placed near a current carrying wire. How will the deflection of the compass get affected if the current in the wire is increased ? What does it indicate ?
- (ii) State Right hand thumb rule.

5. (a) (i) A magnetic compass shows a deflection when placed near a current carrying wire. How will the deflection of the compass get affected if the current in the wire is increased ? What does it indicate ?
- (ii) State Right hand thumb rule.

CBSE 2022 C

2



(a) (i)

- The deflection will increase.
- It indicates that the magnitude of the magnetic field produced at a given point increases as the current through the wire increases.

$\frac{1}{2}$

$\frac{1}{2}$

(ii) If a current carrying conductor is imagined to be held in right hand such that the thumb points along the direction of current then the direction of the wrapped fingers will give the direction of magnetic field lines.

1

(Give full credit if explained by labelled figure)

- (b) List two factors on which the strength of magnetic field at a point due to a current carrying straight conductor depends. State the rule that determines the direction of magnetic field produced in this case.

CBSE 2022 C

- (b) List two factors on which the strength of magnetic field at a point due to a current carrying straight conductor depends. State the rule that determines the direction of magnetic field produced in this case.

CBSE 2022 C

2

(b)	
• Two factors	$\frac{1}{2}$
(i) Strength of current	$\frac{1}{2}$
(ii) Distance of the point from the straight conductor	
• Imagine that you are holding a current carrying straight conductor in your right hand such that the thumb points towards the direction of the current then your fingers will wrap around the conductor in the direction of magnetic lines.	1

- 13.** (a) (i) What is a magnetic needle ? Why does a magnetic needle get deflected when brought near a current carrying straight conductor ?
- (ii) State right hand thumb rule.

CBSE 2022 C

- 13.** (a) (i) What is a magnetic needle ? Why does a magnetic needle get deflected when brought near a current carrying straight conductor ?
- (ii) State right hand thumb rule.

CBSE 2022 C

3

(a) (i) • Magnetic needle is a small bar magnet.	1
• Magnetic needle gets deflected due to the presence of magnetic field produced by current carrying conductor.	1
(ii) Imagine that you are holding a current carrying straight conductor in your right hand such that the thumb points towards the direction of the current then your fingers will wrap around the conductor in the direction of magnetic lines.	1

- (b) (i) The magnetic field in a region is represented by equidistant parallel lines. What do such lines indicate ?
- (ii) Imagine that you are sitting in an examination room with your back to one wall. An electron beam moving horizontally from the back wall towards the front wall is deflected by a strong magnetic field to your right side. What is the direction of the magnetic field ?
- (iii) State the rule which you have applied in determining the direction of the magnetic field in this case.

- (b) (i) The magnetic field in a region is represented by equidistant parallel lines. What do such lines indicate ?
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- (iii) State the rule which you have applied in determining the direction of the magnetic field in this case.

CBSE 2022 C

3

- | | |
|--|---|
| (b) (i) Uniform magnetic field | 1 |
| (ii) Normal to earth surface downwards. | 1 |
| (iii) Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor. | 1 |

3. As shown in the diagram an aluminium rod 'AB' is suspended horizontally between the two poles of a strong horse shoe magnet in such a way that the axis of rod is horizontal and the direction of the magnetic field is vertically upward. The rod is connected in series with a battery and a key.

2

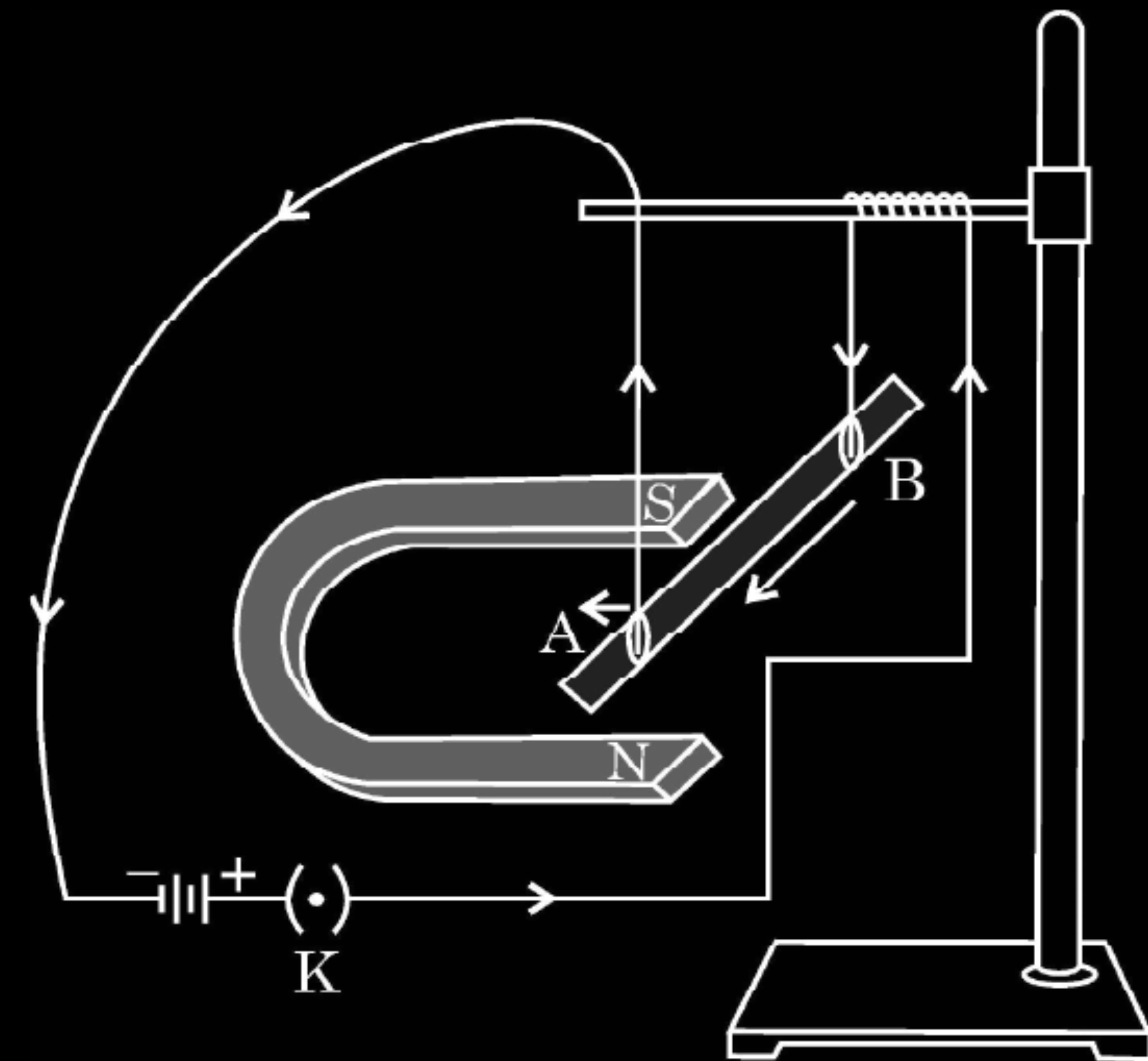
State giving reason :

- (a) What is observed when a current is passed through the aluminum rod from end B to end A ?
- (b) What change is observed in a situation in which the axis of the rod 'AB' is moved and aligned parallel to the magnetic field and current is passed in the rod in the same direction ?

OR

“Magnetic field is a physical quantity that has both direction and magnitude.” How can this statement be proved with the help of magnetic field lines of a bar magnet ?

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(a)

- Rod AB would get displaced
- Reason: When a current carrying conductor is placed in an external magnetic field perpendicularly, it experiences a force.

$\frac{1}{2}$

$\frac{1}{2}$

(b) As current and magnetic field are parallel, there will be no force experienced by the conductor hence no displacement in rod.

1

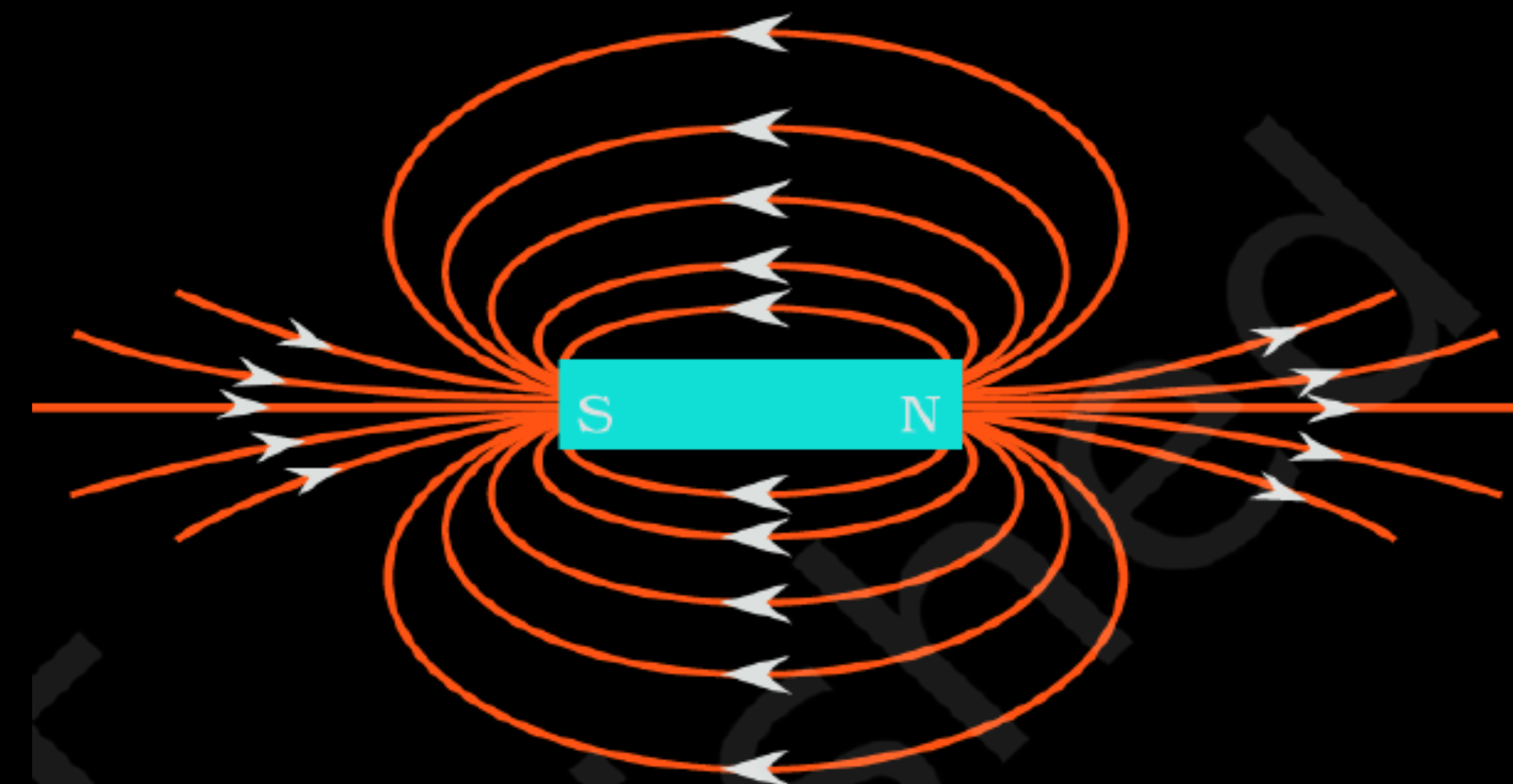
OR

- Field lines emerge from the north pole and merge at south pole/the direction in which a north pole of the compass needle moves inside it.
- The relative strength of the magnetic field is shown by the degree of closeness of the field lines. Crowded are the field lines, stronger is the field.

1

1

Or diagrammatic expression.



5. (a) Name the poles P, Q, R and S of the magnets in the following figures 'a' and 'b' : $\frac{1}{2} + \frac{1}{2} + 1 = 2$

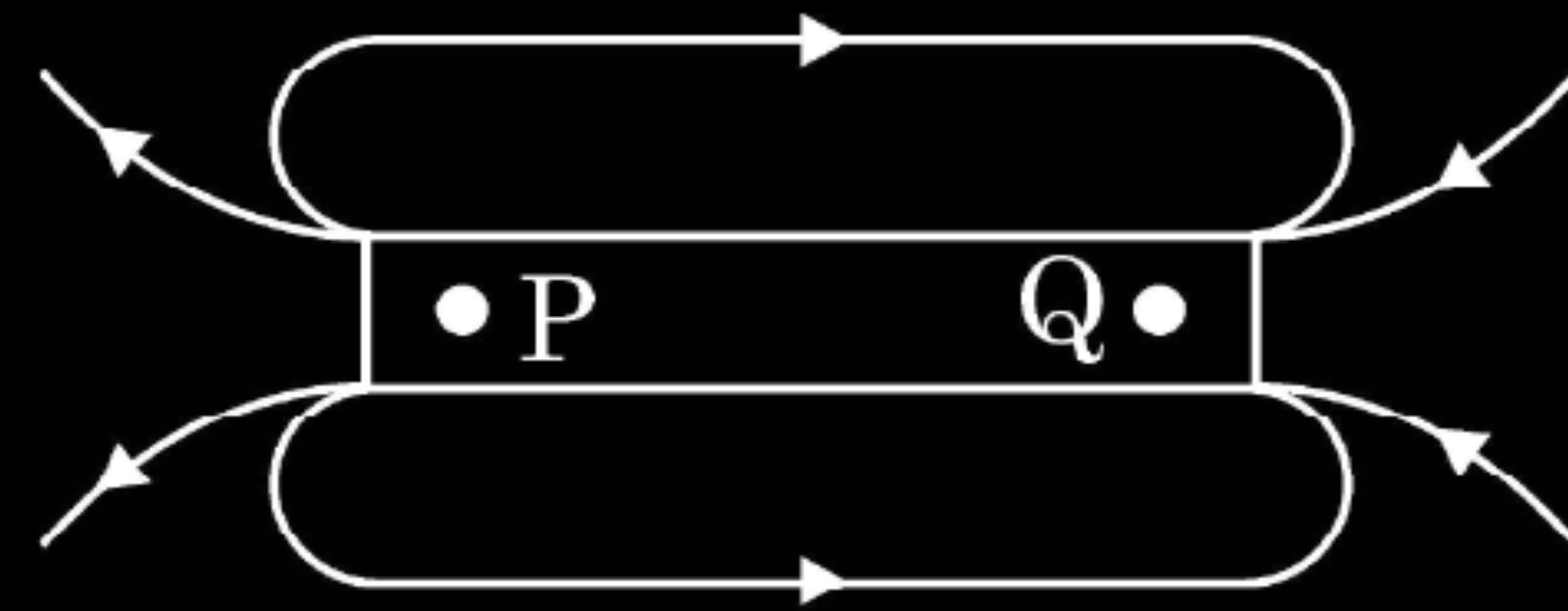


Figure 'a'

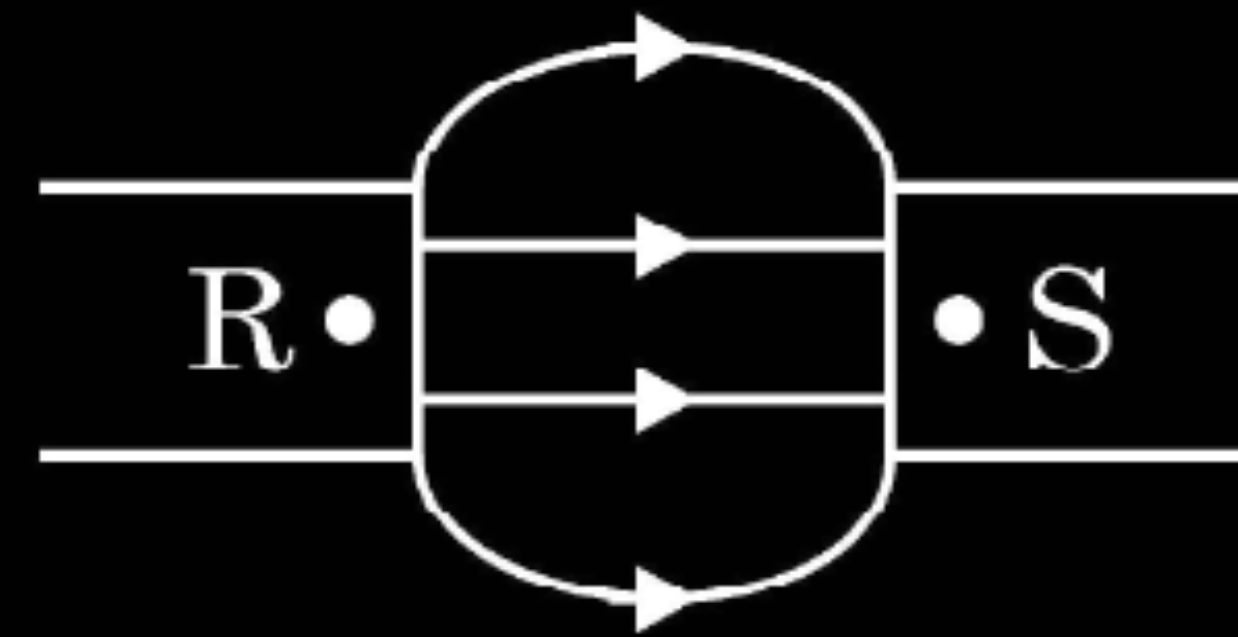
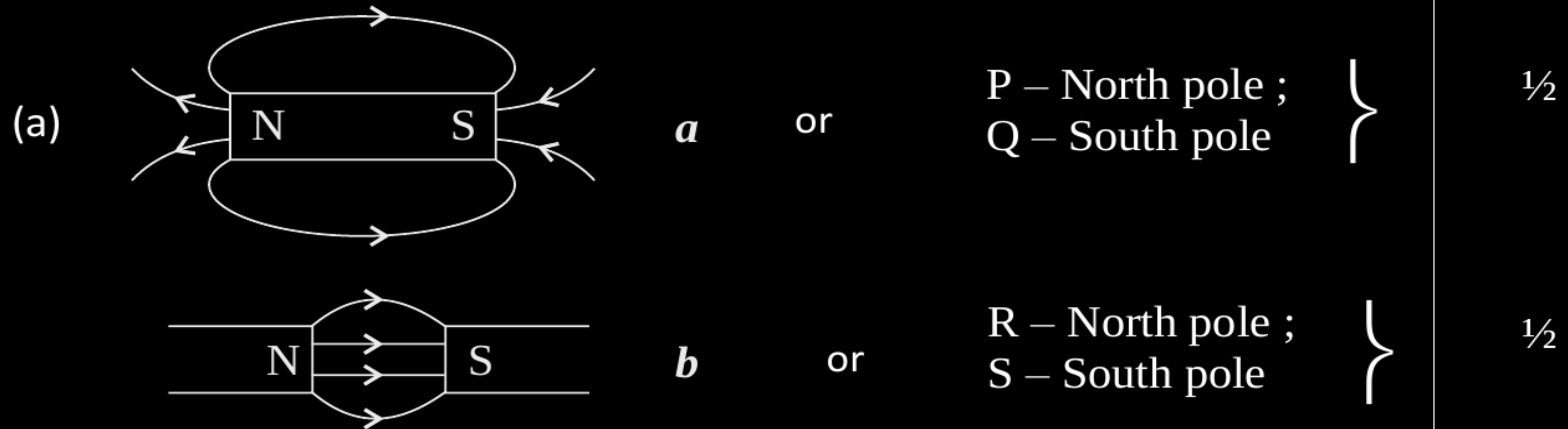


Figure 'b'

- (b) State the inference drawn about the direction of the magnetic field lines on the basis of these diagrams.



(b) The magnetic field lines emerge from the North-pole and merge to South-pole outside the magnet. (Inside the magnet the direction is from South pole to North pole.)

Alternative answer :-

Closed curves which emerge from North pole and merge at the South pole.

When is the force experienced by a current – carrying straight conductor placed in a uniform magnetic field. **1 + 1 = 2**

- (i) Maximum ;
- (ii) Minimum ?

CBSE 2022

When is the force experienced by a current – carrying straight conductor placed in a uniform magnetic field. **1 + 1 = 2**

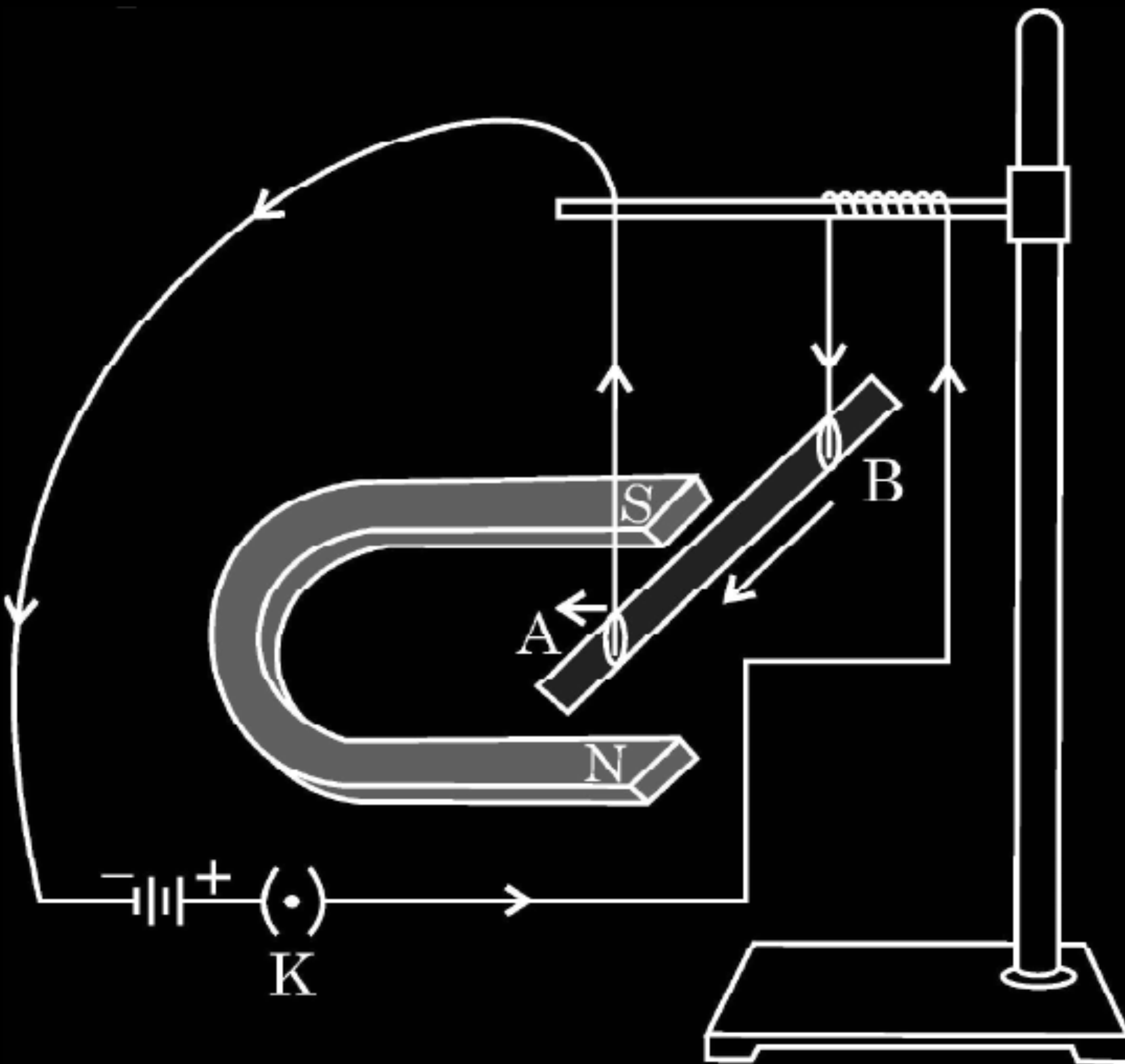
- (i) Maximum ;
- (ii) Minimum ?

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- | | |
|---|---|
| i. Maximum - when the direction of current (current carrying conductor) is perpendicular to the direction of magnetic field. | 1 |
| ii. Minimum - (zero) when the direction of current (current carrying conductor) is parallel / antiparallel / along the direction of magnetic field. | 1 |

14. A student was asked to perform an experiment to study the force on a current carrying conductor in a magnetic field. He took a small aluminum rod AB, a strong horse shoe magnet, some connecting wires, a battery and a switch and connected them as shown. He observed that on passing current, the rod gets displaced. On reversing the direction of current, the direction of displacement also gets reversed. On the basis of your understanding of this phenomenon, answer the following questions :

4



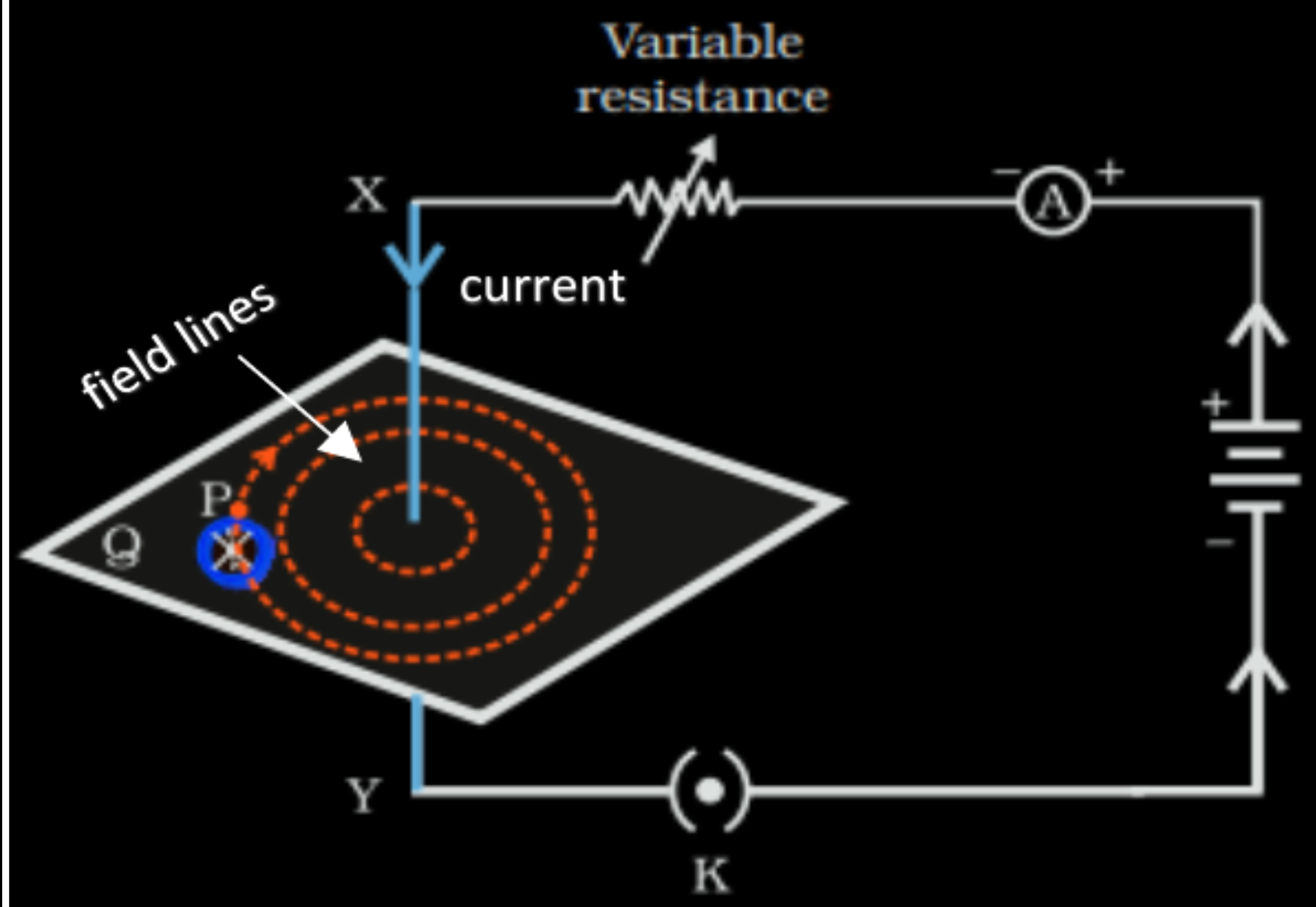
- (a) State the condition under which the displacement of the rod is largest for the same magnitude of current flowing through it.
- (b) State the rule that determines the direction of the force on the conductor AB.
- (c)
 - (i) If the U shaped magnet is held vertically and the aluminum rod is suspended horizontally with its end B towards due north, then on passing current through the rod from B to A as shown, in which direction will the rod be displaced ?
 - (ii) Name any two devices that use current carrying conductors and magnetic field.

OR

Draw the pattern of magnetic field lines produced around a current carrying straight conductor held vertically on a horizontal cardboard. Indicate the direction of the field lines as well as the direction of current flowing through the conductor.

- | | | |
|-----|---|-----------------------------|
| (a) | Direction of current in the rod is perpendicular to the direction of magnetic field. | 1 |
| (b) | Fleming's left-hand rule : Stretch the thumb, forefinger and middle finger of your left-hand such that they are mutually perpendicular. If the first finger points to the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor. | 1 |
| (c) | (i) Towards left or towards west / Into U shape magnet | 1 |
| | (ii) Electric motor/ electric generator/ loudspeakers/ microphones/ electrical measuring instruments (any two) | $\frac{1}{2} + \frac{1}{2}$ |

OR



Direction of current - downward

Direction of field lines – clockwise

Alternatively, if the direction of current is marked upwards then direction of field lines will be anticlockwise.

(Credit full marks if direction of current and field lines are marked in the diagram)

$\frac{1}{2}$
 $\frac{1}{2}$

6. (a) What is a magnetic field ? How is the direction of a magnetic field at a place determined ?

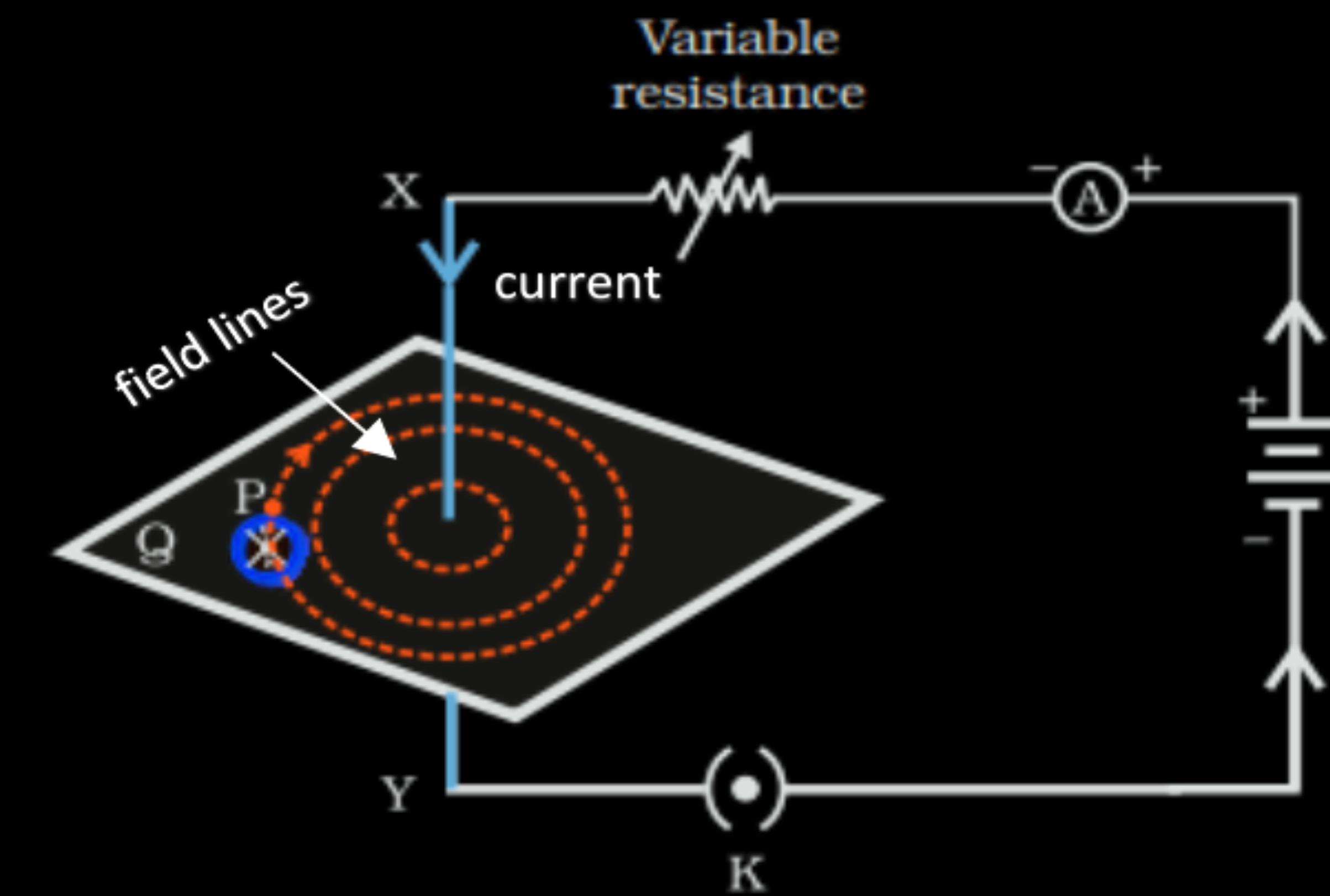
2

CBSE 2022

OR

- (b) What is the pattern of the magnetic field lines produced by a current carrying straight conductor ? State the rule that determines the direction of the magnetic field in this case.

2



6. (a) What is a magnetic field ? How is the direction of a magnetic field at a place determined ? □

2

CBSE 2022

OR

(b) What is the pattern of the magnetic field lines produced by a current carrying straight conductor ? State the rule that determines the direction of the magnetic field in this case.

2

(a)

- Region around a magnet in which its force can be detected.
- The direction of magnetic field at a point is determined by placing a small compass needle. The direction in which its N-pole rests is the direction of magnetic field at that point.

1

1

OR

(b)

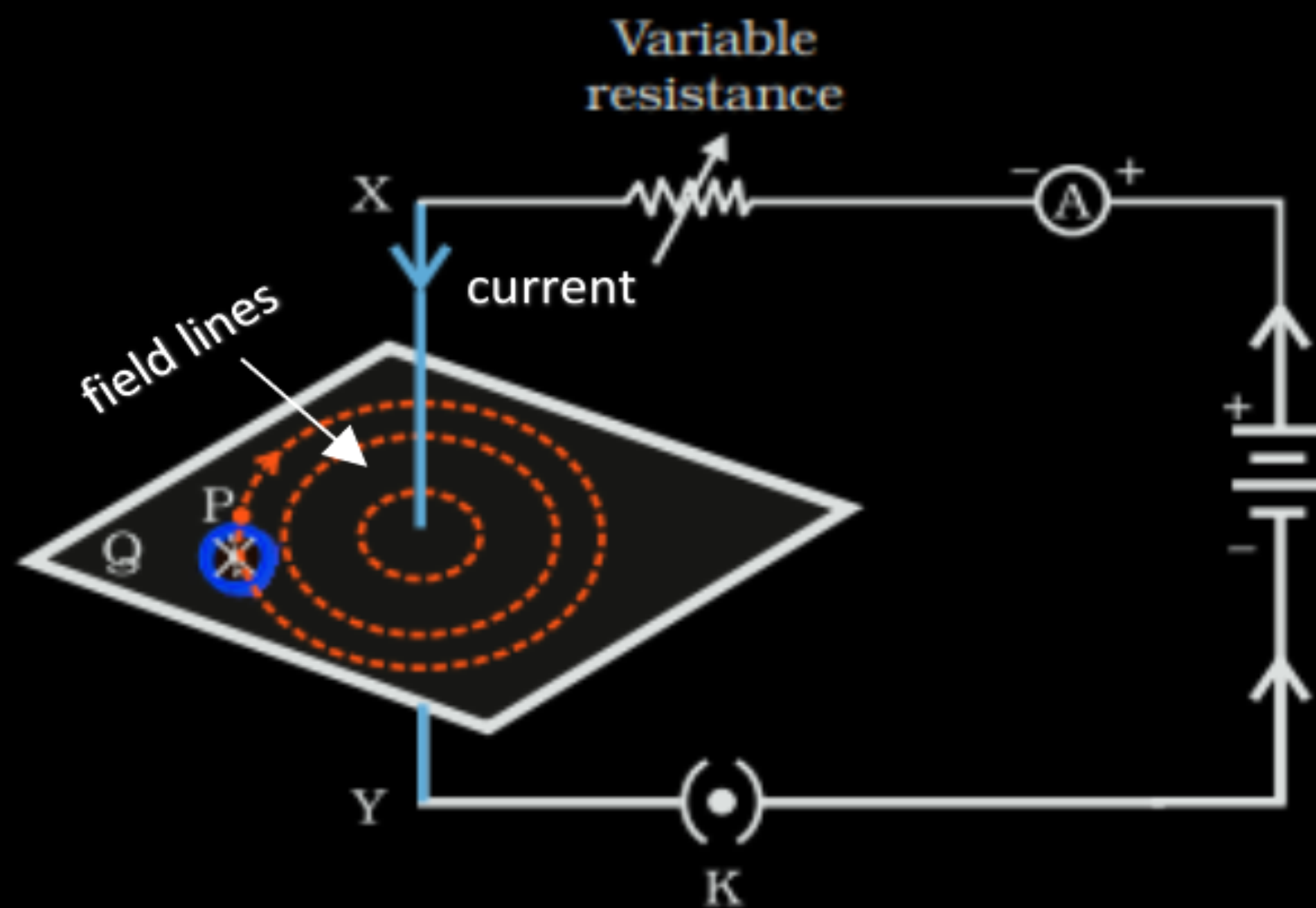
- The magnetic field lines are circles concentric to the straight conductor.
- Right-hand thumb rule:

1

Statement : Imagine that you are holding a current carrying straight conductor in your right hand such that thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of field lines of the magnetic field.

1

Half marks should be deducted if only the name of the law is given and statement not written.



- 11.** (a) (i) What is a Solenoid ? What does the divergence of magnetic field lines near the ends of a current carrying solenoid indicate ?
- (ii) How is an electromagnet made using a current carrying solenoid ?

3

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- 11.** (a) (i) What is a Solenoid ? What does the divergence of magnetic field lines near the ends of a current carrying solenoid indicate ?
- (ii) How is an electromagnet made using a current carrying solenoid ?

3

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(a) (i)

- A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder.
- Divergence of magnetic field lines at the ends indicates that the intensity of magnetic field decreases as we move farther and farther away from the ends.

1

1

(ii) A piece of magnetic material such as soft iron is placed inside a solenoid. When current is switched on, a strong magnetic field produced inside the solenoid magnetises the magnetic material.

1

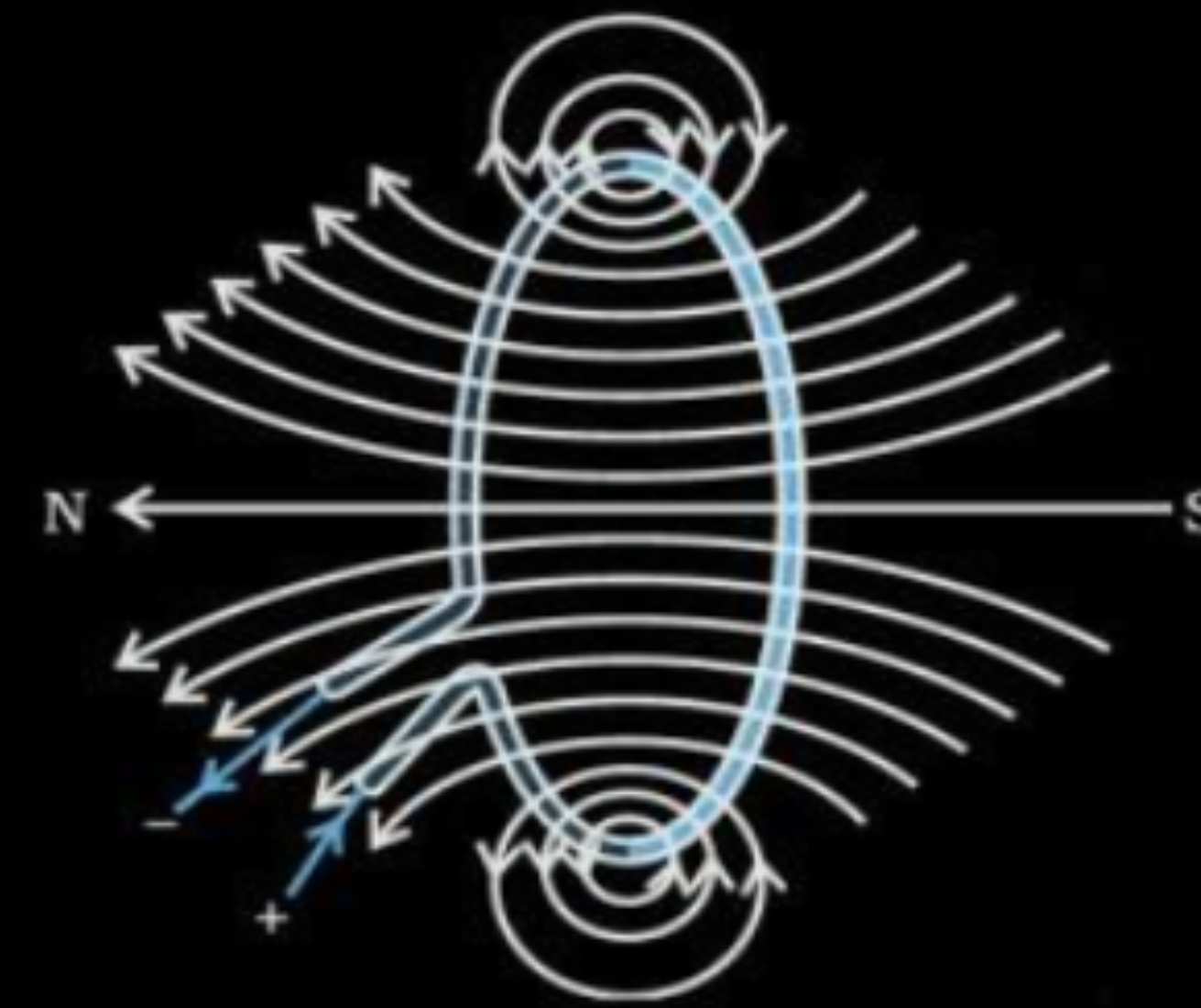
- 12.** (a) Draw the pattern of magnetic field lines produced by a current carrying circular loop showing the direction of current in the loop and the direction of the magnetic field lines.
- (b) State the rule which can be applied to know the direction of magnetic field lines in the above case.

12. (a) Draw the pattern of magnetic field lines produced by a current carrying circular loop showing the direction of current in the loop and the direction of the magnetic field lines.
- (b) State the rule which can be applied to know the direction of magnetic field lines in the above case.

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3

(a) Pattern of magnetic field



1 mark for correct representation of direction of magnetic field lines and current.

(b) Right-hand thumb rule : Hold a current carrying straight conductor in your righthand such that the thumb points towards the direction of current, then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field.

10. (a) Draw the pattern of magnetic field lines around a bar magnet. Mark the position of North Pole, South Pole and the places where the magnetic field is strongest.

(b) Why do the magnetic field lines not intersect each other ?

3

CBSE 2022

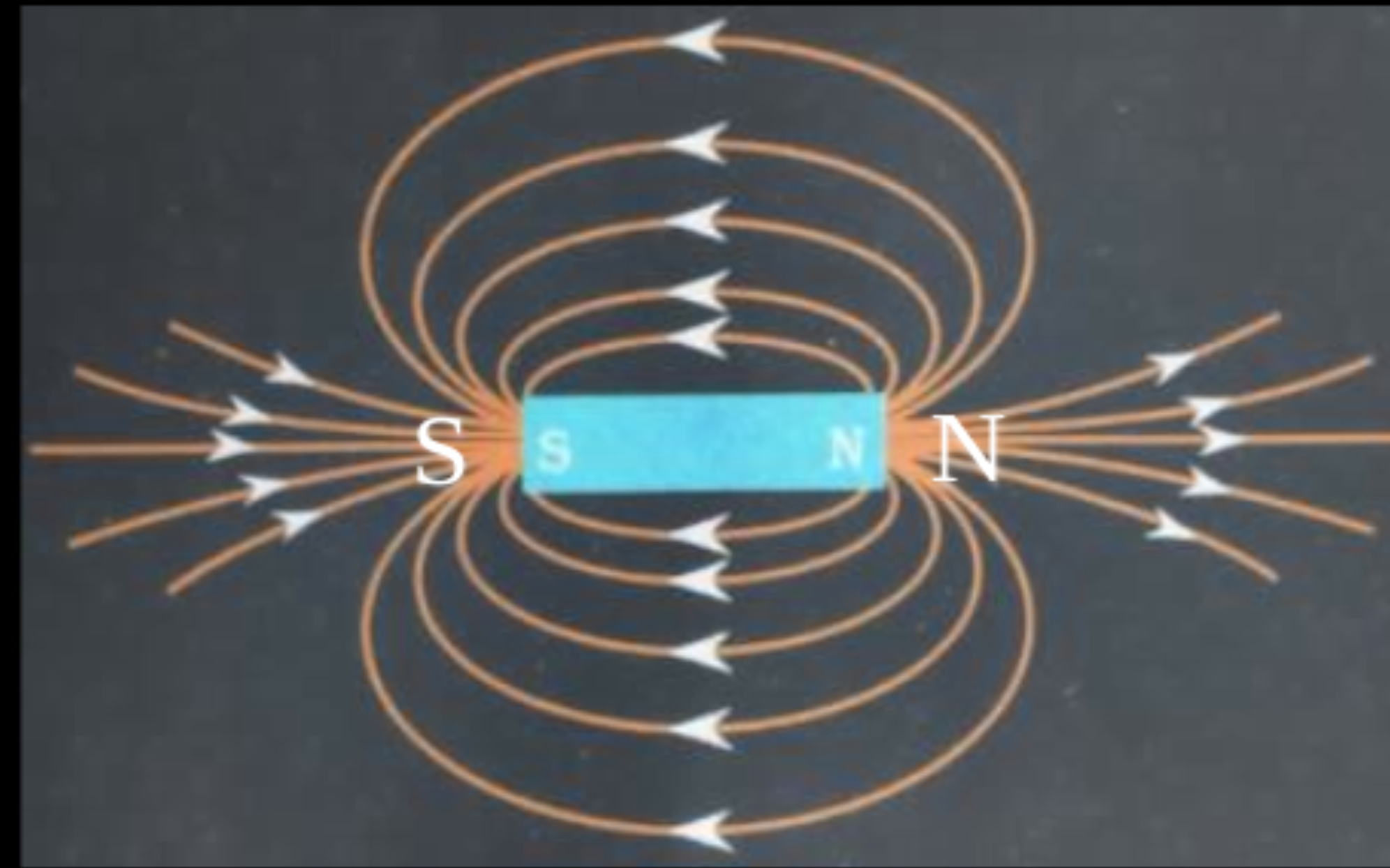
10. (a) Draw the pattern of magnetic field lines around a bar magnet. Mark the position of North Pole, South Pole and the places where the magnetic field is strongest.

(b) Why do the magnetic field lines not intersect each other ?

3

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(a) • Field lines around a bar magnet.



1

- Position of North pole, South pole- to be marked in the diagram
- Field is strongest where lines are crowded / at Poles / or indicated in the diagram.

$\frac{1}{2}$

$\frac{1}{2}$

(b) If two field lines cross each other then at the point of intersection, the compass needle would point towards two directions which is not possible.

1

- 11.** What is a Solenoid ? Draw the pattern of the magnetic field lines around a current carrying solenoid. Mark on the pattern the region where the magnetic field is uniform.

3

CBSE 2022

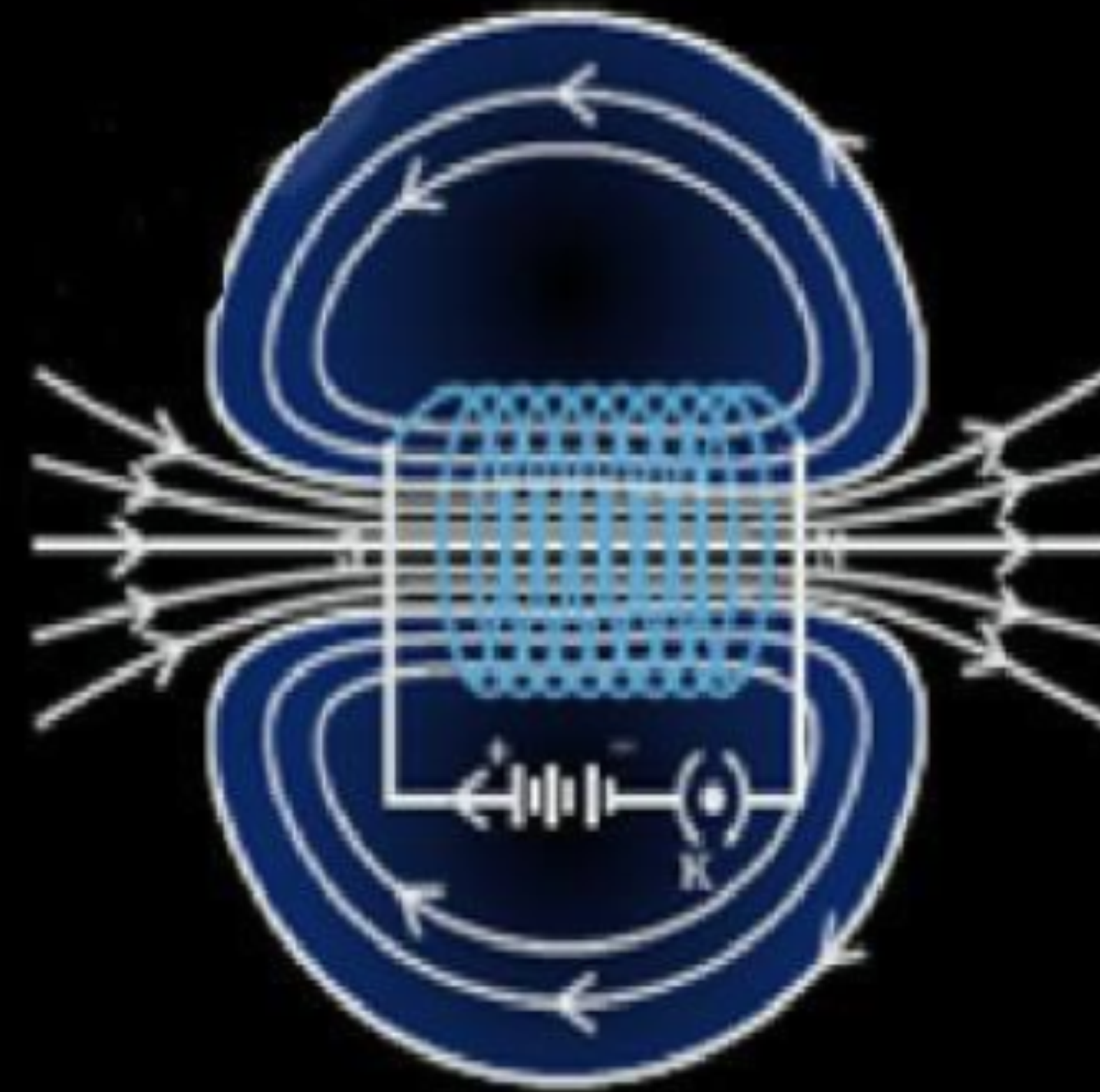
11. What is a Solenoid ? Draw the pattern of the magnetic field lines around a current carrying solenoid. Mark on the pattern the region where the magnetic field is uniform.

3

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- Solenoid: A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder.

- Pattern of field lines:



- Field is uniform only inside the solenoid. (Position of the uniform field to be marked in the figure)

1

1½

½

5. (a) (i) Name and state the rule to determine the direction of force experienced by a current carrying straight conductor placed in a uniform magnetic field which is perpendicular to it.
- (ii) An alpha particle while passing through a magnetic field gets projected towards north. In which direction will an electron project when it passes through the same magnetic field ? 2

5. (a) (i) Name and state the rule to determine the direction of force experienced by a current carrying straight conductor placed in a uniform magnetic field which is perpendicular to it.
- (ii) An alpha particle while passing through a magnetic field gets projected towards north. In which direction will an electron project when it passes through the same magnetic field ?

2

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(a) i) • Fleming's left-hand rule • Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If the first finger points in the direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor. ii) South	$\frac{1}{2}$ 1 $\frac{1}{2}$
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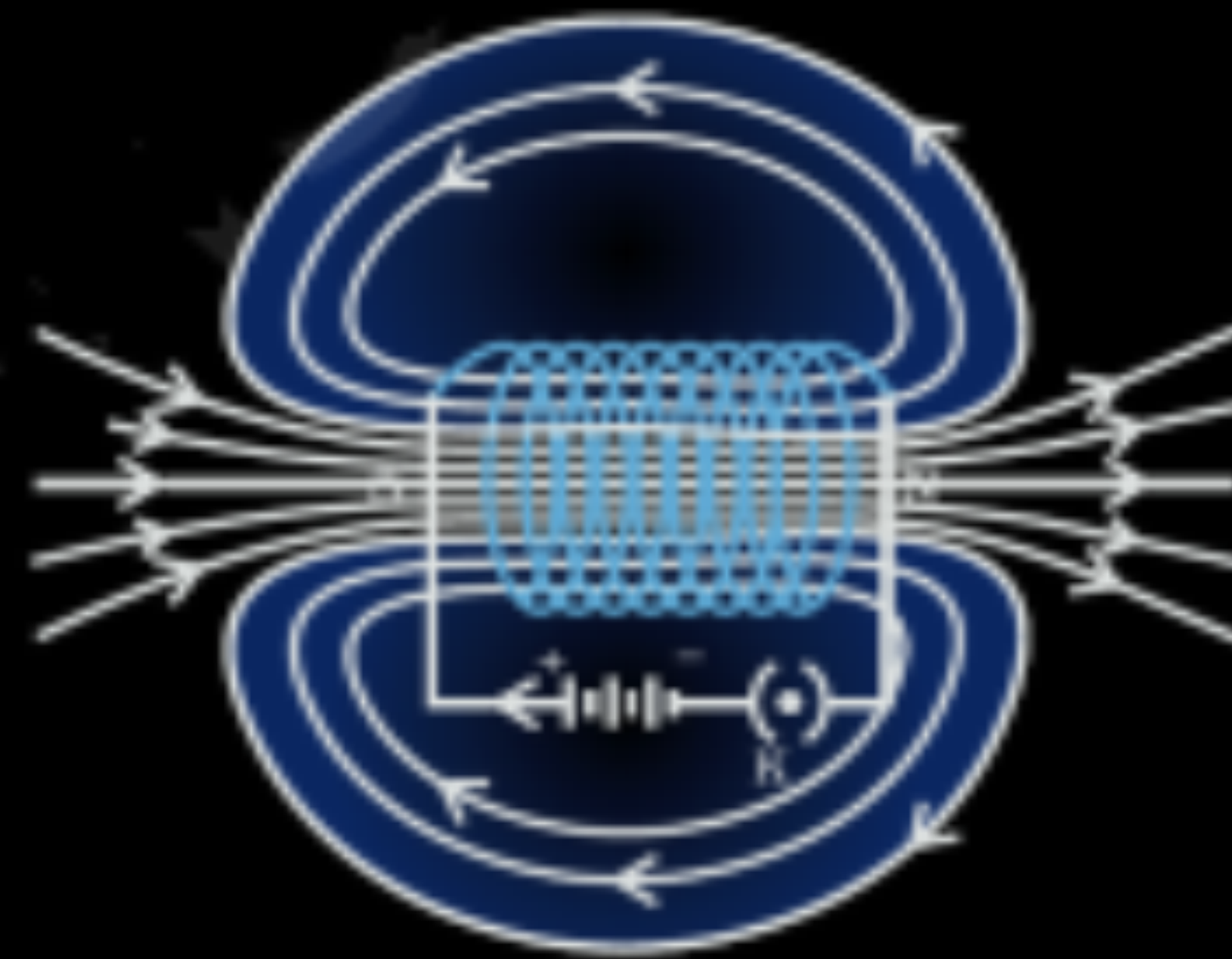
- (b) (i) What is a solenoid ?
- (ii) Draw the pattern of magnetic field lines of the magnetic field produced by a solenoid through which a steady current flows. 2

- (b) (i) What is a solenoid ?
- (ii) Draw the pattern of magnetic field lines of the magnetic field produced by a solenoid through which a steady current flows. 2

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b) i) A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder.

ii)



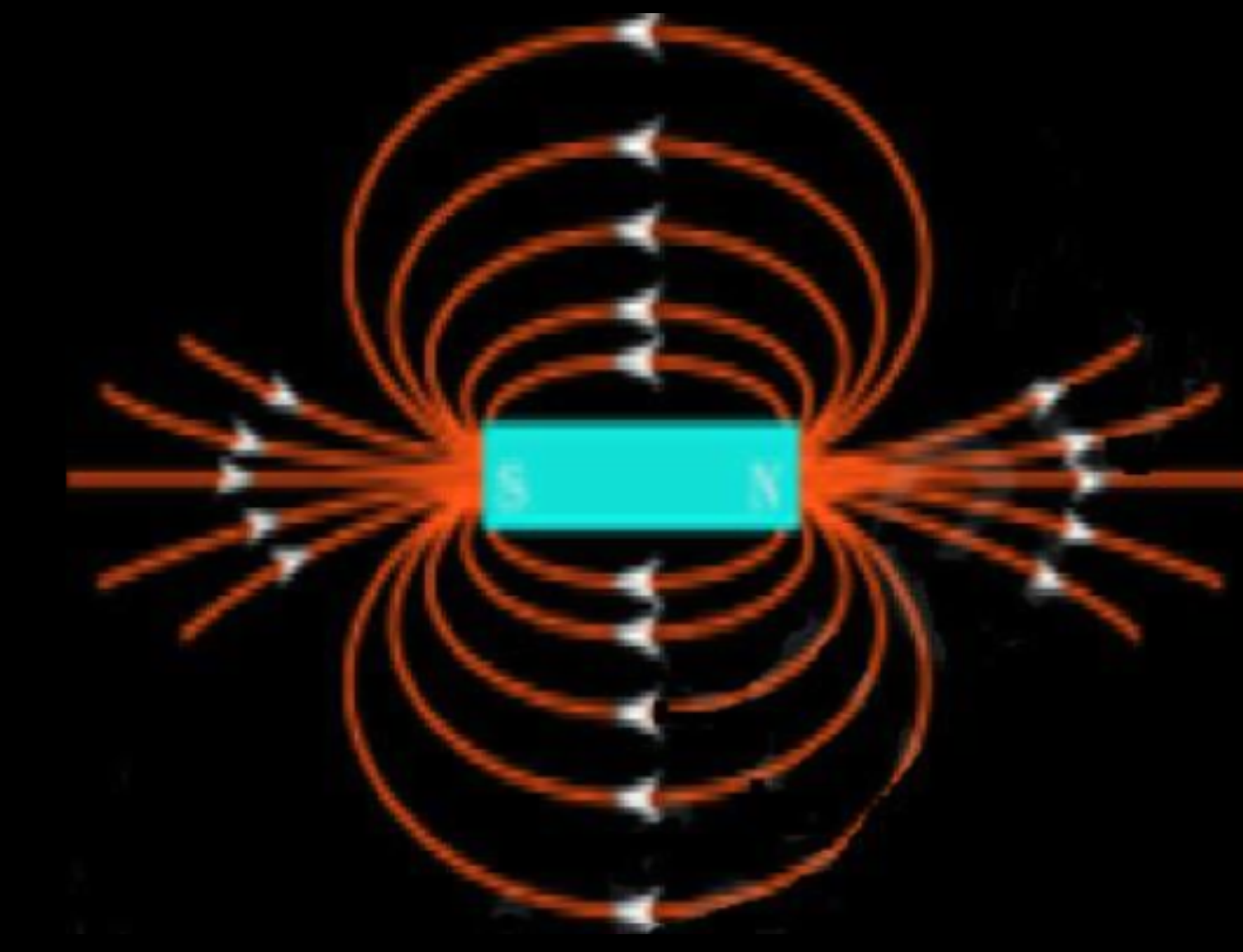
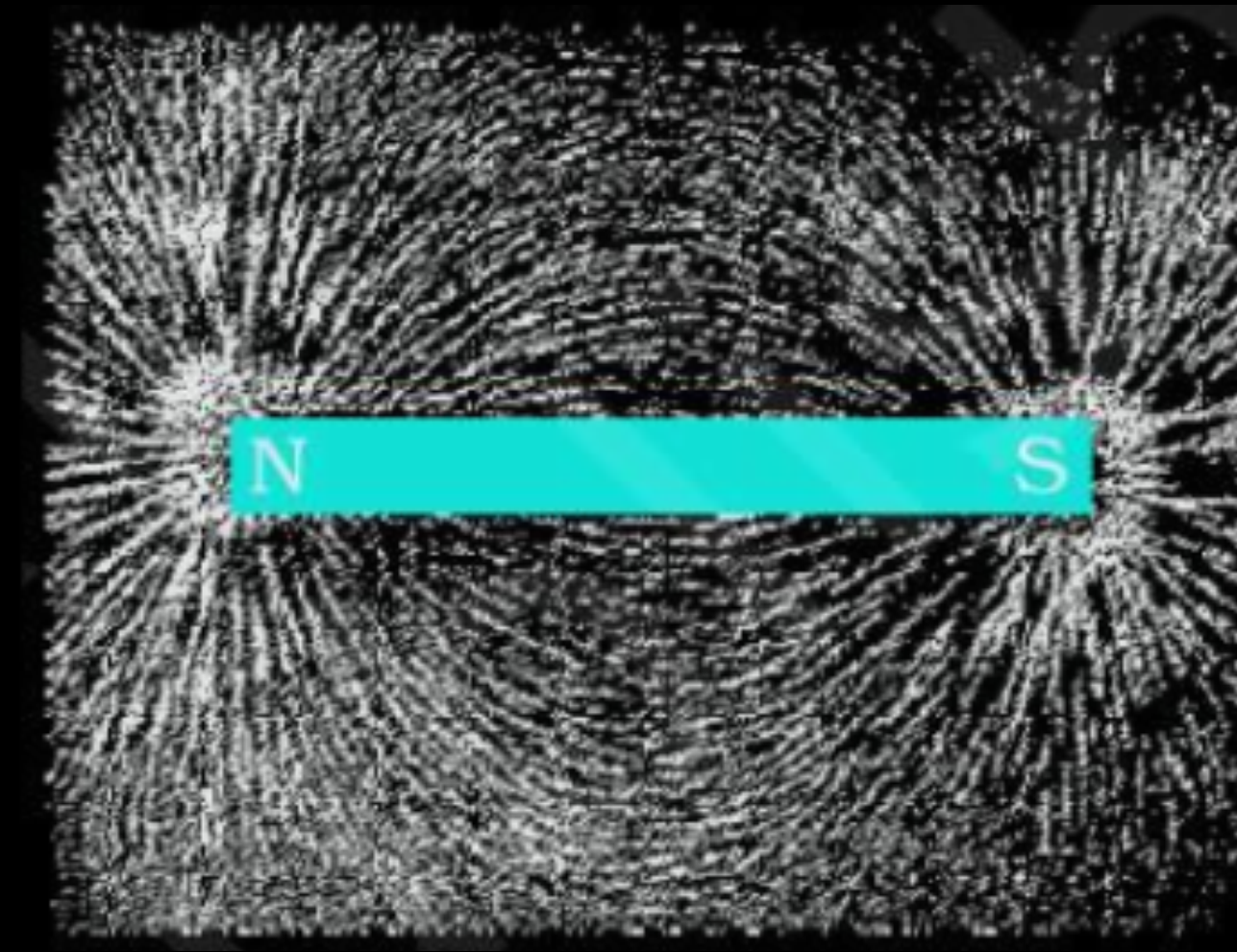
1

14. A student fixes a sheet of white paper on a drawing board using some adhesive materials. She places a bar magnet in the centre of it and sprinkles some iron filings uniformly around the bar magnet using a salt-sprinkler. On tapping the board gently, she observes that the iron filings have arranged themselves in a particular pattern.
- (a) Draw a diagram to show this pattern of iron filings. 1
- (b) What makes iron filings arrange in a definite pattern ? 1
- (c) (i) How is the direction of magnetic field at a point determined using the field lines ? Why do two magnetic field lines not cross each other ? 2

OR

- (ii) How are the magnetic field lines of a bar magnet drawn using a small compass needle ? Draw one magnetic field line each on both sides of the magnet. 2

(a)



1

(c) i) • By placing a compass needle on magnetic field lines, direction of north pole will give direction of magnetic field.

1

- If they cross or intersect , it means that at the point of intersection the compass needle would point into two directions, which is not possible. /

1

If they cross or intersect, it means that at the point of intersection there will be direction of two resultant fields which is not possible.

(c) ii) • Take a small bar magnet, place it in the centre of the drawing sheet fixed on a drawing board and mark its boundary.

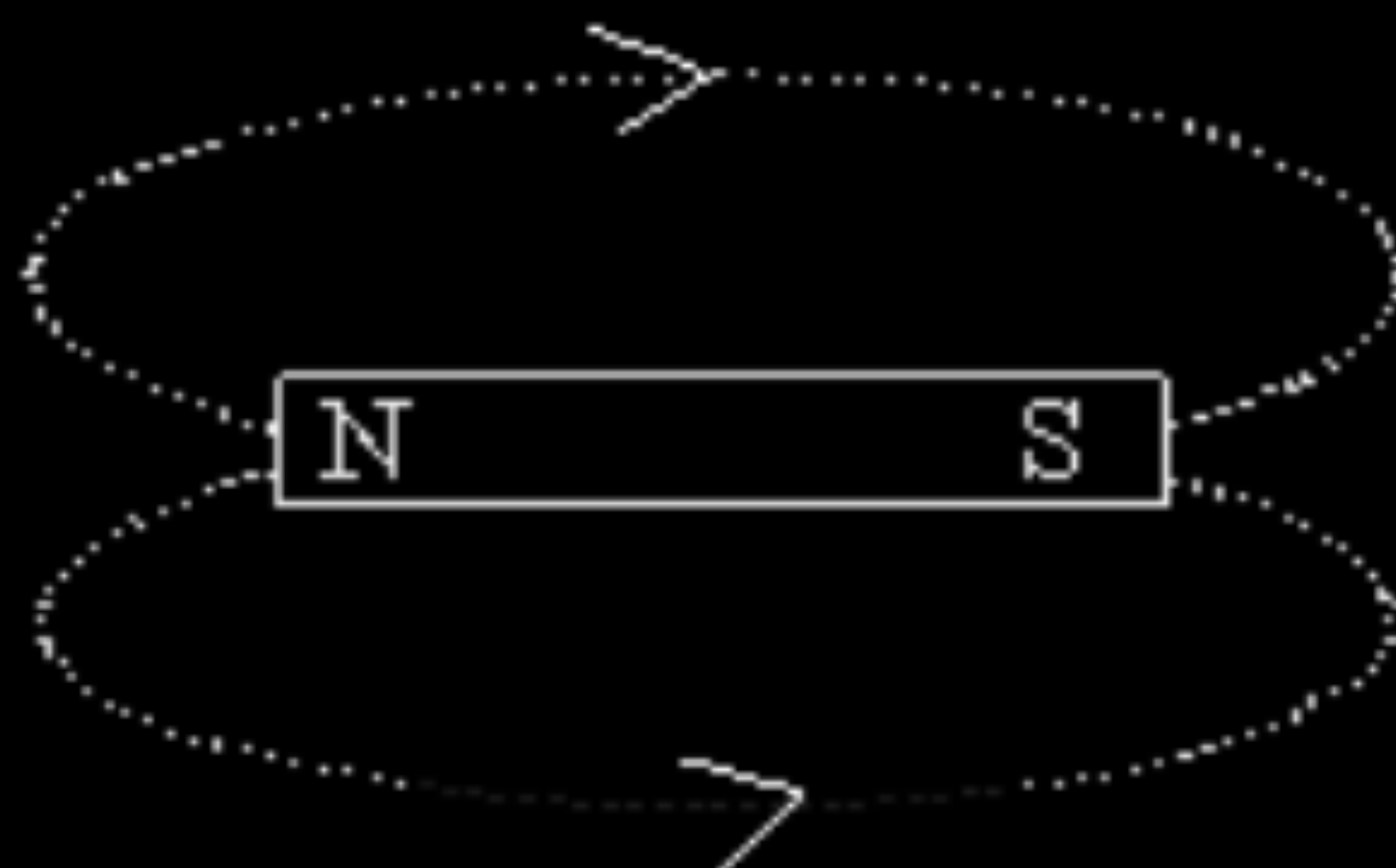
$\frac{1}{2}$

- Place a small compass needle near the north pole of the magnet, south pole of the compass needle points towards the north pole.

$\frac{1}{2}$

- Mark the position of two ends of the needle. Now move the needle to a new position such that the south pole of needle occupies the position previously occupied by the north pole and again mark the new position of the north pole. In this way proceed step by step till you reach the south pole of the magnet. Join the points marked to get a field line. Similarly draw one more field line on the other side of the magnet.

$\frac{1}{2}$



- 30.**
- (a) State Fleming's Left-hand rule.
 - (b) List three characteristic features of the electric current used in our homes.
 - (c) What is a fuse ? Why is it called a safety device ?
 - (d) Why is it necessary to earth metallic electric appliances ?

5

(a)Flemings' Left hand rule: Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If the forefinger points in the direction of magnetic field, middle finger in the direction of current , then the thumb will point in the direction of motion or force acting on the conductor.	1½
(b) Three charactersticfeatures : • Reverses direction periodically. • Frequency of 50 Hz. • Potential difference between live wire and netural wire is about 220V.	½ × 3
(c) • Fuse is a safety device used in a circuit (or appliance) to prevent damage due to overloading/ short circuiting. • It protects the circuit (or appliance) by stopping the flow of any unduly high electric current / If current larger than the specified valueflows through the circuit , due to Joule's heating effect the fuse wire melts and breaks the circuit.	½ ½
(d) It provides a low resistance conducting path for the current and protects the user from electric shock due to leakage of current.	1

23. Give reasons for the following :

- (i) There is either a convergence or a divergence of magnetic field lines near the ends of a current carrying straight solenoid.
- (ii) The current carrying solenoid when suspended freely rests along a particular direction.
- (iii) The burnt out fuse should be replaced by another fuse of identical rating.

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23. Give reasons for the following :

- (i) There is either a convergence or a divergence of magnetic field lines near the ends of a current carrying straight solenoid.
- (ii) The current carrying solenoid when suspended freely rests along a particular direction.
- (iii) The burnt out fuse should be replaced by another fuse of identical rating.

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3

(i) The strength of magnetic field is higher near the poles /ends of solenoid.	1
(ii) A current carrying solenoid behaves as a bar magnet.	1
(iii) If a fuse , with a defined rating , is replaced by one with a larger rating then the fuse wire will not burn even when a current greater than safe limit is flowing. As a result the electrical circuit / appliances will be damaged.	1

10. At the time of short circuit, the electric current in the circuit :

- | | |
|---------------------------|-----------------------|
| (a) vary continuously | (b) does not change |
| (c) reduces substantially | (d) increases heavily |

1

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10. At the time of short circuit, the electric current in the circuit :

- | | |
|---------------------------|-----------------------|
| (a) vary continuously | (b) does not change |
| (c) reduces substantially | (d) increases heavily |

1

CBSE 2020

(d) / Increases heavily

- 30.** (a) What is an electromagnet ? List any two uses.
- (b) Draw a labelled diagram to show how an electromagnet is made.
- (c) State the purpose of soft iron core used in making an electromagnet.
- (d) List two ways of increasing the strength of an electromagnet if the material of the electromagnet is fixed

30. (a) What is an electromagnet ? List any two uses.
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 (d) List two ways of increasing the strength of an electromagnet if the material of the electromagnet is fixed

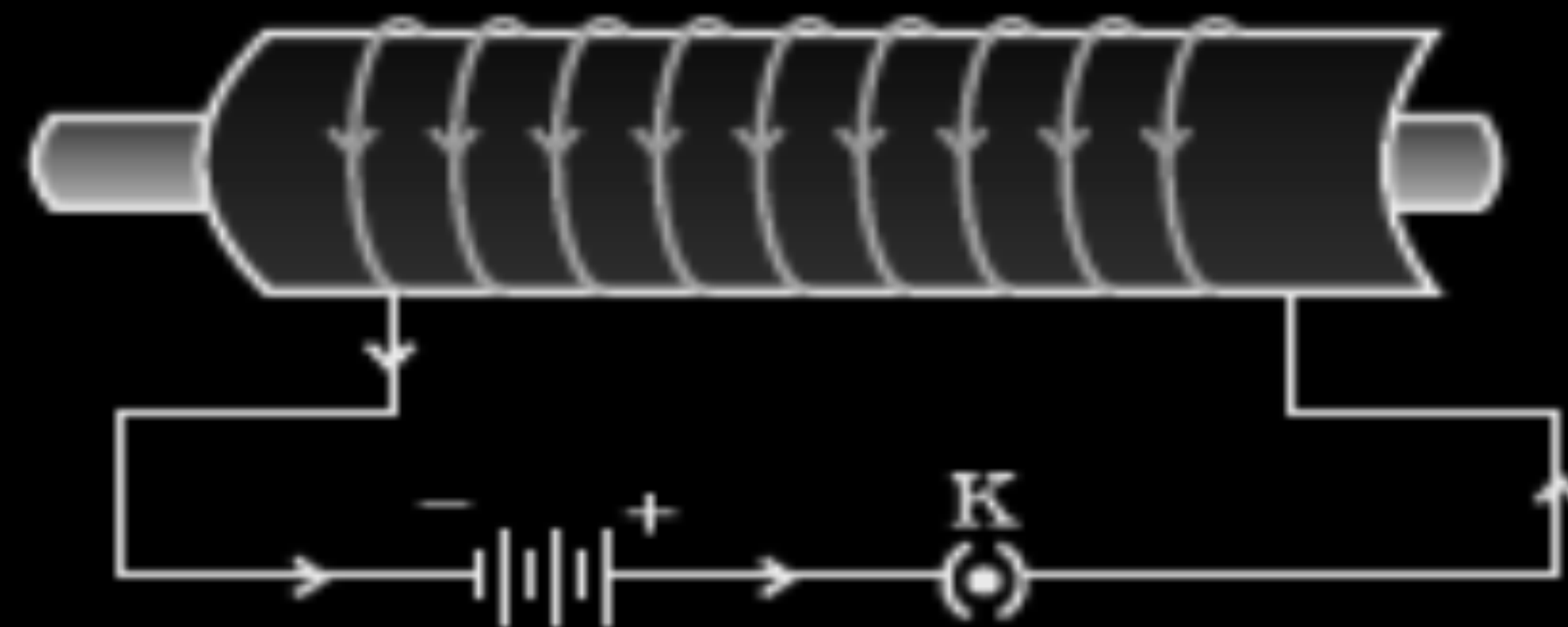
CBSE 2020 5

a) A current carrying solenoid is called an electromagnet /when soft iron is placed inside a solenoid carrying current, the soft iron piece behaves like a magnet so long as electric current passes through it. The magnet so formed is electromagnet.

Uses: In electric motors, electric bells,

(or any other)

b)



(Direction of current)

c) Soft iron core is used to increase the strength/power of the electromagnet.

d) i) By increasing the current

ii) By increasing the number of turns in the coil.

1

$\frac{1}{2} + \frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

Draw the pattern of magnetic field lines produced around a current carrying straight conductor passing perpendicularly through a horizontal cardboard. State and apply right-hand thumb rule to mark the direction of the field lines. How will the strength of the magnetic field change when the point where magnetic field is to be determined is moved away from the straight conductor ? Give reason to justify your answer.

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Draw the pattern of magnetic field lines produced around a current carrying straight conductor passing perpendicularly through a horizontal cardboard. State and apply right-hand thumb rule to mark the direction of the field lines. How will the strength of the magnetic field change when the point where magnetic field is to be determined is moved away from the straight conductor ? Give reason to justify your answer.

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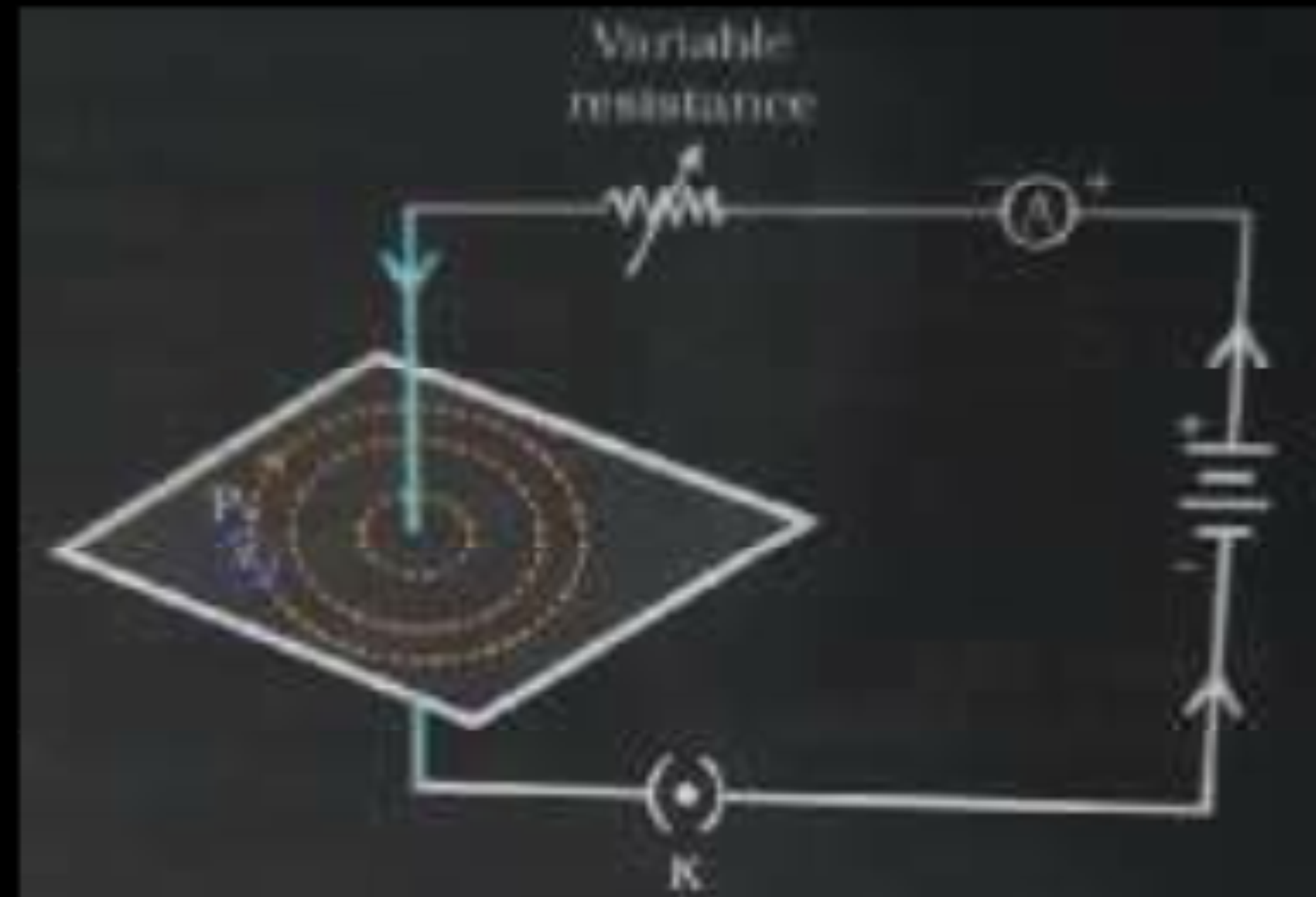


Diagram 1 ½ and direction ½

1+1

Statement of right hand thumb rule.

1

The magnetic field strength decreases with increase of distance from the current carrying conductor.

1

Reason: There is inverse relation between field strength and distance from current carrying conductor.

1

Note: Direction of magnetic field should be in accordance with direction of current

- (a) Name and state the rule to determine the direction of force experienced by a current carrying straight conductor placed in a uniform magnetic field which is perpendicular to it.

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- (a) Name and state the rule to determine the direction of force experienced by a current carrying straight conductor placed in a uniform magnetic field which is perpendicular to it.

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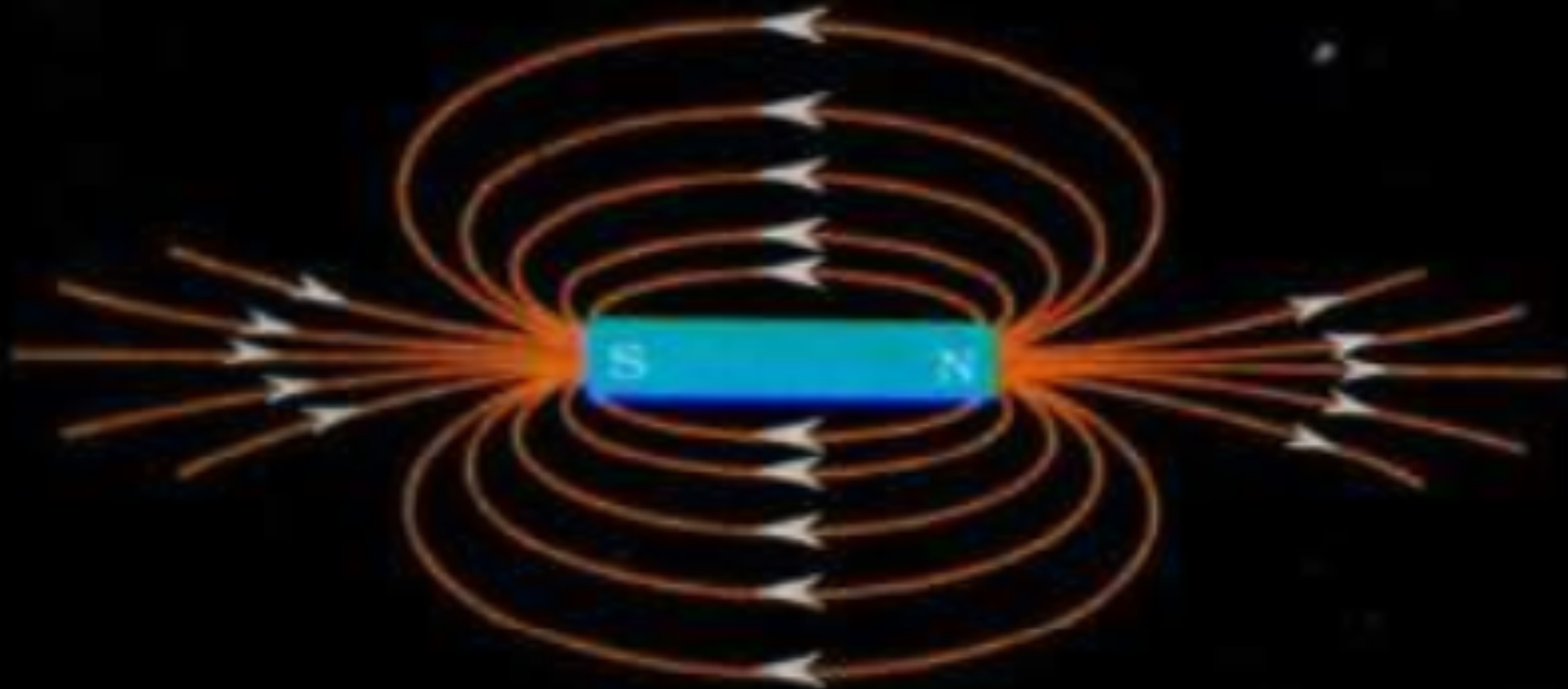
- | | |
|---|---|
| • The rule is Fleming's left hand rule. | 1 |
| • If the first finger points in the direction of the magnetic field and the second finger in the direction of current then the thumb will point in the direction of motion or the force acting on the conductor | 2 |

Draw magnetic field lines around a bar magnet.

CBSE 2019

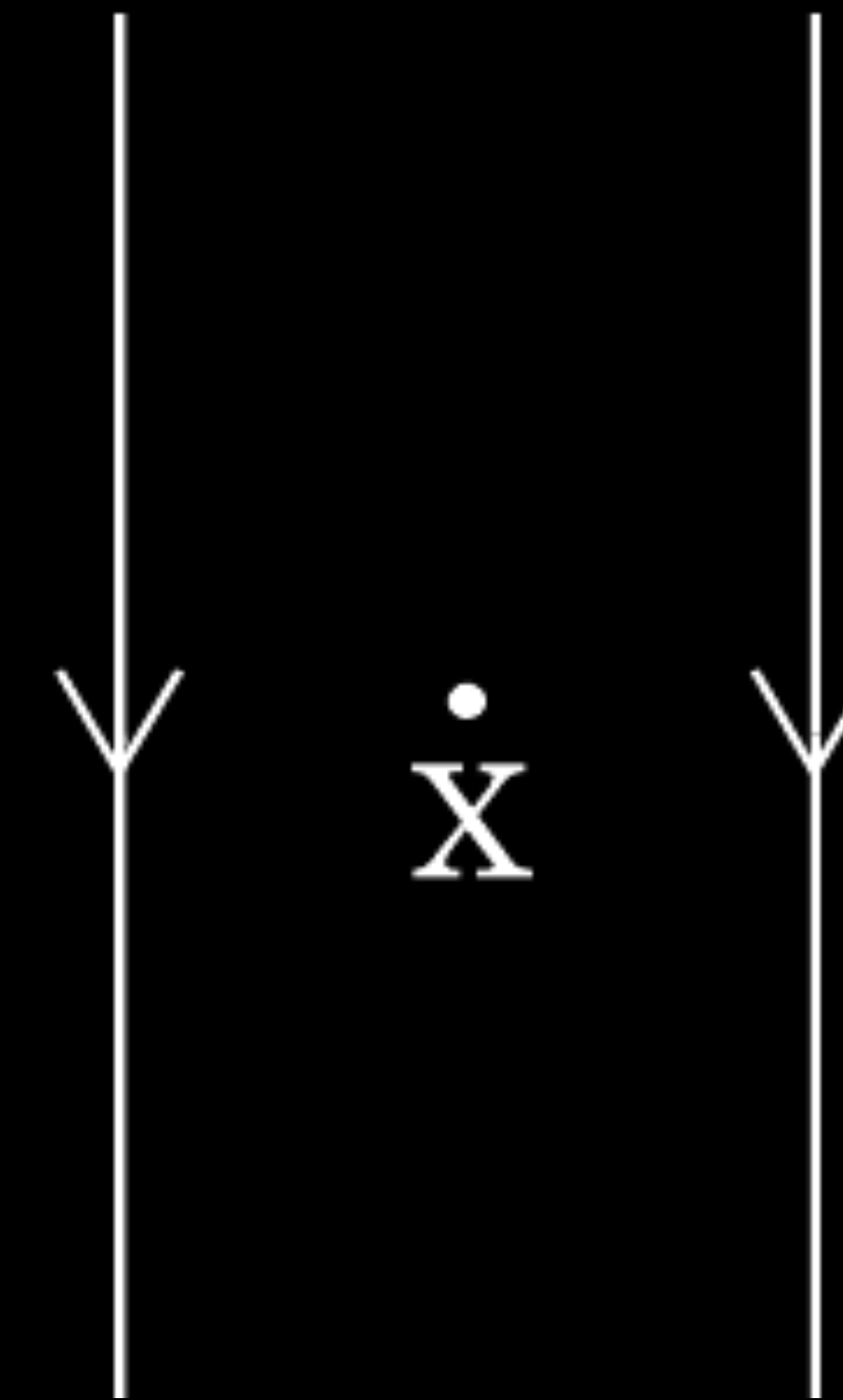
Draw magnetic field lines around a bar magnet.

CBSE 2019

Field pattern  Polarity Direction of field lines	Diagram Labelling 1 1
---	---

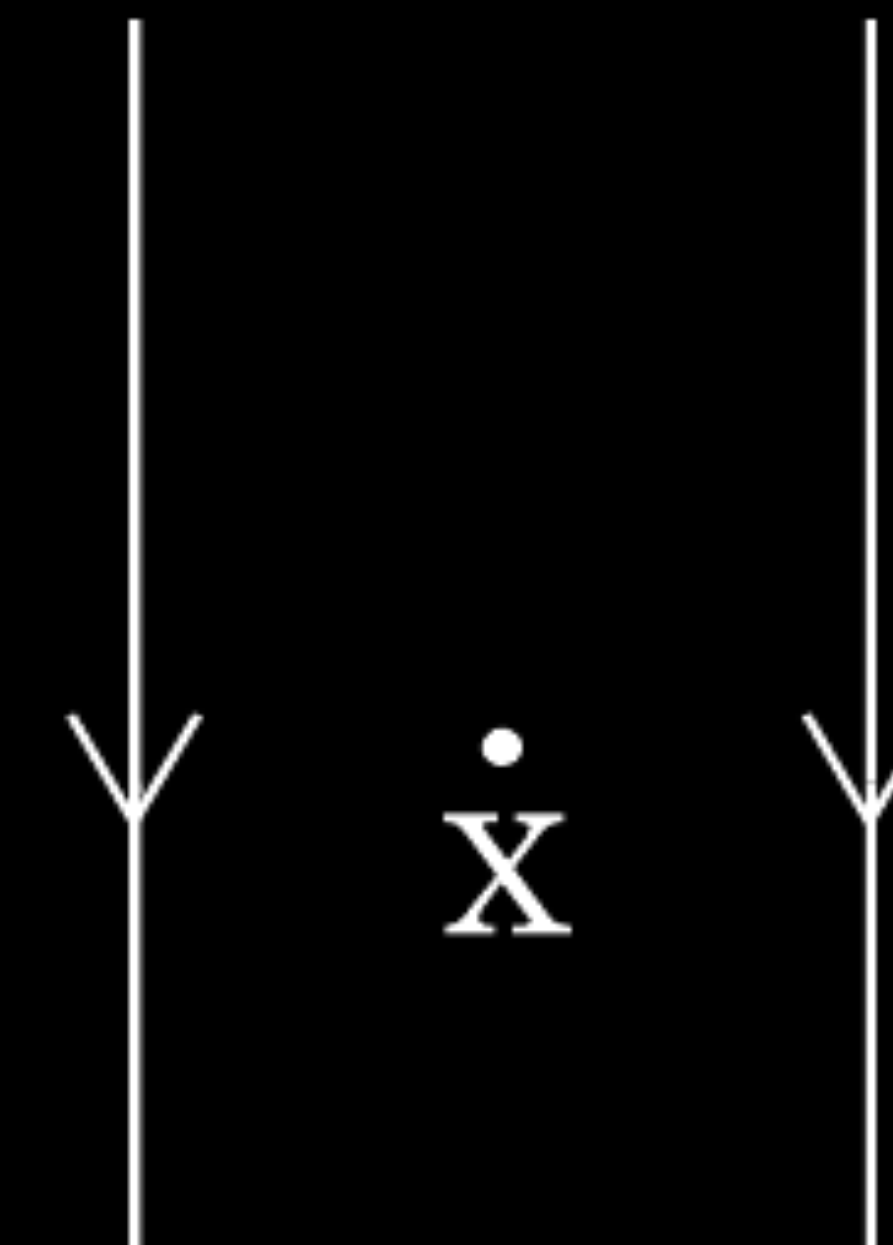
The following diagram shows two parallel straight conductors carrying same current. Copy the diagram and draw the pattern of the magnetic field lines around them showing their directions. What is the magnitude of magnetic field at a point 'X' which is equidistant from the conductors ? Give justification for your answer.

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The following diagram shows two parallel straight conductors carrying same current. Copy the diagram and draw the pattern of the magnetic field lines around them showing their directions. What is the magnitude of magnetic field at a point 'X' which is equidistant from the conductors ? Give justification for your answer.

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Magnetic field will be nullified. B will be zero at X.
 Since, direction of Magnetic Field lines due to both wires opposite to each other.

$\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}$

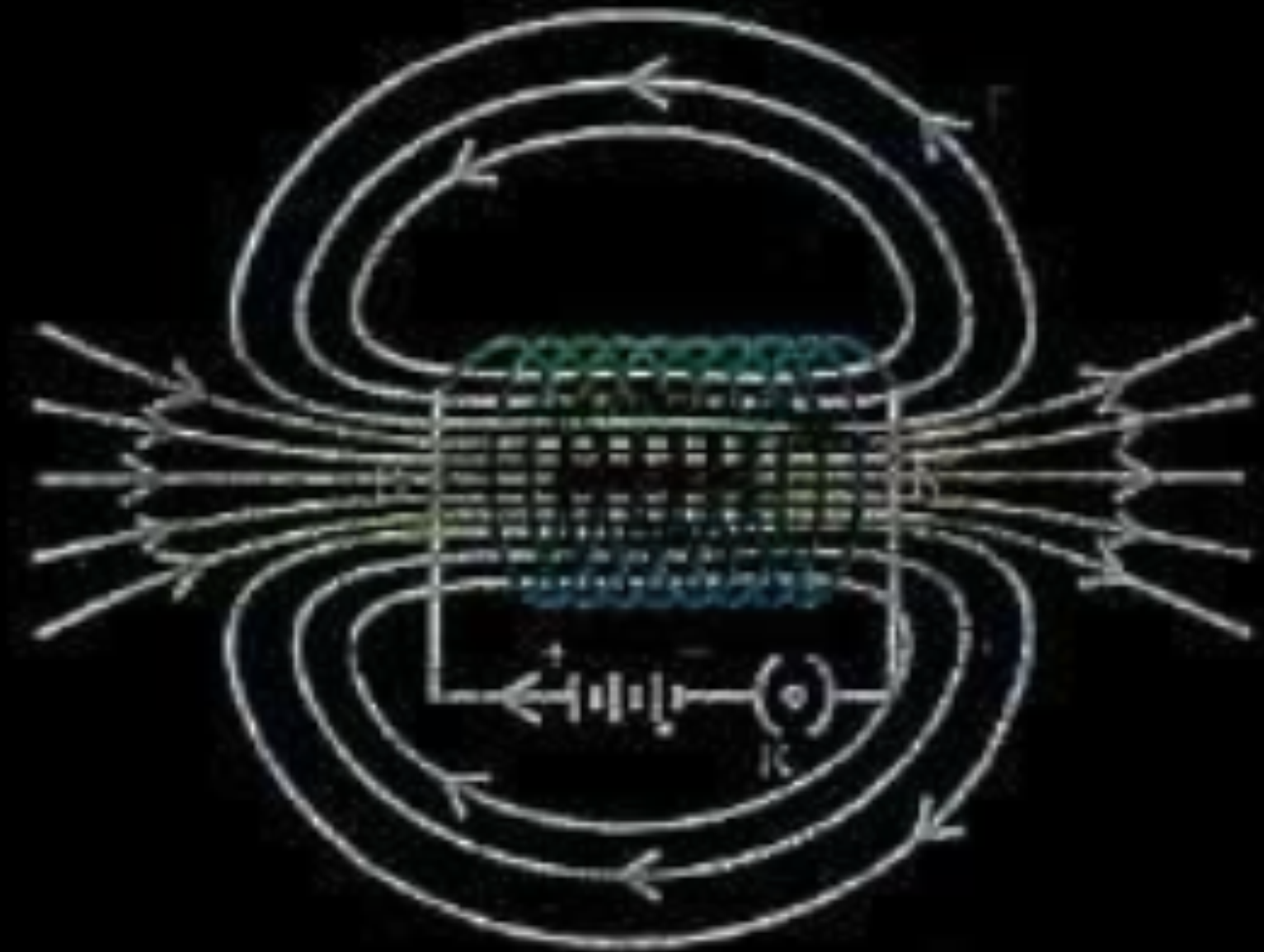
1

Draw magnetic field lines in and around a current carrying straight solenoid.

CBSE 2019

Draw magnetic field lines in and around a current carrying straight solenoid.

CBSE 2019

Field pattern	
Polarity	
Direction of field	

It is established that an electric current through a conductor produces a magnetic field around it. Is there a similar magnetic field produced around a thin beam of moving (i) alpha particles, (ii) neutrons ? Justify your answer in each case.

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It is established that an electric current through a conductor produces a magnetic field around it. Is there a similar magnetic field produced around a thin beam of moving (i) alpha particles, (ii) neutrons ? Justify your answer in each case.

CBSE 2019

• Yes because alpha particles are positively charged and moving charged particles produce magnetic field . • No because neutron has no charge. No magnetic field produced.	$\frac{1}{2} \times 2$ $\frac{1}{2} \times 2$
--	--

State right-hand thumb rule to determine the direction of magnetic field around a current carrying conductor. Apply this rule to find the direction of magnetic field inside and outside a circular loop of wire lying in the plane of a table and current is flowing through it clockwise.

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State right-hand thumb rule to determine the direction of magnetic field around a current carrying conductor. Apply this rule to find the direction of magnetic field inside and outside a circular loop of wire lying in the plane of a table and current is flowing through it clockwise.

CBSE 2019

• Right hand thumb rule: If we hold a current carrying straight conductor in our right hand , such that the thumb gives the direction of electric current then the curled fingers give the direction of the magnetic field.	1
• The magnetic field will be directed perpendicular into the plane of table inside the circular loop.	1
• The magnetic field will be directed perpendicular outside plane of table for a point outside the circular loop. / Diagram: (diagram is not necessary)	1

A compass needle is placed near a current carrying straight conductor. State your observation for the following cases and give reasons for the same in each case :

- (a) Magnitude of electric current is increased.
- (b) The compass needle is displaced away from the conductor.

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A compass needle is placed near a current carrying straight conductor. State your observation for the following cases and give reasons for the same in each case :

- (a) Magnitude of electric current is increased.
- (b) The compass needle is displaced away from the conductor.

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(a) More deflection in compass needle ,Magnetic field is increased.	1
(b) Less deflection in compass needle, Magnetic field is reduced/decreased.	1

List four properties of magnetic field lines.

CBSE 2019

List four properties of magnetic field lines.

CBSE 2019

- | | |
|------|---|
| i. | The field lines emerge from north pole and merge at south pole. |
| ii. | The magnetic field lines are stronger at poles. |
| iii. | The magnetic field lines do not cross /intersect each other. |
| iv. | The magnetic field lines are closed curves. |

$\frac{1}{2} \times 4$

What is a solenoid ? Draw the pattern of magnetic field lines of (i) a current carrying solenoid and (ii) a bar magnet. List two distinguishing features between the two fields.

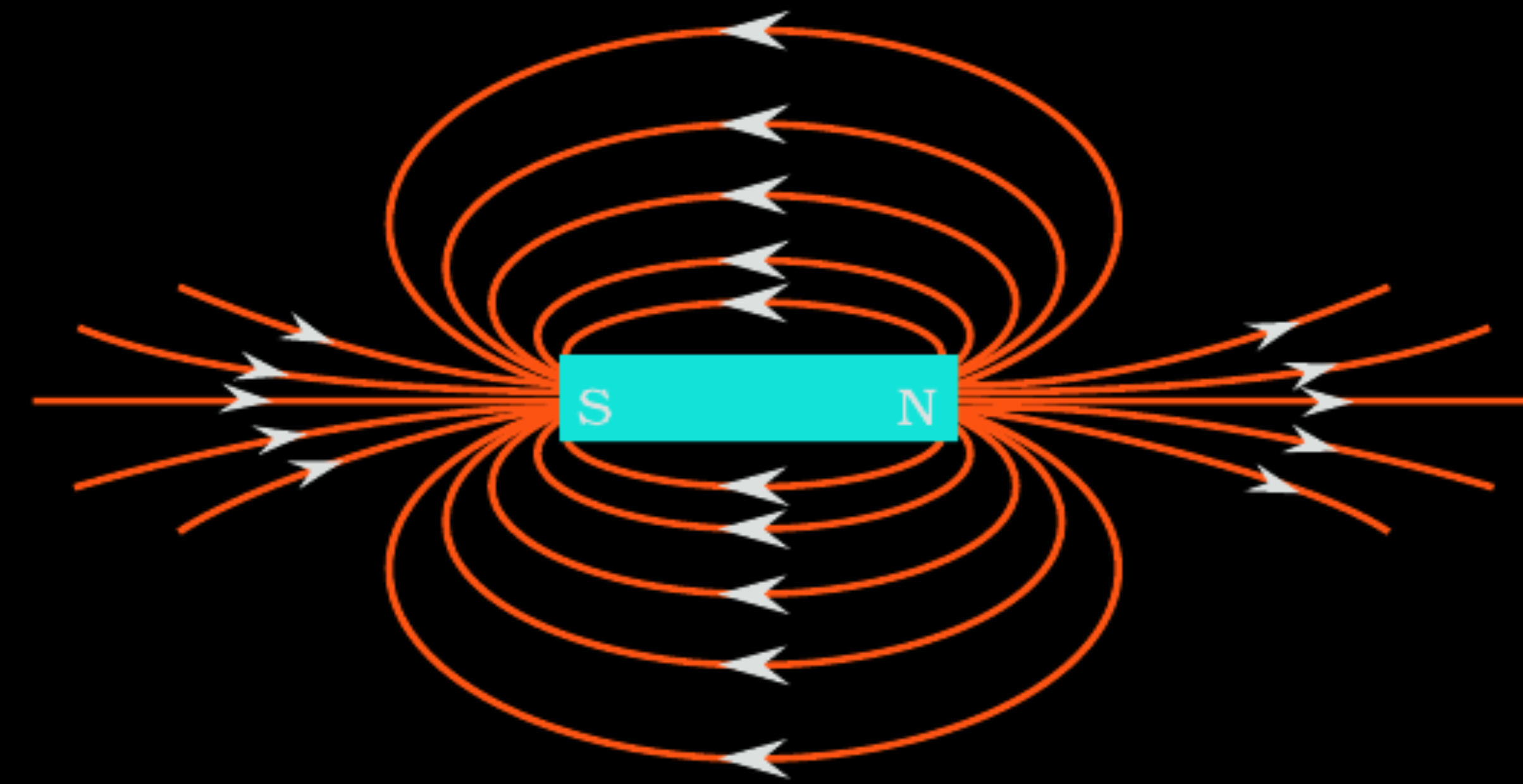
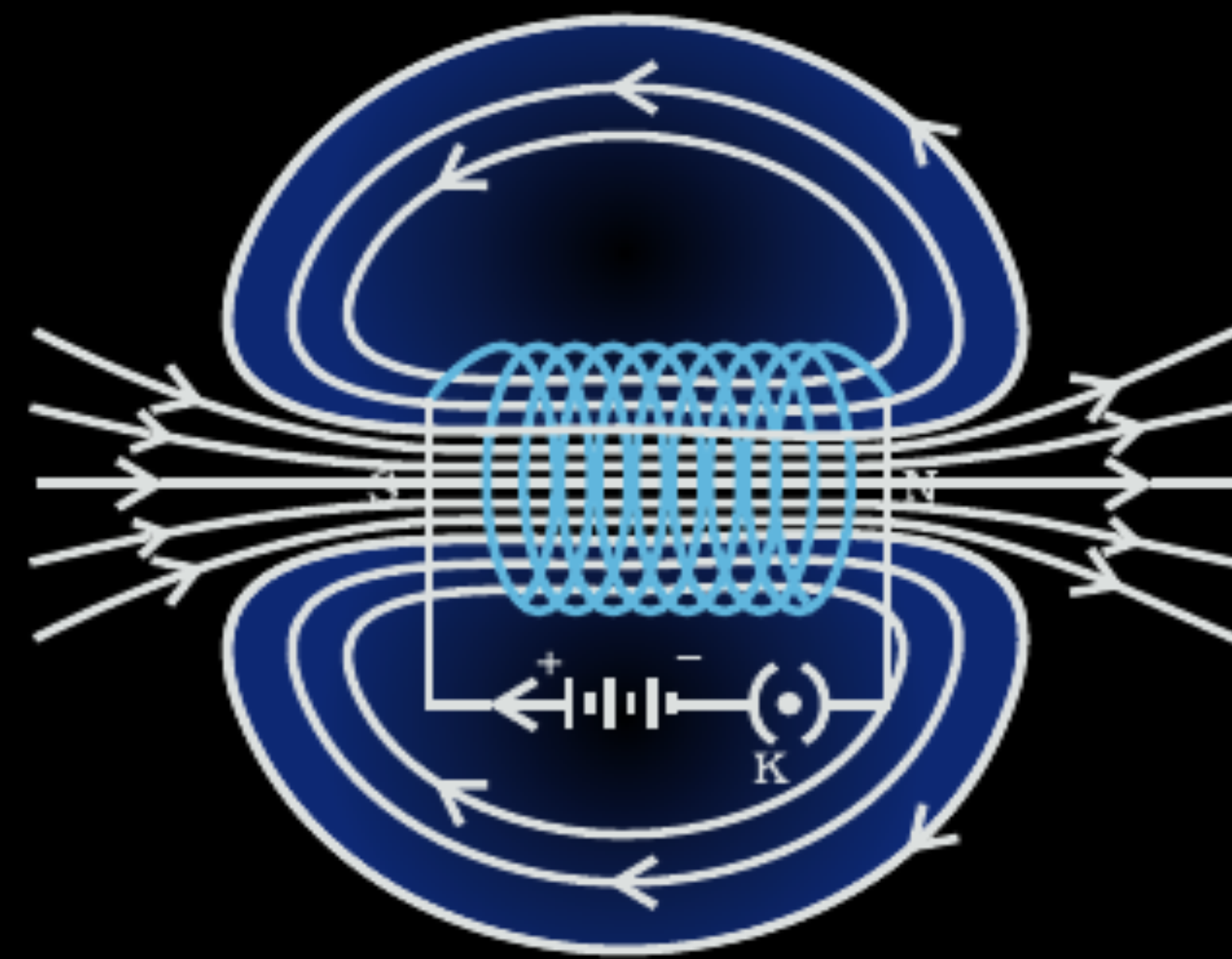
CBSE 2019

What is a solenoid ? Draw the pattern of magnetic field lines of (i) a current carrying solenoid and (ii) a bar magnet. List two distinguishing features between the two fields.

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- A coil of many turns of insulated copper wire wrapped closely in the shape of a cylinder

1



- Distinguishing features –

Solenoid	Bar Magnet
1) Field disappear on stopping the current	1) No effect of current on field.
2) Strength of the field can be changed by changing the current	2) Strength cannot be changed
3) Direction can be reversed by changing the direction of current through it.	3) Direction is fixed and cannot be reversed.

(Any two features)

2

- (a) Draw magnetic field lines produced around a current carrying straight conductor passing through a cardboard. Name, state and apply the rule to mark the direction of these field lines.
- (b) How will the strength of the magnetic field change when the point where magnetic field is to be determined is moved away from the straight wire carrying constant current? Justify your answer.

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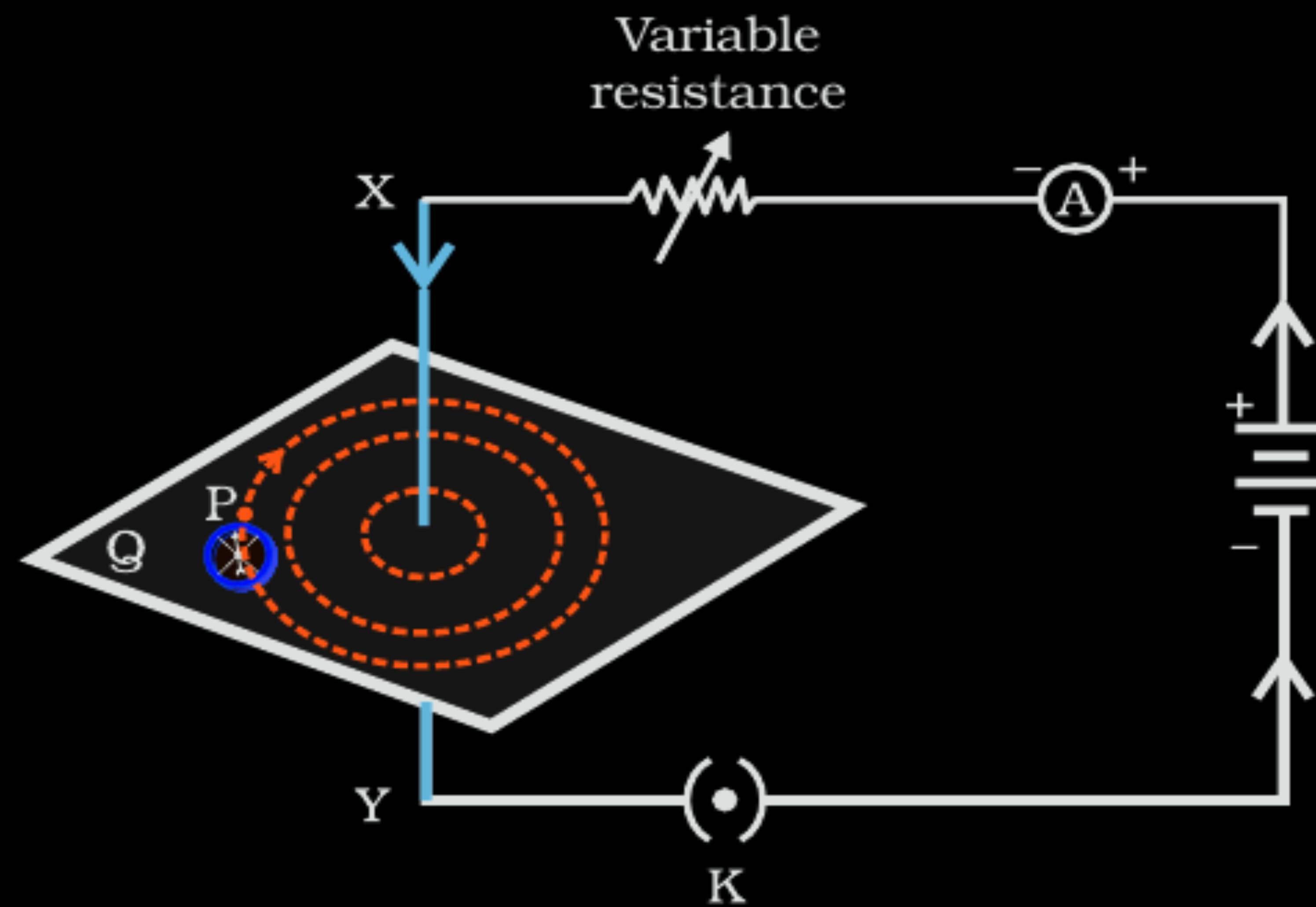


Diagram : 1

Direction : $\frac{1}{2}$

- Rule – Right hand thumb rule

Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field.

1

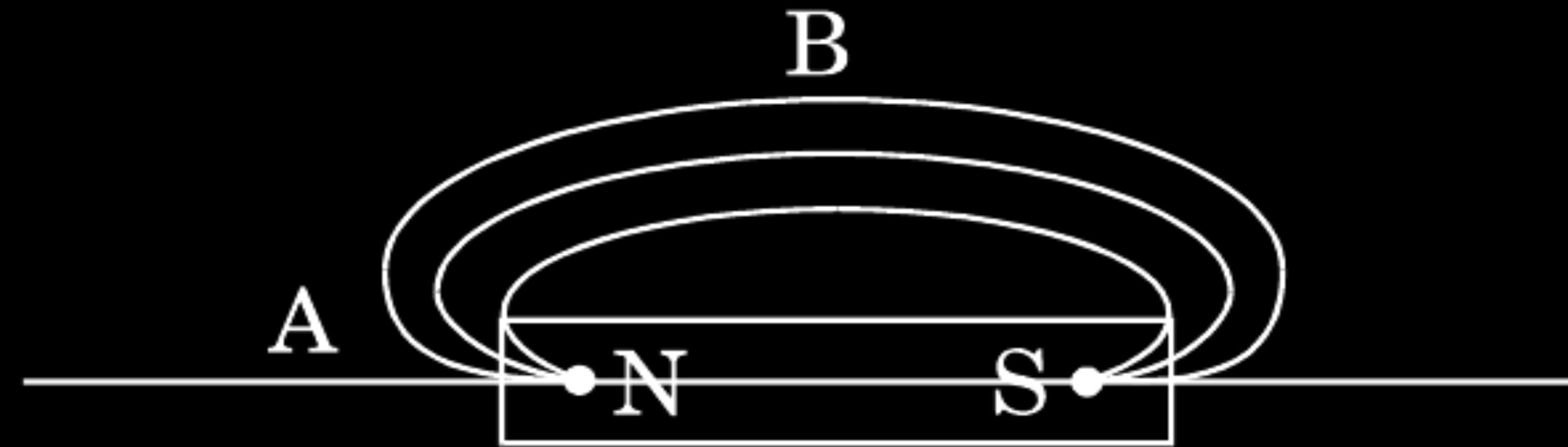
- Strength decreases

1

- Reason - the concentric circles representing the magnetic field around a current-carrying straight wire become larger and longer as the distance increases.

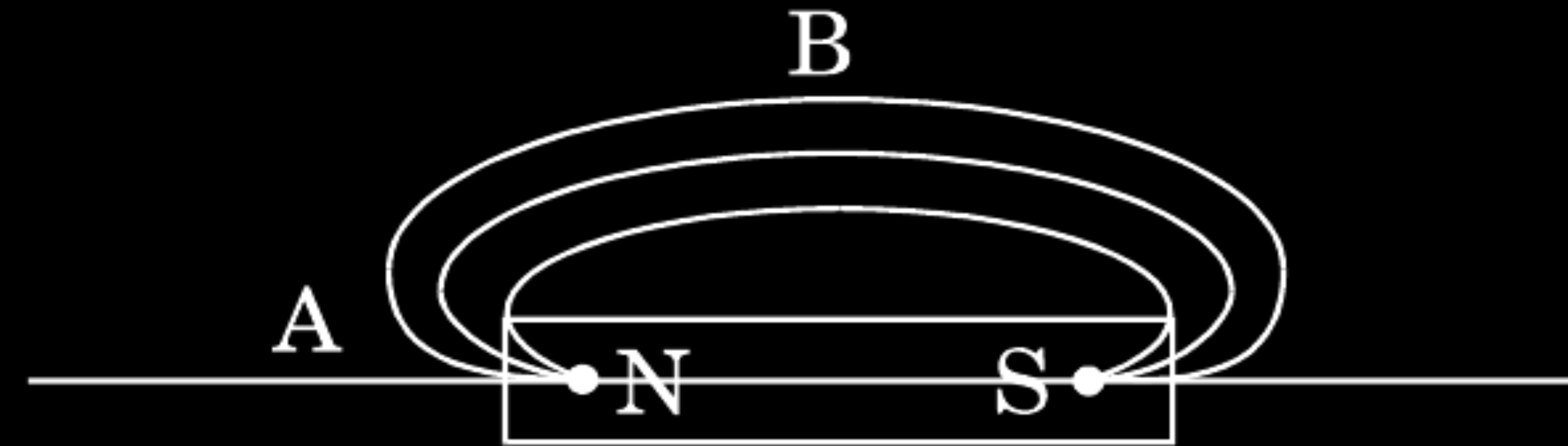
1

Magnetic field lines are shown in the given diagram. A student makes a statement that the magnetic field at 'A' is stronger than at 'B'. Justify this statement. Also redraw the diagram and mark the direction of magnetic field lines.



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Magnetic field lines are shown in the given diagram. A student makes a statement that the magnetic field at 'A' is stronger than at 'B'. Justify this statement. Also redraw the diagram and mark the direction of magnetic field lines.



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- | | |
|---|---|
| • Closeness (crowding) of magnetic field lines is directly related to the strength of the magnetic field. | 1 |
| • Strength of magnetic field at point 'A' (Pole) is more than at point 'B'. | 1 |
| • If the student redraws the diagram and mark the arrows correctly (N to S). | 1 |